Arithmetic Quantum Mechanics Lab Book

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Contents

1 Frontmatter, Status, and Scope

This lab book tracks a research programme connecting arithmetic zeta functions, arithmetic quantum mechanics, supersymmetric quantum mechanics, and topological lattice models such as Kitaev and Levin-Wen systems.

No mathematical or physical claim in this lab book is treated as established until it is marked as *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact.

1.1 Claim Status

The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record. They separate programme-level analogies from results.

1.2 Evidence Chain

Every substantive assertion should be traceable through:

1. a local source manifest under `references/`,
2. a convention entry in `CONVENTIONS.md`,
3. a derivation, script, test, or run bundle when computation is involved,
4. an index row in `INDEX.md`, and
5. a report shard in this lab book.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results.

2.1 Arithmetic And Zeta-Function Side

- Which zeta functions are in scope: local zeta functions over finite fields, Hasse-Weil zeta functions, motivic zeta functions, or other variants?
- Which Frobenius convention and cohomology theory will be used for trace formulas?
- Which finite examples are small enough for exact Sage or GAP checks?

2.2 Quantum-Mechanical Side

- What is meant locally by arithmetic quantum mechanics: a literal Hilbert-space model, a spectral dictionary, or a finite-dimensional approximation?
- Which supersymmetric-QM convention fixes the grading, supercharge, adjoint, Hamiltonian, and Witten-index normalization?
- Which invariants can be checked before proposing an analogy: nilpotence, positivity, trace identities, index stability, or spectral reconstruction?

2.3 Topological-Code Side

- Which lattice model is being used: toric code, quantum double, or Levin-Wen string-net?
- Which boundary conditions, lattice orientation, and star/plaquette conventions are fixed?
- Which category data are required: fusion rules, F-symbols, R-symbols, pivotal or spherical structure, and Drinfeld-center data?

2.4 First Decisions

The first technical session should not start with formulas. It should register sources and fix the minimum conventions needed to compare one finite arithmetic zeta example with one finite quantum or topological model invariant.

3 Lab Log

2026-05-24

Initial lab-book scaffold created.

- Established `report.tex` as the sharded LaTeX master.
- Added report source maps, convention and index files, source-manifest placeholders, run-bundle rules, and a tool runner.
- Recorded that no sources have been acquired and no scientific claims have been checked yet.

2026-05-24 Steane Clifford Ghost Gaussians

The report now includes sourced toric/CSS/Steane stabilizer-supercharge derivations. The latest addition is Section 14: arbitrary Clifford covariance is stated for signed Pauli stabilizer presentations, and presentation changes are lifted by number-preserving fermionic Gaussian operations on the ghosts. The accompanying Steane run checks all standard H/P/CNOT Clifford generators and the elementary GL6 ghost moves.

2026-05-24 Arithmetic Quantum Fields

Section 15 adds the first explicit arithmetic-quantum-field proposal. A finite function space $E \leq \mathrm{Map}(X, V)$ receives a pointwise symplectic form; after quotienting its radical, its Weyl-Heisenberg displacements define the field observable algebra. The run bundle checks finite-set, vector-space, and finite-variety examples over F_3.

2026-05-24 Projective Sheaf Fields

Sections 16 and 17 add the first projective version; Section 18 works out the rational-point stalks; Section 19 records the working dictionary "stalk equals local field germ, fiber equals sampled field value." Local Stacks Project TeX sources now ground the definitions of presheaf, sheaf, stalk, projective space, varieties, twists, and $H^0(\mathbf{P}^n, \mathcal{O}(d))$. The run bundle checks $\mathbf{P}^1_{\mathbb{F}_3}$ with $H^0(\mathcal{O}(d)) \otimes \mathbb{F}_3^2$ for $d = 0, \ldots, 4$, including the $d = 1$ pairing radical, the $d = 4$ evaluation kernel, the local germs whose residues give the evaluation rows, and the full $d = 0, \ldots, 4$ scalar residue table.

2026-05-25 Canonical Boson/Fermion Field Test

Section 20 adds a sourced finite-mode comparison with standard canonical fields. The bosonic test uses $\mathrm{Map}(X, \mathbb{R}_q \oplus \mathbb{R}_p)$ with the summed symplectic form and Stone–von Neumann uniqueness. The fermionic test uses CAR data on $\mathrm{Re}(Z \oplus \overline{Z})$ and the antisymmetric Fock space. The naive proposal that Grassmann numbers are a drop-in target replacement is rejected in this strict Weyl-projective sense; Grassmann variables remain a sourced GAR/fermionic phase-space calculus for later work.

Sections 21 and 22 correct the Grassmann displacement question using Cahill–Glauber's Physical Review A source converted with `marker`. The abstract displacement symbols are represented by $D(\gamma) = \exp(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i))$ inside the Grassmann-extended finite CAR algebra. The derivation shard proves unitarity, the operator shifts $a_i \mapsto a_i + \gamma_i$, and the Grassmann-valued ray composition law, while retaining only the ordinary complex-Hilbert-space caveat as a boundary statement.

Section 23 returns to arithmetic quantum fields in the prime-field case. It separates $\mathrm{Spec}\, \mathbb{F}_p$, which has one point and gives one qudit, from the underlying set $\mathbb{F}_p$, which has p points. For

$X = \mathbb{F}_p$ as a set and $E = \mathrm{Map}(X, \mathbb{F}_p^2)$, the shard proves that the arithmetic Weyl–Heisenberg group is exactly the standard odd-p qudit Pauli group, with the same projective labels and the $p = 2$ fourth-root phase caveat for qubits.

Section 24 then computes the first mixed finite affine spectrum, $R = \mathbb{Z}/6\mathbb{Z}$. The prime ideals are (2) and (3), with residue fields $\mathbb{F}_2$ and $\mathbb{F}_3$. Thus the local factors are a qubit and a qutrit. The label group is their orthogonal direct sum, but the Weyl representation lives on $\mathbb{C}^2 \otimes \mathbb{C}^3$, so the observable algebra is M_6, not $M_2 \oplus M_3$.

Section 25 adds the finite Zariski-open observable-net convention. The open-local assignment $U \mapsto \mathcal{A}(U)$ is proved covariant under inclusions of opens, while the closed-complement assignment $U \mapsto \mathcal{A}(X \setminus U)$ is recorded as a useful but contravariant vanishing-locus convention. The field edge case has only $\emptyset$ and X; for $\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$, the singleton opens give the commuting local subalgebras $M_2 \otimes \mathbf{1}_3$ and $\mathbf{1}_2 \otimes M_3$, which generate M_6.

Section 26 records the directed-system choice needed for a quasi-local algebra. The primary construction is the filtered colimit over quasi-compact Zariski opens, $\varinjlim_U \mathcal{A}(U)$, with inclusions following open inclusion. The closed-complement construction is retained only after reindexing by closed supports ordered by inclusion, as a separate vanishing-locus or defect algebra.

Section 27 begins the presheaf/cosheaf layer. It defines presheaves, sheaves, precosheaves, and cosheaves from local Stacks and Curry sources. The open-local observable assignment is proved to be a precosheaf/local net, but not an ordinary cosheaf of unital $*$-algebras: for the $\mathbb{Z}/6\mathbb{Z}$ two-point cover, the cosheaf coproduct would be $M_2 *_{\mathbb{C}} M_3$, while the physical observable algebra is $M_2 \otimes M_3$. The Weyl label assignment $E(U) = \bigoplus_{x \in U} E_x$ is identified as the cosheaf-like layer.

Section 28 adds the state side. The local state assignment $U \mapsto \mathrm{State}(\mathcal{A}(U))$ is a presheaf of convex sets by restriction of functionals, equivalently by partial trace in finite tensor products. It is not generally a sheaf: the $\mathbb{Z}/6\mathbb{Z}$ two-point system gives two different global density matrices with the same local marginals, and a three-qubit overlap example shows that compatible marginals need not admit any global state. For the algebraic quasi-local algebra, global states are exactly compatible inverse limit families of local states.

Sections 29, 30, and 31 extend the arithmetic-field programme from finite spectra to affine schemes of finite type over $\mathbb{F}_q$. The new convention indexes finite Weyl/qudit sites by closed points, whose residue fields are finite by the Nullstellensatz. For $\mathrm{Spec}\,\mathbb{F}_q[t]$, closed points are monic irreducible polynomials and the point x_π carries a $q^{\deg \pi}$-level qudit. For $\mathrm{Spec}\,\mathbb{F}_q[x, y]$, rational points give a finite q^2-site truncation, but the full quasi-local algebra ranges over all closed points of all degrees. Generic and curve-generic points remain geometric/topological points rather than finite Weyl sites, and regular polynomial profiles are recorded as formal infinite-support fields rather than quasi-local operators.

Sections 32, 33, and 34 add the rational-function Weyl-Heisenberg layer. A new local source manifest registers LCA, locally compact ring, finite Frobenius-ring, finite Clifford/Weil, local-field, and adelic sources. The report now separates the algebraic $K = k(t)$ Heisenberg group, which has an explicit irreducible unitary Schrödinger representation on $\ell^2(K)$, from the full LCA Stone-von Neumann setting $K_{\mathrm{disc}} \times \widehat{K_{\mathrm{disc}}}$ and from local or adelic completions. The $k = \mathbb{F}_3$ examples compute concrete ω-commutation relations for rational-function labels.

Sections 35 and 36 return to concrete finite affine examples. For $R_p = \mathbb{F}_3[t]/(p)$, the report computes the spectrum from the irreducible factorization of p, proves the Chinese-remainder and reduced-quotient descriptions, and gives the complete residue-field Weyl-Heisenberg representation on $\bigotimes_\pi \ell^2(\mathbb{F}_3[t]/(\pi))$. The $t^n = 1$ shard records the characteristic-3 formula $t^n - 1 = (t^m - 1)^{3^{v_3(n)}}$, the cyclotomic degree count via $\mathrm{ord}_d(3)$, and the resulting dimension $\dim \mathcal{H} = 3^m$, with nilpotent multiplicities retained as coordinate-ring data but invisible to the present residue-qudit labels.

Sections 37 and 38 add the nilpotent-sensitive layer. For a local factor $A = \mathbb{F}_3[t]/(\pi^e)$, the report defines the top-coefficient character, proves it is generating, and constructs the projective unitary Weyl representation on $\ell^2(A)$. The Weyl operators span the full matrix algebra, so the representation is irreducible. The $\mathfrak{m}$-adic filtration is interpreted as a jet filtration: position jets of order r pair with momentum jets of order $e-1-r$. For $t^n = 1$, this thickened layer has dimension 3^n, while the reduced residue layer has dimension 3^m for $m = n/3^{v_3(n)}$.

Sections 39 and 40 add the stabilizer local-to-global layer. A phased Lagrangian label subspace is proved to give a rank-one full stabilizer state, while vanishing stabilizer descent defect is the separate condition that the global stabilizer labels are generated from a chosen cover. The two-qutrit example over $\mathbb{F}_3$, generated by $(1, 0, -1, 0)$ and $(0, 1, 0, 1)$, is Lagrangian but has zero intersection with both one-site label spaces, so its descent defect has dimension 2. Thus the broad equivalence between choosing a full stabilizer and satisfying local-to-global gluing is rejected; the corrected statement requires both a global Lagrangian and zero descent defect, plus phase compatibility on overlaps.

Section 41 treats the missing finite product-field example. For $R = k^n$, with $k = \mathbb{F}_q$, the spectrum has n points $\mathfrak{m}_i = \ker(\mathrm{pr}_i)$, each with residue field k, and the idempotents e_i make the topology discrete. Hence the residue-qudit field on $X = \operatorname{Spec} R$ has

$$E(X) = k_X^n \oplus k_Z^n, \qquad \mathcal{H}(X) = \ell^2(k)^{\otimes n}, \qquad \mathcal{A}(X) \cong M_{q^n}(\mathbb{C}).$$

Thus $X = \operatorname{Spec}(k^n)$ gives exactly the ambient n q-level qudit Weyl/Pauli net. Stabilizer codes remain additional data: isotropic additive label subgroups and compatible phases, with the k-linear stabilizers as a subclass when q is not prime.

Section 42 starts the Cech layer. A new Stacks Cohomology source is registered for the standard Cech complex, H^0 gluing, refinements, and ordered Cech cochains, with Curry supplying the dual cosheaf homology convention. The report proves that the full residue-label cosheaf $E(U) = \bigoplus_{x \in U} E_x$ has $\check{H}_0(\mathcal{U}, E) \cong E(U)$ and no higher Cech homology on finite covers, by decomposing pointwise into full simplices. The quantum counterparts are then separated by layer: stabilizer descent is a genuine abelian cokernel of zeroth Cech generation, state descent is the marginal comparison map, and observable descent is the comparison from ordinary algebraic cover colimits to the physical local net.

Section 43 starts the dynamics layer. Gross's arXiv TeX source for finite-dimensional Hudson theory is registered locally. The report fixes the odd-characteristic half-Weyl normalization, then proves that every symplectic morphism of nondegenerate finite label spaces induces a unital $*$-monomorphism of Weyl observable algebras, and that symplectic automorphisms induce Gaussian/quasi-free automorphisms. Gross's Clifford theorem supplies projective unitary implementations, while Gross-Hudson identifies pure finite Gaussian states with stabilizer states.

Section 44 starts the finite-field scale or continuum-limit layer. For $X_0/\mathbb{F}_q$, the rank-r scale is $X_r = X_0 \times_{\mathbb{F}_q} \mathbb{F}_{q^r}$. The report proves that base change of schemes is canonical but naive diagonal Weyl-label embeddings multiply absolute-trace commutators by s/r. Trace-normalized embeddings repair this and are canonical on the prime-to-p subtower. For $\operatorname{Spec} \mathbb{F}_3$, the current canonical continuum algebra is the prime-to-3 filtered colimit of $M_{3^r}(\mathbb{C})$, and Frobenius acts by $W(q, p) \mapsto W(q^3, p^3)$.

Section 45 proves that the remaining full-tower trace choices are coherent gauge data rather than an obstruction. The key object is a trace-density system $a_r \in \mathbb{F}_{q^r}$ satisfying $\mathrm{Tr}_{\mathbb{F}_{q^s}/\mathbb{F}_{q^r}}(a_s) = a_r$ for every $r \mid s$. Such systems exist by a factorial-tower trace-surjectivity construction, and they define coherent normalizers $c_{s,r} = a_s/a_r$. Frobenius acts on this space of gauges; for $p \mid s/r$, no trace-one scalar fixed by the relative Frobenius group exists.

Section 46 builds the first arithmetic base-space field over $\operatorname{Spec} \mathbb{Z}$. The closed-prime layer has one ℓ-level qudit at each rational prime and an algebraic infinite tensor product over finite prime supports. The generic point is separated as a $\mathbb{Q}$-Weyl sector rather than a finite qudit. The report rejects $z \mapsto z+1$ as a scheme automorphism of $\operatorname{Spec}\mathbb{Z}$, then defines the internal residue-translation action $n \mapsto \alpha_n$ on the prime-qudit algebra, proves its profinite local periods, and shows that α_n for $n \neq 0$ is not implemented in the $|0\rangle$ incomplete tensor sector.

2026-05-26 Affine Line and Regular Functions

Sections 47–49 prove the base-preserving symmetry group $x \mapsto ax+b$, build its closed-point Weyl-net action, and separate the full $\mathbb{F}_3$ net from rational finite-support truncations. Sections 50–53 then identify regular functions as residue profiles, finite equations h as support selectors, and $W_h^{\mathrm{red}}(F,G)$ as the actual finite Weyl smearing. The affine action is $g_\# F = F \circ g^{-1}$, it transports reduced smearings by $\alpha_g W_h^{\mathrm{red}}(F,G) = W_{g_\# h}^{\mathrm{red}}(g_\# F, g_\# G)$, and repeated factors remain Artin-ring thickness rather than extra closed points.

2026-05-26 Minimal Fermion Catalogue

Section 54 registers sources for supergeometry, spin structures, and logarithmic forms, then fixes the first fermion catalogue. Ordinary $\operatorname{Spec} k$ has no forced fermion mode; the minimal sourced entries are an odd superpoint, $\Pi\Omega^1$ of an even dual-number point, and a log-spin curve such as $(\mathbf{P}^1_{\mathbb{F}_3}, \{0,\infty\}, \mathcal{O})$.

4 Reproducibility Map

Generated artifacts live in run bundles:

```
runs/<YYYY-MM-DD>-<slug>/
  README.md
  data/
  figures/
```

Each producer must be listed in `scripts/run_all.jl` and `INDEX.md`. Each CSV must be documented in `data/SCHEMA.md`. Each report claim that uses generated data must point back to the producer and run bundle.

4.1 Current Validation Commands

```
make check-report-shards
make report
julia --project=. -e 'using Pkg; Pkg.test()'
julia --project=. scripts/run_all.jl --fast
```

The registered producers are indexed in `INDEX.md`; the canonical local driver is `scripts/run_all.jl`.

5 Source Queue and Open Questions

5.1 Source Queue

Registered sources now cover toric-code stabilizers, symplectic QECC conventions, finite Hudson/Gaussian stabilizer theory, canonical CCR and CAR field conventions, Fock spaces, Weyl-Heisenberg and Weil representations over LCA groups, locally compact rings, finite Frobenius rings, finite Heisenberg groups, and local/adelic fields, and the algebraic-geometry definitions used for spectra, Zariski topology, affine-scheme functoriality, sheaves, cosheaves, Cech cohomology, projective space, separable trace pairing, and absolute/relative Frobenius. Before adding technical content in the remaining directions, register sources for:

- Weil conjectures, zeta functions, Frobenius traces, and cohomological normalization.
- Arithmetic quantum mechanics and spectral interpretations of arithmetic data.
- Adelic self-dual models, Bost–Connes-style arithmetic C^*-dynamical systems, and any zeta-producing time evolution proposed for the $\operatorname{Spec}\mathbb{Z}$ prime-qudit algebra.
- Supersymmetric quantum mechanics, Witten index, and Hodge/de Rham model conventions.
- Levin-Wen string-net models and categorical input data beyond the toric stabilizer/CSS layer.

5.2 Open Questions

1. What is the smallest finite arithmetic example whose zeta data can be computed exactly and traced through the proposed dictionary?
2. Which quantum-mechanical invariant is the first honest comparator: trace, index, spectrum, ground-state degeneracy, or projector rank?
3. Which parts of the comparison are literal constructions and which parts are analogy or bookkeeping?
4. What is the correct generic rational-function sector for $\mathbf{A}^1_{\mathbb{F}_q}$: a discrete $k(x)$ Weyl system with a dual-label affine action, a local-field completion at infinity, or an adelic self-dual model?

6 Toric Code Supercharge From Local Checks

6.1 Question

Can one formulate the Kitaev toric code concretely as

$$H = QQ^* + Q^*Q$$

for a supercharge Q, and can the code space be realized by the elementary "Q-cohomology" construction?

The answer recorded here is:

> Yes, after adding one auxiliary fermion, or ghost, for each local stabilizer check. This gives a local-check supersymmetric package. It is not yet a purely physical-qubit supercharge.

The toric-code stabilizer convention is sourced from the local TeX file `references/toric_code/kitaev_quant_ph_9707021_source/anyons.tex`. Lines 189–222 define qubits on edges, star operators A_s, plaquette operators B_p, and the protected subspace; lines 328–337 define Kitaev's Hamiltonian and state that its ground state is the protected subspace.

6.2 Elementary Construction

Let V be the physical Hilbert space of the toric-code qubits. For every star or plaquette stabilizer S_i, define the violated-check projector

$$P_i = \frac{1 - S_i}{2}.$$

Thus $P_i v = 0$ means that the state v satisfies check i, while $P_i v = v$ means that this component violates check i. The shifted positive toric-code Hamiltonian is

$$H_{\mathrm{TC}} = \sum_i P_i.$$

Now add one auxiliary fermion for each check. Concretely, this means we allow formal labels e_i, one per check, and impose the usual rule that swapping two labels changes the sign. Let c_i^* be the operation "attach the label e_i", and let c_i be the operation "remove the label e_i", with zero result if the label is absent. These operations satisfy

$$c_i c_j^* + c_j^* c_i = \delta_{ij}, \qquad c_i c_j + c_j c_i = 0, \qquad c_i^* c_j^* + c_j^* c_i^* = 0.$$

Define the cochain supercharge

$$Q = \sum_i c_i^* P_i.$$

This operator sends a physical state to the formal list of its violated local checks. Because the projectors P_i commute, and because two fermion creations anticommute, $Q^2 = 0$. The adjoint $Q^* = \sum_i c_i P_i$ removes one ghost label. The cross-terms in $QQ^* + Q^*Q$ cancel in pairs, again using commutation of the projectors and anticommutation of the fermions, so

$$QQ^* + Q^*Q = \sum_i P_i = H_{\mathrm{TC}}$$

on the physical ghost-vacuum sector, and $H_{\mathrm{TC}} \otimes I$ on the full ghost-extended space.

6.3 What The Cohomology Means

Here is the promised elementary definition. Degree zero means "no ghost label attached", so a degree-zero state is just a physical toric-code state $v \in V$. The degree-zero Q-cohomology is:

- closed degree-zero states: those v with $Qv = 0$;
- exact degree-zero states: those of the form Qw where w has degree -1.

There are no degree -1 states in this construction, so no nonzero exact degree-zero states. Also, $Qv = \sum_i e_i \otimes P_i v$, and the one-ghost labels e_i are independent. Therefore

$$Qv = 0 \quad \iff \quad P_i v = 0 \text{ for every check } i \quad \iff \quad S_i v = v \text{ for every check } i.$$

Thus the degree-zero Q-cohomology is exactly the toric-code code space. The qualifier "degree-zero" is essential: on a code state all P_i vanish, so the full ghost-extended cohomology also contains ghost-labelled copies of the code space in higher degrees. The physical code is the no-ghost part.

Equivalently, using the boundary operator $Q^* = \sum_i c_i P_i$, degree-zero homology is the quotient of all physical states by the span of states with at least one local-check violation. Since the P_i are commuting orthogonal projectors, this quotient has the protected subspace as its canonical representative.

6.4 Algebraic Validation

The validation script is `scripts/lattice_codes/toric_supercharge_validation.jl`. It checks the construction algebraically, not by building the full Hilbert matrices. For the $k = 4$ torus it verifies:

- $n = 2k^2 = 32$ physical qubits and 32 local checks;
- each check has weight 4;
- the binary symplectic commutator of every pair of checks is zero;
- the stabilizer rank is $2k^2 - 2 = 30$;
- the code dimension is $2^{32-30} = 4$;
- the CAR/projector algebra certifies $Q^2 = 0$ and $QQ^* + Q^*Q = H_{\mathrm{TC}}$;
- the degree-zero Q-cohomology dimension is 4.

The generated CSV lives in the run bundle `runs/2026-05-24-toric-supercharge/` as `data/toric_supercharge_summary.csv`. Its first row is a sentinel noting that no full physical or ghost Hilbert matrices are built.

6.5 Boundary Maps and the Same Construction

There is a second, more topological description of the toric code. It starts from the cellular chain complex over $\mathbb{F}_2$

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0,$$

where C_2 is spanned by faces, C_1 by edges, and C_0 by vertices. The local Bombin–Martin-Delgado TeX source listed in `references/toric_code/SOURCES.md` defines these maps for 2-complexes in lines 2050–2140 and records $\partial^2 = 0$. The local QECC Zoo source in the same manifest records the toric-code family as a CSS code with star and plaquette stabilizers in lines 20–39.

The cellular description does suggest a different supercharge first. Define the cochain maps

$$\delta^0 = \partial_1^T : C^0 \to C^1, \qquad \delta^1 = \partial_2^T : C^1 \to C^2.$$

On the finite graded vector space $C^0 \oplus C^1 \oplus C^2$, the cellular supercharge

$$Q_{\text{cell}} = \delta^0 + \delta^1$$

is nilpotent because

$$Q_{\text{cell}}^2 = \delta^1 \delta^0 = (\partial_1 \partial_2)^T = 0.$$

Its middle cohomology means the following elementary quotient: take all edge bit strings $a \in C^1$ with $\delta^1 a = 0$, and identify two such strings when they differ by $\delta^0 b$ for some vertex bit string $b \in C^0$. In symbols,

$$H^1_{\text{cell}} = \ker(\delta^1)/\operatorname{im}(\delta^0).$$

The quantum CSS unification is then:

$$H_X = \partial_1, \qquad H_Z = \partial_2^T.$$

The rows of H_X are the vertex-star supports for the X-checks. The rows of H_Z, equivalently the columns of ∂_2, are the face-boundary supports for the Z-checks. The CSS commutation condition is not extra:

$$H_X H_Z^T = \partial_1 \partial_2 = 0.$$

The precise chain-to-ghost statement is proved in Section 7. It gives an explicit map from cellular cohomology classes to toric-code states and derives the ghost anticommutator from the CAR relations and the commuting stabilizer projectors. The operator-level comparison of Q_{cell} and Q_{gh} is in Section 8.

Concretely, the boundary map $\partial_2 : C_2 \to C_1$ and $\partial_1 : C_1 \to C_0$ act on classical $\mathbb{F}_2$-chains: formal sums of faces, edges, and vertices. They classify error strings, check supports, and logical operators. The boundary data determine the stabilizer checks S_i, and those checks determine the projectors $P_i = (1 - S_i)/2$ used in Q.

The chain complex also explains the number of encoded qubits. The first homology group is

$$H_1 = \ker(\partial_1)/\operatorname{im}(\partial_2).$$

For the torus this has dimension 2 over $\mathbb{F}_2$, so the code space has dimension $2^2 = 4$. In the Z-basis one sees the dual statement: the plaquette constraints select 1-cocycles, and the star operators identify cocycles differing by a coboundary. Hence the code basis is naturally labelled by H^1, while the logical string operators are labelled by the paired H_1/H^1 data.

The validation script is `scripts/lattice_codes/toric_chain_ghost_unification.jl`. For $k = 4$, it verifies $\partial_1 \partial_2 = 0$, checks that rows of ∂_1 match the star supports and columns of ∂_2 match the plaquette supports, and computes

$$\operatorname{rank} \partial_1 = 15, \quad \operatorname{rank} \partial_2 = 15, \quad \dim H_1 = 32 - 15 - 15 = 2.$$

It also records that $Q_{\text{cell}}^2 = 0$, that the middle cellular cohomology has 2 binary generators, that the Z-basis cocycle space has $2^{17} = 131072$ elements, and that quotienting by the $2^{15} = 32768$ star-generated coboundaries leaves 4 classes. The run output is the CSV `toric_chain_ghost_unification.csv` in the `2026-05-24-toric-chain-ghost-unification` run bundle.

7 Proof of the Toric Chain-to-Ghost Lift

This section replaces the slogan "the chain complex supplies the supports" by an explicit finite-dimensional statement.

7.1 Data

Fix the $k \times k$ periodic square cellulation of the torus, $k \geq 2$. All chain groups in this subsection are vector spaces over $\mathbb{F}_2$:

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0.$$

The bases are faces, edges, and vertices. Thus

$$|C_0| = k^2, \qquad |C_1| = 2k^2, \qquad |C_2| = k^2.$$

Let

$$\delta^0 = \partial_1^T : C^0 \to C^1, \qquad \delta^1 = \partial_2^T : C^1 \to C^2.$$

The cellular supercharge is the degree $+1$ map

$$Q_{\text{cell}} = \delta^0 + \delta^1 \quad \text{on} \quad C^0 \oplus C^1 \oplus C^2.$$

Since $\partial_1 \partial_2 = 0$, one has

$$Q_{\text{cell}}^2 = \delta^1 \delta^0 = (\partial_1 \partial_2)^T = 0.$$

The degree-one cellular cohomology is the quotient

$$H^1_{\text{cell}} = \ker(\delta^1) / \operatorname{im}(\delta^0).$$

In elementary terms, its elements are edge-bit strings a whose parity around every plaquette is zero, with two such strings identified if they differ by the edge-bit string produced by applying star flips at some set of vertices.

7.2 From Boundary Matrices to Pauli Checks

The physical Hilbert space is

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes C_1}.$$

Its computational Z-basis is written $\{|a\rangle : a \in C^1\}$, where $a_e \in \mathbb{F}_2$ is the bit on edge e.

For an edge-bit string $r \in C^1$, define Pauli monomials by

$$X^r |a\rangle = |a + r\rangle, \qquad Z^r |a\rangle = (-1)^{r \cdot a} |a\rangle,$$

where $r \cdot a = \sum_e r_e a_e$ is computed in $\mathbb{F}_2$. They obey

$$X^r Z^s = (-1)^{r \cdot s} Z^s X^r.$$

For a vertex v, let $r_v = \delta^0 v \in C^1$, the row of ∂_1 supported on edges incident to v. Define the star check

$$A_v = X^{r_v}.$$

For a face f, let $s_f = \partial_2 f \in C_1$, identified with its edge-bit string in C^1. Define the plaquette check

$$B_f = Z^{s_f}.$$

Then

$$A_v B_f = (-1)^{r_v \cdot s_f} B_f A_v.$$

But $r_v \cdot s_f$ is exactly the coefficient of v in $\partial_1 \partial_2 f$, hence it is zero. Therefore every A_v commutes with every B_f. Checks of the same Pauli type commute automatically, and each check squares to the identity.

7.3 The Code Space as a Linearized Cohomology Set

The toric-code space is

$$\mathcal{C} = \{|\psi\rangle \in \mathcal{H} : A_v|\psi\rangle = |\psi\rangle,\ B_f|\psi\rangle = |\psi\rangle \text{ for all } v, f\}.$$

We now construct a basis of $\mathcal{C}$ from $\ker(\delta^1)/\operatorname{im}(\delta^0)$.

First, for a basis vector $|a\rangle$,

$$B_f|a\rangle = (-1)^{s_f \cdot a}|a\rangle.$$

The condition $B_f|a\rangle = |a\rangle$ for every f is therefore

$$s_f \cdot a = 0 \quad \text{for all } f,$$

which is precisely $\delta^1 a = 0$. Thus the simultaneous $+1$ eigenspace of all plaquettes is spanned by $|a\rangle$ with $a \in \ker(\delta^1)$.

Second, a star acts by

$$A_v|a\rangle = |a + \delta^0 v\rangle.$$

Thus the star group translates a by elements of $\operatorname{im}(\delta^0)$. Because $\delta^1\delta^0 = 0$, these translations preserve $\ker(\delta^1)$.

For each class $[a] \in H^1_{\text{cell}}$, define

$$|[a]\rangle = \frac{1}{\sqrt{|\operatorname{im}\delta^0|}} \sum_{u \in \operatorname{im}(\delta^0)} |a + u\rangle.$$

This vector is well-defined: replacing a by $a + u_0$ just reindexes the sum. It is fixed by every star A_v, because A_v translates the summation set by $\delta^0 v$. It is fixed by every plaquette B_f, because every $a + u$ in the sum lies in $\ker(\delta^1)$. Hence $|[a]\rangle \in \mathcal{C}$.

If $[a] \neq [b]$, the two sums have disjoint computational-basis supports, so $|[a]\rangle$ and $|[b]\rangle$ are orthogonal. Conversely, any vector in the plaquette $+1$ eigenspace is a linear combination of basis states with $a \in \ker(\delta^1)$, and imposing all star equations forces its coefficients to be constant on each coset of $\operatorname{im}(\delta^0)$. Therefore the vectors $|[a]\rangle$ form a basis of $\mathcal{C}$, and there is a concrete isomorphism

$$\mathbb{C}[H^1_{\text{cell}}] \cong \mathcal{C}, \qquad [a] \mapsto |[a]\rangle.$$

For the periodic square torus this gives four states. Indeed, suppose $\sum_v \alpha_v(\text{row } v \text{ of } \partial_1) = 0$. For each edge with endpoints v, w, the coefficient of that edge gives $\alpha_v + \alpha_w = 0$. The graph is connected, so all α_v are equal. Thus the only row relation is the sum of all rows, and $\operatorname{rank}\partial_1 = k^2 - 1$.

Similarly, suppose $\sum_f \beta_f \partial_2 f = 0$. For each edge separating adjacent faces f, g, the coefficient of that edge gives $\beta_f + \beta_g = 0$. The dual graph is connected, so all β_f are equal. Thus the only column relation is the sum of all face boundaries, and $\operatorname{rank}\partial_2 = k^2 - 1$. Hence

$$\dim_{\mathbb{F}_2} H^1_{\text{cell}} = 2k^2 - (k^2 - 1) - (k^2 - 1) = 2.$$

Thus $\dim_{\mathbb{C}} \mathcal{C} = |H^1_{\text{cell}}| = 2^2 = 4.$

7.4 The Ghost Supercharge Is the Koszul Lift of These Checks

Let the list of stabilizer checks be

$$\{S_i\}_{i\in I} = \{A_v : v \in C_0\} \cup \{B_f : f \in C_2\}.$$

The previous subsection proved that all S_i commute and that $S_i^2 = I$. Define the violated-check projectors

$$P_i = \frac{I - S_i}{2}.$$

They are self-adjoint idempotents, and they commute with each other.

Now attach one auxiliary fermion to each $i \in I$. Let ϵ_i create the i-ghost and ι_i annihilate it. These operations act on the exterior algebra on the ghost labels and satisfy

$$\iota_i\epsilon_j + \epsilon_j\iota_i = \delta_{ij}, \qquad \epsilon_i\epsilon_j + \epsilon_j\epsilon_i = 0, \qquad \iota_i\iota_j + \iota_j\iota_i = 0.$$

They commute with the physical projectors P_i. Define

$$Q_{\mathrm{gh}} = \sum_{i\in I} \epsilon_i P_i, \qquad Q^*_{\mathrm{gh}} = \sum_{i\in I} \iota_i P_i.$$

Nilpotence is immediate but worth writing out:

$$Q^2_{\mathrm{gh}} = \sum_{i,j} \epsilon_i\epsilon_j P_i P_j = \sum_{i<j} (\epsilon_i\epsilon_j + \epsilon_j\epsilon_i) P_i P_j = 0.$$

For the anticommutator,

$$\begin{aligned} Q_{\mathrm{gh}} Q^*_{\mathrm{gh}} + Q^*_{\mathrm{gh}} Q_{\mathrm{gh}} &= \sum_{i,j} (\epsilon_i\iota_j + \iota_j\epsilon_i) P_i P_j \\ &= \sum_i P_i^2 = \sum_i P_i. \end{aligned}$$

This is the shifted toric-code Hamiltonian on $\mathcal{H}$, tensored with the identity on the ghost exterior algebra.

Finally, degree-zero means "no ghost present". A degree-zero vector $|\psi\rangle \in \mathcal{H}$ is Q_{gh}-closed exactly when

$$Q_{\mathrm{gh}}|\psi\rangle = \sum_i \epsilon_i P_i |\psi\rangle = 0,$$

which, by independence of the one-ghost labels, is equivalent to $P_i|\psi\rangle = 0$ for every i. This is equivalent to $S_i|\psi\rangle = |\psi\rangle$ for every star and plaquette. There are no degree -1 states, so no degree-zero exact states. Therefore

$$H^0(Q_{\mathrm{gh}}) = \mathcal{C} \cong \mathbb{C}[H^1_{\mathrm{cell}}].$$

This is the concrete unification. Q_{cell} is the cellular differential on finite $\mathbb{F}_2$-cochains. Q_{gh} is a complex-linear operator on the physical Hilbert space tensored with ghosts. The first one supplies the quotient set H^1_{cell}; the second one has the toric-code Hamiltonian as its anticommutator and selects the complex vector space freely spanned by that quotient. The sharper operator-level relation is derived in Section 8.

8 Operator Relation Between Cellular and Ghost Supercharges

The honest operator-level answer is: Q_{cell} is not the restriction of Q_{gh} to a low-ghost-number subspace of the check-ghost Fock space. The precise relation is induced, not restrictive: the cellular maps δ^0, δ^1 define translations and syndrome characters on the physical edge-bit Hilbert space, and Q_{gh} is the Koszul differential for the corresponding commuting projector equations.

8.1 Why It Is Not a Low-Ghost Restriction

Let

$$E = C^1$$

be the $\mathbb{F}_2$-vector space of edge-bit strings, and let

$$\mathcal{H} = \mathbb{C}[E]$$

be the complex vector space with basis $\{|a\rangle : a \in E\}$. The check-ghost space used by Q_{gh} is

$$G_{\text{check}} = \mathbb{C}\{\eta_v : v \in C_0\} \oplus \mathbb{C}\{\theta_f : f \in C_2\}.$$

The full ghost-extended space is

$$\mathcal{H}_{\text{gh}} = \mathcal{H} \otimes \Lambda(G_{\text{check}}).$$

Its low ghost-number pieces are therefore

$$\mathcal{H} \otimes \Lambda^0 G_{\text{check}} = \mathcal{H},$$
$$\mathcal{H} \otimes \Lambda^1 G_{\text{check}} = \mathcal{H} \otimes \big(\mathbb{C}\{v\text{-ghosts}\} \oplus \mathbb{C}\{f\text{-ghosts}\}\big).$$

There is no edge one-particle sector here. But the cellular differential has the form

$$Q_{\text{cell}} : \; C^0 \to C^1 \to C^2.$$

Its middle degree is the edge space C^1. In contrast,

$$Q_{\text{gh}} : \; \mathcal{H} \otimes \Lambda^r G_{\text{check}} \longrightarrow \mathcal{H} \otimes \Lambda^{r+1} G_{\text{check}}$$

adds a vertex-check or face-check ghost and keeps the same physical Hilbert space. On ghost degree zero it maps

$$\mathcal{H} \longrightarrow \mathcal{H} \otimes (\mathbb{C}\{v\text{-ghosts}\} \oplus \mathbb{C}\{f\text{-ghosts}\}),$$

not $C^0 \to C^1$. On ghost degree one it maps to two-check ghosts, not to faces. Thus Q_{cell} is not literally Q_{gh} on any canonical low-ghost subspace of $\mathcal{H}_{\text{gh}}$.

If one wants a Fock-space operator whose one-particle restriction is Q_{cell}, one must build a different Fock space. Choose an oriented signed cellular boundary over $\mathbb{Z}$,

$$C_2^{\mathbb{Z}} \xrightarrow{\partial_2^{\mathbb{Z}}} C_1^{\mathbb{Z}} \xrightarrow{\partial_1^{\mathbb{Z}}} C_0^{\mathbb{Z}}, \qquad \partial_1^{\mathbb{Z}} \partial_2^{\mathbb{Z}} = 0.$$

Let $W = W_0 \oplus W_1 \oplus W_2$ be the complex vector space with basis the oriented cells, and let a_σ^*, a_σ be creation and annihilation operators for cell fermions. The second-quantized cellular differential is

$$\widehat{Q}_{\text{cell}} = \sum_{e,v} (\delta^0_{\mathbb{Z}})_{ev} a_e^* a_v + \sum_{f,e} (\delta^1_{\mathbb{Z}})_{fe} a_f^* a_e.$$

It preserves fermion number and raises cell degree. Its square is zero because $\delta^1_{\mathbb{Z}} \delta^0_{\mathbb{Z}} = 0$, and on the one-particle sector W it is exactly the signed cellular cochain differential. This is a cell-fermion Fock space, not the check-ghost Fock space used by Q_{gh}. Reducing the signed incidence matrices modulo 2 gives the CSS supports used below.

8.2 The Induced Operator Construction

The actual bridge starts with the mod-2 cellular maps

$$C^0 \xrightarrow{\delta^0} E \xrightarrow{\delta^1} C^2, \qquad \delta^1\delta^0 = 0.$$

The edge-bit space E is an abelian group under addition. For each $u \in E$, define the translation operator on $\mathcal{H} = \mathbb{C}[E]$ by

$$T_u|a\rangle = |a+u\rangle.$$

For each $s \in E$, define the character multiplication operator by

$$M_s|a\rangle = (-1)^{s\cdot a}|a\rangle.$$

These obey

$$T_u M_s = (-1)^{s\cdot u} M_s T_u.$$

Now feed the boundary maps into these two representations of E. For a vertex v, set

$$A_v = T_{\delta^0 v}.$$

For a face f, set $s_f = \partial_2 f \in E$ and

$$B_f = M_{s_f}.$$

Then

$$A_v B_f = (-1)^{s_f\cdot\delta^0 v} B_f A_v.$$

The exponent $s_f \cdot \delta^0 v$ is the f-coordinate of $\delta^1\delta^0 v$, equivalently the v-coefficient of $\partial_1\partial_2 f$. It is zero. Hence all star and plaquette checks commute.

The violated-check projectors are therefore

$$P_v^X = \frac{I - T_{\delta^0 v}}{2}, \qquad P_f^Z = \frac{I - M_{s_f}}{2}.$$

Their action on the computational basis is concrete:

$$P_v^X|a\rangle = \frac{|a\rangle - |a+\delta^0 v\rangle}{2},$$

and

$$P_f^Z|a\rangle = \frac{1-(-1)^{s_f\cdot a}}{2}|a\rangle = (\delta^1 a)_f|a\rangle,$$

where the bit $(\delta^1 a)_f \in \mathbb{F}_2$ is embedded as 0 or 1 in $\mathbb{C}$. Thus the plaquette part of Q_{gh} reads off the cellular coboundary $\delta^1 a$ as a face-ghost syndrome, while the star part is a finite-difference operator along the $\operatorname{im}\delta^0$ orbit.

8.3 The Ghost Differential

Let exterior multiplication by η_v and θ_f be denoted by the same symbols. Define

$$Q_X = \sum_v \eta_v P_v^X, \qquad Q_Z = \sum_f \theta_f P_f^Z, \qquad Q_{\text{gh}} = Q_X + Q_Z.$$

The projectors commute because the checks commute, and exterior multiplication by ghost labels anticommutes. Therefore

$$Q_{\text{gh}}^2 = \sum_{i,j} \gamma_i \gamma_j P_i P_j = \sum_{i<j} (\gamma_i \gamma_j + \gamma_j \gamma_i) P_i P_j = 0,$$

where γ_i ranges over all η_v, θ_f. With ghost annihilation operators γ_i^* satisfying $\gamma_i^* \gamma_j + \gamma_j \gamma_i = \delta_{ij}$, one also has

$$Q_{\text{gh}} Q_{\text{gh}}^* + Q_{\text{gh}}^* Q_{\text{gh}} = \sum_i P_i.$$

Thus Q_{gh} is the Koszul differential for the equations

$$T_{\delta^0 v} \psi = \psi, \qquad M_{s_f} \psi = \psi.$$

Those equations are exactly the stabilizer equations.

8.4 Cohomology-Level Relation

Let

$$\psi = \sum_{a \in E} \psi_a |a\rangle \in \mathcal{H}$$

be a no-ghost state. The plaquette equations are:

$$P_f^Z \psi = 0 \text{ for all } f \quad \Longleftrightarrow \quad \psi_a = 0 \text{ unless } \delta^1 a = 0.$$

Indeed, P_f^Z multiplies the coefficient of $|a\rangle$ by the bit $(\delta^1 a)_f$. The star equations are:

$$P_v^X \psi = 0 \text{ for all } v \quad \Longleftrightarrow \quad \psi_a = \psi_{a + \delta^0 v} \text{ for all } a, v.$$

Thus a no-ghost state is Q_{gh}-closed exactly when its coefficients are supported on $\ker \delta^1$ and are constant on cosets of $\operatorname{im} \delta^0$. Since there are no degree -1 states, there are no no-ghost exact states. Therefore

$$H^0(Q_{\text{gh}}) \cong \mathbb{C}[\ker \delta^1 / \operatorname{im} \delta^0] = \mathbb{C}[H^1(Q_{\text{cell}})].$$

This is the concrete relation. Q_{cell} does not act inside the check-ghost Fock space. Instead, its two arrows induce two kinds of operators on $\mathbb{C}[C^1]$: δ^0 gives translations, and δ^1 gives syndrome multiplications. The ghost supercharge is the Koszul operator for the projector constraints built from those induced operators. The equality is therefore a cohomology-level identification, not an operator restriction:

$$H^0(Q_{\text{gh}}) \simeq \mathbb{C}[H^1(Q_{\text{cell}})].$$

9 Chain Complexes as Symplectic CSS Stabilizers

This section answers the symplectic-geometry question. The conclusion is:

> The chain/cochain CSS construction is not an alternative to the symplectic description. It is a functorial way to manufacture an isotropic stabilizer subspace L inside the Pauli symplectic vector space. The usual homology and cohomology groups appear as the two halves of the symplectic reduction $L^{\perp}/L$.

The local source anchors are as follows. Gottesman's TeX source records CSS commutation as row orthogonality in `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 1227–1246, and the binary symplectic stabilizer form in lines 1293–1341. Ashikhmin–Knill's local TeX source `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313, gives the nonbinary shift/phase commutator; lines 351–407 relate nonbinary stabilizers to self-orthogonal classical data. The Bombin–Martin-Delgado local source `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 1206–1259, writes qudit Pauli operators with a symplectic commutator; lines 2050–2136 define the chain/cochain pairing; lines 2383–2452 state the homological-code construction as a symplectic code.

9.1 The Symplectic Stabilizer Side

First take prime qudit dimension p. Put one qudit on each of n sites. The Pauli label space is

$$V = \mathbb{F}_p^n \oplus \mathbb{F}_p^n.$$

We write an element as (x, z), where x is the vector of X-exponents and z is the vector of Z-exponents. With $\zeta = \exp(2\pi i/p)$, define

$$X^x|a\rangle = |a + x\rangle, \qquad Z^z|a\rangle = \zeta^{z\cdot a}|a\rangle,$$

where $a, x, z \in \mathbb{F}_p^n$. Then

$$X^x Z^z X^{x'} Z^{z'} = \zeta^{z\cdot x' - x\cdot z'} X^{x'} Z^{z'} X^x Z^z.$$

Thus the commutator is controlled by the symplectic form

$$\omega((x, z), (x', z')) = z \cdot x' - x \cdot z'.$$

A stabilizer specified by exponent labels $L \subset V$ is commuting exactly when

$$\omega(\ell, \ell') = 0 \quad \text{for all } \ell, \ell' \in L.$$

That is, L is isotropic. If $\dim L = r$, the joint eigenspace of the corresponding independent stabilizers has dimension

$$p^{n-r}.$$

The logical Pauli labels are not V/L, because Pauli operators must preserve the code space. They are

$$L^{\perp}/L, \qquad L^{\perp} = \{v \in V : \omega(v, \ell) = 0 \text{ for all } \ell \in L\}.$$

This is the finite symplectic reduction. Its dimension is $2(n - r)$.

9.2 The CSS Case Inside the Symplectic Module

A CSS stabilizer is specified by two subspaces

$$R_X, R_Z \subset \mathbb{F}_p^n.$$

The X-type labels are $(x, 0)$ with $x \in R_X$, and the Z-type labels are $(0, z)$ with $z \in R_Z$. Hence

$$L = (R_X \oplus 0) + (0 \oplus R_Z) \subset V.$$

The same-type commutators vanish automatically. The mixed commutator is

$$\omega((x, 0), (0, z)) = -x \cdot z.$$

Therefore

$$L \text{ isotropic} \quad \Longleftrightarrow \quad x \cdot z = 0 \text{ for all } x \in R_X,\ z \in R_Z.$$

If H_X and H_Z are matrices whose rows span R_X and R_Z, this condition is exactly

$$H_X H_Z^T = 0.$$

So the CSS row-orthogonality condition is literally the isotropy condition in the Pauli symplectic vector space.

9.3 Feed in a Chain Complex

Now let

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

be a finite chain complex over $\mathbb{F}_p$, so

$$\partial_1 \partial_2 = 0.$$

Put the physical qudits on the chosen basis of C_1, so $n = \dim C_1$. Let

$$\delta^0 = \partial_1^T : C^0 \to C^1, \qquad \delta^1 = \partial_2^T : C^1 \to C^2.$$

After identifying C^1 with coordinate row vectors on the edge basis, define

$$R_X = \operatorname{im} \delta^0, \qquad R_Z = \operatorname{im} \partial_2.$$

Thus the stabilizer subspace is

$$L = (\operatorname{im} \delta^0 \oplus 0) + (0 \oplus \operatorname{im} \partial_2) \subset C^1 \oplus C_1.$$

We now derive isotropy, not just state it. Take $\alpha \in C^0$ and $\beta \in C_2$. The corresponding X- and Z-type labels are $(\delta^0\alpha, 0)$ and $(0, \partial_2\beta)$. Their symplectic pairing is

$$\begin{aligned}
\omega((\delta^0\alpha, 0), (0, \partial_2\beta)) &= -\langle \delta^0\alpha, \partial_2\beta \rangle \\
&= -\langle \alpha, \partial_1\partial_2\beta \rangle \\
&= 0.
\end{aligned}$$

The first line is the Pauli symplectic form. The second line is the definition of the coboundary as the transpose of the boundary. The last line is exactly $\partial_1\partial_2 = 0$. This is the clean bridge: the chain-complex identity is the CSS isotropy identity.

9.4 The Exact Sequence Hidden in Symplectic Geometry

The stronger statement identifies the symplectic reduction. Let

$$B^1 = \operatorname{im}\delta^0, \quad Z^1 = \ker\delta^1, \quad B_1 = \operatorname{im}\partial_2, \quad Z_1 = \ker\partial_1.$$

Then $L = B^1 \oplus B_1$, using the CSS embedding into $C^1 \oplus C_1$.

Compute $L^\perp$. A general Pauli label is $(x, z) \in C^1 \oplus C_1$. It is orthogonal to every $(u, 0)$ with $u \in B^1$ iff

$$0 = \omega((x, z), (u, 0)) = z \cdot u \quad \text{for all } u \in \operatorname{im}\delta^0.$$

This says z annihilates $\operatorname{im}\delta^0$, equivalently

$$z \in (\operatorname{im}\delta^0)^\perp = \ker\partial_1 = Z_1.$$

It is orthogonal to every $(0, b)$ with $b \in B_1$ iff

$$0 = \omega((x, z), (0, b)) = -x \cdot b \quad \text{for all } b \in \operatorname{im}\partial_2.$$

This says

$$x \in (\operatorname{im}\partial_2)^\perp = \ker\delta^1 = Z^1.$$

Therefore

$$L^\perp = Z^1 \oplus Z_1.$$

Quotienting by $L = B^1 \oplus B_1$ gives the short exact sequence

$$0 \longrightarrow B^1 \oplus B_1 \longrightarrow Z^1 \oplus Z_1 \longrightarrow H^1 \oplus H_1 \longrightarrow 0,$$

and hence

$$L^\perp/L \cong H^1 \oplus H_1.$$

This is the precise connection. The chain/cochain short exact sequence is the CSS presentation of the symplectic reduction of an isotropic stabilizer subspace.

9.5 Lagrangians and Logical Bases

If L has dimension $n - k$, then $L^\perp/L$ has dimension $2k$. The code encodes k qudits. A stabilizer code itself is therefore represented by an isotropic subspace L, not usually a Lagrangian subspace of V. It becomes Lagrangian only when $k = 0$, i.e. for a stabilizer state.

The Lagrangians appear one level down, in the logical symplectic module

$$L^\perp/L \cong H^1 \oplus H_1.$$

For example,

$$\Lambda_Z = Z^1 \oplus B_1, \qquad \Lambda_X = B^1 \oplus Z_1$$

are isotropic subspaces of $L^\perp$ containing L. Their quotients are

$$\Lambda_Z/L \cong H^1, \qquad \Lambda_X/L \cong H_1.$$

These are complementary Lagrangian subspaces of the logical Pauli module, and the induced pairing is the natural evaluation pairing

$$H^1 \times H_1 \longrightarrow \mathbb{F}_p, \qquad ([x], [z]) \mapsto x(z).$$

Choosing a logical computational basis is choosing one of these Lagrangian polarizations.

9.6 Qudit Generalization

For prime p, the derivation above is literal. For prime powers $q = p^m$, replace $\mathbb{F}_p$ by $\mathbb{F}_q$ and use the additive character

$$\chi(t) = \exp(2\pi i \ \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t)/p).$$

The Pauli commutator is then controlled by the trace-symplectic form

$$\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}\big(z \cdot x' - x \cdot z'\big).$$

The CSS chain-complex condition over $\mathbb{F}_q$ is still

$$\partial_1\partial_2 = 0,$$

and the same calculation proves isotropy. Thus the chain/cochain construction does generalize cleanly to field-valued qudits.

The caveat is composite non-field dimension D. One can work over $\mathbb{Z}_D$, but then the clean vector-space statements about rank, dimension, and duality become module statements. Torsion can matter. This is why the field symplectic formulation is the robust one: the chain complex is safe when interpreted as producing the isotropic subspace L, while the logical information is recovered from the symplectic reduction $L^\perp/L$.

9.7 Exact Validation

The validation script is `symplectic_css_bridge_validation.jl` under `scripts/lattice_codes/`. It builds oriented boundary matrices for the $k = 4$ periodic square torus over $\mathbb{F}_p$ for $p = 2, 3, 5$. It verifies:

$$\partial_1\partial_2 = 0, \qquad L \subseteq L^\perp,$$

$$\operatorname{rank} \partial_1 = 15, \qquad \operatorname{rank} \partial_2 = 15, \qquad \dim H_1 = 32 - 15 - 15 = 2.$$

The corresponding stabilizer subspace has rank 30, so the symplectic count gives $32 - 30 = 2$ encoded qudits and Hilbert dimension p^2. The generated CSV is `symplectic_css_bridge_summary.csv` in the `2026-05-24-symplectic-css-bridge` run bundle.

10 Symplectic Dictionary for CSS Supercharges

This section relates the supercharge operators to the symplectic stabilizer geometry of Section 9. The result is not that the supercharge is itself a symplectic form. Rather, for any CSS stabilizer subspace L, the supercharge is the Koszul differential for the commuting projector equations defining the L-invariant subspace.

The source anchors are registered in the local symplectic-QECC and toric-code source manifests. The locators used here are Gottesman `Thesis.tex`, lines 1293–1341; Ashikhmin–Knill `n9.tex`, lines 249–375; and Bombin–Martin-Delgado `HomologicalErrorCorrection6.tex`, lines 1206–1259.

10.1 CSS Symplectic Data

Work first over a prime field $\mathbb{F}_p$. A CSS code on n qudits is given by two row spaces

$$R_X, R_Z \subset \mathbb{F}_p^n$$

with

$$x \cdot z = 0 \quad \text{for all } x \in R_X,\ z \in R_Z.$$

The Pauli label space is

$$V = \mathbb{F}_p^n \oplus \mathbb{F}_p^n, \qquad \omega((x,z),(x',z')) = z \cdot x' - x \cdot z'.$$

The CSS stabilizer label subspace is

$$L = (R_X \oplus 0) + (0 \oplus R_Z) \subset V.$$

The row-orthogonality condition is exactly $L \subseteq L^{\perp}$.

Let

$$G = (\ell_1, \dots, \ell_r)$$

be an ordered $\mathbb{F}_p$-basis of L. This basis is extra data. It is needed to write a small generator-based supercharge, but the code subspace depends only on L.

10.2 Projectors From Symplectic Labels

For $v = (x,z) \in V$, write $W(v) = X^x Z^z$. In the whole Pauli label space,

$$W(v)W(w) = \zeta^{\omega(v,w)} W(w)W(v), \qquad \zeta = e^{2\pi i/p}.$$

For the CSS subspace $L = (R_X \oplus 0) + (0 \oplus R_Z)$, the canonical lifts are stronger than commuting. If $\ell = (x,z)$ and $\ell' = (x',z')$ lie in L, then $z \cdot x' = x \cdot z' = 0$, because $R_X \perp R_Z$. Hence

$$W(\ell)W(\ell') = W(\ell + \ell'), \qquad W(\ell)^* = W(-\ell).$$

Thus W restricts to an honest representation of the additive group L.

For each generator ℓ_i, define the stabilizer-line averaging projector

$$\Pi_i = \frac{1}{p} \sum_{a \in \mathbb{F}_p} W(a\ell_i).$$

This is the projector onto states fixed by the one-parameter stabilizer subgroup $\{W(a\ell_i) : a \in \mathbb{F}_p\}$. To check this directly,

$$\begin{aligned}\Pi_i^2 &= \frac{1}{p^2}\sum_{a,b} W((a+b)\ell_i)\\ &= \frac{1}{p}\sum_{c} W(c\ell_i) = \Pi_i,\end{aligned}$$

since for each c there are p pairs (a,b) with $a+b=c$. Also $\Pi_i^* = \Pi_i$, because $W(a\ell_i)^* = W(-a\ell_i)$. Set

$$P_i = I - \Pi_i.$$

Then P_i is a self-adjoint projector onto the subspace violating the ℓ_i-stabilizer equation. The projectors P_i commute with each other because the line averages commute.

For qubits, $p=2$, this reduces to the previous formula:

$$\Pi_i = \frac{I + W(\ell_i)}{2}, \qquad P_i = \frac{I - W(\ell_i)}{2}.$$

10.3 The Generator-Based Supercharge

Attach one fermionic ghost ϵ_i to each chosen generator ℓ_i. Exterior multiplication by ϵ_i and contraction ι_i satisfy

$$\iota_i\epsilon_j + \epsilon_j\iota_i = \delta_{ij}, \qquad \epsilon_i\epsilon_j + \epsilon_j\epsilon_i = 0, \qquad \iota_i\iota_j + \iota_j\iota_i = 0.$$

They commute with all physical Pauli projectors. Define

$$Q_G = \sum_{i=1}^{r} \epsilon_i P_i, \qquad Q_G^* = \sum_{i=1}^{r} \iota_i P_i.$$

Now compute:

$$\begin{aligned}Q_G^2 &= \sum_{i,j} \epsilon_i\epsilon_j P_i P_j\\ &= \sum_{i<j} (\epsilon_i\epsilon_j + \epsilon_j\epsilon_i) P_i P_j = 0.\end{aligned}$$

The anticommutator is

$$\begin{aligned}Q_G Q_G^* + Q_G^* Q_G &= \sum_{i,j} (\epsilon_i\iota_j + \iota_j\epsilon_i) P_i P_j\\ &= \sum_i P_i^2 = \sum_i P_i.\end{aligned}$$

Thus the supersymmetric Hamiltonian is the stabilizer-violation penalty for the chosen generator basis.

Degree zero means no ghost. A physical vector ψ is degree-zero closed iff

$$Q_G\psi = \sum_i \epsilon_i P_i \psi = 0.$$

The one-ghost labels are independent, so this is equivalent to $P_i\psi = 0$ for every i, equivalently $\Pi_i\psi = \psi$ for every i. Since the ℓ_i form a basis of L, this means

$$W(\ell)\psi = \psi \quad \text{for every } \ell \in L.$$

There are no degree -1 states, so no degree-zero exact states. Therefore

$$H^0(Q_G) = \{\psi : W(\ell)\psi = \psi \text{ for all } \ell \in L\},$$

the CSS stabilizer code space. If $r = \dim L$, its dimension is p^{n-r}.

10.4 Logical Paulis Commute With the Supercharge

Let $v \in L^\perp$. Then $\omega(v, \ell_i) = 0$ for every i, so $W(v)$ commutes with every $W(a\ell_i)$, hence with every Π_i, P_i, and

$$Q_G = \sum_i \epsilon_i P_i.$$

Therefore $W(v)$ acts on $H^0(Q_G)$.

If $v \in L$, then $W(v)$ acts as the identity on $H^0(Q_G)$ in the trivial-character sector. Hence the action factors through

$$L^\perp / L.$$

This is the exact point where symplectic geometry enters the supercharge cohomology: the degree-zero cohomology carries the logical Pauli action of the symplectic reduction.

If $v \notin L^\perp$, then some ℓ_i has $\omega(\ell_i, v) \neq 0$. For $\psi \in H^0(Q_G)$,

$$W(a\ell_i)W(v)\psi = \zeta^{a\omega(\ell_i,v)} W(v)\psi.$$

Averaging over a gives $\Pi_i W(v)\psi = 0$. Thus

$$P_i W(v)\psi = W(v)\psi, \qquad Q_G W(v)\psi \text{ has a nonzero } \epsilon_i\text{-component}.$$

So Pauli labels outside $L^\perp$ are precisely those detected by the supercharge as nonzero stabilizer syndrome.

10.5 Basis Dependence and Basis-Free Variant

The operator Q_G depends on the chosen basis G of L. Changing generators changes the individual projectors P_i and therefore changes the positive Hamiltonian $\sum_i P_i$. The degree-zero cohomology does not change: it is always the common invariant subspace for the whole L.

There is also a basis-free, but redundant, version. Let $\mathbb{P}(L)$ be the set of one-dimensional subspaces of L. For a line $\lambda = \mathbb{F}_p \ell$, define

$$\Pi_\lambda = \frac{1}{p} \sum_{a \in \mathbb{F}_p} W(a\ell), \qquad P_\lambda = I - \Pi_\lambda.$$

This is independent of the nonzero representative ℓ. Attach one ghost ϵ_λ to each line and set

$$Q_L^{\mathrm{all}} = \sum_{\lambda \in \mathbb{P}(L)} \epsilon_\lambda P_\lambda.$$

The same CAR/projector proof gives $(Q_L^{\mathrm{all}})^2 = 0$. Its degree-zero cohomology is again the CSS code space. The price is redundancy: if $\dim L = r$, then

$$|\mathbb{P}(L)| = \frac{p^r - 1}{p - 1}.$$

The generator-based Q_G is smaller; the projective-line Q_L^{all} is intrinsic to L.

10.6 General CSS Dictionary

For a CSS code with check matrices H_X, H_Z over $\mathbb{F}_p$, the dictionary is summarized by the table below. In the cellular toric-code case, this means

$$Q_{\text{cell}} = \delta^0 + \delta^1 \quad \rightsquigarrow \quad R_X = \operatorname{im}\delta^0, \quad R_Z = \operatorname{im}\partial_2 \quad \rightsquigarrow \quad L \quad \rightsquigarrow \quad Q_G.$$

The cellular differential supplies the two CSS support spaces. The symplectic step packages them as L. The ghost/Koszul step turns a chosen basis of L into commuting projectors and the operator Q_G.

CSS/symplectic object	supercharge object
R_X, R_Z	X- and Z-type stabilizer equations
$L = (R_X \oplus 0) + (0 \oplus R_Z)$	common invariant equations
$L \subseteq L^\perp$	commuting projectors P_i
basis G of L	minimal ghost list
$P_i = I - \frac{1}{p}\sum_a W(a\ell_i)$	violated-check projector
$Q_G = \sum_i \epsilon_i P_i$	Koszul supercharge
$L^\perp$	Paulis commuting with Q_G
$L^\perp/L$	logical Paulis on $H^0(Q_G)$

Thus the supercharge is a homological-algebra package built on top of an isotropic subspace. Symplectic geometry supplies L, its normalizer $L^\perp$, and the logical quotient; the supercharge resolves the common stabilizer equations defining the code space.

10.7 Prime-Power Note

For $q = p^m$, replace $\mathbb{F}_p$ by $\mathbb{F}_q$, and use the additive character

$$\chi(t) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t)/p).$$

For a line $\mathbb{F}_q\ell$, the invariant projector is

$$\Pi_\ell = \frac{1}{q}\sum_{a\in\mathbb{F}_q} W(a\ell).$$

The same calculation proves $\Pi_\ell^2 = \Pi_\ell$, commuting projectors for isotropic L, $Q^2 = 0$, and

$$H^0(Q) \cong \text{the } L\text{-stabilizer code.}$$

For composite non-field qudit dimension, one must replace vector spaces by modules over $\mathbb{Z}_D$, and the clean count $p^{n-\dim L}$ need not be imported without checking torsion.

10.8 Exact Validation

The script `css_supercharge_symplectic_dictionary.jl` in `scripts/lattice_codes/` validates the dictionary without building Hilbert matrices. It checks the binary Steane CSS code and a qutrit CSS toy code. The CSV `css_supercharge_symplectic_dictionary.csv` records CSS isotropy, stabilizer rank, code dimension, logical symplectic dimension, ghost counts, and the Q^2 and anticommutator certificates in the `2026-05-24-css-supercharge-symplectic-dictionary` run bundle.

11 Steane Code Symplectic Data in Molecular Detail

This section fixes one concrete Steane-code presentation and writes all finite-vector-space objects explicitly. The local source anchor is Gottesman's `Thesis.tex`: lines 1249–1267 give the seven-qubit CSS stabilizer table, and lines 1269–1284 give the two displayed encoded codewords.

11.1 Binary Check Space

Work over $\mathbb{F}_2$ with qubits ordered $1, \ldots, 7$. The three binary rows read from Gottesman's X-check table are

$$\begin{aligned} g_1 &= 1111000, \\ g_2 &= 1100110, \\ g_3 &= 1010101. \end{aligned}$$

Equivalently,

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

Let

$$D = \mathrm{row}(H) = \langle g_1, g_2, g_3 \rangle \subset \mathbb{F}_2^7.$$

The eight elements of D are

$$\begin{gathered} 0000000,\ 1111000,\ 1100110,\ 1010101, \\ 0011110,\ 0101101,\ 0110011,\ 1001011. \end{gathered}$$

The row dot products are

$$HH^T = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_2.$$

Thus $D \subseteq D^\perp$.

Put

$$C = D^\perp = \{a \in \mathbb{F}_2^7 : Ha^T = 0\}.$$

Since $\operatorname{rank} H = 3$, $\dim C = 4$. Let

$$u = 0000111.$$

Then $u \in C$, but $u \notin D$. Therefore

$$C = D \sqcup (u + D).$$

The second coset is

$$\begin{gathered} 0000111,\ 1111111,\ 1100001,\ 1010010, \\ 0011001,\ 0101010,\ 0110100,\ 1001100. \end{gathered}$$

These are exactly the two cosets supporting Gottesman's displayed $|\overline{0}\rangle$ and $|\overline{1}\rangle$ states.

11.2 The Stabilizer Subspace

The Pauli label space is
$$V = \mathbb{F}_2^7 \oplus \mathbb{F}_2^7.$$
We write a label as (x, z), corresponding to $X^x Z^z$. The binary symplectic form is
$$\omega((x, z), (x', z')) = z \cdot x' + x \cdot z',$$
where the sign from the odd-prime formula disappears in characteristic 2.

The six stabilizer labels are
$$\begin{aligned} \ell_1 &= (g_1, 0), \quad \ell_2 = (g_2, 0), \quad \ell_3 = (g_3, 0), \\ \ell_4 &= (0, g_1), \quad \ell_5 = (0, g_2), \quad \ell_6 = (0, g_3). \end{aligned}$$
Their span is
$$L = (D \oplus 0) + (0 \oplus D) \subset V.$$
This is an isotropic subspace. Indeed, same-type pairings vanish automatically. For a mixed pair,
$$\omega((d, 0), (0, d')) = d \cdot d' = 0 \qquad (d, d' \in D),$$
because $D \subseteq D^\perp$. Therefore $L \subseteq L^\perp$.

The dimensions and sizes are
$$\dim L = 6, \qquad |L| = 2^6 = 64.$$

11.3 The Centralizer

Let $(x, z) \in V$. It is orthogonal to every $(d, 0) \in D \oplus 0$ exactly when
$$z \cdot d = 0 \quad \text{for all } d \in D,$$
that is, $z \in C$. It is orthogonal to every $(0, d) \in 0 \oplus D$ exactly when
$$x \cdot d = 0 \quad \text{for all } d \in D,$$
that is, $x \in C$. Hence
$$L^\perp = C \oplus C.$$
Consequently
$$\dim L^\perp = 8, \qquad |L^\perp| = 2^8 = 256.$$

11.4 The Logical Symplectic Quotient

The logical Pauli label space is the finite symplectic reduction
$$L^\perp / L = (C \oplus C)/(D \oplus D) \cong (C/D) \oplus (C/D).$$
Since $C = D \sqcup (u + D)$, the quotient C/D is one-dimensional. Thus
$$\dim(L^\perp / L) = 2.$$
Choose the representatives
$$\overline{X} = (u, 0), \qquad \overline{Z} = (0, u).$$

Their symplectic pairing is

$$\omega(\overline{X}, \overline{Z}) = u \cdot u = 1 + 1 + 1 = 1 \quad \text{in } \mathbb{F}_2.$$

Thus the corresponding Pauli operators anticommute. The four logical Pauli classes are represented by

$$(0,0), \quad (u,0), \quad (0,u), \quad (u,u).$$

This is the full symplectic anatomy of the code: D supplies the CSS check space, $L = D \oplus D$ is the isotropic stabilizer subspace, $L^{\perp} = C \oplus C$ is the Pauli centralizer, and $L^{\perp}/L$ is the one-qubit logical symplectic module.

12 Steane Code Fermions, Supercharge, and Cohomology

Keep the notation of Section 11. This section constructs the actual fermions and supercharge for the same six Gottesman-table stabilizer generators.

12.1 Physical Stabilizer Projectors

The physical Hilbert space is

$$\mathcal{H} = \mathbb{C}\{|a\rangle : a \in \mathbb{F}_2^7\}.$$

For $d \in \mathbb{F}_2^7$, define

$$X(d)|a\rangle = |a + d\rangle, \qquad Z(d)|a\rangle = (-1)^{d \cdot a}|a\rangle.$$

The six stabilizer operators are

$$\begin{aligned} S_1 &= X(g_1), \quad S_2 = X(g_2), \quad S_3 = X(g_3), \\ S_4 &= Z(g_1), \quad S_5 = Z(g_2), \quad S_6 = Z(g_3). \end{aligned}$$

The violated-check projectors are

$$P_i = \frac{I - S_i}{2}, \qquad \Pi_i = \frac{I + S_i}{2}.$$

Since the labels span the isotropic subspace L, all S_i, hence all P_i, commute.

12.2 The Two Codeword Cosets

The two normalized Steane basis states in this convention are

$$|\overline{0}\rangle = \frac{1}{\sqrt{8}} \sum_{d \in D} |d\rangle, \qquad |\overline{1}\rangle = \frac{1}{\sqrt{8}} \sum_{d \in D} |u + d\rangle.$$

For an X-type check $X(g_j)$, adding $g_j \in D$ permutes D and also permutes $u + D$. Hence

$$X(g_j)|\overline{0}\rangle = |\overline{0}\rangle, \qquad X(g_j)|\overline{1}\rangle = |\overline{1}\rangle.$$

For a Z-type check $Z(g_j)$, every vector in D is orthogonal to g_j, and u is also orthogonal to g_j. Therefore

$$Z(g_j)|d\rangle = |d\rangle, \qquad Z(g_j)|u + d\rangle = |u + d\rangle.$$

So both displayed states are fixed by all six stabilizers.

Conversely, let

$$|\psi\rangle = \sum_{a \in \mathbb{F}_2^7} \psi_a |a\rangle.$$

The equations $Z(g_j)|\psi\rangle = |\psi\rangle$ force $\psi_a = 0$ unless $a \in C = D^\perp$. The equations $X(g_j)|\psi\rangle = |\psi\rangle$ force $\psi_{a+g_j} = \psi_a$, so the amplitudes are constant on D-cosets inside C. There are exactly two such cosets, D and $u + D$. Thus the common stabilizer $+1$ space is

$$\mathcal{C} = \operatorname{span}\{|\overline{0}\rangle, |\overline{1}\rangle\}, \qquad \dim \mathcal{C} = 2.$$

12.3 Six Fermions

Let

$$E = \mathrm{span}_{\mathbb{C}}\{\epsilon_1, \dots, \epsilon_6\}.$$

The ghost Fock space is the exterior algebra ΛE. Exterior multiplication by ϵ_i and contraction ι_i obey

$$\iota_i\epsilon_j + \epsilon_j\iota_i = \delta_{ij}, \qquad \epsilon_i\epsilon_j + \epsilon_j\epsilon_i = 0, \qquad \iota_i\iota_j + \iota_j\iota_i = 0.$$

The ghost-extended Hilbert space is

$$\mathcal{K} = \mathcal{H} \otimes \Lambda E, \qquad \dim \mathcal{K} = 2^7\, 2^6 = 8192.$$

Define

$$Q = \sum_{i=1}^{6} \epsilon_i P_i, \qquad Q^* = \sum_{i=1}^{6} \iota_i P_i.$$

Written without abbreviations,

$$\begin{aligned} Q = \epsilon_1 \frac{I - X(g_1)}{2} + \epsilon_2 \frac{I - X(g_2)}{2} + \epsilon_3 \frac{I - X(g_3)}{2} \\ + \epsilon_4 \frac{I - Z(g_1)}{2} + \epsilon_5 \frac{I - Z(g_2)}{2} + \epsilon_6 \frac{I - Z(g_3)}{2}. \end{aligned}$$

Because the P_i commute and the ϵ_i anticommute,

$$Q^2 = \sum_{i<j} (\epsilon_i\epsilon_j + \epsilon_j\epsilon_i) P_i P_j = 0.$$

The supersymmetric Hamiltonian is

$$\begin{aligned} QQ^* + Q^*Q &= \sum_{i,j} (\epsilon_i\iota_j + \iota_j\epsilon_i) P_i P_j \\ &= \sum_i P_i. \end{aligned}$$

Thus it is the Steane stabilizer-violation Hamiltonian, shifted and scaled so that the code sector has energy 0.

12.4 Cohomology, Defined and Computed

Ghost degree is exterior degree. The degree-q subspace is

$$\mathcal{K}^q = \mathcal{H} \otimes \Lambda^q E.$$

The degree-q cohomology of Q is

$$H^q(Q) = \frac{\{\phi \in \mathcal{K}^q : Q\phi = 0\}}{\{Q\eta : \eta \in \mathcal{K}^{q-1}\}}.$$

For $q = 0$, the denominator is zero because $\mathcal{K}^{-1} = 0$.

Decompose the physical Hilbert space into stabilizer-syndrome sectors

$$\mathcal{H} = \bigoplus_{\lambda \in \mathbb{F}_2^6} \mathcal{H}_\lambda, \qquad S_i\psi = (-1)^{\lambda_i}\psi.$$

On $\mathcal{H}_\lambda$, the projector P_i acts as multiplication by λ_i. Therefore

$$Q|_{\mathcal{H}_\lambda\otimes\Lambda E} = \epsilon(\alpha_\lambda), \qquad \alpha_\lambda = \sum_i \lambda_i \epsilon_i,$$

where $\epsilon(\alpha)$ means exterior multiplication by α.

If $\lambda \neq 0$, choose a contraction h with $h(\alpha_\lambda) = 1$. Then

$$hQ + Qh = I$$

on $\mathcal{H}_\lambda \otimes \Lambda E$, so that syndrome block is acyclic: it contributes no cohomology. If $\lambda = 0$, then $Q = 0$, and $\mathcal{H}_0 = \mathcal{C}$ has dimension 2. Hence

$$H^q(Q) \cong \mathcal{C} \otimes \Lambda^q E, \qquad \dim H^q(Q) = 2\binom{6}{q}.$$

The dimensions are

q	0	1	2	3	4	5	6
$\dim H^q(Q)$	2	12	30	40	30	12	2.

The physical Steane code is the no-ghost part:

$$H^0(Q) = \mathcal{C} = \operatorname{span}\{|\bar{0}\rangle, |\bar{1}\rangle\}.$$

The full Q-cohomology has ghost-labelled copies of the same code space, with total dimension $2\cdot 2^6 = 128$. The ground space of $\{Q, Q^*\}$ has the same ghost degeneracy. The no-ghost sector is therefore the physical stabilizer code; the higher ghost-degree classes are auxiliary Koszul bookkeeping.

12.5 What the Higher Ghost Classes Mean

The word "bookkeeping" in the previous paragraph should be read narrowly. It means that higher ghost degree does not add new Steane logical qubits. It does not mean that the Koszul cohomology is meaningless.

The six equations defining the code are

$$P_1\psi = \cdots = P_6\psi = 0.$$

The Koszul differential remembers these as six separate equations. On a nonzero syndrome sector $\lambda \neq 0$, at least one equation is violated and the complex is contractible by the homotopy h above. On the zero-syndrome sector, all six functions P_i vanish. Therefore Q becomes the zero operator on

$$\mathcal{C} \otimes \Lambda E,$$

and every exterior ghost monomial survives.

Thus the higher cohomology is exactly

$$H^{>0}(Q) = \mathcal{C} \otimes \Lambda^{>0} E.$$

It records the exterior algebra on the six chosen stabilizer equations over the constraint surface. Equivalently, it is the finite-dimensional analogue of conormal or derived-intersection data for the equations $P_i = 0$. This data is useful if the question is about the Koszul resolution itself, or about how the constraint presentation changes when generators are added, removed, or made redundant.

It is not, however, an invariant of the Steane code alone. If we choose a different independent generating set for the same L, the ghost basis changes. If we add redundant stabilizer generators, the higher exterior algebra gets larger while the physical code space is unchanged. The intrinsic QECC content is therefore the no-ghost code $\mathcal{C} = H^0(Q)$ together with the logical symplectic quotient $L^\perp/L$. The full Q-cohomology is a presentation-dependent derived enhancement of the stabilizer equations.

12.6 Exact Validation

The producer `steane_supercharge_molecular.jl` writes three CSV files in the Steane molecular run bundle. The summary CSV records $\dim D = 3$, $\dim C = 4$, $\dim L = 6$, $\dim L^{\perp} = 8$, $|\mathcal{K}| = 8192$, $\dim H^0(Q) = 2$, and full cohomology dimension 128. The vector CSV records the two codeword cosets and the logical representatives $(u, 0)$, $(0, u)$. The cohomology CSV records the seven dimensions displayed above.

13 Steane Clifford Morphisms of the Koszul Supercharge

Here "symplectic morphism" means a Clifford operation: a unitary U whose conjugation action on Paulis induces a symplectic automorphism F_U of the binary label space. Gottesman's local source anchors are `Thesis.tex`: lines 2271–2305 for valid encoded Clifford operations preserving the stabilizer, lines 2307–2348 for the Hadamard, phase, and CNOT Pauli-label actions, and lines 2439–2445 for bitwise Hadamard and phase on the seven-qubit code.

13.1 General Clifford Lift

Let $G = (\ell_1, \ldots, \ell_6)$ be the Steane generator list from Section 11, and let

$$Q_G = \sum_i \epsilon_i P_{\ell_i}, \qquad P_{\ell_i} = \frac{I - W(\ell_i)}{2}.$$

If a Clifford U sends $W(\ell)$ to $W(F_U \ell)$, up to the stabilizer phase convention, then

$$U P_{\ell_i} U^\dagger = P_{F_U \ell_i}.$$

Let $G' = F_U(G)$, give G' ghosts ϵ'_i, and define

$$Q_{G'} = \sum_i \epsilon'_i P_{F_U \ell_i}.$$

The map

$$\Phi_U = U \otimes \Lambda T, \qquad T(\epsilon_i) = \epsilon'_i,$$

satisfies

$$\Phi_U Q_G = Q_{G'} \Phi_U.$$

This is a chain isomorphism, hence it induces isomorphisms

$$H^q(Q_G) \cong H^q(Q_{G'}).$$

If G' is just a permutation of G, and T is the same permutation of ghosts, this is an automorphism of the same written operator Q_G. If G' is a different independent basis of the same L, it is instead an isomorphism to a different presentation of the same stabilizer constraints.

13.2 Transversal Hadamard

For bitwise Hadamard $R^{\otimes 7}$,

$$F_H(x, z) = (z, x).$$

Therefore

$$(g_i, 0) \mapsto (0, g_i), \qquad (0, g_i) \mapsto (g_i, 0).$$

So

$$\ell_1 \leftrightarrow \ell_4, \quad \ell_2 \leftrightarrow \ell_5, \quad \ell_3 \leftrightarrow \ell_6.$$

With the ghost swap

$$\epsilon_1 \leftrightarrow \epsilon_4, \quad \epsilon_2 \leftrightarrow \epsilon_5, \quad \epsilon_3 \leftrightarrow \epsilon_6,$$

we get the exact chain automorphism

$$(R^{\otimes 7} \otimes \Lambda T_H) Q_G = Q_G (R^{\otimes 7} \otimes \Lambda T_H).$$

On the logical symplectic quotient this sends

$$\overline{X} = (u, 0) \mapsto (0, u) = \overline{Z}, \qquad \overline{Z} = (0, u) \mapsto (u, 0) = \overline{X}.$$

On the no-ghost code basis,

$$R^{\otimes 7}|\overline{0}\rangle = \frac{|\overline{0}\rangle + |\overline{1}\rangle}{\sqrt{2}}, \qquad R^{\otimes 7}|\overline{1}\rangle = \frac{|\overline{0}\rangle - |\overline{1}\rangle}{\sqrt{2}}.$$

Thus the induced map on $H^0(Q_G)$ is the encoded Hadamard, and on $H^q(Q_G) \cong \mathcal{C} \otimes \Lambda^q E$ it is encoded Hadamard tensored with the ghost swap.

13.3 Transversal Phase

For bitwise phase $P^{\otimes 7}$, ignoring Pauli phases in the binary label,

$$F_P(x, z) = (x, z + x).$$

Hence

$$(g_i, 0) \mapsto (g_i, g_i) = \ell_i + \ell_{i+3}, \qquad (0, g_i) \mapsto (0, g_i) = \ell_{i+3}.$$

These six images are an independent basis of the same L, but they are not the original generator list and not a permutation of it. Therefore

$$P^{\otimes 7} \otimes \Lambda T_P$$

is a chain isomorphism

$$(\mathcal{H} \otimes \Lambda E, Q_G) \cong (\mathcal{H} \otimes \Lambda E', Q_{F_P(G)}),$$

not literally an automorphism of the original written Q_G.

To compare $Q_{F_P(G)}$ back with Q_G, decompose into the same physical syndrome sectors. The zero-syndrome sector is the Steane code. Every nonzero syndrome sector is contractible, as in Section 12. Both complexes therefore deformation retract to their zero-syndrome cohomology model

$$\mathcal{C} \otimes \Lambda E$$

after choosing the corresponding ghost basis. This is the precise homotopy equivalence: it is presentation-dependent on ghosts, but identical on the physical code sector. On logical labels,

$$\overline{X} = (u, 0) \mapsto (u, u) = \overline{X} + \overline{Z}, \qquad \overline{Z} = (0, u) \mapsto (0, u) = \overline{Z}.$$

13.4 Conclusion

For Steane, Clifford symmetry gives more than an abstract equivalence relation. It gives a chain isomorphism from the Koszul complex of one stabilizer presentation to the Koszul complex of the transformed presentation. When the Clifford permutes the chosen generators, as transversal Hadamard does here, this is an exact automorphism of Q_G. When the Clifford changes the independent generator basis, as transversal phase does here, the complexes are homotopy equivalent through their common zero-syndrome deformation retract, but the full higher-ghost cohomology remains presentation-labelled.

13.5 Exact Validation

The producer `steane_clifford_koszul_morphisms.jl` writes the Steane Clifford/Koszul run bundle. It records the Hadamard generator permutation, the phase generator-basis change, the logical label actions, the 63 contractible nonzero syndrome sectors, and the cohomology dimensions listed in Section 12.

14 Clifford Group Ghost-Gaussian Theorem

The previous section treated two transversal Steane Cliffords. The correct general statement is stronger, but it needs one extra convention: binary Pauli labels alone are not enough. Gottesman's local source says that Clifford normalizer elements conjugate Pauli products to Pauli products `Thesis.tex`, lines 2292–2305 and 2307–2348; the binary stabilizer language drops phases at lines 1303–1327. The source manifest records the full local path. The supercharge projectors do not allow that drop.

14.1 Signed Pauli Convention

For n qubits define the Hermitian representative

$$W(x,z) = i^{x\cdot z} \bigotimes_{j=1}^{n} X_j^{x_j} Z_j^{z_j}, \qquad x,z \in \mathbb{F}_2^n.$$

A signed Pauli stabilizer generator is

$$S = \tau W(x,z), \qquad \tau \in \{\pm 1\}.$$

The violated-check projector is

$$P_S = \frac{I - S}{2}.$$

Thus if a Clifford gives

$$UW(x,z)U^\dagger = \tau W(x', z'),$$

then the transported projector is

$$UP_{W(x,z)}U^\dagger = \frac{I - \tau W(x',z')}{2}.$$

When $\tau = -1$, this is the opposite syndrome projector for the unsigned binary label (x', z'). This is why the theorem below is stated for signed Pauli operators, not only for symplectic binary labels.

14.2 The Koszul Complex Attached to a Signed Presentation

Let

$$G = (S_1, \dots, S_r)$$

be an ordered independent list of commuting signed Hermitian Pauli operators, with no product equal to $-I$. Let $E = \mathrm{span}_{\mathbb{C}}\{\epsilon_1, \dots, \epsilon_r\}$, and let $c_i^\dagger$ denote exterior multiplication by ϵ_i on ΛE. The ghost-extended complex is

$$\mathcal{K}(G) = \mathcal{H} \otimes \Lambda E, \qquad Q_G = \sum_{i=1}^{r} c_i^\dagger P_{S_i}.$$

This is exactly the earlier supercharge notation $\sum_i \epsilon_i P_{S_i}$, now with the creation operator written explicitly. Since the P_{S_i} commute and the $c_i^\dagger$ anticommute, $Q_G^2 = 0$.

14.3 Theorem: Arbitrary Clifford Covariance

Let U be any n-qubit Clifford. Define the transported signed presentation

$$U(G) = \left(US_1U^\dagger, \dots, US_rU^\dagger\right)$$

and give it ghosts $E^U = \mathrm{span}\{\epsilon_1^U, \dots, \epsilon_r^U\}$. Let $T_U : E \to E^U$ be the linear map $T_U(\epsilon_i) = \epsilon_i^U$. Then

$$\Phi_U = U \otimes \Lambda T_U$$

is a chain isomorphism

$$\Phi_U Q_G = Q_{U(G)} \Phi_U.$$

Equivalently,

$$Q_{U(G)} = (U \otimes \Lambda T_U) Q_G (U^\dagger \otimes \Lambda T_U^{-1}).$$

Proof. For each i,

$$UP_{S_i}U^\dagger = U\frac{I - S_i}{2}U^\dagger = \frac{I - US_iU^\dagger}{2} = P_{US_iU^\dagger}.$$

Now apply both sides to a pure tensor $\psi \otimes \omega$:

$$\begin{aligned}\Phi_U Q_G(\psi \otimes \omega) &= \sum_i U(P_{S_i}\psi) \otimes \epsilon_i^U \wedge (\Lambda T_U)\omega \\ &= \sum_i P_{US_iU^\dagger}(U\psi) \otimes \epsilon_i^U \wedge (\Lambda T_U)\omega \\ &= Q_{U(G)}\Phi_U(\psi \otimes \omega).\end{aligned}$$

The map is invertible because U and T_U are invertible. This proves the chain isomorphism and hence gives canonical isomorphisms

$$H^q(Q_G) \cong H^q(Q_{U(G)})$$

for every ghost degree q.

14.4 Where the Ghost Gaussian Enters

The covariance theorem uses the transported ordered list $U(G)$, so the ghost map is just $\epsilon_i \mapsto \epsilon_i^U$. The genuinely nontrivial ghost operation appears when we rewrite a signed stabilizer group in a different independent generator basis.

First consider a row swap: exchange S_a and S_b. The supercharge is unchanged after the same swap of ghost creation operators,

$$c_a^\dagger \leftrightarrow c_b^\dagger.$$

This is a number-preserving fermionic Gaussian operation on the ghost Fock space.

Now consider the elementary row shear

$$S_a \longmapsto S_aS_b, \qquad S_i \longmapsto S_i \quad (i \neq a).$$

The key projector identity is not linear over $\mathbb{C}$ in the syndrome bits; it contains the old stabilizer operator:

$$\begin{aligned}P_{S_aS_b} &= \frac{I - S_aS_b}{2} \\ &= \frac{I - S_a}{2} + S_a\frac{I - S_b}{2} \\ &= P_{S_a} + S_aP_{S_b}.\end{aligned}$$

Therefore the sheared supercharge is

$$\begin{aligned}Q' &= c_a^\dagger P_{S_a S_b} + c_b^\dagger P_{S_b} + \sum_{i\neq a,b} c_i^\dagger P_{S_i} \\ &= c_a^\dagger P_{S_a} + (c_b^\dagger + S_a c_a^\dagger) P_{S_b} + \sum_{i\neq a,b} c_i^\dagger P_{S_i}.\end{aligned}$$

Thus the ghost creation operators transform by

$$c_b^\dagger \longmapsto c_b^\dagger + S_a c_a^\dagger, \qquad c_i^\dagger \longmapsto c_i^\dagger \quad (i \neq b).$$

This is a number-preserving fermionic Gaussian with an even operator-valued coefficient:

$$\Gamma_{ab} = \exp(S_a c_a^\dagger c_b).$$

Indeed S_a commutes with all projectors and ghost operators, and the commutator gives

$$\Gamma_{ab} c_b^\dagger \Gamma_{ab}^{-1} = c_b^\dagger + S_a c_a^\dagger, \qquad \Gamma_{ab} c_i^\dagger \Gamma_{ab}^{-1} = c_i^\dagger \quad (i \neq b).$$

Consequently

$$Q' = \Gamma_{ab} Q_G \Gamma_{ab}^{-1}.$$

Row swaps and row shears generate $\mathrm{GL}_r(\mathbb{F}_2)$. Therefore every independent presentation

$$S_a' = \prod_i S_i^{A_{ai}}, \qquad A \in \mathrm{GL}_r(\mathbb{F}_2),$$

is obtained by composing these elementary ghost Gaussians. The closed projector formula for a row $I = (i_1, \ldots, i_m)$ is the telescoping identity

$$P_{\prod_{\nu=1}^m S_{i_\nu}} = \sum_{\mu=1}^m \left(\prod_{\nu<\mu} S_{i_\nu} \right) P_{S_{i_\mu}}.$$

This formula is the concrete reason that a scalar exterior $\mathrm{GL}_r(\mathbb{F}_2)$ operation on ghosts is not enough on the full physical Hilbert space. The syndrome parity $\sigma_a \oplus \sigma_b$ is represented over $\mathbb{C}$ by

$$\sigma_a + (-1)^{\sigma_a} \sigma_b,$$

not by $\sigma_a + \sigma_b$. The coefficient $(-1)^{\sigma_a}$ is exactly the eigenvalue of S_a.

On the code subspace all $S_i = +1$. Hence every row shear reduces there to the ordinary exterior linear map

$$c_b^\dagger \mapsto c_b^\dagger + c_a^\dagger.$$

So the higher-ghost cohomology is functorial under the exterior representation of the chosen presentation change on the zero-syndrome sector, while the full chain-level comparison remembers operator-valued syndrome signs.

14.5 Steane: All Clifford Generators

For the Steane code, $r = 6$. The standard Clifford generating gates are the seven single-qubit H's, seven single-qubit P's, and the 42 ordered CNOT gates. This is the generating set described in the local Gottesman source at lines 2292–2348. The producer `scripts/lattice_codes/steane_all_clifford_ghost_gaussians.jl` applies these 56 gates to the six sourced Steane stabilizer generators from Section 11.

The run records 336 signed generator images. Every one of the 56 image lists has binary rank 6 and is isotropic for the binary symplectic form. Since the listed H, P, and CNOT gates generate the Clifford group, any Clifford word is handled by composing these signed Pauli-label updates and then applying the covariance theorem above.

The same run records the 45 elementary ghost-Gaussian presentation moves for the Steane six-generator group: 15 row swaps and 30 directed row shears. For each shear it checks all 64 Steane syndrome sectors, i.e.

$$30 \cdot 64 = 1920$$

instances of

$$P(S_a S_b) = P(S_a) + S_a P(S_b).$$

If a Clifford U lies in the Steane stabilizer normalizer, then

$$U S_i U^\dagger = \prod_j S_j^{A_{ij}}$$

for some $A \in \mathrm{GL}_6(\mathbb{F}_2)$. In that case

$$(U \otimes I) Q_G (U^\dagger \otimes I) = Q_{A(G)} = \Gamma_A Q_G \Gamma_A^{-1},$$

where Γ_A is any product of the elementary ghost Gaussians above that realizes the row-reduction path for A. If U is an arbitrary Clifford not preserving the Steane stabilizer group, the first equality still gives a strict supercharge for the transported code $U\mathcal{C}$, but no ghost-only operation can turn its physical projectors back into the original Steane projectors.

14.6 Conclusion

The precise answer is therefore Clifford conjugation on physical qubits plus a fermionic Gaussian change of ghost presentation. No Bogoliubov mixing of creation and annihilation operators is needed for the cohomological complex; the transformations preserve ghost degree. The only non-scalar feature is essential: when a product stabilizer is rewritten in terms of old violated-check projectors, the Gaussian coefficients are even operators generated by the old stabilizers themselves.

15 Arithmetic Quantum Fields

15.1 Status and Source Anchors

This section introduces a proposal. The finite-field Weyl facts are source-local: Ashikhmin–Knill define p^m-ary shift/phase operators and their commutator in `n9.tex`, lines 249–313, and Bombin–Martin-Delgado record the qudit Pauli symplectic commutator in `HomologicalErrorCorrection6.tex`, lines 1206–1259. The new content below is the checked construction of field label spaces from finite function spaces.

15.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & X \text{ is a finite physical space.} \\
D2 & : & (V, \omega_V) \text{ is a finite symplectic } F\text{-vector space.} \\
D3 & : & E \leq \mathrm{Map}(X, V) \text{ is a chosen } F\text{-linear field space.} \\
D4 & : & \Omega_E(\phi, \psi) = \sum_{x \in X} w_x \omega_V(\phi(x), \psi(x)). \\
L1 & : & \mathrm{Map}(X, V) \text{ is an } F\text{-vector space pointwise.} \\
L2 & : & \Omega_E \text{ is alternating and bilinear.} \\
L3 & : & \Omega \text{ is nondegenerate on all maps if } \omega_V \text{ is and } w_x \neq 0. \\
L4 & : & E \text{ is symplectic iff } \mathrm{rad}(E) = 0. \\
L5 & : & E_{\mathrm{red}} = E/\mathrm{rad}(E) \text{ is symplectic.} \\
L6 & : & W : E_{\mathrm{red}} \to U(\mathcal{H}) \text{ is a projective Weyl representation.} \\
T1 & : & \mathcal{A}_{X,E} = \mathrm{span}_{\mathbb{C}}\{W(e)\} \text{ is the field-observable algebra.}
\end{array}$$

The numbering is used only to check that no later construction is invoked before its inputs are defined.

15.3 Definitions D1–D4

Let $F = \mathbb{F}_q$, $q = p^m$. A finite physical space is first only a finite set X. If X has additional structure, for example an F-vector-space or variety structure, then $\mathrm{Hom}(X, V)$ means a specified class of structure-preserving maps. Without that extra choice the neutral object is

$$\mathrm{Map}(X, V) = \{\phi : X \to V\}.$$

The proposed arithmetic field label space is not automatically all maps. It is a chosen F-linear subspace

$$E \leq \mathrm{Map}(X, V).$$

Examples include all functions, linear maps, affine maps, and polynomial functions up to a degree bound, all interpreted after evaluation on the finite point set.

Fix nonzero weights $w_x \in F^\times$. In this shard and in the validation run all weights are $w_x = 1$. Define

$$\Omega_E(\phi, \psi) = \sum_{x \in X} w_x \, \omega_V(\phi(x), \psi(x)).$$

15.4 Pointwise Linear Algebra L1–L5

Lemma $L1$. $\mathrm{Map}(X, V)$ is an F-vector space under pointwise operations:

$$(a\phi + b\psi)(x) = a\phi(x) + b\psi(x).$$

Proof. The right side lies in V, so it defines a map $X \to V$. Associativity, commutativity of addition, distributivity, scalar identity, and additive inverses hold after evaluating at each x, where they are exactly the vector space axioms in V. Thus the function set is a vector space. Any subset closed under these operations is an F-linear field subspace E.

Lemma $L2$. Ω_E is F-bilinear and alternating. *Proof.* For $a, b \in F$,

$$\begin{aligned}\Omega_E(a\phi + b\psi, \eta) &= \sum_x w_x \omega_V(a\phi(x) + b\psi(x), \eta(x)) \\ &= a\Omega_E(\phi, \eta) + b\Omega_E(\psi, \eta).\end{aligned}$$

Linearity in the second argument is identical. Also

$$\Omega_E(\phi, \phi) = \sum_x w_x \omega_V(\phi(x), \phi(x)) = 0$$

because ω_V is alternating. Hence Ω_E is an alternating bilinear form.

Lemma $L3$. If $E = \mathrm{Map}(X, V)$, ω_V is nondegenerate, and every $w_x \neq 0$, then Ω_E is nondegenerate. *Proof.* Let $\phi \neq 0$. Choose x_0 with $\phi(x_0) \neq 0$. Since ω_V is nondegenerate, choose $v \in V$ with $\omega_V(\phi(x_0), v) \neq 0$. Define $\psi(x_0) = w_{x_0}^{-1} v$ and $\psi(x) = 0$ for $x \neq x_0$. Then

$$\Omega_E(\phi, \psi) = \omega_V(\phi(x_0), v) \neq 0.$$

Thus no nonzero ϕ is orthogonal to all maps.

Lemma $L4$. For a subspace E, define

$$\mathrm{rad}(E) = \{\phi \in E : \Omega_E(\phi, \psi) = 0 \text{ for all } \psi \in E\}.$$

Then (E, Ω_E) is symplectic iff $\mathrm{rad}(E) = 0$. *Proof.* By definition, nondegeneracy means the only vector pairing to zero with every element of E is 0. That set is exactly $\mathrm{rad}(E)$. Together with Lemma $L2$, this is precisely the definition of a symplectic vector space.

Lemma $L5$. The quotient $E_{\mathrm{red}} = E/\mathrm{rad}(E)$ is symplectic. *Proof.* If $r \in \mathrm{rad}(E)$, then $\Omega_E(\phi + r, \psi) = \Omega_E(\phi, \psi)$, so the form descends to the quotient. If $[\phi]$ pairs to zero with every $[\psi]$, then ϕ lies in the radical, so $[\phi] = 0$. Hence the descended form is nondegenerate.

15.5 Weyl-Heisenberg Field Displacements L6

Choose a symplectic basis of E_{red}, so

$$E_{\mathrm{red}} \cong F^n \oplus F^n, \qquad e = (q, p).$$

Let $\chi(t) = \exp(2\pi i \, \mathrm{Tr}_{F/\mathbb{F}_p}(t)/p)$. On $\mathcal{H} = \mathbb{C}^{F^n}$, set

$$T_q|a\rangle = |a + q\rangle, \qquad R_p|a\rangle = \chi(p \cdot a)|a\rangle, \qquad W(q, p) = T_q R_p.$$

Then

$$W(q, p)W(q', p') = \chi(p \cdot q')\, W(q + q', p + p')$$

and therefore

$$W(e)W(f) = \chi(\Omega(e, f))W(f)W(e).$$

The operators $W(e)$ are unitary because T_q permutes the basis and R_p is diagonal with unit-modulus entries. The multiplication law shows that $e \mapsto W(e)$ is a projective unitary representation of the additive group of E_{red}, with cocycle $c(e, f) = \chi(p \cdot q')$.

15.6 Definition: Arithmetic Quantum Field

An arithmetic quantum field datum is

$$(X, V, E, \Omega_E, W_E)$$

where X is finite, V is finite symplectic over F_q, $E \leq \mathrm{Map}(X, V)$ is a chosen finite field-label space, and W_E is the Weyl projective representation of E_{red}. The field displacement operators are

$$\mathsf{A}_{X,E} = \{W_E(e) : e \in E_{\mathrm{red}}\}.$$

The observable algebra of this finite arithmetic quantum field is

$$\mathcal{A}_{X,E} = \mathrm{span}_{\mathbb{C}}\, \mathsf{A}_{X,E} \subseteq \mathrm{End}(\mathcal{H}).$$

When $\dim_F E_{\mathrm{red}} = 2n$, the local Weyl source says these operators form an operator basis, so $\mathcal{A}_{X,E} \cong M_{q^n}(\mathbb{C})$.

15.7 Molecular F3 Examples

For the run we take $V = F_3^2$ with $\omega((q,p),(q',p')) = pq' - qp'$. A scalar function space $\mathcal{U} \leq \mathrm{Map}(X, F_3)$ gives

$$E = \mathcal{U}q \oplus \mathcal{U}p, \qquad B(f,g) = \sum_{x \in X} f(x)g(x),$$

$$\Omega((q,p),(q',p')) = B(p,q') - B(q,p').$$

If $m = \dim \mathcal{U}$ and $r = \mathrm{rank}\, B$, then

$$\dim E = 2m, \quad \mathrm{rank}\, \Omega = 2r, \quad \dim \mathrm{rad}(E) = 2(m-r).$$

example	$\lvert X \rvert$	$\dim \mathcal{U}$	r	$\dim \mathrm{rad}$	$\lvert E_{\mathrm{red}} \rvert$
two points, all maps	2	2	2	0	81
three points, all maps	3	3	3	0	729
F_3, $\langle t \rangle$	3	1	1	0	9
F_3, $\langle 1, t \rangle$	3	2	1	2	9
F_3, $\langle 1, t, t^2 \rangle$	3	3	3	0	729
F_3^2, $\langle x, y \rangle$	9	2	0	4	1
F_3^2, all maps	9	9	9	0	387420489
$y = x^2$, $\langle 1 \rangle$	3	1	0	2	1
$y = x^2$, $\langle x, y \rangle$	3	2	2	0	81
$y = x^2$, $\langle 1, x, y \rangle$	3	3	3	0	729
$G_m(F_3)$, $\langle 1, t \rangle$	2	2	2	0	81

The negative controls are meaningful. For affine fields on F_3, the constant function has zero norm because $1+1+1 = 0$ in F_3, so constants lie in the radical. For $\langle x, y \rangle$ on F_3^2, every sum $\sum x^2$, $\sum xy$, and $\sum y^2$ vanishes, so the entire field space is radical and the reduced Weyl label group is trivial.

For the parabola $X = \{(0,0),(1,1),(2,1)\}$, the checked basis values are

$$1 = (1,1,1), \qquad x = (0,1,2), \qquad y = (0,1,1),$$

with Gram matrix

$$\begin{pmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix},$$

which has rank 3. Thus degree-one ambient coordinate functions already give the full nondegenerate three-dimensional scalar function algebra on these three points.

15.8 Exact Validation

The producer is `arithmetic_quantum_fields.jl` under `scripts/bridges/`. It writes the 2026-05-24 arithmetic-quantum-fields run bundle. The summary CSV records point counts, scalar Gram ranks, symplectic ranks, radicals, reduced Weyl label counts, Hilbert dimensions, and observable-algebra dimensions. The basis CSV records every scalar basis vector and every scalar Gram row used in the examples above.

16 Projective Sheaf Field Definitions

16.1 Status and Ground Truth

This shard is definition-building, not a classification theorem. It is source-local where it recalls standard algebraic geometry and checked where it proves the finite linear-algebra statements. The local ground truth is the Stacks Project TeX snapshot registered in `references/algebraic_geometry/SOURCES.md`. We use:

- presheaves and sections: `stacks_project_sheaves.tex`, lines 73–123;
- sheaves: `stacks_project_sheaves.tex`, lines 500–525;
- Proj and its sheaves: `stacks_project_constructions.tex`, lines 939–1225;
- twists $\mathcal{O}(n)$: `stacks_project_constructions.tex`, lines 1695–1705;
- projective space: `stacks_project_constructions.tex`, lines 2775–2886;
- varieties and projective varieties: `stacks_project_varieties.tex`, lines 53–58 and 4843–4865;
- $H^0(\mathbf{P}^n, \mathcal{O}(d))$: `stacks_project_coherent.tex`, lines 1557–1622.

16.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & X \text{ is a topological space.} \\
D2 & : & \mathcal{F} \text{ is a presheaf on } X. \\
D3 & : & \mathcal{F} \text{ is a sheaf if compatible local sections glue uniquely.} \\
D4 & : & \Gamma(U,\mathcal{F}) = \mathcal{F}(U) = H^0(U,\mathcal{F}). \\
D5 & : & X \text{ is a variety, projective variety, or projective space over } F_q. \\
D6 & : & \mathcal{O}_X(d) \text{ is the } d\text{th twist on a projective space.} \\
D7 & : & (V,\omega_V) \text{ is a finite symplectic } F_q\text{-vector space.} \\
D8 & : & E(X,\mathcal{L},V) = H^0(X,\mathcal{L}) \otimes_{F_q} V. \\
D9 & : & \Omega \text{ is the finite rational-point pairing after choosing evaluations.} \\
L1 & : & E(X,\mathcal{L},V) \text{ is an } F_q\text{-vector space.} \\
L2 & : & \Omega \text{ is bilinear and alternating.} \\
L3 & : & E/\operatorname{rad}(E) \text{ is the honest Weyl label space.}
\end{array}$$

16.3 Presheaves

Let X be a topological space. A presheaf of sets $\mathcal{F}$ on X consists of two pieces of data.

First, every open subset $U \subset X$ is assigned a set $\mathcal{F}(U)$. Its elements are called sections over U. Second, every inclusion of open sets $V \subset U$ is assigned a restriction map

$$\rho^U_V : \mathcal{F}(U) \longrightarrow \mathcal{F}(V).$$

The maps must satisfy two identities:

$$\rho^U_U = \mathrm{id}_{\mathcal{F}(U)}, \qquad \rho^U_W = \rho^V_W \circ \rho^U_V \quad \text{when } W \subset V \subset U.$$

For $s \in \mathcal{F}(U)$, write

$$s|_V = \rho^U_V(s).$$

Thus $s|_W = (s|_V)|_W$. The notation

$$\Gamma(U, \mathcal{F}) = \mathcal{F}(U) = H^0(U, \mathcal{F})$$

means exactly the same set of sections over U; the last two symbols do not add a new construction in degree zero.

For a fixed field F_q, a presheaf of F_q-vector spaces is the same data with each $\mathcal{F}(U)$ an F_q-vector space and each restriction map F_q-linear. Equivalently, it is a presheaf of sets whose section sets carry compatible vector-space operations.

16.4 Sheaves

A sheaf is a presheaf satisfying a locality-and-gluing axiom. Let

$$U = \bigcup_{i \in I} U_i$$

be an open cover. Suppose $s_i \in \mathcal{F}(U_i)$ is a section on each member of the cover. The family is compatible if for every pair i, j,

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}.$$

The presheaf $\mathcal{F}$ is a sheaf if every compatible family has one and only one glued section $s \in \mathcal{F}(U)$ such that

$$s|_{U_i} = s_i \qquad \text{for all } i.$$

This says precisely: a sheaf is an object whose sections are determined locally, and whose locally compatible data assemble globally.

16.5 Varieties, Projective Space, and Twists

Let k be a field. In the Stacks convention, a variety over k is a scheme X over k such that X is integral and the structure morphism

$$X \longrightarrow \operatorname{Spec}(k)$$

is separated and of finite type. A projective variety is a variety for which this structure morphism is projective; equivalently in the cited Stacks source, it embeds as a closed subscheme of some $\mathbf{P}^n_k$.

Projective n-space over a ring R is

$$\mathbf{P}^n_R = \operatorname{Proj}(R[T_0, \dots, T_n]),$$

where every T_i has degree 1. Its standard opens are

$$D_+(T_i), \qquad i = 0, \dots, n,$$

and these cover $\mathbf{P}^n_R$. For $n = 1$, the two standard opens

$$D_+(X), \qquad D_+(Y)$$

are affine lines with coordinates Y/X and X/Y, respectively.

For $X = \operatorname{Proj}(S)$, the dth twist of the structure sheaf is

$$\mathcal{O}_X(d) = \widetilde{S(d)}.$$

In the special case $X = \mathbf{P}^n_R$, the local source computes

$$H^0(\mathbf{P}^n_R, \mathcal{O}(d)) = \begin{cases} R[T_0, \ldots, T_n]_d, & d \geq 0, \\ 0, & d < 0, \end{cases}$$

where $R[T_0, \ldots, T_n]_d$ is the degree-d homogeneous part. Thus

$$H^0(\mathbf{P}^1_{F_q}, \mathcal{O}(d)) = F_q\text{-span}\{X^{d-i}Y^i : 0 \leq i \leq d\}$$

for $d \geq 0$.

16.6 Why Naive Hom Is Not the Projective Default

For a finite set X, the previous shard used $E \leq \mathrm{Map}(X, V)$. For a projective variety this is too naive if $\mathrm{Hom}(X, V)$ is read as regular maps to an affine vector space. A regular map to affine space is controlled by global regular functions, and the Stacks source records that a proper geometrically integral variety has

$$H^0(X, \mathcal{O}_X) = k.$$

So for connected projective spaces this would see only constants.

The projective replacement is therefore sheaf-theoretic:

$$E(X, \mathcal{L}, V) = H^0(X, \mathcal{L}) \otimes_{F_q} V,$$

where $\mathcal{L}$ is an explicitly chosen sheaf on X. The sheaf is part of the quantum-field datum. Different choices $\mathcal{L} = \mathcal{O}(d)$ give different finite field-label spaces.

16.7 Finite Symplectic Target and Pointwise Form

Let V be a finite-dimensional F_q-vector space. A symplectic form on V is an F_q-bilinear pairing

$$\omega_V : V \times V \to F_q$$

such that $\omega_V(v, v) = 0$ for all v and the only vector pairing to zero with every vector is 0. The standard two-dimensional example is

$$V = F_q^2, \qquad \omega_V((a, b), (c, d)) = bc - ad.$$

To turn $E(X, \mathcal{L}, V)$ into a finite Weyl label space we also need a finite scalar pairing on $H^0(X, \mathcal{L})$. Let $S \subset X(F_q)$ be a finite set of rational points, let $w_P \in F_q$, and choose an evaluation coordinate

$$\mathrm{ev}_P : H^0(X, \mathcal{L}) \to F_q$$

for every $P \in S$. Then define

$$B(s, t) = \sum_{P \in S} w_P \, \mathrm{ev}_P(s) \, \mathrm{ev}_P(t).$$

This evaluation choice is automatic for honest scalar functions, but for sections of a nontrivial line bundle it is a convention: one has chosen local trivializations or fixed projective representatives at the points.

For pure tensors set

$$\Omega(s \otimes v, t \otimes w) = B(s, t)\omega_V(v, w),$$

and extend bilinearly to all of $E(X, \mathcal{L}, V)$.

Lemma $L1$. $E(X, \mathcal{L}, V)$ is an F_q-vector space.

Proof. The source definition of a sheaf of vector spaces gives $H^0(X, \mathcal{L}) = \mathcal{L}(X)$ as an F_q-vector space. The tensor product of two F_q-vector spaces is an F_q-vector space, with addition and scalar multiplication inherited from the tensor construction. Hence $H^0(X, \mathcal{L}) \otimes_{F_q} V$ is an F_q-vector space.

Lemma $L2$. Ω is F_q-bilinear and alternating.

Proof. The map B is bilinear because it is a sum of products of linear evaluation maps. The form ω_V is bilinear by definition. Therefore their tensor product pairing is bilinear on pure tensors and hence bilinear after extension. Also

$$\Omega(s \otimes v, s \otimes v) = B(s, s)\omega_V(v, v) = 0,$$

and the same alternating identity follows for arbitrary sums by bilinearity and the alternating property of ω_V.

Lemma $L3$. Let

$$\operatorname{rad}(E) = \{\phi \in E : \Omega(\phi, \psi) = 0 \text{ for every } \psi \in E\}.$$

Then $E_{\text{red}} = E/\operatorname{rad}(E)$ has a nondegenerate alternating form induced by Ω.

Proof. If $r \in \operatorname{rad}(E)$, then $\Omega(\phi + r, \psi) = \Omega(\phi, \psi)$, so the form descends to the quotient. If a quotient class $[\phi]$ pairs to zero with every quotient class, then ϕ pairs to zero with every element of E, hence $\phi \in \operatorname{rad}(E)$ and $[\phi] = 0$. Thus the descended form is nondegenerate.

The Weyl-Heisenberg field displacement algebra is built from E_{red}, not from unreduced E, exactly as in Shard AQM-14.

17 Projective Line Sheaf Field Calculation

17.1 Status and Local Evidence

This shard is self-contained for the example

$$X = \mathbf{P}^1_{\mathbb{F}_3}, \qquad V = \mathbb{F}_3^2.$$

The algebraic geometry inputs are the local Stacks TeX files cited in Shard AQM-15. The exact finite calculation is checked by `scripts/bridges/projective_line_sheaf_fields.jl`, with run bundle `runs/2026-05-24-projective-line-sheaf-fields/`. The output files are `data/projective_line_sheaf_field_summary.csv` and `data/projective_line_sheaf_field_basis_rows.csv`.

17.2 The Field and the Points

The field is

$$\mathbb{F}_3 = \{0, 1, 2\}, \qquad 2 = -1, \qquad 1 + 1 + 1 = 0.$$

The projective line has rational points given by nonzero pairs $(a, b) \in \mathbb{F}_3^2 \setminus \{(0, 0)\}$ modulo nonzero scalar multiplication:

$$[a : b] = [\lambda a : \lambda b] \quad (\lambda \in \mathbb{F}_3^\times).$$

There are four classes. The chosen representatives and order are

$$P_\infty = [1 : 0], \quad P_0 = [0 : 1], \quad P_1 = [1 : 1], \quad P_2 = [2 : 1].$$

17.3 The Sheaves and Their Sections

For every $d \geq 0$, the source computation gives

$$H^0(\mathbf{P}^1_{\mathbb{F}_3}, \mathcal{O}(d)) = \mathbb{F}_3[X, Y]_d.$$

Thus a basis is

$$X^d,\ X^{d-1}Y,\ X^{d-2}Y^2,\ \ldots,\ Y^d.$$

This is a statement about sections of $\mathcal{O}(d)$, not about global regular functions when $d > 0$.

For this finite rational-point calculation, evaluate $X^{d-i}Y^i$ on the fixed representatives above:

$$X^{d-i}Y^i(P_\infty) = \begin{cases} 1, & i = 0, \\ 0, & i > 0, \end{cases}$$

and

$$X^{d-i}Y^i(P_t) = t^{d-i} \qquad (t = 0, 1, 2).$$

This is the explicit chart-coordinate evaluation convention from CONVENTIONS (r). It is not silently canonical for every line bundle.

17.4 Scalar Pairing and Symplectic Extension

For scalar sections s, t, define

$$B(s,t) = \sum_{P \in \mathbf{P}^1(\mathbb{F}_3)} s(P)t(P).$$

All weights are 1. The target is $V = \mathbb{F}_3^2$ with $\omega((q,p),(q',p')) = pq' - qp'$. The field space is

$$E_d = H^0(\mathbf{P}^1, \mathcal{O}(d)) \otimes_{\mathbb{F}_3} V.$$

If $U_d = H^0(\mathbf{P}^1, \mathcal{O}(d))$, write

$$E_d = U_d q \oplus U_d p.$$

For $(q,p),(q',p') \in U_d q \oplus U_d p$,

$$\Omega_d((q,p),(q',p')) = B(p,q') - B(q,p').$$

If the scalar Gram matrix $G_d = (B(e_i, e_j))$ has rank r, then

$$\dim E_d = 2(d+1), \quad \operatorname{rank} \Omega_d = 2r, \quad \dim \operatorname{rad}(E_d) = 2(d+1-r).$$

The Weyl label space is $E_{d,\mathrm{red}} = E_d / \operatorname{rad}(E_d)$.

17.5 Degree 0: Constants

The basis is 1. Its value row is

$$1 : \quad (1,1,1,1).$$

Therefore

$$B(1,1) = 1+1+1+1 = 4 = 1 \in \mathbb{F}_3.$$

So

$$G_0 = [1].$$

The scalar rank is 1, hence Ω_0 has rank 2. The reduced field has dimension 2, the Weyl Hilbert dimension is 3, and the Weyl operator basis has 9 elements.

17.6 Degree 1: Linear Sections

The basis is X, Y. The value rows are

$$X : \ (1,0,1,2), \qquad Y : \ (0,1,1,1).$$

Compute every scalar product:

$$\begin{aligned} B(X,X) &= 1^2 + 0^2 + 1^2 + 2^2 = 1+0+1+4 = 6 = 0, \\ B(X,Y) &= 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 + 2 \cdot 1 = 3 = 0, \\ B(Y,Y) &= 0^2 + 1^2 + 1^2 + 1^2 = 3 = 0. \end{aligned}$$

Thus

$$G_1 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The two sections evaluate independently at the four points, but the chosen all-weights-one scalar pairing is totally zero on their span. Consequently

$$\operatorname{rank} \Omega_1 = 0, \qquad \operatorname{rad}(E_1) = E_1.$$

The reduced Weyl label space is zero-dimensional. This is the first concrete warning that evaluation rank and symplectic rank are different questions.

17.7 Degree 2: Quadratic Sections

The basis is X^2, XY, Y^2. The value rows are

	P_∞	P_0	P_1	P_2
X^2	1	0	1	1
XY	0	0	1	2
Y^2	0	1	1	1

The scalar products give

$$G_2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

For example,

$$B(X^2, Y^2) = 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 2,$$

and

$$B(XY, XY) = 0 + 0 + 1 + 4 = 5 = 2.$$

The determinant is

$$\det(G_2) = 1 \in \mathbb{F}_3,$$

so G_2 has rank 3. Hence Ω_2 has rank 6, no radical, and

$$\dim E_2 = 6, \qquad \dim \mathcal{H} = 3^3 = 27, \qquad |\{W(e)\}| = 3^6 = 729.$$

17.8 Degree 3: Cubic Sections

The basis rows and Gram matrix are

	P_∞	P_0	P_1	P_2
X^3	1	0	1	2
X^2Y	0	0	1	1
XY^2	0	0	1	2
Y^3	0	1	1	1

and

$$G_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \end{pmatrix}.$$

The validator computes rank 4, equivalently $\det(G_3) = 1$. Thus

$$\dim E_3 = 8, \quad \operatorname{rank} \Omega_3 = 8, \quad \dim \mathcal{H} = 3^4 = 81.$$

17.9 Degree 4: First Evaluation Kernel

The scalar section dimension is 5, but there are only four rational points. The value rows include

$$X^3Y: \ (0, 0, 1, 2), \qquad XY^3: \ (0, 0, 1, 2),$$

so

$$X^3Y - XY^3$$

evaluates to zero at all four chosen representatives. Therefore the scalar evaluation map has a one-dimensional kernel. The run confirms

$$\operatorname{rank}(\mathrm{ev}) = 4, \qquad \operatorname{rank}(G_4) = 4.$$

The scalar radical is exactly one-dimensional, and after tensoring with $V = \mathbb{F}_3^2$,

$$\dim \operatorname{rad}(E_4) = 2.$$

Thus

$$\dim E_4 = 10, \qquad \dim E_{4,\mathrm{red}} = 8, \qquad \dim \mathcal{H} = 3^4 = 81.$$

17.10 Run Summary

d	$\dim U_d$	rank(ev)	$\operatorname{rank} G_d$	$\dim \operatorname{rad} E_d$	$\dim E_{d,\mathrm{red}}$	$\dim \mathcal{H}$
0	1	1	1	0	2	3
1	2	2	0	4	0	1
2	3	3	3	0	6	27
3	4	4	4	0	8	81
4	5	4	4	2	8	81

The lessons are concrete: the projective sheaf field is controlled by $H^0(\mathcal{O}(d))$, the finite rational-point pairing is an additional choice and can introduce radicals, and the Weyl-Heisenberg observable algebra is formed after quotienting by this radical.

18 Projective Line Stalks

18.1 Status and Local Evidence

This shard continues Shard AQM-16. The phrase "each point in the example" means the four rational points used by the finite pairing: $[1:0]$, $[0:1]$, $[1:1]$, and $[2:1]$. The scheme has more points than these four; a general formula for those other stalks is recorded at the end.

The source definition of a stalk is local: `references/algebraic_geometry/stacks_project_sheaves.tex`, lines 894–944. The source formula for stalks on $\mathrm{Proj}(S)$ is `references/algebraic_geometry/stacks_project_constructions.tex`, lines 1091–1116 and 1182–1207. The checked rows are produced by `scripts/bridges/projective_line_sheaf_fields.jl`.

18.2 What a Stalk Is

Let X be a topological space, $x \in X$, and $\mathcal{F}$ a presheaf. The stalk at x is

$$\mathcal{F}_x = \operatorname*{colim}_{x \in U} \mathcal{F}(U),$$

where U runs over open neighbourhoods of x. Concretely, an element is represented by (U, s), with $x \in U$ and $s \in \mathcal{F}(U)$, and $(U, s) \sim (U', s')$ when some open neighbourhood $U'' \subset U \cap U'$ of x satisfies $s|_{U''} = s'|_{U''}$. Thus a stalk element is a germ: it remembers the section only arbitrarily near the point.

18.3 The Proj Stalk Formula

Let

$$S = \mathbb{F}_3[X, Y], \qquad \deg X = \deg Y = 1, \qquad X_{\mathrm{sch}} = \mathrm{Proj}(S) = \mathbf{P}^1_{\mathbb{F}_3}.$$

A point of X_{sch} is a homogeneous prime $\mathfrak{p} \subset S$ not containing the irrelevant ideal (X, Y). For a graded S-module M, the associated sheaf $\widetilde{M}$ has stalk

$$(\widetilde{M})_{\mathfrak{p}} = M_{(\mathfrak{p})}.$$

The right hand side consists of homogeneous fractions

$$a/b, \qquad a, b \in M \text{ homogeneous}, \quad \deg a = \deg b, \quad b \notin \mathfrak{p}.$$

For the structure sheaf, this gives

$$\mathcal{O}_{X_{\mathrm{sch}}, \mathfrak{p}} = S_{(\mathfrak{p})}.$$

For the twist $\mathcal{O}(d) = \widetilde{S(d)}$,

$$\mathcal{O}(d)_{\mathfrak{p}} = S(d)_{(\mathfrak{p})}.$$

18.4 The Two Standard Charts

On the chart $D_+(X)$, set $u = Y/X$. Then

$$D_+(X) = \operatorname{Spec} \mathbb{F}_3[u].$$

On this chart we use the local frame $e_X^{(d)}$ of $\mathcal{O}(d)$ corresponding to X^d. The section

$$X^{d-i} Y^i$$

has local expression

$$u^i e_X^{(d)}.$$

On the chart $D_+(Y)$, set $v = X/Y$. Then

$$D_+(Y) = \operatorname{Spec} \mathbb{F}_3[v].$$

On this chart we use the local frame $e_Y^{(d)}$ corresponding to Y^d. The same section has local expression

$$v^{d-i} e_Y^{(d)}.$$

On the overlap $D_+(XY)$, $u = 1/v$, and the two frames satisfy

$$e_X^{(d)} = v^d e_Y^{(d)}, \qquad e_Y^{(d)} = u^d e_X^{(d)}.$$

This is why both local expressions describe the same germ where both charts are available.

18.5 Point P-Infinity: [1:0]

The homogeneous prime is

$$\mathfrak{p}_\infty = (Y).$$

Since $X \notin \mathfrak{p}_\infty$, this point lies in $D_+(X)$. In the coordinate $u = Y/X$, the corresponding prime of $\mathbb{F}_3[u]$ is

$$(u).$$

Therefore

$$\mathcal{O}_{P_\infty} = \mathbb{F}_3[u]_{(u)}, \qquad \mathfrak{m}_{P_\infty} = (u), \qquad \kappa(P_\infty) = \mathbb{F}_3[u]_{(u)}/(u) = \mathbb{F}_3.$$

The $\mathcal{O}(d)$-stalk is free of rank one:

$$\mathcal{O}(d)_{P_\infty} = \mathbb{F}_3[u]_{(u)} e_X^{(d)}.$$

For the monomial $X^{d-i}Y^i$,

$$(X^{d-i}Y^i)_{P_\infty} = u^i e_X^{(d)}.$$

Reducing modulo (u) gives

$$u^i e_X^{(d)} \longmapsto \begin{cases} 1 \cdot \bar{e}_X^{(d)}, & i = 0, \\ 0, & i > 0. \end{cases}$$

This is exactly the evaluation rule used earlier at $[1:0]$.

18.6 Point P0: [0:1]

The homogeneous prime is

$$\mathfrak{p}_0 = (X).$$

Since $Y \notin \mathfrak{p}_0$, use $D_+(Y)$ and $v = X/Y$. The local prime is (v). Hence

$$\mathcal{O}_{P_0} = \mathbb{F}_3[v]_{(v)}, \qquad \mathfrak{m}_{P_0} = (v), \qquad \kappa(P_0) = \mathbb{F}_3.$$

The twist stalk is

$$\mathcal{O}(d)_{P_0} = \mathbb{F}_3[v]_{(v)} e_Y^{(d)}.$$

The monomial $X^{d-i}Y^i$ has germ

$$v^{d-i} e_Y^{(d)}.$$

Modulo (v),

$$v^{d-i} e_Y^{(d)} \longmapsto \begin{cases} 1 \cdot \bar{e}_Y^{(d)}, & i = d, \\ 0, & i < d. \end{cases}$$

18.7 Point P1: [1:1]

The homogeneous prime is

$$\mathfrak{p}_1 = (X - Y).$$

On $D_+(Y)$, it becomes

$$(v - 1) \subset \mathbb{F}_3[v].$$

Therefore

$$\mathcal{O}_{P_1} = \mathbb{F}_3[v]_{(v-1)}, \qquad \mathfrak{m}_{P_1} = (v - 1), \qquad \kappa(P_1) = \mathbb{F}_3.$$

The twist stalk is

$$\mathcal{O}(d)_{P_1} = \mathbb{F}_3[v]_{(v-1)} e_Y^{(d)}.$$

The germ of $X^{d-i}Y^i$ is

$$v^{d-i} e_Y^{(d)}.$$

Modulo $(v - 1)$, we set $v = 1$, so

$$v^{d-i} e_Y^{(d)} \longmapsto 1 \cdot \bar{e}_Y^{(d)}$$

for every i. This explains why every monomial evaluates to 1 at $[1 : 1]$.

18.8 Point P2: [2:1]

The homogeneous prime is

$$\mathfrak{p}_2 = (X - 2Y) = (X + Y).$$

On $D_+(Y)$, it becomes

$$(v - 2) \subset \mathbb{F}_3[v].$$

Thus

$$\mathcal{O}_{P_2} = \mathbb{F}_3[v]_{(v-2)}, \qquad \mathfrak{m}_{P_2} = (v - 2), \qquad \kappa(P_2) = \mathbb{F}_3.$$

The twist stalk is

$$\mathcal{O}(d)_{P_2} = \mathbb{F}_3[v]_{(v-2)} e_Y^{(d)}.$$

The germ of $X^{d-i}Y^i$ is

$$v^{d-i} e_Y^{(d)}.$$

Modulo $(v - 2)$, we set $v = 2 = -1$. Therefore

$$v^{d-i} e_Y^{(d)} \longmapsto 2^{d-i} \bar{e}_Y^{(d)}.$$

The residue is 1 when $d - i$ is even and 2 when $d - i$ is odd.

18.9 Target-Valued Stalks

The arithmetic field sheaf used in the example is

$$\mathcal{O}(d) \otimes_{\mathbb{F}_3} V, \qquad V = \mathbb{F}_3^2.$$

At any rational point P, its stalk is

$$(\mathcal{O}(d) \otimes V)_P = \mathcal{O}(d)_P \otimes_{\mathbb{F}_3} V.$$

Because $\mathcal{O}(d)_P$ is a free rank-one $\mathcal{O}_P$-module, this is two copies of the local ring:

$$(\mathcal{O}(d) \otimes V)_P \cong \mathcal{O}_P q \oplus \mathcal{O}_P p.$$

The residue at P is

$$(\mathcal{O}(d) \otimes V)_P \longrightarrow \kappa(P) \otimes_{\mathbb{F}_3} V = V,$$

and this is the pointwise value used in the finite symplectic pairing.

18.10 Other Scheme Points

The finite field algebra in Shard AQM-16 uses only rational points, but the scheme has more stalks. The generic point has $\mathcal{O}_\eta = \mathbb{F}_3(v)$ and $\mathcal{O}(d)_\eta = \mathbb{F}_3(v)e_Y^{(d)}$. A non-rational closed point on $D_+(Y)$ is given by an irreducible $h(v) \in \mathbb{F}_3[v]$ of degree > 1; then

$$\mathcal{O}_h = \mathbb{F}_3[v]_{(h)}, \quad \mathcal{O}(d)_h = \mathbb{F}_3[v]_{(h)}e_Y^{(d)}, \quad \kappa(h) = \mathbb{F}_3[v]/(h).$$

These stalks are not sampled by the current rational-point pairing.

19 Stalks as Field Germs

19.1 Answer to the Terminology Question

It is useful, but slightly dangerous, to say that a stalk is the "field at the point." The precise dictionary used here is:

$$\text{stalk} = \text{local field germ at the point}, \qquad \text{fiber} = \text{actual value after residue reduction}.$$

The source definition of a stalk as germs is `references/algebraic_geometry/stacks_project_sheaves.tex`, lines 894–944. The source Proj stalk formula is `references/algebraic_geometry/stacks_project_constructions.tex`, lines 1091–1116 and 1182–1207. The checked finite table is `runs/2026-05-24-projective-line-sheaf-fields/data/projective_line_stalk_rows.csv`.

Let $x \in X$ and let $\mathcal{E}$ be a sheaf of $\mathcal{O}_X$-modules. The stalk $\mathcal{E}_x$ consists of germs of sections near x. If X is a scheme, $\mathcal{O}_x$ is a local ring, with maximal ideal $\mathfrak{m}_x$, and residue field

$$\kappa(x) = \mathcal{O}_x/\mathfrak{m}_x.$$

The fiber of $\mathcal{E}$ at x is

$$\mathcal{E}(x) = \mathcal{E}_x \otimes_{\mathcal{O}_x} \kappa(x).$$

Thus $\mathcal{E}_x$ remembers infinitesimal or algebraic variation near x, while $\mathcal{E}(x)$ is the value obtained after setting the local coordinates equal to their value at x.

19.2 The Projective-Line Field Sheaf

In the current example

$$X = \mathbf{P}^1_{\mathbb{F}_3}, \qquad V = \mathbb{F}_3 q \oplus \mathbb{F}_3 p, \qquad \mathcal{E}_d = \mathcal{O}(d) \otimes_{\mathbb{F}_3} V.$$

The rational points are ordered as

$$P_\infty = [1:0], \quad P_0 = [0:1], \quad P_1 = [1:1], \quad P_2 = [2:1].$$

For every rational point P, the residue field is $\kappa(P) = \mathbb{F}_3$. Hence

$$\mathcal{E}_d(P) = \mathcal{E}_{d,P} \otimes_{\mathcal{O}_P} \kappa(P) \cong \mathbb{F}_3 q \oplus \mathbb{F}_3 p = V.$$

This is the exact sense in which the arithmetic quantum field has a point-value in V.

The stalks themselves are larger local modules. With

$$u = Y/X \quad \text{on } D_+(X), \qquad v = X/Y \quad \text{on } D_+(Y),$$

and local frames $e_X^{(d)}$, $e_Y^{(d)}$, the four rational-point stalks are:

P	$\mathcal{O}_P$	$\mathcal{O}(d)_P$	$\mathcal{E}_{d,P}$
P_∞	$\mathbb{F}_3[u]_{(u)}$	$\mathbb{F}_3[u]_{(u)} e_X^{(d)}$	$\mathbb{F}_3[u]_{(u)} e_X^{(d)} \otimes V$
P_0	$\mathbb{F}_3[v]_{(v)}$	$\mathbb{F}_3[v]_{(v)} e_Y^{(d)}$	$\mathbb{F}_3[v]_{(v)} e_Y^{(d)} \otimes V$
P_1	$\mathbb{F}_3[v]_{(v-1)}$	$\mathbb{F}_3[v]_{(v-1)} e_Y^{(d)}$	$\mathbb{F}_3[v]_{(v-1)} e_Y^{(d)} \otimes V$
P_2	$\mathbb{F}_3[v]_{(v-2)}$	$\mathbb{F}_3[v]_{(v-2)} e_Y^{(d)}$	$\mathbb{F}_3[v]_{(v-2)} e_Y^{(d)} \otimes V.$

19.3 Germ-to-Value Maps

Let

$$s_{d,i} = X^{d-i}Y^i, \qquad 0 \le i \le d,$$

and let $w = aq + bp \in V$. The vector-valued section is

$$\Phi_{d,i,w} = s_{d,i} \otimes w.$$

Its stalk germ and fiber value at the four rational points are:

P	germ in $\mathcal{E}_{d,P}$	fiber value in V
P_∞	$u^i e_X^{(d)} \otimes w$	$\begin{cases} w, & i=0, \\ 0, & i>0, \end{cases}$
P_0	$v^{d-i} e_Y^{(d)} \otimes w$	$\begin{cases} w, & i=d, \\ 0, & i<d, \end{cases}$
P_1	$v^{d-i} e_Y^{(d)} \otimes w$	w
P_2	$v^{d-i} e_Y^{(d)} \otimes w$	$2^{d-i}w.$

The last line uses $v = 2 = -1$ in $\mathbb{F}_3$. If $w = (a,b)$ in the ordered basis (q,p), multiplication by the scalar $\lambda \in \mathbb{F}_3$ means

$$\lambda w = (\lambda a, \lambda b).$$

These maps are not extra choices: they are the quotient maps $\mathcal{E}_{d,P} \to \mathcal{E}_d(P)$ induced by $\mathcal{O}_P \to \kappa(P)$.

19.4 All Scalar Residues for Degrees 0 Through 4

The following table gives the scalar $\lambda_{d,i}(P)$ such that

$$\Phi_{d,i,w}(P) = \lambda_{d,i}(P)\, w.$$

It is exactly the stalk-residue layer underlying the evaluation matrices in Shard AQM-16.

d	$s_{d,i}$	P_∞	P_0	P_1	P_2
0	1	1	1	1	1
1	X	1	0	1	2
1	Y	0	1	1	1
2	X^2	1	0	1	1
2	XY	0	0	1	2
2	Y^2	0	1	1	1
3	X^3	1	0	1	2
3	X^2Y	0	0	1	1
3	XY^2	0	0	1	2
3	Y^3	0	1	1	1
4	X^4	1	0	1	1
4	X^3Y	0	0	1	2
4	X^2Y^2	0	0	1	1
4	XY^3	0	0	1	2
4	Y^4	0	1	1	1.

19.5 Concrete Reading of the Table

For $d = 4$, the section $X^3Y \otimes (q + 2p)$ has:

P	germ	value
P_∞	$ue_X^{(4)} \otimes (q + 2p)$	0
P_0	$v^3e_Y^{(4)} \otimes (q + 2p)$	0
P_1	$v^3e_Y^{(4)} \otimes (q + 2p)$	$q + 2p$
P_2	$v^3e_Y^{(4)} \otimes (q + 2p)$	$2q + p.$

The P_2 value is $2(q + 2p) = 2q + 4p = 2q + p$ in $\mathbb{F}_3$.

Thus a stalk is the correct local object if one wants to discuss the field near a point. The finite Weyl label construction samples the residue/fiber values, not the whole local stalk module.

20 Canonical Boson and Fermion Fields as Field-Label Tests

20.1 Status and Source Anchors

This shard is a kinematic comparison. It does not introduce dynamics, Hamiltonians, continuum limits, locality cones, or equations of motion. The question is only whether the field-label construction from Shard AQM-14 reproduces the standard finite-mode canonical fields.

The CCR and CAR ground truth is local; paths and hashes are registered in `references/canonical_fields/SOURCES.md`. The line anchors used here are: Derezinski's `derezinski.tex`, lines 1078–1135 for Weyl CCR, 1197–1284 for Schrodinger/Stone–von Neumann, 1437–1524 for CAR, 1612–1696 for finite CAR representations, 1803–1964 and 2098–2131 for Fock and creation operators, 2251–2301 for bosonic Fock CCR fields, and 2629–2662 for fermionic Fock CAR fields; Bekka's `CCR-GeneralRings-v5.tex`, lines 220–231, 253–273, 287–350, and 603–633; and Keyl–Schlingemann's `fidelity_arXiv.tex`, lines 253–260, 270–355, 411–492, and 496–531.

20.2 Lamport Dependency Skeleton

$$
\begin{aligned}
D1 &: X \text{ is a finite set of sites.}\\
D2 &: V_B = \mathbb{R}_q \oplus \mathbb{R}_p, \quad \omega_B((q,p),(q',p')) = pq' - p'q.\\
D3 &: E_B = \mathrm{Map}(X, V_B), \quad \Omega_B = \textstyle\sum_{x\in X} \omega_B.\\
D4 &: W_B : E_B \to U(\mathcal{H}) \text{ satisfies Weyl-form CCR.}\\
L1 &: (E_B, \Omega_B) \text{ is a real symplectic vector space.}\\
L2 &: \text{Irreducible regular } W_B \text{ is Schrodinger by Stone–von Neumann.}\\
T1 &: D1\text{–}D4 \text{ reproduce the finite standard bosonic field.}\\
D5 &: Z = \mathrm{Map}(X, K) \text{ is a finite complex one-particle space.}\\
D6 &: Y_F = \mathrm{Re}(Z \oplus \overline{Z}), \quad \alpha((z,\bar z),(w,\bar w)) = \mathrm{Re}\langle z, w\rangle.\\
D7 &: \Gamma_a(Z) = \textstyle\bigoplus_k \wedge^k Z, \quad \phi(z,\bar z) = a^*(z) + a(z).\\
L3 &: a, a^* \text{ satisfy the finite exterior-algebra CAR.}\\
L4 &: \{\phi(y), \phi(y')\} = 2\alpha(y, y').\\
T2 &: D5\text{–}D7 \text{ reproduce the finite standard fermionic field.}\\
L5 &: \text{Grassmann target replacement is not the same construction as } T2.
\end{aligned}
$$

20.3 Bosonic Field Labels D1–D4

Let X be finite. Write

$$\mathcal{Q}_X = \mathrm{Map}(X, \mathbb{R})$$

for the real configuration space. Its algebraic dual is identified with $\mathrm{Map}(X, \mathbb{R})$ by the finite sum

$$p(q) = \sum_{x\in X} p_x q_x.$$

Thus

$$E_B = \mathrm{Map}(X, \mathbb{R}_q \oplus \mathbb{R}_p) \cong \mathcal{Q}_X \oplus \mathcal{Q}_X^*.$$

For $\phi = (q,p)$ and $\psi = (q',p')$, define

$$\Omega_B(\phi, \psi) = \sum_{x\in X} \big(p_x q'_x - p'_x q_x\big).$$

A Weyl-form CCR representation of E_B is a map

$$W_B : E_B \to U(\mathcal{H})$$

such that

$$W_B(\phi)W_B(\psi) = \exp\left(-\frac{i}{2}\Omega_B(\phi,\psi)\right) W_B(\phi+\psi).$$

This is Derezinski's convention with $\eta = p$ and $q = q$.

Lemma $L1$. (E_B, Ω_B) is a finite-dimensional real symplectic vector space. *Proof.* Pointwise addition and scalar multiplication make E_B a real vector space. The formula for Ω_B is a finite sum of bilinear expressions, hence is bilinear. Also

$$\Omega_B(\phi,\phi) = \sum_x (p_x q_x - p_x q_x) = 0,$$

so it is alternating. If $\phi = (q,p) \neq 0$, then either some $q_{x_0} \neq 0$ or some $p_{x_0} \neq 0$. In the first case take ψ supported at x_0 with $p'_{x_0} = 1$ and $q'_{x_0} = 0$; then $\Omega_B(\phi,\psi) = -q_{x_0} \neq 0$. In the second case take ψ supported at x_0 with $q'_{x_0} = 1$ and $p'_{x_0} = 0$; then $\Omega_B(\phi,\psi) = p_{x_0} \neq 0$. Thus the radical is zero.

Lemma $L2$. Every irreducible regular representation of the Weyl-form CCR over (E_B, Ω_B) is unitarily equivalent to the Schrodinger representation on

$$L^2(\mathcal{Q}_X) \cong L^2(\mathbb{R}^{|X|}).$$

Proof. Choose an ordering of X. Then $\mathcal{Q}_X \cong \mathbb{R}^N$ for $N = |X|$, and Lemma $L1$ identifies E_B with $\mathcal{Q}_X^* \oplus \mathcal{Q}_X$ carrying the canonical symplectic form. Derezinski's finite-dimensional Stone–von Neumann theorem states that a regular CCR representation over such a finite symplectic space is a Schrodinger representation up to multiplicity, and irreducibility is exactly the one-copy case. Bekka's source records the same local-field uniqueness principle and the irreducibility of the Schrodinger pair.

Theorem $T1$. For finite X, the AQM-14 field-label recipe with real target V_B recovers the standard finite bosonic canonical field. *Proof.* The input is exactly $D1$–$D3$: a finite physical set, a target vector space with a scalar-valued alternating form, and a field-label space of maps. Lemma $L1$ checks that the summed pointwise form is symplectic. Definition $D4$ is the Weyl quantization step. By Lemma $L2$, the irreducible regular output is, up to unitary equivalence, the Schrodinger representation on $L^2(\mathbb{R}^{|X|})$. Derezinski's bosonic Fock source gives the equivalent finite-mode Fock realization by creation and annihilation operators. Therefore the finite standard bosonic field is a direct real analogue of the arithmetic finite-field Weyl construction, with $\mathbb{F}_q$ replaced by $\mathbb{R}$.

20.4 Fermionic Field Labels D5–D7

Now let K be a finite-dimensional complex internal one-particle Hilbert space and set

$$Z = \mathrm{Map}(X, K).$$

Its inner product is the finite sum

$$\langle z, w\rangle_Z = \sum_{x\in X} \langle z_x, w_x\rangle_K.$$

The real CAR label space is

$$Y_F = \mathrm{Re}(Z \oplus \overline{Z}) = \{(z, \bar{z}) : z \in Z\},$$

equipped with

$$\alpha((z,\bar{z}),(w,\bar{w})) = \mathrm{Re}\langle z,w\rangle_Z.$$

The fermionic Hilbert space is the antisymmetric Fock space

$$\Gamma_a(Z) = \bigoplus_{k=0}^{\dim Z} \wedge^k Z.$$

For $z \in Z$, define creation by $a^*(z)\Psi = z \wedge \Psi$. Define annihilation $a(z)$ as the adjoint of $a^*(z)$, equivalently on a decomposable wedge by

$$a(z)(w_1 \wedge \cdots \wedge w_k) = \sum_{j=1}^{k} (-1)^{j-1} \langle z, w_j\rangle\, w_1 \wedge \cdots \widehat{w_j} \cdots \wedge w_k.$$

The real fermion field operator is

$$\phi(z,\bar{z}) = a^*(z) + a(z).$$

Lemma $L3$. The exterior creation and annihilation operators satisfy

$$\{a^*(z), a^*(w)\} = 0, \qquad \{a(z), a(w)\} = 0, \qquad \{a(z), a^*(w)\} = \langle z,w\rangle I.$$

Proof. The first identity is the antisymmetry of the wedge product:

$$z \wedge w \wedge \Psi + w \wedge z \wedge \Psi = 0.$$

The second follows from the displayed contraction formula, since deleting two vectors in opposite orders gives opposite signs. For the mixed relation, use the contraction identity

$$a(z)(w \wedge \Psi) = \langle z,w\rangle \Psi - w \wedge a(z)\Psi.$$

Therefore

$$\begin{aligned} a(z)a^*(w)\Psi + a^*(w)a(z)\Psi &= a(z)(w \wedge \Psi) + w \wedge a(z)\Psi \\ &= \langle z,w\rangle \Psi. \end{aligned}$$

This proves the three CAR identities.

Lemma $L4$. For $y = (z,\bar{z})$ and $y' = (w,\bar{w})$,

$$\{\phi(y), \phi(y')\} = 2\alpha(y,y')I.$$

Proof. Expand the anticommutator and apply Lemma $L3$:

$$\begin{aligned} \{\phi(y), \phi(y')\} &= \{a^*(z) + a(z), a^*(w) + a(w)\} \\ &= \langle z,w\rangle I + \langle w,z\rangle I \\ &= 2\,\mathrm{Re}\langle z,w\rangle I = 2\alpha(y,y')I. \end{aligned}$$

Theorem $T2$. For finite X, the standard finite fermionic canonical field is the CAR construction $D5$–$D7$, not a Weyl-Heisenberg quantization of a Grassmann-valued target.

Proof. Derezinski's CAR definition uses a real Hilbert label space with positive scalar product and field operators satisfying an anticommutator relation. Definitions $D5$–$D7$ instantiate this with $Z = \mathrm{Map}(X,K)$. Lemma $L4$ proves the CAR relation on the antisymmetric Fock space, and Derezinski's Fock source identifies this as the fermionic Fock representation of the CAR. Thus the finite fermionic field arises from the same first step "choose site fields", but the quantization functor changes: symmetric symplectic/Weyl data give bosons, while positive-symmetric Clifford/CAR data give fermions.

20.5 Grassmann Target Test L5

Lemma *L*5. Replacing V_B by a Grassmann algebra is not a direct construction of the standard finite fermion field in the AQM-14 sense.

Proof. The AQM-14 Weyl construction requires an ordinary vector-space label group, a scalar-valued alternating form, and a projective unitary representation whose cocycle is a complex phase or finite-field character. The standard fermion field in $T2$ instead satisfies the CAR relation $\{\phi(y), \phi(y')\} = 2\alpha(y, y')I$, with a symmetric positive scalar product. This is already a different algebraic input.

Keyl–Schlingemann show that Grassmann variables do enter fermionic phase-space methods inside GAR. Cahill–Glauber give the sharper finite-mode object: Grassmann-valued displacement operators built from concrete CAR generators. Their multiplier is valued in the even Grassmann algebra, not generally in the complex phases. Therefore this is not, without extending the scalar algebra, the same as the ordinary projective unitary representation of the additive group $\mathrm{Map}(X, V)$ used in the bosonic and arithmetic Weyl constructions. The naive target-replacement conjecture is rejected only in that strict scalar-projective sense.

20.6 Current Dictionary

The comparison is therefore compact:

- finite boson: labels $\mathrm{Map}(X, \mathbb{R}_q \oplus \mathbb{R}_p)$, alternating symplectic form, CCR/Weyl quantization;
- finite arithmetic field: labels $E \leq \mathrm{Map}(X, V_{\mathbb{F}_q})$, alternating symplectic form after radical reduction, finite Weyl quantization;
- finite fermion: labels $\mathrm{Re}(Z \oplus \overline{Z})$ with $Z = \mathrm{Map}(X, K)$, positive symmetric form, CAR/Clifford quantization.

Thus the bosonic conjecture is checked at finite kinematic level. The fermionic conjecture is repaired: standard fermions first enter this programme through CAR/Clifford data on an antisymmetric Fock space, and Grassmann variables then give the Cahill–Glauber displacement calculus in the Grassmann-extended CAR algebra. Section 21 proves that positive representation theorem and records the ordinary complex-Hilbert-space caveat.

21 Grassmann-Weyl Fermion Displacements

21.1 Status and Source Anchors

This shard corrects the Grassmann question left open by Section 20. The intended object is not a homomorphism sending a Grassmann generator to a CAR annihilator. The intended object is the fermionic displacement operator of Cahill–Glauber:

$$D(\gamma) = \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i)\right),$$

where the $a_i, a_i^\dagger$ are concrete CAR operators and the γ_i, γ_i^* are Grassmann coefficients.

The canonical local source is `references/canonical_fields/PhysRevA.59.1538.pdf`, converted with `marker` to `references/canonical_fields/physrevA_59_1538_marker/PhysRevA.59.1538/PhysRevA.59.1538.md`. The marker line anchors are: lines 37–45 for the fermion CAR and vacuum, 49–66 for Grassmann variables, cross anticommutation, and the adjoint convention; 72–81 for the definition and finite expansion of $D(\gamma)$; 83–101 for the displacement of a_n and $a_n^\dagger$; 103–133 for normal ordering and the ray-composition law; 350–362 for displacement operator completeness; and 364–368 for the caveat that physical quantum operators themselves do not involve Grassmann variables.

21.2 Lamport Dependency Skeleton

$D1$	:	I is a finite mode set.
$D2$	:	$\mathrm{CAR}(I)$ is generated by odd $a_i, a_i^\dagger$ with $\{a_i, a_j^\dagger\} = \delta_{ij}$.
$D3$	:	Λ_I is generated by odd γ_i, γ_i^* with all Grassmann anticommutators zero.
$D4$	:	$\mathcal{A}_I = \mathrm{CAR}(I)\widehat{\otimes}\Lambda_I$ is the graded tensor product.
$D5$	:	$\sigma(\alpha, \beta) = \sum_i (\beta_i^* \alpha_i - \alpha_i^* \beta_i)$.
$L1$	:	σ is an even central skew-adjoint Grassmann cocycle form.
$D6$	:	D_I is the abstract Grassmann displacement ray algebra.
$D7$	:	$K(\gamma) = \sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i), \quad D(\gamma) = e^{K(\gamma)}$.
$L2$	:	$K(\gamma)^\dagger = -K(\gamma)$, hence $D(\gamma)$ is unitary.
$L3$	:	$D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n$ and similarly for $a_n^\dagger$.
$L4$	:	$[K(\alpha), K(\beta)] = \sigma(\alpha, \beta)$.
$T1$	:	$\gamma \mapsto D(\gamma)$ is a unitary representation of D_I in $\mathcal{A}_I$.
$T2$	:	A faithful ordinary complex-scalar Hilbert representation of this Grassmann cocycle is not the theorem.

21.3 The Concrete Target Algebra D1–D4

Let $I = \{1, \dots, N\}$. The finite CAR algebra $\mathrm{CAR}(I)$ is the complex unital $*$-algebra generated by odd symbols $a_i, a_i^\dagger$, $i \in I$, subject to

$$\{a_i, a_j^\dagger\} = \delta_{ij} 1, \qquad \{a_i, a_j\} = 0, \qquad \{a_i^\dagger, a_j^\dagger\} = 0.$$

Here $\{x, y\} = xy + yx$. The adjoint satisfies $(a_i)^\dagger = a_i^\dagger$ and $(a_i^\dagger)^\dagger = a_i$.

Let Λ_I be the Grassmann $*$-algebra generated by odd symbols γ_i, γ_i^*, $i \in I$, with

$$\{\gamma_i, \gamma_j\} = 0, \qquad \{\gamma_i^*, \gamma_j\} = 0, \qquad \{\gamma_i^*, \gamma_j^*\} = 0.$$

The symbols γ_i and γ_i^* are independent generators related by the chosen adjoint. Following the source, adjunction reverses the order of all odd quantities:

$$(x_1 x_2 \cdots x_k)^\dagger = x_k^\dagger \cdots x_2^\dagger x_1^\dagger$$

when the x_r are fermionic operators or Grassmann variables.

Define the Grassmann-extended CAR algebra

$$\mathcal{A}_I = \mathrm{CAR}(I) \widehat{\otimes} \Lambda_I$$

as the unital $*$-algebra generated by the two preceding sets, with the additional cross anticommutation rules

$$\{\gamma_i, a_j\} = \{\gamma_i, a_j^\dagger\} = \{\gamma_i^*, a_j\} = \{\gamma_i^*, a_j^\dagger\} = 0.$$

An element is even if it is a sum of monomials with even total fermionic degree, and odd if it has odd total degree. Even Grassmann elements commute with all CAR generators. This is the algebra in which Cahill–Glauber's unitary displacement operators live.

21.4 The Abstract Ray Algebra D5–D6

A Grassmann displacement parameter is an I-tuple $\gamma = (\gamma_i)_{i \in I}$ of odd Grassmann elements, with adjoint tuple $\gamma^* = (\gamma_i^*)_{i \in I}$. The additive law is componentwise.

For two parameters α, β, define the even Grassmann form

$$\sigma(\alpha, \beta) = \sum_{i \in I} (\beta_i^* \alpha_i - \alpha_i^* \beta_i)$$

and its exponential

$$c(\alpha, \beta) = \exp\left(\frac{1}{2}\sigma(\alpha, \beta)\right).$$

The exponential is a finite polynomial because every Grassmann monomial is nilpotent.

Lemma *L*1. σ is even, central in $\mathcal{A}_I$, skew-adjoint, and gives a multiplicative two-cocycle c:

$$c(\alpha, \beta) c(\alpha + \beta, \delta) = c(\beta, \delta) c(\alpha, \beta + \delta).$$

Proof. Each summand $\beta_i^* \alpha_i - \alpha_i^* \beta_i$ is a product of two odd Grassmann elements, hence is even. Even Grassmann elements commute with all Grassmann generators and with all CAR generators, so σ is central. The adjoint convention gives

$$(\beta_i^* \alpha_i)^\dagger = \alpha_i^* \beta_i, \qquad (\alpha_i^* \beta_i)^\dagger = \beta_i^* \alpha_i,$$

so $\sigma(\alpha, \beta)^\dagger = -\sigma(\alpha, \beta)$. Bilinearity gives

$$\begin{aligned} &\sigma(\alpha, \beta) + \sigma(\alpha + \beta, \delta) \\ &\quad = \sigma(\alpha, \beta) + \sigma(\alpha, \delta) + \sigma(\beta, \delta) \\ &\quad = \sigma(\beta, \delta) + \sigma(\alpha, \beta + \delta). \end{aligned}$$

Because the exponents are central, exponentials multiply by adding exponents. This proves the displayed two-cocycle identity.

Define the abstract Grassmann displacement ray algebra D_I to be the unital $*$-algebra over the even Grassmann scalars generated by symbols $\mathsf{D}(\gamma)$, one for each displacement parameter, with relations

$$\mathsf{D}(0) = 1, \qquad \mathsf{D}(\alpha)\mathsf{D}(\beta) = \mathsf{D}(\alpha + \beta) c(\alpha, \beta), \qquad \mathsf{D}(\gamma)^\dagger = \mathsf{D}(-\gamma).$$

Lemma *L*1 is exactly the associativity check for the product relation.

21.5 The Cahill–Glauber Representation D7–T1

For a parameter γ, define

$$K(\gamma) = \sum_{i \in I} (a_i^\dagger \gamma_i - \gamma_i^* a_i), \qquad D(\gamma) = e^{K(\gamma)} \in \mathcal{A}_I.$$

Each term $a_i^\dagger \gamma_i$ and $\gamma_i^* a_i$ is even, so $K(\gamma)$ is even and the ordinary commutator is the relevant bracket.

Theorem $T1$: representation to be proved. The assignment

$$\Pi : \mathsf{D}_I \longrightarrow \mathcal{A}_I^\times, \qquad \Pi(\mathsf{D}(\gamma)) = D(\gamma) = \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i) \right),$$

is a unitary representation of the abstract Grassmann displacement ray algebra in the Grassmann-extended finite CAR algebra. Its product law is

$$D(\alpha)D(\beta) = D(\alpha + \beta) \exp\left(\frac{1}{2} \sum_i (\beta_i^* \alpha_i - \alpha_i^* \beta_i) \right).$$

21.6 Conclusion

The unitary representation of the abstract fermion displacement objects is:

$$\boxed{\mathsf{D}(\gamma) \mapsto \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i) \right)}$$

inside the Grassmann-extended finite CAR algebra. It is unitary, it displaces the concrete CAR generators by $a_i \mapsto a_i + \gamma_i$ and $a_i^\dagger \mapsto a_i^\dagger + \gamma_i^*$, and it obeys the Grassmann-valued ray product of Cahill–Glauber. The derivations of unitarity, the displacement identities, the product law, and the ordinary Hilbert-space boundary are in Section 22.

22 Grassmann-Weyl Fermion Displacement Derivations

22.1 Status and Dependencies

This shard proves the Lamport items $L2$–$T2$ stated in Section 21. It uses the same Cahill–Glauber source anchors: marker lines 72–81 for $D(\gamma)$, lines 83–101 for the displacement identities, and lines 103–133 for the Baker–Hausdorff and ray-composition formula.

$D1$–$D7$	:	definitions from Section 21.
$L1$	:	σ is the central Grassmann two-cocycle form.
$L2$	:	$K(\gamma)^\dagger = -K(\gamma)$, so $D(\gamma)$ is unitary.
$L3$	:	$D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n$ and similarly for $a_n^\dagger$.
$L4$	:	$[K(\alpha), K(\beta)] = \sigma(\alpha, \beta)$.
$T1$	:	$D(\gamma)$ represents the abstract displacement ray algebra.
$T2$	:	The theorem is Grassmann-extended CAR-unitary, not ordinary scalar-projective Hilbert-unitary.

22.2 Unitarity L2

Lemma $L2$.

$$K(\gamma)^\dagger = -K(\gamma), \qquad D(\gamma)^\dagger D(\gamma) = D(\gamma)D(\gamma)^\dagger = 1.$$

Proof. Using the adjoint convention from $D3$,

$$(a_i^\dagger \gamma_i)^\dagger = \gamma_i^* a_i, \qquad (\gamma_i^* a_i)^\dagger = a_i^\dagger \gamma_i.$$

Therefore

$$K(\gamma)^\dagger = \sum_i (\gamma_i^* a_i - a_i^\dagger \gamma_i) = -K(\gamma).$$

Hence $D(\gamma)^\dagger = e^{-K(\gamma)} = D(-\gamma)$, and multiplying $e^{-K(\gamma)}e^{K(\gamma)}$ or $e^{K(\gamma)}e^{-K(\gamma)}$ gives 1.

22.3 Displacement of the CAR Generators L3

Lemma $L3$. For every $n \in I$,

$$D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n, \qquad D(\gamma)^\dagger a_n^\dagger D(\gamma) = a_n^\dagger + \gamma_n^*.$$

Proof. The conjugation identity $e^{-K}Be^{K} = B + [B,K] + \frac{1}{2}[[B,K],K] + \cdots$ applies because K is even. For $i \neq n$, the i-summand of $K(\gamma)$ commutes with a_n: the sign from moving a_n through the CAR generator is cancelled by the sign from moving it through the Grassmann coefficient. For $i = n$,

$$\begin{aligned}
[a_n, a_n^\dagger \gamma_n - \gamma_n^* a_n] &= a_n a_n^\dagger \gamma_n - a_n^\dagger \gamma_n a_n \\
&= (1 - a_n^\dagger a_n)\gamma_n - a_n^\dagger \gamma_n a_n \\
&= \gamma_n - a_n^\dagger a_n \gamma_n + a_n^\dagger a_n \gamma_n \\
&= \gamma_n.
\end{aligned}$$

The element γ_n commutes with $K(\gamma)$, so all higher commutators vanish. Thus $D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n$. The same calculation with $a_n^\dagger$ gives

$$\begin{aligned}
[a_n^\dagger, a_n^\dagger \gamma_n - \gamma_n^* a_n] &= -a_n^\dagger \gamma_n^* a_n + \gamma_n^* a_n a_n^\dagger \\
&= \gamma_n^* a_n^\dagger a_n + \gamma_n^*(1 - a_n^\dagger a_n) \\
&= \gamma_n^*,
\end{aligned}$$

and again higher commutators vanish.

22.4 The Grassmann Commutator L4

Lemma *L*4. For all parameters α, β,

$$[K(\alpha), K(\beta)] = \sigma(\alpha, \beta).$$

Proof. It is enough to compute one mode, since different modes commute after the same double sign cancellation used in Lemma $L3$. Put

$$p = a^\dagger \alpha, \quad q = \alpha^* a, \quad r = a^\dagger \beta, \quad s = \beta^* a.$$

Then $K(\alpha) = p - q$ and $K(\beta) = r - s$. The products pr, rp, qs, sq vanish because $(a^\dagger)^2 = a^2 = 0$. The mixed commutators are

$$\begin{aligned}
[p, s] &= a^\dagger \alpha \beta^* a - \beta^* a a^\dagger \alpha \\
&= -\beta^* \alpha\, a^\dagger a - \beta^* (1 - a^\dagger a) \alpha \\
&= -\beta^* \alpha,
\end{aligned}$$

and

$$\begin{aligned}
[q, r] &= \alpha^* a a^\dagger \beta - a^\dagger \beta \alpha^* a \\
&= \alpha^* (1 - a^\dagger a) \beta + \alpha^* \beta\, a^\dagger a \\
&= \alpha^* \beta.
\end{aligned}$$

Therefore

$$[K(\alpha), K(\beta)] = [p - q, r - s] = -[p, s] - [q, r] = \beta^* \alpha - \alpha^* \beta.$$

Summing over $i \in I$ gives the displayed formula.

22.5 Representation Theorem T1

Theorem *T*1. The assignment

$$\Pi(\mathsf{D}(\gamma)) = \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i) \right)$$

is a unitary representation of the abstract Grassmann displacement ray algebra in the Grassmann-extended finite CAR algebra.

Proof. Lemma $L2$ proves unitarity and the $*$-relation $D(\gamma)^\dagger = D(-\gamma)$. Lemma $L4$ says that the commutator $[K(\alpha), K(\beta)]$ is the central element $\sigma(\alpha, \beta)$. The Baker–Hausdorff identity for central commutator gives

$$\begin{aligned}
e^{K(\alpha)} e^{K(\beta)} &= e^{K(\alpha)+K(\beta)} e^{\frac{1}{2}[K(\alpha), K(\beta)]} \\
&= e^{K(\alpha+\beta)} e^{\frac{1}{2}\sigma(\alpha,\beta)} \\
&= D(\alpha + \beta) c(\alpha, \beta).
\end{aligned}$$

This is exactly the defining product relation of D_I. Lemma $L1$ supplies associativity of the abstract relation, so Π is a well-defined representation.

22.6 Boundary of the Statement T2

Theorem $T2$. The positive theorem $T1$ is not an ordinary projective representation $\gamma \mapsto U_\gamma \in U(\mathcal{H})$ with complex $U(1)$ multiplier. It is a unitary representation in $\mathcal{A}_I$, the CAR algebra extended by Grassmann coefficients.

Proof. The multiplier in $T1$ is

$$c(\alpha, \beta) = \exp\left(\frac{1}{2}\sum_i (\beta_i^* \alpha_i - \alpha_i^* \beta_i)\right),$$

which is an even Grassmann unit. Cahill–Glauber's notation records that no nonzero Grassmann monomial is an ordinary complex number, and it later separates physical quantum operators from Grassmann variables. Therefore unless the Grassmann coefficient algebra is included in the target, the multiplier is not a complex phase.

More explicitly, if $\rho : \Lambda_I \to B(\mathcal{H})$ is a Hilbert-space $*$-representation of the Grassmann coefficient algebra, then $\rho(\gamma_i) = 0$ for each i. Indeed, with $b = \rho(\gamma_i)$,

$$b^2 = 0, \qquad bb^\dagger + b^\dagger b = 0,$$

and hence for every $v \in \mathcal{H}$,

$$0 = \langle v, (bb^\dagger + b^\dagger b)v \rangle = \|b^\dagger v\|^2 + \|bv\|^2,$$

so $b = 0$. Thus the ordinary complex Hilbert representation collapses the Grassmann cocycle. The representation found in $T1$ avoids this collapse by keeping the Grassmann variables as formal odd coefficients anticommuting with the concrete CAR generators.

23 Prime-Field Arithmetic Fields and Qudit Paulis

23.1 Status and Source Anchors

This shard stays below schemes and sheaves. The only commutative rings are prime fields $\mathbb{F}_p$. The local Pauli/Weyl sources are Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313 for T, R, their product, and the tensor product rule; Gottesman's thesis, `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 1627–1681 for qudits, nice error bases, phases, and $C^a D^b$ commutation; and Bombin–Martin–Delgado, `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 1208–1259 for qudit Paulis, the symplectic commutator, the Pauli group, and the projection to $\mathbb{Z}_D^{2n}$.

23.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & p \text{ is prime and } R = \mathbb{F}_p. \\
L1 & : & R \text{ has one prime ideal and } \operatorname{Frac}(R) = R. \\
D2 & : & X_{\mathrm{sp}} = \operatorname{Spec} R, \quad X_{\mathrm{el}} = R \text{ as a finite set.} \\
D3 & : & V = R_X \oplus R_Z, \quad \omega_V((x,z),(x',z')) = zx' - xz'. \\
D4 & : & E_{\mathrm{el}} = \operatorname{Map}(X_{\mathrm{el}}, V). \\
D5 & : & \Omega((\xi,\zeta),(\xi',\zeta')) = \sum_{u \in X_{\mathrm{el}}} (\zeta(u)\xi'(u) - \xi(u)\zeta'(u)). \\
D6 & : & \mathcal{H}_X = \ell^2(R^{X_{\mathrm{el}}}), \quad W(\xi,\zeta) = T_\xi R_\zeta. \\
L2 & : & (V,\omega_V) \text{ and } (E_{\mathrm{el}},\Omega) \text{ are symplectic.} \\
L3 & : & W(\xi,\zeta)W(\xi',\zeta') = \chi(\zeta\cdot\xi')W(\xi+\xi',\zeta+\zeta'). \\
T1 & : & \langle \chi(c)I, W(E_{\mathrm{el}})\rangle \text{ is the standard odd-}p \text{ qudit Pauli group on } p \text{ qudits.} \\
T2 & : & \text{Stabilizers require an extra isotropic label subspace and phases.} \\
L4 & : & p = 2 \text{ has the same projective labels but needs the usual } \mu_4 \text{ Pauli phases for Hermitian } Y.
\end{array}
$$

23.3 Prime Field Points L1–D2

Lemma *L*1. The ring $R = \mathbb{F}_p$ has exactly one prime ideal, namely (0), and $\operatorname{Frac}(R) = R$.

Proof. A field has only two ideals, (0) and the whole field. The whole field is not prime because prime ideals are proper. The zero ideal is prime because the quotient $R/(0) = R$ is an integral domain. Thus $\operatorname{Spec} R = \{(0)\}$. The fraction field consists of symbols a/b with $b \neq 0$, and the map $a/b \mapsto ab^{-1}$ is a well-defined field isomorphism $\operatorname{Frac}(R) \cong R$.

There are therefore two concrete point sets in play. The spectrum

$$X_{\mathrm{sp}} = \operatorname{Spec} \mathbb{F}_p$$

has one point. The underlying set of field elements

$$X_{\mathrm{el}} = \mathbb{F}_p$$

has p points. The one-point choice gives one qudit. The p-point choice is the one that can give the p-qudit Pauli group.

23.4 The All-Map Arithmetic Field D3–L2

Let

$$V = R_X \oplus R_Z$$

with symplectic form
$$\omega_V((x,z),(x',z')) = zx' - xz'.$$
For $X = X_{\mathrm{el}}$, define
$$E_{\mathrm{el}} = \mathrm{Map}(X,V) \cong R^X \oplus R^X.$$
Write an element of E_{el} as (ξ,ζ), where $\xi,\zeta : X \to R$. Define
$$\Omega((\xi,\zeta),(\xi',\zeta')) = \sum_{u\in X} \big(\zeta(u)\xi'(u) - \xi(u)\zeta'(u)\big).$$

Lemma *L*2. (V,ω_V) and (E_{el},Ω) are symplectic R-vector spaces. Moreover $E_{\mathrm{el}} \cong R^p \oplus R^p$.

Proof. The form ω_V is bilinear. It is alternating because
$$\omega_V((x,z),(x,z)) = zx - xz = 0.$$
If $(x,z) \neq 0$, then $\omega_V((x,z),(1,0)) = z$ and $\omega_V((x,z),(0,1)) = -x$, so some pairing is nonzero. Thus V is symplectic.

Since X has p elements, choosing any ordering of X identifies R^X with R^p. Hence $E_{\mathrm{el}} \cong R^p \oplus R^p$. The form Ω is the direct sum of the pointwise symplectic forms. If $(\xi,\zeta) \neq 0$, choose u_0 with $(\xi(u_0),\zeta(u_0)) \neq 0$ and use the previous paragraph to choose a value at u_0 pairing nontrivially, with all other values zero. Thus Ω is nondegenerate and alternating.

Map versus Hom. The p-qudit statement uses all maps. If instead one chooses the structure-preserving space
$$\mathrm{Hom}_R(R,V),$$
where R is viewed as a one-dimensional R-vector space, then evaluation at 1 gives $\mathrm{Hom}_R(R,V) \cong V$. That smaller choice is the one-qudit label space, not the p-qudit label space.

23.5 Weyl Operators L3

Let $\chi(t) = \exp(2\pi i\,\tilde{t}/p)$, where $\tilde{t} \in \{0,\dots,p-1\}$ represents $t \in R$. On the Hilbert space
$$\mathcal{H}_X = \ell^2(R^X),$$
with basis $\{|s\rangle : s : X \to R\}$, define
$$T_\xi|s\rangle = |s+\xi\rangle, \qquad R_\zeta|s\rangle = \chi\left(\sum_{u\in X} \zeta(u)s(u)\right)|s\rangle,$$
and
$$W(\xi,\zeta) = T_\xi R_\zeta.$$

Lemma *L*3. For $e = (\xi,\zeta)$ and $f = (\xi',\zeta')$,
$$W(e)W(f) = \chi(\zeta\cdot\xi')W(e+f),$$
where $\zeta\cdot\xi' = \sum_u \zeta(u)\xi'(u)$. Hence
$$W(e)W(f) = \chi(\Omega(e,f))W(f)W(e).$$

Proof. The translation operators add and the phase operators add:

$$T_\xi T_{\xi'} = T_{\xi+\xi'}, \qquad R_\zeta R_{\zeta'} = R_{\zeta+\zeta'}.$$

Also

$$\begin{aligned} R_\zeta T_{\xi'}|s\rangle &= R_\zeta|s+\xi'\rangle \\ &= \chi(\zeta \cdot s + \zeta \cdot \xi')|s+\xi'\rangle \\ &= \chi(\zeta \cdot \xi')T_{\xi'}R_\zeta|s\rangle. \end{aligned}$$

Therefore

$$T_\xi R_\zeta T_{\xi'} R_{\zeta'} = \chi(\zeta \cdot \xi')T_{\xi+\xi'}R_{\zeta+\zeta'}.$$

Swapping e and f gives the commutator factor

$$\chi(\zeta \cdot \xi' - \zeta' \cdot \xi) = \chi(\Omega(e,f)).$$

23.6 Identification with Qudit Paulis T1–T2

For each $u \in X$, let $\delta_u \in R^X$ be the delta function. Define single-site operators

$$X_u = W(\delta_u, 0), \qquad Z_u = W(0, \delta_u).$$

Let $\mathcal{P}_p^\chi(X)$ be the group generated by all X_u, Z_u and the central character phases $\chi(c)I$, $c \in R$. For odd p, this is the standard prime-dimensional qudit Pauli group in the source convention: one p-dimensional qudit for each $u \in X$, shifts generated by X_u, and phases generated by Z_u. For $p = 2$, it is the unphased Weyl group whose projective labels are the usual qubit Pauli labels; the conventional Hermitian Y convention is recorded in $L4$.

Theorem $T1$**.** For $X = \mathbb{F}_p$ as a set of elements,

$$\langle \chi(c)I,\ W(e) : c \in R,\ e \in E_{\mathrm{el}}\rangle = \mathcal{P}_p^\chi(X).$$

Thus, for odd p, the all-map arithmetic quantum field on $X = \mathbb{F}_p$ with target $\mathrm{Frac}(\mathbb{F}_p)^2 = \mathbb{F}_p^2$ gives the ambient Gottesman/Ashikhmin–Knill p-qudit Pauli group for p qudits. For $p = 2$, the projective statement is the same and the full conventional Pauli phase convention is $L4$.

Proof. The operators $W(\delta_u, 0)$ and $W(0, \delta_u)$ are among the arithmetic field Weyl operators, so $\mathcal{P}_p^\chi(X)$ is contained in the generated arithmetic Weyl group. Conversely, for any (ξ, ζ),

$$T_\xi = \prod_{u \in X} X_u^{\xi(u)}, \qquad R_\zeta = \prod_{u \in X} Z_u^{\zeta(u)},$$

with any fixed site order, because different X_u's commute and different Z_u's commute. Thus $W(\xi, \zeta) = T_\xi R_\zeta$ lies in $\mathcal{P}_p^\chi(X)$. The two generated groups are equal.

Theorem $T2$**.** The arithmetic field datum gives the ambient Pauli/Weyl group. A stabilizer code is extra data: an isotropic label subspace $L \le E_{\mathrm{el}}$, plus a compatible choice of phases/eigenvalue character.

Proof. If $S \le \mathcal{P}_p^\chi(X)$ is Abelian, its image L in the projective label space E_{el} satisfies $\Omega(L, L) = 0$ by Lemma $L3$. Conversely, if p is odd and $L \le E_{\mathrm{el}}$ is isotropic, define

$$\widetilde{W}(\xi, \zeta) = \chi\left(\frac{1}{2}\,\zeta \cdot \xi\right) W(\xi, \zeta).$$

Using Lemma $L3$,

$$\widetilde{W}(e)\widetilde{W}(f) = \chi\left(\frac{1}{2}\Omega(e,f)\right)\widetilde{W}(e+f).$$

On an isotropic L, this becomes $\widetilde{W}(e)\widetilde{W}(f) = \widetilde{W}(e+f)$. Multiplying this lift by a character $L \to \mathbb{C}^\times$ changes the stabilizer eigenvalues. Thus the arithmetic field supplies the ambient labels and commutation form; the choice of stabilizer is a later choice of isotropic subspace and phases.

23.7 The p=2 Edge Case L4

For $p = 2$, the same label space E_{el}, the same Hilbert space, and the same commutator criterion

$$W(e)W(f) = (-1)^{\Omega(e,f)}W(f)W(e)$$

remain valid. The edge is the phase convention. There is no element $1/2 \in \mathbb{F}_2$, and the unphased operator $W(1,1) = XZ$ is not the Hermitian Pauli Y. The standard qubit Pauli group uses fourth roots of unity and the relabeling

$$\widehat{W}(\xi,\zeta) = i^{\sum_u \xi(u)\zeta(u)}W(\xi,\zeta),$$

so $\widehat{W}(\delta_u,\delta_u) = Y_u$. Hence $p = 2$ has the same projective arithmetic field labels and the same isotropy criterion, but the full conventional Pauli group has center μ_4, not only the character phases μ_2.

23.8 Conclusion

The conjecture is correct with one precision. If $X = \operatorname{Spec}\mathbb{F}_p$, the construction gives one qudit. If $X = \mathbb{F}_p$ is the underlying p-element set and $E = \mathrm{Map}(X, \mathbb{F}_p^2)$, the Weyl–Heisenberg arithmetic field gives exactly the standard odd-p qudit Pauli group, and for $p = 2$ gives the same projective Pauli labels with the usual fourth-root phase adjustment. In every prime case the projective labels are

$$E \cong \mathbb{F}_p^p \oplus \mathbb{F}_p^p.$$

Actual stabilizer codes are not automatic fields; they are isotropic subspaces of this field label space together with phase/eigenvalue choices.

24 Spec Z/6Z and Residue Qudit Factorisation

24.1 Status and Source Anchors

This shard records the morally preferred convention suggested by Section 23: for a field $\mathbb{F}_p$, the arithmetic point space is $\operatorname{Spec}\mathbb{F}_p$, so the local quantum system is one p-level qudit. For a non-field finite ring there is no single quotient field. The concrete replacement used here is one local phase space over each residue field.

The ring calculation for $\mathbb{Z}/6\mathbb{Z}$ is proved directly below. The Weyl/Pauli facts use the same local sources as AQM-22: Ashikhmin–Knill `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313 for finite-field shift/phase bases and product laws, and Bombin–Martin–Delgado `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrect tex`, lines 1208–1259 for qudit Paulis, symplectic commutators, and projective label maps.

24.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & R = \mathbb{Z}/6\mathbb{Z}. \\
L1 & : & R \cong \mathbb{F}_2 \times \mathbb{F}_3 \text{ by Chinese remainders.} \\
L2 & : & \operatorname{Spec} R = \{\mathfrak{p}_2 = (2), \mathfrak{p}_3 = (3)\}. \\
D2 & : & \kappa(\mathfrak{p}_2) = \mathbb{F}_2, \quad \kappa(\mathfrak{p}_3) = \mathbb{F}_3. \\
D3 & : & E_R = \kappa(\mathfrak{p}_2)^2 \oplus \kappa(\mathfrak{p}_3)^2. \\
D4 & : & \Omega_R = \Omega_2 \oplus \Omega_3, \text{ with each } \Omega_i \text{ the standard } zx' - xz'. \\
L3 & : & E_R \text{ is an orthogonal direct sum of local symplectic groups.} \\
D5 & : & \mathcal{H}_R = \ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3). \\
L4 & : & W_R(e_2, e_3) = W_2(e_2) \otimes W_3(e_3) \text{ has the product Weyl law.} \\
T1 & : & \operatorname{span}\{W_R(e) : e \in E_R\} \cong M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}). \\
T2 & : & \text{The Hilbert space is a tensor product, not } \mathbb{C}^2 \oplus \mathbb{C}^3.
\end{array}
$$

24.3 The Spectrum L1–D2

Let $R = \mathbb{Z}/6\mathbb{Z}$. Define

$$\Phi : R \longrightarrow \mathbb{F}_2 \times \mathbb{F}_3, \qquad \Phi(\bar{n}) = (\bar{n} \bmod 2, \bar{n} \bmod 3).$$

Lemma $L1$. Φ is a ring isomorphism.

Proof. It is a well-defined ring homomorphism because congruence modulo 6 implies congruence modulo 2 and modulo 3. Its kernel consists of residue classes divisible by both 2 and 3, hence by 6, so the kernel is zero. Both rings have six elements, so an injective map is bijective.

Lemma $L2$.

$$\operatorname{Spec} R = \{\mathfrak{p}_2, \mathfrak{p}_3\}, \qquad \mathfrak{p}_2 = (2) = \{0, 2, 4\}, \quad \mathfrak{p}_3 = (3) = \{0, 3\}.$$

Moreover $R/\mathfrak{p}_2 \cong \mathbb{F}_2$ and $R/\mathfrak{p}_3 \cong \mathbb{F}_3$.

Proof. Under Φ, the ideal $\mathfrak{p}_2 = (2)$ maps to $0 \times \mathbb{F}_3$, and $\mathfrak{p}_3 = (3)$ maps to $\mathbb{F}_2 \times 0$. The quotients are therefore $\mathbb{F}_2$ and $\mathbb{F}_3$, respectively, so both ideals are prime.

Every ideal J of $\mathbb{F}_2 \times \mathbb{F}_3$ is $I_2 \times I_3$: if $(a, b) \in J$, then $(a, 0) = (a, b)(1, 0) \in J$ and $(0, b) = (a, b)(0, 1) \in J$. Thus the proper ideals are 0×0, $0 \times \mathbb{F}_3$, and $\mathbb{F}_2 \times 0$. The quotient by 0×0 is $\mathbb{F}_2 \times \mathbb{F}_3$, which is not a domain because $(1, 0)(0, 1) = 0$. Hence 0×0 is not prime, and the two ideals listed above are the only prime ideals.

The residue fields are

$$\kappa(\mathfrak{p}_2) = R/\mathfrak{p}_2 \cong \mathbb{F}_2, \qquad \kappa(\mathfrak{p}_3) = R/\mathfrak{p}_3 \cong \mathbb{F}_3.$$

24.4 Residue-Field Phase Space D3–L3

For a point $\mathfrak{p}$, define its local phase space by

$$V_{\mathfrak{p}} = \kappa(\mathfrak{p})_X \oplus \kappa(\mathfrak{p})_Z, \qquad \Omega_{\mathfrak{p}}((x,z),(x',z')) = zx' - xz'.$$

For $R = \mathbb{Z}/6\mathbb{Z}$, the global label group is

$$E_R = V_{\mathfrak{p}_2} \oplus V_{\mathfrak{p}_3} = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2.$$

This is a direct sum of finite abelian groups, not a vector space over one field. Define the alternating bicharacter

$$B_R(e,f) = (-1)^{\Omega_{\mathfrak{p}_2}(e_2,f_2)} \exp\left(\frac{2\pi i}{3}\Omega_{\mathfrak{p}_3}(e_3,f_3)\right),$$

where $e = (e_2, e_3)$ and $f = (f_2, f_3)$.

Lemma *L*3. The two local summands are orthogonal and nondegenerate: if $B_R(e,f) = 1$ for all $f \in E_R$, then $e = 0$.

Proof. Orthogonality is built into the formula: e_2 only pairs with f_2, and e_3 only pairs with f_3. The local form is nondegenerate because a nonzero (x,z) pairs nontrivially with either $(1,0)$ or $(0,1)$. If $B_R(e,f) = 1$ for all f, first take $f_3 = 0$, forcing $e_2 = 0$. Then take $f_2 = 0$, forcing $e_3 = 0$. Thus $e = 0$.

24.5 Weyl Representation L4–T1

Let

$$\mathcal{H}_2 = \ell^2(\mathbb{F}_2), \qquad \mathcal{H}_3 = \ell^2(\mathbb{F}_3), \qquad \mathcal{H}_R = \mathcal{H}_2 \otimes \mathcal{H}_3.$$

For $i = 2, 3$, let $W_i(x,z) = T_i(x)R_i(z)$ be the usual local shift-phase Weyl operator on $\ell^2(\mathbb{F}_i)$. Define

$$W_R(e_2, e_3) = W_2(e_2) \otimes W_3(e_3).$$

Lemma *L*4. For $e = (e_2, e_3)$ and $f = (f_2, f_3)$,

$$W_R(e)W_R(f) = c_R(e,f)W_R(e+f),$$

where

$$c_R(e,f) = (-1)^{z_2 x_2'} \exp\left(\frac{2\pi i}{3} z_3 x_3'\right)$$

if $e_i = (x_i, z_i)$ and $f_i = (x_i', z_i')$. Consequently

$$W_R(e)W_R(f) = B_R(e,f)W_R(f)W_R(e).$$

Proof. The local Weyl laws give

$$W_i(e_i)W_i(f_i) = c_i(e_i,f_i)W_i(e_i+f_i).$$

Tensoring the two equations gives the displayed product law with $c_R = c_2c_3$. Dividing the product law for e, f by the one for f, e gives the displayed commutator bicharacter.

Theorem $T1$**.**

$$\mathrm{span}_{\mathbb{C}}\{W_R(e) : e \in E_R\} = \mathrm{End}(\mathcal{H}_2) \otimes \mathrm{End}(\mathcal{H}_3) \cong M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}).$$

Proof. For a single p-level qudit, the local source records that the p^2 operators $T^x R^z$ form an operator basis. Thus $\{W_2(e_2)\}$ is a basis of $M_2(\mathbb{C})$, and $\{W_3(e_3)\}$ is a basis of $M_3(\mathbb{C})$. Tensor products of bases form a basis of the tensor product vector space, so the $4 \cdot 9 = 36$ operators $W_R(e_2, e_3)$ span $M_2(\mathbb{C}) \otimes M_3(\mathbb{C})$. The last isomorphism is the standard Kronecker product identification.

24.6 Direct Sum or Tensor Product? T2

Theorem $T2$**.** The quantum system attached to $\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$ by the residue-field Weyl rule is a qubit tensor a qutrit,

$$\mathcal{H}_R \cong \mathbb{C}^2 \otimes \mathbb{C}^3 \cong \mathbb{C}^6,$$

not the Hilbert direct sum $\mathbb{C}^2 \oplus \mathbb{C}^3$.

Proof. The direct sum occurs in the label group:

$$E_R = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2.$$

Weyl quantization sends an orthogonal direct sum of independent phase spaces to tensor products of the local Weyl representations, as shown explicitly in Lemma $L4$.

A Hilbert direct sum would have operator algebra

$$\mathrm{End}(\mathbb{C}^2) \oplus \mathrm{End}(\mathbb{C}^3) \cong M_2(\mathbb{C}) \oplus M_3(\mathbb{C}),$$

whose complex dimension is $4 + 9 = 13$ and whose center is two-dimensional. The Weyl operators from E_R instead give 36 independent operators and the full matrix algebra $M_6(\mathbb{C})$, whose center is one-dimensional. Thus the direct-sum Hilbert interpretation is incompatible with the Weyl label count and with Theorem $T1$.

24.7 Conclusion

The expectation "qutrit plus qubit" is correct only at the level of local factors. The precise statement is

$$\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z}) = \{\mathfrak{p}_2, \mathfrak{p}_3\}, \qquad \kappa(\mathfrak{p}_2) = \mathbb{F}_2, \quad \kappa(\mathfrak{p}_3) = \mathbb{F}_3,$$

so the local systems are one qubit and one qutrit. The global Weyl label space is their orthogonal direct sum, but the Hilbert space is their tensor product. Equivalently, the observable algebra is $M_2 \otimes M_3 \cong M_6$, not $M_2 \oplus M_3$.

25 Zariski Opens and Observable Nets

25.1 Status and Source Anchors

This shard stays at the concrete finite-spectrum level. The Zariski topology definitions are sourced from the Stacks Project snapshot `references/algebraic_geometry/stacks_project_algebra.tex`: the spectrum, $V(T)$, and $D(f)$ are defined in lines 2972–2983; the Zariski topology and standard opens are defined in lines 3104–3117. The finite Weyl local systems are those of AQM-22 and AQM-23.

The main conclusion is a convention choice. If the objective is a net whose inclusions follow inclusions of open sets, then an open U should carry the algebra acting nontrivially on U itself. The assignment $U \mapsto \mathcal{A}(X \setminus U)$ is also meaningful, but it is a closed-support or vanishing-locus assignment and its inclusions run in the opposite direction.

25.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & X = \operatorname{Spec} R, \quad V(T) = \{\mathfrak{p} : T \subset \mathfrak{p}\}, \quad D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}. \\
D2 & : & \text{For each } x \in X \text{ choose } \mathcal{H}_x = \ell^2(\kappa(x)),\ \mathcal{A}_x = \operatorname{End}(\mathcal{H}_x). \\
D3 & : & E(S) = \bigoplus_{x \in S} \kappa(x)^2, \quad \mathcal{H}(S) = \bigotimes_{x \in S} \mathcal{H}_x, \quad \mathcal{A}(S) = \bigotimes_{x \in S} \mathcal{A}_x. \\
L1 & : & S \subset T \Rightarrow \iota_{S,T} : \mathcal{A}(S) \hookrightarrow \mathcal{A}(T) \text{ is a functorial unital } *\text{-inclusion.} \\
L2 & : & \iota_{S,T} \text{ is the inclusion of Weyl algebras induced by } E(S) \hookrightarrow E(T). \\
T1 & : & U \mapsto \mathcal{A}(U) \text{ is the covariant open-local net.} \\
T2 & : & U \mapsto \mathcal{A}(X \setminus U) \text{ is contravariant in opens.} \\
E1 & : & X = \operatorname{Spec} \mathbb{F}_p \text{ has only } \emptyset \text{ and } X. \\
E2 & : & X = \operatorname{Spec}(\mathbb{Z}/6\mathbb{Z}) \text{ has two clopen points and algebras } M_2, M_3, M_6.
\end{array}$$

25.3 Definitions D1–D3

Let R be a commutative ring and let

$$X = \operatorname{Spec} R$$

be the set of prime ideals of R. For a subset $T \subset R$, define

$$V(T) = \{\mathfrak{p} \in X : T \subset \mathfrak{p}\}.$$

The Zariski closed subsets are the subsets $V(T)$. The open subsets are their complements. For $f \in R$, define the standard open

$$D(f) = \{\mathfrak{p} \in X : f \notin \mathfrak{p}\} = X \setminus V(\{f\}).$$

In this shard X is finite. Fix an ordering of X only to make tensor products strict. For $x = \mathfrak{p} \in X$, set

$$\mathcal{H}_x = \ell^2(\kappa(x)), \qquad \mathcal{A}_x = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_x),$$

where $\kappa(x)$ is the residue field at x. The local phase label group is $\kappa(x)^2$ with the standard alternating form

$$\omega_x((q,p),(q',p')) = pq' - qp'.$$

For any subset $S \subset X$, define

$$E(S) = \bigoplus_{x \in S} \kappa(x)^2, \qquad \mathcal{H}(S) = \bigotimes_{x \in S} \mathcal{H}_x, \qquad \mathcal{A}(S) = \bigotimes_{x \in S} \mathcal{A}_x.$$

The empty direct sum is 0, and the empty tensor product is $\mathbb{C}$.

25.4 Functorial Inclusions L1–L2

Lemma $L1$. If $S \subset T \subset X$, then

$$\iota_{S,T} : \mathcal{A}(S) \longrightarrow \mathcal{A}(T), \qquad a \longmapsto a \otimes \mathbf{1}_{\mathcal{H}(T\setminus S)}$$

is an injective unital $*$-homomorphism. If $S \subset T \subset U$, then

$$\iota_{T,U} \circ \iota_{S,T} = \iota_{S,U}.$$

Proof. The map preserves multiplication, adjoints, and the unit because tensoring with an identity operator preserves these operations. It is injective: if $a \otimes \mathbf{1} = 0$, then applying the operator to a simple tensor $v \otimes w$ with $w \neq 0$ gives $av \otimes w = 0$ for every v, hence $av = 0$ for every v, and $a = 0$. The composition identity is

$$(a \otimes \mathbf{1}_{T\setminus S}) \otimes \mathbf{1}_{U\setminus T} = a \otimes \mathbf{1}_{U\setminus S}.$$

Lemma $L2$. Let $W_S(e)$ denote the tensor product of local Weyl operators for $e \in E(S)$. Under the extension-by-zero inclusion $E(S) \hookrightarrow E(T)$,

$$\iota_{S,T}(W_S(e)) = W_T(e).$$

Consequently the finite Weyl-Heisenberg observable algebra generated by $E(S)$ is included in the one generated by $E(T)$.

Proof. For $x \in S$, the x-factor of $W_T(e)$ is the same local Weyl operator as in $W_S(e)$. For $x \in T \setminus S$, extension by zero gives the local label 0, and $W_x(0) = \mathbf{1}_{\mathcal{H}_x}$. Thus

$$W_T(e) = W_S(e) \otimes \mathbf{1}_{\mathcal{H}(T\setminus S)} = \iota_{S,T}(W_S(e)).$$

Taking complex spans gives the stated Weyl algebra inclusion.

25.5 Open-Local Versus Closed-Complement T1–T2

Theorem $T1$. The assignment

$$U \longmapsto \mathcal{A}_{\text{open}}(U) := \mathcal{A}(U)$$

from Zariski open subsets $U \subset X$ to finite-dimensional $*$-algebras is covariant under inclusion: if $U \subset V$, then

$$\mathcal{A}_{\text{open}}(U) \hookrightarrow \mathcal{A}_{\text{open}}(V)$$

by $\iota_{U,V}$. This is the natural choice when one wants observable algebras with inclusions following open inclusions.

Proof. An inclusion of opens $U \subset V$ is an inclusion of subsets of the finite set X. Lemma $L1$ gives the unital $*$-inclusion, and Lemma $L2$ identifies it with the corresponding inclusion of Weyl-Heisenberg algebras.

Theorem $T2$**.** The assignment

$$U \longmapsto \mathcal{A}_{\mathrm{cl}}(U) := \mathcal{A}(X \setminus U)$$

acts nontrivially only on the closed complement of U, but it is contravariant in opens. If $U \subset V$, then

$$X \setminus V \subset X \setminus U,$$

so the natural inclusion is

$$\mathcal{A}_{\mathrm{cl}}(V) \hookrightarrow \mathcal{A}_{\mathrm{cl}}(U).$$

Proof. The complement of an open set is closed by definition of open. Complementing the inclusion $U \subset V$ reverses it, giving $X \setminus V \subset X \setminus U$. Lemma $L1$ then gives the displayed inclusion. Thus this construction is valid as a closed-support or vanishing-locus algebra, but it is not the open-local net of Theorem $T1$.

25.6 Edge Case: A Field E1

Let $R = \mathbb{F}_p$. AQM-22 proves that

$$X = \operatorname{Spec} \mathbb{F}_p = \{\eta\}, \qquad \eta = (0).$$

The only Zariski closed subsets are $V(0) = X$ and $V(1) = \emptyset$, hence the only opens are $\emptyset$ and X. The open-local algebras are

$$\mathcal{A}_{\mathrm{open}}(\emptyset) = \mathbb{C}, \qquad \mathcal{A}_{\mathrm{open}}(X) = \operatorname{End}(\ell^2(\mathbb{F}_p)) \cong M_p(\mathbb{C}),$$

with the scalar inclusion $\lambda \mapsto \lambda \mathbf{1}$. The closed-complement assignment reverses these:

$$\mathcal{A}_{\mathrm{cl}}(\emptyset) = M_p(\mathbb{C}), \qquad \mathcal{A}_{\mathrm{cl}}(X) = \mathbb{C}.$$

Thus the one-point field case has no nontrivial intermediate localization; it only tests the direction of the convention.

25.7 The Two-Point Case $\mathbb{Z}/6\mathbb{Z}$ E2

Let $R = \mathbb{Z}/6\mathbb{Z}$. AQM-23 proves

$$X = \operatorname{Spec} R = \{\mathfrak{p}_2, \mathfrak{p}_3\}, \qquad \mathfrak{p}_2 = (2), \quad \mathfrak{p}_3 = (3),$$

with residue fields $\kappa(\mathfrak{p}_2) = \mathbb{F}_2$ and $\kappa(\mathfrak{p}_3) = \mathbb{F}_3$. Since $2 \in \mathfrak{p}_2$ and $2 \notin \mathfrak{p}_3$,

$$D(2) = \{\mathfrak{p}_3\}, \qquad V(2) = \{\mathfrak{p}_2\}.$$

Since $3 \notin \mathfrak{p}_2$ and $3 \in \mathfrak{p}_3$,

$$D(3) = \{\mathfrak{p}_2\}, \qquad V(3) = \{\mathfrak{p}_3\}.$$

The standard opens therefore contain both singletons, so the topology is discrete:

$$\emptyset, \quad \{\mathfrak{p}_2\} = D(3), \quad \{\mathfrak{p}_3\} = D(2), \quad X.$$

With the order $\mathfrak{p}_2, \mathfrak{p}_3$,

$$\mathcal{H}(X) = \mathbb{C}^2 \otimes \mathbb{C}^3, \qquad \mathcal{A}(X) = M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}).$$

The open-local and closed-complement algebras are:

U	$\mathcal{A}_{\mathrm{open}}(U) \subset \mathcal{A}(X)$	$\mathcal{A}_{\mathrm{cl}}(U) = \mathcal{A}(X \setminus U)$
$\emptyset$	$\mathbb{C}\,\mathbf{1}$	$M_2 \otimes M_3$
$D(3) = \{\mathfrak{p}_2\}$	$M_2 \otimes \mathbf{1}_3$	$\mathbf{1}_2 \otimes M_3$
$D(2) = \{\mathfrak{p}_3\}$	$\mathbf{1}_2 \otimes M_3$	$M_2 \otimes \mathbf{1}_3$
X	$M_2 \otimes M_3$	$\mathbb{C}\,\mathbf{1}$

Thus $D(3)$, the open where 3 does not vanish, carries the qubit algebra in the open-local convention and the qutrit algebra in the closed-complement convention. Likewise $D(2)$ carries the qutrit algebra open-locally and the qubit algebra as its closed complement.

Finally,

$$[M_2 \otimes \mathbf{1}_3, \mathbf{1}_2 \otimes M_3] = 0,$$

and the two singleton open algebras generate

$$(M_2 \otimes \mathbf{1}_3)(\mathbf{1}_2 \otimes M_3) = M_2 \otimes M_3.$$

This is the finite-spectrum analogue of locality: disjoint point supports commute, and the union of the supports generates the tensor-product algebra.

25.8 Conclusion

The best correspondence for the present programme is

$$\boxed{U \text{ open} \quad \rightsquigarrow \quad \mathcal{A}(U) \text{ acting nontrivially on } U.}$$

This gives isotony of observable algebras:

$$U \subset V \quad \Rightarrow \quad \mathcal{A}(U) \subset \mathcal{A}(V).$$

The alternative

$$U \rightsquigarrow \mathcal{A}(X \setminus U)$$

should be named explicitly as a closed-complement or vanishing-locus assignment. It is useful when one wants observables supported where a function vanishes, but its functorial direction is opposite to open inclusion.

26 Quasi-Local Zariski Algebra

26.1 Status and Source Anchors

This shard records the answer to the directed-system question left by AQM-24. The Zariski-topology source remains `references/algebraic_geometry/stacks_project_algebra.tex`: standard opens form a basis in lines 3104–3117, $\operatorname{Spec} R$ is quasi-compact in lines 3251–3273, and the topology has a quasi-compact open basis with quasi-compact intersections in lines 3275–3298.

No analytic completion convention is fixed here. The object constructed below is the algebraic quasi-local $*$-algebra, i.e. the filtered colimit of the local finite-region $*$-algebras. A later C^*-completion should be recorded as an additional convention.

26.2 Lamport Dependency Skeleton

$$\begin{array}{rcl} D1 & : & \text{A directed set is a preorder in which any two elements have a common upper bound.} \\ D2 & : & \mathsf{O}(X) = \{\text{Zariski opens of } X\}, \quad \mathsf{K}(X) = \{\text{quasi-compact Zariski opens of } X\}. \\ D3 & : & U \in \mathsf{K}(X) \mapsto \mathcal{A}(U) \text{ is the open-local algebra of AQM-24.} \\ L1 & : & \mathsf{O}(X) \text{ is directed under inclusion.} \\ L2 & : & \mathsf{K}(X) \text{ is directed under inclusion.} \\ T1 & : & (\mathcal{A}(U))_{U\in\mathsf{K}(X)} \text{ is the natural quasi-local directed system.} \\ T2 & : & \text{Closed complements are naturally indexed by closed supports.} \end{array}$$

26.3 Directed Region Posets D1–L2

A preorder I is called directed if for every $i, j \in I$ there exists $k \in I$ such that $i \leq k$ and $j \leq k$. A directed system of $*$-algebras over I is a family $(A_i)_{i\in I}$ with structure maps $\alpha_{ij} : A_i \to A_j$ for $i \leq j$, satisfying

$$\alpha_{ii} = \mathrm{id}, \qquad \alpha_{jk}\alpha_{ij} = \alpha_{ik}.$$

Let $X = \operatorname{Spec} R$. Write $\mathsf{O}(X)$ for the set of Zariski open subsets ordered by inclusion. Write $\mathsf{K}(X)$ for the quasi-compact open subsets ordered by inclusion. At the present concrete level, $\mathsf{K}(X)$ can be represented by finite unions of standard opens

$$D(f_1) \cup \cdots \cup D(f_n),$$

using the Stacks source cited above.

Lemma $L1$**.** $\mathsf{O}(X)$ is directed under inclusion.

Proof. If $U, V \subset X$ are open, then $U \cup V$ is open and contains both. Thus $U \cup V$ is a common upper bound.

Lemma $L2$**.** $\mathsf{K}(X)$ is directed under inclusion.

Proof. If U, V are quasi-compact opens, then $U \cup V$ is open. To show it is quasi-compact, let (W_a) be an open cover of $U \cup V$. It restricts to an open cover of U and to an open cover of V. Since U and V are quasi-compact, finitely many W_a's cover U, and finitely many W_a's cover V. The union of those two finite subfamilies covers $U \cup V$. Hence $U \cup V \in \mathsf{K}(X)$, and it is a common upper bound.

26.4 The Quasi-Local System T1

For $U \in \mathsf{K}(X)$, let $\mathcal{A}(U)$ be the open-local Weyl observable algebra from AQM-24. For $U \subset V$, use the inclusion

$$\iota_{U,V} : \mathcal{A}(U) \hookrightarrow \mathcal{A}(V), \qquad a \mapsto a \otimes \mathbf{1}_{V \setminus U}.$$

Theorem $T1$**.** $(\mathcal{A}(U), \iota_{U,V})_{U \in \mathsf{K}(X)}$ is a directed system of $*$-algebras. Its algebraic filtered colimit

$$\mathcal{A}_{\mathrm{qloc}}(X) := \varinjlim_{U \in \mathsf{K}(X)} \mathcal{A}(U)$$

is the natural Zariski quasi-local algebra for the present programme.

Proof. The index category is directed by Lemma $L2$. For every inclusion $U \subset V$, AQM-24 proves that $\iota_{U,V}$ is an injective unital $*$-homomorphism. The same lemma proves functoriality:

$$\iota_{V,W} \circ \iota_{U,V} = \iota_{U,W}$$

for $U \subset V \subset W$. These are exactly the axioms of a directed system. The displayed $\varinjlim$ is therefore well-defined as the algebraic colimit of this system.

26.5 Closed Supports T2

The closed-complement assignment from AQM-24 sends an open U to

$$\mathcal{A}(X \setminus U).$$

With opens ordered by inclusion, this assignment is contravariant. Therefore it is not the primary quasi-local system if "local region grows" is meant to match $U \subset V$.

The same construction becomes covariant if it is indexed directly by closed supports. Let $\mathsf{Z}(X)$ be the set of Zariski closed subsets ordered by inclusion. Since a finite union of closed subsets is closed, $\mathsf{Z}(X)$ is directed: $Z, Z' \subset Z \cup Z'$. The assignment

$$Z \longmapsto \mathcal{A}(Z)$$

with the same tensor-by-identity inclusions is then a directed system. This is best read as a closed-support or vanishing-locus quasi-local algebra, not as the open-local observable algebra.

26.6 Finite Checks

For $X = \operatorname{Spec} \mathbb{F}_p$, the space has a largest open X. Hence the directed system has a terminal object and

$$\mathcal{A}_{\mathrm{qloc}}(X) = \mathcal{A}(X) \cong M_p(\mathbb{C}).$$

For $X = \operatorname{Spec}(\mathbb{Z}/6\mathbb{Z})$, all opens are quasi-compact and X is again terminal. The quasi-local algebra is

$$\mathcal{A}_{\mathrm{qloc}}(X) = M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}).$$

The proper open-local subalgebras are the two commuting singleton algebras

$$M_2(\mathbb{C}) \otimes \mathbf{1}_3, \qquad \mathbf{1}_2 \otimes M_3(\mathbb{C}),$$

and their common upper bound is the full open X.

26.7 Conclusion

For the quasi-local algebra, the natural general convention is

$$\boxed{\mathcal{A}_{\mathrm{qloc}}(X) = \varinjlim_{U \in \mathsf{K}(X)} \mathcal{A}(U)}$$

with $\mathsf{K}(X)$ the quasi-compact Zariski opens and with inclusions following open inclusion. The closed-complement possibility is retained as a separate closed-support construction, useful for vanishing loci and defects, but not as the primary open-local quasi-local algebra.

27 Presheaves, Cosheaves, and Observable Nets

27.1 Status and Source Anchors

Stacks Project `references/algebraic_geometry/stacks_project_sheaves.tex` defines presheaves in lines 73–123, sheaves of sets in lines 500–525, and category-valued sheaves by an equalizer diagram in lines 719–736. It also records that $U \mapsto \bigoplus_{x\in U} M_x$ is generally not a sheaf in lines 622–637.

Curry's cosheaf source `references/cosheaves/curry_1303_3255_source/ch_abstract_sheaves.tex` defines pre-sheaves and pre-cosheaves in lines 89–92, sheaves and cosheaves by cover limits and colimits in lines 102–118, and rewrites the cosheaf condition as a coequalizer in lines 184–218. The vector-space exact sequence form is in lines 243–259. Curry's category source `references/cosheaves/curry_1303_3255_source/ch_categories.tex` defines colimits in lines 337–358 and records coproducts and unions as colimits in lines 366–384.

27.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & \mathsf{O}(X) \text{ is the category of open subsets ordered by inclusion.}\\
D2 & : & \text{A presheaf is a functor } \mathsf{O}(X)^{op} \to \mathcal{C}.\\
D3 & : & \text{A sheaf is a presheaf satisfying cover equalizer gluing.}\\
D4 & : & \text{A precosheaf is a functor } \mathsf{O}(X) \to \mathcal{C}.\\
D5 & : & \text{A cosheaf is a precosheaf satisfying cover coequalizer gluing.}\\
D6 & : & U \mapsto E(U) = \bigoplus_{x\in U} E_x \text{ is the label precosheaf.}\\
D7 & : & U \mapsto \mathcal{A}(U) \text{ is the open-local observable assignment.}\\
L1 & : & E \text{ satisfies the finite-cover cosheaf axiom in abelian groups.}\\
L2 & : & \mathcal{A} \text{ is a precosheaf of unital } *\text{-algebras.}\\
L3 & : & \mathcal{A} \text{ satisfies net additivity inside the ambient algebra.}\\
T1 & : & \mathcal{A} \text{ is not an ordinary cosheaf of unital } *\text{-algebras.}\\
T2 & : & \text{Quantization turns label cosheaf gluing into tensor-local net gluing.}
\end{array}
$$

27.3 Presheaves and Sheaves D1–D3

Let X be a topological space. The category $\mathsf{O}(X)$ has open sets $U \subset X$ as objects and a unique morphism $U \to V$ when $U \subset V$.

Let $\mathcal{C}$ be a category. A $\mathcal{C}$-valued presheaf on X is a contravariant functor

$$F : \mathsf{O}(X)^{op} \longrightarrow \mathcal{C}.$$

Equivalently, for $V \subset U$ it has restriction morphisms

$$\rho^U_V : F(U) \longrightarrow F(V)$$

with $\rho^U_U = \mathrm{id}$ and $\rho^U_W = \rho^V_W \rho^U_V$ for $W \subset V \subset U$.

Assume $\mathcal{C}$ has the products and equalizers needed for the displayed cover diagram. A presheaf F is a sheaf if for every open cover $U = \bigcup_i U_i$ the diagram

$$F(U) \longrightarrow \prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j)$$

is an equalizer. In sets this says that compatible local sections glue to a unique section over U.

27.4 Precosheaves and Cosheaves D4–D5

We use "precosheaf" and "copresheaf" synonymously. A $\mathcal{C}$-valued precosheaf on X is a covariant functor

$$G : \mathsf{O}(X) \longrightarrow \mathcal{C}.$$

For $V \subset U$ it has extension morphisms

$$r_{U,V} : G(V) \longrightarrow G(U)$$

with $r_{U,U} = \mathrm{id}$ and $r_{U,W} = r_{U,V} r_{V,W}$ for $W \subset V \subset U$.

Assume $\mathcal{C}$ has the coproducts and coequalizers needed for the displayed cover diagram. A precosheaf G is a cosheaf if for every open cover $U = \bigcup_i U_i$ the diagram

$$\coprod_{i,j} G(U_i \cap U_j) \rightrightarrows \coprod_i G(U_i) \longrightarrow G(U)$$

is a coequalizer. In an additive category such as abelian groups or vector spaces, the finite-cover form is the exact sequence

$$\bigoplus_{i,j} G(U_i \cap U_j) \longrightarrow \bigoplus_i G(U_i) \longrightarrow G(U) \longrightarrow 0.$$

The arrows identify data that have been counted twice on overlaps.

27.5 The Label Cosheaf D6–L1

At the finite residue-qudit level, each point $x \in X$ has a local Weyl label group $E_x = \kappa(x)^2$, regarded as an abelian group. Define

$$E(U) = \bigoplus_{x \in U} E_x.$$

For $V \subset U$, the extension map $E(V) \to E(U)$ is extension by zero outside V.

Lemma *L*1. For finite covers, E satisfies the cosheaf axiom in abelian groups.

Proof. Let $U = \bigcup_i U_i$ be a finite open cover. The map

$$\bigoplus_i E(U_i) \longrightarrow E(U)$$

adds the extension-by-zero images. It is surjective because every summand E_x with $x \in U$ lies in at least one U_i.

The only relations needed are overlap relations. If $x \in U_i \cap U_j$, then the two copies of E_x inside $E(U_i) \oplus E(U_j)$ have the same image in $E(U)$, and their difference is exactly in the image of $E(U_i \cap U_j)$. Conversely, if a finite sum maps to zero in $E(U)$, then for each point x the sum of its finitely many copies over the index set $\{i : x \in U_i\}$ is zero. This index set is connected through overlaps, because every pair i, j in it has $x \in U_i \cap U_j$. Hence the pointwise zero relation is generated by the pairwise overlap relations. Since the direct sum over points separates these pointwise calculations, the displayed sequence is exact.

27.6 The Observable Precosheaf D7–L3

Let $\mathcal{A}(U)$ be the open-local observable algebra from AQM-24. For $V \subset U$, define

$$\iota_{V,U} : \mathcal{A}(V) \hookrightarrow \mathcal{A}(U), \qquad a \mapsto a \otimes \mathbf{1}_{U \setminus V}.$$

Lemma $L2$. $U \mapsto \mathcal{A}(U)$ is a precosheaf of unital $*$-algebras.

Proof. The identity and composition laws are exactly Lemma $L1$ of AQM-24:

$$\iota_{U,U} = \mathrm{id}, \qquad \iota_{V,W}\iota_{U,V} = \iota_{U,W}$$

for $U \subset V \subset W$. Thus $\mathcal{A}$ is a covariant functor from $\mathsf{O}(X)$ to unital $*$-algebras.

Lemma $L3$. For finite unions,

$$\mathcal{A}(U \cup V) = \mathcal{A}(U) \vee \mathcal{A}(V),$$

where $\vee$ denotes the unital $*$-subalgebra generated by the two images inside $\mathcal{A}(U \cup V)$. If $U \cap V = \emptyset$, the two images commute and the generated algebra is the tensor product.

Proof. Every tensor factor of $\mathcal{A}(U \cup V)$ belongs to U or to V. Hence the images of $\mathcal{A}(U)$ and $\mathcal{A}(V)$ generate all local matrix factors in the union. If $U \cap V = \emptyset$, the two images act on disjoint tensor factors, so elementary tensors commute factor by factor. The generated algebra is then the tensor product of the two local algebras.

27.7 Why This Is Not an Ordinary Cosheaf T1

Theorem $T1$. The observable precosheaf $\mathcal{A}$ is not, in general, a cosheaf valued in the category of unital $*$-algebras with ordinary categorical colimits.

Proof. It is enough to use $X = \mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$. Let

$$U = \{(2)\}, \qquad V = \{(3)\}.$$

Then $U \cap V = \emptyset$, $U \cup V = X$, and AQM-24 gives

$$\mathcal{A}(U) = M_2, \qquad \mathcal{A}(V) = M_3, \qquad \mathcal{A}(X) = M_2 \otimes M_3.$$

Since $\mathcal{A}(\emptyset) = \mathbb{C}$, the cosheaf coequalizer for this cover is the coproduct of unital $*$-algebras amalgamated over the common unit:

$$M_2 *_{\mathbb{C}} M_3.$$

This coproduct is not $M_2 \otimes M_3$. Indeed, if it were the tensor product with the canonical embeddings, then every pair of unital $*$-homomorphisms $M_2 \to B$ and $M_3 \to B$ would have commuting images, because the two canonical tensor factors commute.

Take $B = M_6(\mathbb{C})$, written as 3×3 blocks in a 2×2 matrix. Let

$$f(a) = a \otimes I_3, \qquad g(b) = \begin{bmatrix} b & 0 \\ 0 & VbV^* \end{bmatrix},$$

where $V \in M_3(\mathbb{C})$ swaps the first two basis vectors. These are unital $*$-homomorphisms. For the matrix units $e_{12} \in M_2$ and $E_{11} \in M_3$,

$$f(e_{12})g(E_{11}) = \begin{bmatrix} 0 & E_{22} \\ 0 & 0 \end{bmatrix}, \qquad g(E_{11})f(e_{12}) = \begin{bmatrix} 0 & E_{11} \\ 0 & 0 \end{bmatrix}.$$

These are unequal. Therefore the universal object for arbitrary unital $*$-homomorphisms cannot be the tensor product. The cosheaf colimit in unital $*$-algebras is too free: it does not impose locality commutation.

27.8 Quantization and the Correct Dictionary T2

Theorem $T2$**.** The correct next-layer dictionary is:

$$\begin{array}{rcl} E & : & \text{a cosheaf-like label object in abelian groups or vector spaces,} \\ \mathcal{A} & : & \text{a precosheaf/local net of observable algebras,} \\ & & \text{with additivity and locality imposed inside the ambient algebra.} \end{array}$$

Proof. Lemma $L1$ proves that labels glue by the expected additive colimit. Lemmas $L2$ and $L3$ prove that observable algebras form a covariant local net and satisfy the physical additivity relation

$$\mathcal{A}(U \cup V) = \mathcal{A}(U) \vee \mathcal{A}(V).$$

Theorem $T1$ proves that this additivity is not the ordinary cosheaf axiom in unital $*$-algebras: categorical coproducts give free products, whereas quantized independent labels give commuting tensor factors. Thus quantization does not preserve the raw cosheaf colimit from labels to observable algebras; it turns additive label gluing into tensor-local algebra gluing.

27.9 Conclusion

The open-local observable assignment is naturally a precosheaf:

$$U \subset V \quad \Rightarrow \quad \mathcal{A}(U) \hookrightarrow \mathcal{A}(V).$$

It should not be called an ordinary cosheaf of unital $*$-algebras unless the target category or gluing notion is changed. The label layer E is the cosheaf-like layer. The observable layer is the AQFT-style net: isotone, local on disjoint opens, and additive by generated subalgebras.

28 State Presheaf of the Quasi-Local Algebra

28.1 Status and Source Anchors

This shard adds the state side of AQM-26. Derezinski records that normal states on $B(\mathcal{H})$ are given by trace-class functionals $\psi(A) = \mathrm{Tr}(\gamma A)$, with positive trace-one operators γ called density matrices, in `references/canonical_fields/derezinski_math_ph_0511030_source/derezinski.tex`, lines 3141–3149. Gottesman records the finite quantum-information convention that density matrices are positive trace-one operators and that a subsystem density matrix is obtained by tracing out the remaining degrees of freedom in `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 404–411.

28.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & \mathrm{State}(A) = \{\omega : A \to \mathbb{C} \mid \omega(1) = 1,\ \omega(a^*a) \geq 0\}. \\
D2 & : & \text{For } A = \mathrm{End}(\mathcal{H}),\ \omega(a) = \mathrm{Tr}(\rho a), \quad \rho \geq 0,\ \mathrm{Tr}\,\rho = 1. \\
D3 & : & \mathsf{S}(U) = \mathrm{State}(\mathcal{A}(U)). \\
L1 & : & U \subset V \Rightarrow \mathsf{S}(V) \to \mathsf{S}(U),\ \omega \mapsto \omega \circ \iota_{U,V},\ \text{is a state restriction.} \\
T1 & : & \mathsf{S} \text{ is a presheaf of convex sets.} \\
T2 & : & \mathsf{S} \text{ is not generally separated, hence not a sheaf.} \\
T3 & : & \text{Compatible overlapping marginals need not glue to a global state.} \\
T4 & : & \mathrm{State}(\mathcal{A}_{\mathrm{qloc}}(X)) \cong \varprojlim_U \mathrm{State}(\mathcal{A}(U)) \text{ for the algebraic filtered colimit.}
\end{array}
$$

28.3 State Spaces D1–D3

Let A be a unital $*$-algebra. A state on A is a linear functional $\omega : A \to \mathbb{C}$ such that

$$\omega(1) = 1, \qquad \omega(a^*a) \geq 0 \quad \text{for all } a \in A.$$

For a finite-dimensional Hilbert space $\mathcal{H}$ and $A = \mathrm{End}(\mathcal{H})$, we use the density-matrix representation

$$\omega_\rho(a) = \mathrm{Tr}(\rho a), \qquad \rho \geq 0, \quad \mathrm{Tr}\,\rho = 1.$$

The state space is convex: if ω_0, ω_1 are states and $0 \leq t \leq 1$, then $t\omega_0 + (1-t)\omega_1$ is again normalized and positive.

For the open-local observable net $U \mapsto \mathcal{A}(U)$, define

$$\mathsf{S}(U) = \mathrm{State}(\mathcal{A}(U)).$$

28.4 Restriction Is a Presheaf L1–T1

Lemma $L1$. If $U \subset V$, then

$$\rho^V_U : \mathsf{S}(V) \longrightarrow \mathsf{S}(U), \qquad \rho^V_U(\omega) = \omega \circ \iota_{U,V}$$

is a well-defined affine map.

Proof. The inclusion $\iota_{U,V}$ is a unital $*$-homomorphism by AQM-24. Thus

$$\rho^V_U(\omega)(1) = \omega(\iota_{U,V}(1)) = \omega(1) = 1.$$

For $a \in \mathcal{A}(U)$,

$$\rho^V_U(\omega)(a^*a) = \omega(\iota_{U,V}(a^*a)) = \omega(\iota_{U,V}(a)^*\iota_{U,V}(a)) \geq 0.$$

Hence $\rho_U^V(\omega)$ is a state. Affineness follows from linearity of composition with $\iota_{U,V}$.

When $V = U \sqcup W$, the same restriction is the density-matrix partial trace. If ω_ρ is a state on $\mathcal{A}(U) \otimes \mathcal{A}(W)$, then the restricted state on $\mathcal{A}(U)$ is represented by

$$\rho_U = \mathrm{Tr}_W(\rho),$$

because $\mathrm{Tr}(\rho(a \otimes 1_W)) = \mathrm{Tr}(\rho_U a)$.

Theorem $T1$**.** S is a presheaf of convex sets on X.

Proof. For $U \subset V \subset W$,

$$\rho_U^W(\omega) = \omega \circ \iota_{U,W} = \omega \circ \iota_{V,W} \circ \iota_{U,V} = \rho_U^V(\rho_V^W(\omega)),$$

using functoriality of ι from AQM-24. Also $\rho_U^U = \mathrm{id}$. The restriction maps are affine by Lemma $L1$, so this is a contravariant functor from opens to convex sets.

28.5 Failure of Separatedness T2

Theorem $T2$**.** S is not generally a sheaf. In fact it is not generally separated: two global states can have the same restrictions to an open cover.

Proof. Use $X = \mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$, with the two singleton opens

$$U = \{(2)\}, \qquad V = \{(3)\}, \qquad X = U \sqcup V.$$

Then $\mathcal{A}(X) = M_2 \otimes M_3$. Choose bases $|0\rangle, |1\rangle$ of $\mathbb{C}^2$ and $|0\rangle, |1\rangle, |2\rangle$ of $\mathbb{C}^3$, and set

$$|\psi\rangle = \frac{|0,0\rangle + |1,1\rangle}{\sqrt{2}}, \qquad \rho = |\psi\rangle\langle\psi|.$$

Its restrictions are

$$\rho_U = \frac{I_2}{2}, \qquad \rho_V = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2}.$$

Now define the product density matrix

$$\sigma = \rho_U \otimes \rho_V.$$

The density matrices ρ and σ have the same restrictions to U and to V by construction. They are not equal: ρ has the off-diagonal matrix entry $|0,0\rangle\langle 1,1|/2$, while σ has no such entry. Thus the two global states agree on the cover $\{U, V\}$ but differ globally. The uniqueness part of the sheaf condition fails.

28.6 Failure of Existence for Overlapping Marginals T3

Theorem $T3$**.** Even when local states agree on an overlap, a global extending state need not exist.

Proof. Consider a discrete three-point finite spectrum with qubit factors labelled A, B, C. For example, one may take $\mathrm{Spec}(\mathbb{F}_2 \times \mathbb{F}_2 \times \mathbb{F}_2)$. Let

$$U = \{A, B\}, \qquad V = \{B, C\}, \qquad U \cap V = \{B\}.$$

On AB, take the Bell state

$$\rho_{AB} = |\Phi_{AB}\rangle\langle\Phi_{AB}|, \qquad |\Phi_{AB}\rangle = \frac{|0,0\rangle + |1,1\rangle}{\sqrt{2}}.$$

On BC, take

$$\rho_{BC} = |\Phi_{BC}\rangle\langle\Phi_{BC}|, \qquad |\Phi_{BC}\rangle = \frac{|0,0\rangle + |1,1\rangle}{\sqrt{2}}.$$

Both restrict to $I_2/2$ on the overlap B.

Assume a density matrix θ_{ABC} extends both. Since $\mathrm{Tr}_C\,\theta_{ABC} = \rho_{AB}$ is the rank-one projection $P_{AB} = |\Phi_{AB}\rangle\langle\Phi_{AB}|$, positivity forces θ_{ABC} to be supported on $\mathrm{ran}(P_{AB}) \otimes \mathbb{C}^2_C$. Indeed,

$$\mathrm{Tr}\big(((1 - P_{AB}) \otimes 1_C)\theta_{ABC}\big) = \mathrm{Tr}((1 - P_{AB})\rho_{AB}) = 0,$$

and a positive operator with zero trace is zero on that positive part. Therefore

$$\theta_{ABC} = P_{AB} \otimes \tau_C$$

for some density matrix τ_C. Its BC-marginal is then

$$\mathrm{Tr}_A(\theta_{ABC}) = \frac{I_2}{2} \otimes \tau_C,$$

which is a product state. It cannot equal the pure entangled Bell state ρ_{BC}. Hence the compatible pair (ρ_{AB}, ρ_{BC}) does not glue to a state on ABC.

28.7 Global States as an Inverse Limit T4

Let $\mathsf{K}(X)$ be the directed poset of quasi-compact opens used in AQM-25, and let

$$\mathcal{A}_{\mathrm{qloc}}(X) = \varinjlim_{U \in \mathsf{K}(X)} \mathcal{A}(U)$$

be the algebraic quasi-local $*$-algebra. Since the structure maps are injective, we may regard $\mathcal{A}_{\mathrm{qloc}}(X)$ as the algebraic union of the local algebras.

Theorem *T4.* There is a natural affine bijection

$$\mathrm{State}(\mathcal{A}_{\mathrm{qloc}}(X)) \cong \varprojlim_{U \in \mathsf{K}(X)} \mathrm{State}(\mathcal{A}(U)).$$

Proof. Given a state ω on $\mathcal{A}_{\mathrm{qloc}}(X)$, restrict it to each $\mathcal{A}(U)$. These restrictions are compatible by functoriality of restriction, so they define an element of the inverse limit.

Conversely, let $(\omega_U)_U$ be a compatible family of local states. If $a \in \mathcal{A}_{\mathrm{qloc}}(X)$, choose U and $a_U \in \mathcal{A}(U)$ representing it, and define

$$\omega(a) = \omega_U(a_U).$$

This is independent of the representative: if a_U and a_V represent the same quasi-local element, choose $W \supset U, V$; their images in $\mathcal{A}(W)$ agree, and compatibility gives the same value. Linearity follows after moving two representatives to a common W. Normalization is $\omega(1) = 1$. Positivity holds because if $a \in \mathcal{A}(U)$, then $a^*a \in \mathcal{A}(U)$ and

$$\omega(a^*a) = \omega_U(a^*a) \geq 0.$$

The two constructions are inverse to each other.

28.8 Conclusion

The state side reverses the observable-net arrows:

$$U \subset V \quad \Rightarrow \quad \mathrm{State}(\mathcal{A}(V)) \longrightarrow \mathrm{State}(\mathcal{A}(U)).$$

Thus local quantum states form a presheaf of convex sets. They do not form a sheaf in general: uniqueness fails because marginals do not determine correlations, and existence fails because compatible overlapping marginals may not have a joint quantum state. The correct quasi-local replacement is the inverse-limit statement of Theorem $T4$, not an ordinary sheaf gluing axiom.

29 Closed-Point Affine Arithmetic Fields

29.1 Status and Source Anchors

This shard records the convention needed before treating $\operatorname{Spec}\mathbb{F}_q[t]$ and $\operatorname{Spec}\mathbb{F}_q[x,y]$. The finite-spectrum rule of AQM-23 and AQM-24 attached one finite Weyl system to each point because all residue fields in those examples were finite. For infinite affine schemes over $\mathbb{F}_q$, not every point has finite residue field. The finite qudit layer is therefore indexed by closed points only.

The topology and residue-field definitions are source-local from the Stacks Project snapshot: `references/algebraic_geometry/stacks_project_algebra.tex`, lines 120–158, 250–274, 2972–2983, 3104–3117, and 3310–3350. The Nullstellensatz finite-residue statement is in the same file, lines 6520–6665. The finite Weyl laws use Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313, and the Pauli symplectic commutator source used in AQM-22.

29.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & k=\mathbb{F}_q,\ R \text{ a finite-type } k\text{-algebra},\ X=\operatorname{Spec}R. \\
D2 & : & |X|_{\mathrm{cl}}=\operatorname{Max}R, \quad d(x)=[\kappa(x):k] \text{ for } x\in|X|_{\mathrm{cl}}. \\
L1 & : & \kappa(x) \text{ is finite and } |\kappa(x)|=q^{d(x)} \text{ for closed } x. \\
D3 & : & S\Subset|X|_{\mathrm{cl}} \text{ is a finite closed support.} \\
D4 & : & E(S)=\bigoplus_{x\in S}\kappa(x)^2, \quad \mathcal{H}(S)=\bigotimes_{x\in S}\ell^2(\kappa(x)). \\
L2 & : & E(S) \text{ has a finite Weyl projective representation } W_S. \\
D5 & : & C(U)=U\cap|X|_{\mathrm{cl}},\ E(U)=\bigoplus_{x\in C(U)}\kappa(x)^2,\ \mathcal{A}(U)=\varinjlim_{S\Subset C(U)}\mathcal{A}(S). \\
T1 & : & U\mapsto\mathcal{A}(U) \text{ is an open-local net.} \\
T2 & : & \operatorname{State}(\mathcal{A}(U))\cong\varprojlim_{S\Subset C(U)}\operatorname{State}(\mathcal{A}(S)). \\
T3 & : & \text{Regular-function profiles are generally not quasi-local operators.}
\end{array}
$$

29.3 Closed Finite-Residue Sites D1–L1

Let $k=\mathbb{F}_q$, $q=p^r$, and let R be a finite-type k-algebra. Put $X=\operatorname{Spec}R$. A closed point of X means a maximal ideal $x=\mathfrak{m}\subset R$. Its residue field is

$$\kappa(x)=R_{\mathfrak{m}}/\mathfrak{m}R_{\mathfrak{m}}\cong R/\mathfrak{m}.$$

The last isomorphism holds because $\mathfrak{m}$ is maximal.

Lemma *L*1. If $x\in|X|_{\mathrm{cl}}$, then $\kappa(x)$ is a finite field. More precisely, if $d(x)=[\kappa(x):k]$, then

$$|\kappa(x)|=q^{d(x)}.$$

Proof. The Stacks Nullstellensatz source says that for a maximal ideal in a finite type k-algebra, the residue field is a finite extension of k. Hence $\kappa(x)$ is a $d(x)$-dimensional vector space over the q-element field k, so it has $q^{d(x)}$ elements.

Non-closed points are not discarded from X; they remain topological points. They are simply not finite Weyl sites in this convention unless an additional infinite-residue representation convention is later fixed.

29.4 Finite Closed Supports D3–L2

For a finite subset $S \Subset |X|_{\mathrm{cl}}$, define

$$E(S) = \bigoplus_{x \in S} \kappa(x)^2.$$

Write $e_x = (q_x, p_x)$. Define the local alternating form

$$\omega_x(e_x, f_x) = p_x q'_x - q_x p'_x$$

and the finite-support form

$$\Omega_S(e, f) = (\omega_x(e_x, f_x))_{x \in S}.$$

This is not a form into one field when the residue degrees vary. The representation uses the associated alternating bicharacter. Let

$$\psi_x(a) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{\kappa(x)/\mathbb{F}_p}(a)\right).$$

Set

$$B_S(e, f) = \prod_{x \in S} \psi_x(p_x q'_x - q_x p'_x).$$

Let

$$\mathcal{H}_x = \ell^2(\kappa(x)), \qquad \mathcal{H}(S) = \bigotimes_{x \in S} \mathcal{H}_x.$$

On the basis $|a\rangle$, $a \in \kappa(x)$, define

$$T_x(q)|a\rangle = |a + q\rangle, \qquad R_x(p)|a\rangle = \psi_x(pa)|a\rangle, \qquad W_x(q, p) = T_x(q)R_x(p).$$

For $e \in E(S)$, put $W_S(e) = \bigotimes_{x \in S} W_x(e_x)$.

Lemma *L2.* For $e, f \in E(S)$,

$$W_S(e)W_S(f) = c_S(e, f)W_S(e + f), \qquad c_S(e, f) = \prod_{x \in S} \psi_x(p_x q'_x),$$

and

$$W_S(e)W_S(f) = B_S(e, f)W_S(f)W_S(e).$$

The complex span

$$\mathcal{A}(S) = \operatorname{span}_{\mathbb{C}}\{W_S(e) : e \in E(S)\}$$

is

$$\mathcal{A}(S) \cong \bigotimes_{x \in S} \operatorname{End}(\mathcal{H}_x).$$

Proof. At one site,

$$R_x(p)T_x(q')|a\rangle = \psi_x(p(a + q'))|a + q'\rangle = \psi_x(pq')T_x(q')R_x(p)|a\rangle.$$

Thus $W_x(e_x)W_x(f_x) = \psi_x(p_x q'_x)W_x(e_x + f_x)$, and dividing by the same formula with e_x, f_x reversed gives the local commutator $\psi_x(p_x q'_x - q_x p'_x)$. Tensoring over the finite set S gives the two displayed laws. The local Weyl source says the finite-field operators $T^a R^b$ form an operator basis; tensor products of bases are bases of tensor products.

29.5 Open-Local Algebra D5–T1

For an open set $U \subset X$, define its closed-point support

$$C(U) = U \cap |X|_{\mathrm{cl}}.$$

The field-label group on U is the finite-support direct sum

$$E(U) = \bigoplus_{x \in C(U)} \kappa(x)^2 = \varinjlim_{S \Subset C(U)} E(S).$$

The observable algebra is the algebraic quasi-local algebra over finite closed supports:

$$\mathcal{A}(U) = \varinjlim_{S \Subset C(U)} \mathcal{A}(S) = \mathrm{span}_{\mathbb{C}}\{W_U(e) : e \in E(U)\}.$$

If $S \subset T$, the structure map is

$$a \longmapsto a \otimes \mathbf{1}_{\mathcal{H}(T \setminus S)}.$$

Theorem *T*1. $U \mapsto \mathcal{A}(U)$ is an open-local net: it is isotone, local on disjoint closed supports, and additive for open covers.

Proof. If $U \subset V$, then $C(U) \subset C(V)$. The finite-support inclusions therefore induce a unital injective $*$-homomorphism $\mathcal{A}(U) \hookrightarrow \mathcal{A}(V)$. If two labels have disjoint closed supports, then every local commutator factor in $B(e, f)$ is 1, so their Weyl operators commute. If $U = \bigcup_i U_i$ and $e \in E(U)$, the support of e is finite. Assign each support point to one U_i that contains it. Then e is a finite sum of labels $e_i \in E(U_i)$, and $W_U(e)$ is a scalar times a product of the corresponding $W_{U_i}(e_i)$. Thus $\mathcal{A}(U)$ is generated by the images of the $\mathcal{A}(U_i)$.

29.6 States T2

A state on a unital $*$-algebra is a normalized positive linear functional. For finite S, states on $\mathcal{A}(S)$ are density matrices on $\mathcal{H}(S)$. For an open U, define

$$\mathsf{S}(U) = \mathrm{State}(\mathcal{A}(U)).$$

Theorem *T*2. There is a natural affine bijection

$$\mathrm{State}(\mathcal{A}(U)) \cong \varprojlim_{S \Subset C(U)} \mathrm{State}(\mathcal{A}(S)).$$

Proof. The algebra $\mathcal{A}(U)$ is the filtered algebraic colimit of the finite matrix algebras $\mathcal{A}(S)$. Restricting a state to each finite $\mathcal{A}(S)$ gives a compatible inverse-limit family. Conversely, a compatible family (ω_S) defines $\omega(a) = \omega_S(a)$ whenever $a \in \mathcal{A}(S)$. If a has two finite-support representatives, move both to a common larger finite support; compatibility gives the same value. Linearity and positivity are checked in a common finite support. The two constructions are inverse.

If local density matrices ρ_x are chosen for all closed $x \in C(U)$, then

$$\rho_S = \bigotimes_{x \in S} \rho_x$$

defines a compatible product state. Taking each ρ_x to be the normalized identity gives the product tracial state, whose value on $W_U(e)$ is 0 for $e \neq 0$, because some nontrivial local Weyl operator has trace 0.

29.7 Regular Profiles Are Not Local Operators T3

For $a, b \in R$, reduction modulo each closed point gives a closed-point field profile

$$e_{a,b}(x) = (a \bmod x,\ b \bmod x) \in \kappa(x)^2.$$

This is a genuine arithmetic profile over all closed points. It is an element of $E(U)$, however, only when it has finite support in $C(U)$. Otherwise the formal product $W(e_{a,b})$ is an infinite-volume field symbol, not an element of the algebraic quasi-local algebra $\mathcal{A}(U)$.

Thus the present finite-residue theory has two layers:

$$\text{quasi-local Weyl observables} \quad \text{and} \quad \text{formal regular-function field profiles.}$$

A later analytic or representation-theoretic convention may relate them, but the current report does not silently identify them.

30 The Arithmetic Field on $\operatorname{Spec}\mathbb{F}_q[t]$

30.1 Status and Source Anchors

This shard instantiates AQM-28 for $R_1 = k[t]$, $k = \mathbb{F}_q$. The spectrum and $D(f)$ notation are sourced from the Stacks Project snapshot, `references/algebraic_geometry/stacks_project_algebra.tex`, lines 2972–2983 and 3104–3117. Residue fields are sourced from lines 250–274 and 3310–3350. The polynomial-ring classification below is proved directly from the Euclidean property of $k[t]$. The Weyl layer is the finite closed-point construction of AQM-28.

30.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & R_1 = k[t], \quad X_1 = \operatorname{Spec} R_1. \\
L1 & : & X_1 = \{\eta = (0)\} \cup \{x_\pi = (\pi) : \pi \text{ monic irreducible}\}. \\
L2 & : & \kappa(\eta) = k(t), \quad \kappa(x_\pi) = k[t]/(\pi), \quad |\kappa(x_\pi)| = q^{\deg \pi}. \\
L3 & : & D(g) = \emptyset \text{ if } g = 0, \text{ and for } g \neq 0 \text{ it omits exactly the closed points dividing } g. \\
D2 & : & E_1(U) = \bigoplus_{x_\pi \in U} \kappa(x_\pi)^2, \quad \mathcal{A}_1(U) = \varinjlim_{S \Subset C(U)} \mathcal{A}(S). \\
T1 & : & \mathcal{A}_1(U) \text{ is the open-local Weyl algebra on closed points of } U. \\
T2 & : & \operatorname{State}(\mathcal{A}_1(U)) \text{ is the inverse limit of finite-volume density-matrix states.} \\
T3 & : & \text{Nonzero polynomial field profiles have infinite closed-point support.}
\end{array}$$

30.3 Points of the Affine Line L1–L2

Let $\mathcal{P}_k$ be the set of monic irreducible polynomials in $k[t]$. The zero ideal is prime because $k[t]$ is a domain; write

$$\eta = (0)$$

for the generic point.

Lemma $L1$. The remaining prime ideals are exactly

$$x_\pi = (\pi), \qquad \pi \in \mathcal{P}_k.$$

Proof. Since $k[t]$ is Euclidean, every nonzero ideal has a unique monic generator. Let $\mathfrak{p} \neq (0)$ be prime and write $\mathfrak{p} = (g)$ with g monic. If $g = ab$ with nonconstant a, b, then $ab \in \mathfrak{p}$, so primality gives $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, contradicting minimal degree of the monic generator g. Thus g is irreducible. Conversely, if π is irreducible, then $k[t]/(\pi)$ is a field, so (π) is maximal and therefore prime.

Lemma $L2$.

$$\kappa(\eta) = k(t), \qquad \kappa(x_\pi) = k[t]/(\pi).$$

If $d = \deg \pi$, then $\kappa(x_\pi)$ has q^d elements. Thus the closed point x_π carries one q^d-level finite-field qudit.

Proof. The residue field at the zero prime is the fraction field of $k[t]$, namely $k(t)$. For the maximal ideal (π), localization does not change the quotient residue field, so $\kappa(x_\pi) = k[t]/(\pi)$. The residue classes of $1, t, \ldots, t^{d-1}$ form a k-basis, hence the quotient has q^d elements.

The generic point η has infinite residue field $k(t)$. It is not a finite Weyl site under the closed-point convention of AQM-28.

30.4 Zariski Opens L3

Let $g \in k[t]$. If $g = 0$, then $D(g) = \emptyset$. If

$$g = u \prod_{i=1}^{r} \pi_i^{n_i}$$

with $u \in k^\times$ and distinct monic irreducibles π_i, then

$$D(g) = \{\eta\} \cup \{x_\pi : \pi \in \mathcal{P}_k,\ \pi \notin \{\pi_1, \dots, \pi_r\}\}.$$

Derivation. The generic prime (0) contains no nonzero g, so $\eta \in D(g)$. The closed prime (π) contains g iff π divides g. Therefore $D(g)$ is exactly the displayed set. In particular every nonempty open contains η, and its closed-point support is cofinite in $\mathcal{P}_k$.

For a closed point x_π,

$$\{x_\pi\} = V(\pi), \qquad X_1 \setminus \{x_\pi\} = D(\pi).$$

Thus closed singletons are not open, but their complements are standard opens.

30.5 Weyl-Heisenberg Field Net D2–T1

For an open $U \subset X_1$, define

$$C_1(U) = U \cap \{x_\pi : \pi \in \mathcal{P}_k\}.$$

The affine-line field-label group is

$$E_1(U) = \bigoplus_{x_\pi \in C_1(U)} \kappa(x_\pi)^2.$$

For $D(g)$ with $g \neq 0$,

$$E_1(D(g)) = \bigoplus_{\pi \nmid g} \big(k[t]/(\pi)\big)^2.$$

The corresponding algebra is

$$\mathcal{A}_1(D(g)) = \varinjlim_{S \Subset \{\pi : \pi \nmid g\}} \bigotimes_{\pi \in S} M_{q^{\deg \pi}}(\mathbb{C}).$$

For $g = 1$, this is the global algebra

$$\mathcal{A}_1(X_1) = \varinjlim_{S \Subset \mathcal{P}_k} \bigotimes_{\pi \in S} M_{q^{\deg \pi}}(\mathbb{C}).$$

Theorem *T*1. The assignment $U \mapsto \mathcal{A}_1(U)$ is the open-local Weyl-Heisenberg net on $\operatorname{Spec} k[t]$.

Proof. This is AQM-28 applied to $R_1 = k[t]$. If $U \subset V$, then $C_1(U) \subset C_1(V)$, so finite closed supports in U are finite closed supports in V. The algebra inclusion tensors by identities on the extra closed points. Additivity for open covers follows because every Weyl label has finite closed support.

At a single closed point x_π, let $K_\pi = k[t]/(\pi)$. For $a, b \in K_\pi$, write

$$X_\pi(a) = W_\pi(a, 0), \qquad Z_\pi(b) = W_\pi(0, b).$$

Then

$$Z_\pi(b) X_\pi(a) = \psi_\pi(ab) X_\pi(a) Z_\pi(b),$$

while operators at distinct closed points commute.

30.6 States T2

For every open U,

$$\mathrm{State}(\mathcal{A}_1(U)) \cong \varprojlim_{S \Subset C_1(U)} \mathrm{State}\left(\bigotimes_{x_\pi \in S} M_{q^{\deg \pi}}(\mathbb{C}) \right).$$

The restriction maps are partial traces over omitted closed points. The product tracial state is obtained by taking the normalized trace at every closed point; it sends every nontrivial finite-support Weyl operator to 0.

The state presheaf is not separated for the Zariski topology. Let

$$x_0 = (t), \qquad x_1 = (t-1), \qquad U = D(t), \quad V = D(t-1).$$

Then $U \cup V = X_1$. On the two q-dimensional factors x_0, x_1, take the maximally entangled vector

$$|\Phi\rangle = q^{-1/2} \sum_{a \in k} |a, a\rangle.$$

Let $\rho = |\Phi\rangle\langle\Phi|$, and let $\sigma = (I_q/q) \otimes (I_q/q)$. Tensor both with the product tracial state on all other closed points. The two global states restrict equally to U and to V, because each restriction traces out one of x_0, x_1. They differ globally, since $\rho \neq \sigma$ on the two-site subalgebra.

30.7 Polynomial Field Profiles T3

A pair of polynomials $a, b \in k[t]$ defines a closed-point profile

$$e_{a,b}(x_\pi) = (a \bmod \pi,\ b \bmod \pi).$$

There are infinitely many monic irreducibles in $k[t]$: if $\pi_1, \dots, \pi_n$ were all of them, then $P = \pi_1 \cdots \pi_n$ is nonconstant and $P + 1$ has an irreducible divisor which divides none of the π_i, a contradiction. If $a \neq 0$, then $a \bmod \pi = 0$ only for the finitely many irreducible divisors of a. Hence a is nonzero at infinitely many closed points. The same holds for b. Therefore

$$e_{a,b} \in E_1(X_1) \iff a = b = 0.$$

Thus the coordinate function t, and every nonzero polynomial profile, is a formal infinite-support field profile. Its finite restrictions

$$e_{a,b}|_S \in \bigoplus_{x_\pi \in S} \kappa(x_\pi)^2$$

give finite-volume Weyl operators, but the full profile does not define an element of the algebraic quasi-local algebra.

30.8 Conclusion

$\operatorname{Spec} k[t]$ produces a countable closed-point array of finite-field qudits: one q^d-level qudit for each monic irreducible polynomial of degree d. Nonempty Zariski opens remove finitely many closed points and always contain the generic point, but the generic point has residue field $k(t)$ and is not a finite Weyl site in the current convention. The quasi-local observables are finite closed-support Weyl operators; polynomial fields are formal infinite-volume profiles until a completion convention is added.

31 The Arithmetic Field on $\operatorname{Spec}\mathbb{F}_q[x,y]$

31.1 Status and Source Anchors

This shard instantiates AQM-28 for $R_2 = k[x,y]$, $k = \mathbb{F}_q$. The Stacks Project source describes $\operatorname{Spec} k[x,y]$ in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 4889–4927: besides (0), height-one primes are generated by irreducible polynomials, and non-principal primes are maximal of the displayed quotient form. The finite residue field statement for maximal ideals is the Nullstellensatz source, lines 6520–6665. Weyl and state conventions are those of AQM-28.

31.2 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: R_2 = k[x,y], \quad X_2 = \operatorname{Spec} R_2.\\
L1 &: X_2 \text{ has } (0),\ \text{height-one points } (f),\ \text{and closed maximal ideals } \mathfrak{m}.\\
L2 &: \kappa(0) = k(x,y),\ \kappa((f)) = \operatorname{Frac}(k[x,y]/(f)),\ \kappa(\mathfrak{m})/k \text{ finite}.\\
D2 &: C_2(U) = U \cap \operatorname{Max} R_2, \quad E_2(U) = \textstyle\bigoplus_{\mathfrak{m}\in C_2(U)} \kappa(\mathfrak{m})^2.\\
T1 &: \mathcal{A}_2(U) = \varinjlim_{S \Subset C_2(U)} \textstyle\bigotimes_{\mathfrak{m}\in S} M_{q^{d(\mathfrak{m})}}(\mathbb{C}).\\
T2 &: \text{The rational-point truncation is a finite } q^2\text{-site subtheory, not the full Spec theory.}\\
T3 &: \operatorname{State}(\mathcal{A}_2(U)) \text{ is an inverse limit over finite closed supports.}\\
T4 &: \text{Coordinate profiles such as } x, y \text{ are formal infinite-support fields.}
\end{aligned}$$

31.3 Point Types L1–L2

Let $X_2 = \operatorname{Spec} k[x,y]$.

Lemma $L1$. The points of X_2 have the following forms.

$$\begin{array}{ll}
\text{generic point:} & (0),\\
\text{curve-generic points:} & (f), \quad f \in k[x,y] \text{ irreducible},\\
\text{closed points:} & \mathfrak{m} \subset k[x,y] \text{ maximal.}
\end{array}$$

Proof. The Stacks example cited above records that (0) is prime, that a principal ideal generated by an irreducible polynomial is prime because $k[x,y]$ is a UFD, and that the remaining non-principal primes are maximal after reducing to a prime of a one-variable polynomial ring over a finite algebraic extension of k. This is exactly the displayed trichotomy.

Lemma $L2$. The residue fields are

$$\kappa((0)) = k(x,y),$$

$$\kappa((f)) = \operatorname{Frac}(k[x,y]/(f)) \quad \text{for irreducible } f,$$

and, for a closed point $\mathfrak{m}$,

$$\kappa(\mathfrak{m}) = k[x,y]/\mathfrak{m} \quad \text{is a finite extension of } k.$$

If $d(\mathfrak{m}) = [\kappa(\mathfrak{m}) : k]$, then $|\kappa(\mathfrak{m})| = q^{d(\mathfrak{m})}$.

Proof. The first two formulas are the definition of residue field as the fraction field of $R/\mathfrak{p}$. The maximal-ideal formula follows because localizing at a maximal ideal does not change the quotient residue field. The Nullstellensatz source gives that this residue field is finite over k, and the cardinality follows from its k-vector-space dimension.

The generic point and curve-generic points are not finite Weyl sites in the current convention. The generic residue $k(x, y)$ is infinite. If $f \in k[x]$ is irreducible, then

$$k[x, y]/(f) \cong (k[x]/(f))[y],$$

so $\kappa((f)) \cong (k[x]/(f))(y)$, also infinite. If f has positive degree in y, then $k[x] \hookrightarrow k[x, y]/(f)$, since a nonzero polynomial in $k[x]$ cannot be divisible by such an f. Hence the quotient and its fraction field are infinite. After swapping x and y if necessary, this covers every irreducible f.

31.4 Closed Points as Finite Qudit Sites

A rational point $(a, b) \in k^2$ gives the maximal ideal

$$\mathfrak{m}_{a,b} = (x - a, y - b),$$

with residue field k. Hence it is one q-level site.

Closed points of larger degree are equally concrete. Let $\pi(x) \in k[x]$ be monic irreducible of degree d, let $K = k[x]/(\pi)$, and choose $\beta \in K$ represented by $\widetilde{\beta}(x) \in k[x]$. Then

$$\mathfrak{m}_{\pi,\beta} = (\pi(x),\ y - \widetilde{\beta}(x))$$

is maximal, with residue field K. It contributes one q^d-level finite-field qudit.

Derivation. The quotient by $(\pi(x), y - \widetilde{\beta}(x))$ sends x to the class of x in K and y to β, giving a quotient isomorphic to K. Since K is a field, the ideal is maximal.

31.5 Zariski Opens and Closed-Point Supports

For $h \in k[x, y]$,

$$D(h) = \{\mathfrak{p} : h \notin \mathfrak{p}\}.$$

The closed-point support used by the quantum net is

$$C_2(D(h)) = \{\mathfrak{m} \in \operatorname{Max} R_2 : h \notin \mathfrak{m}\}.$$

Equivalently, if $\bar{x}, \bar{y}$ are the residue classes in $\kappa(\mathfrak{m})$, then

$$\mathfrak{m} \in C_2(D(h)) \quad \Longleftrightarrow \quad h(\bar{x}, \bar{y}) \neq 0.$$

Examples:

$$C_2(D(x)) = \{\mathfrak{m} : x \notin \mathfrak{m}\},$$

so $D(x)$ removes the closed points on the vertical line $x = 0$. The open

$$D(x) \cup D(y) = X_2 \setminus V(x, y)$$

removes only the rational origin (x, y) from the closed-point support.

31.6 Weyl-Heisenberg Net T1

For an open $U \subset X_2$, define

$$E_2(U) = \bigoplus_{\mathfrak{m} \in C_2(U)} \kappa(\mathfrak{m})^2.$$

For a finite support $S \Subset C_2(U)$,

$$\mathcal{A}(S) = \bigotimes_{\mathfrak{m} \in S} \operatorname{End}_{\mathbb{C}} \ell^2(\kappa(\mathfrak{m})) \cong \bigotimes_{\mathfrak{m} \in S} M_{q^{d(\mathfrak{m})}}(\mathbb{C}).$$

The open-local algebra is

$$\mathcal{A}_2(U) = \varinjlim_{S \Subset C_2(U)} \mathcal{A}(S) = \operatorname{span}_{\mathbb{C}}\{W_U(e) : e \in E_2(U)\}.$$

Theorem *T*1. $U \mapsto \mathcal{A}_2(U)$ is the affine-plane open-local Weyl-Heisenberg net.

Proof. This is AQM-28 with $R = R_2$. Inclusion of opens gives inclusion of closed supports, hence tensor-by-identity inclusions of finite matrix algebras and an inclusion of their filtered colimits. The Weyl commutator at a closed point $\mathfrak{m}$ is

$$Z_{\mathfrak{m}}(b)X_{\mathfrak{m}}(a) = \psi_{\mathfrak{m}}(ab)X_{\mathfrak{m}}(a)Z_{\mathfrak{m}}(b),$$

and different closed points commute.

31.7 Rational-Point Truncation T2

Let

$$S_{\text{rat}} = \{\mathfrak{m}_{a,b} : a, b \in k\}.$$

This finite support has q^2 closed points, each with residue field k. The associated finite subtheory is

$$E(S_{\text{rat}}) = \bigoplus_{(a,b) \in k^2} k^2,$$

$$\mathcal{H}(S_{\text{rat}}) = \bigotimes_{(a,b) \in k^2} \ell^2(k), \qquad \mathcal{A}(S_{\text{rat}}) \cong \bigotimes_{(a,b) \in k^2} M_q(\mathbb{C}).$$

This is the finite k-rational-point truncation, matching the finite-set construction on k^2. It is not the full $\operatorname{Spec} k[x, y]$ theory, because the full closed-point set also contains all non-rational closed points of degrees $d > 1$.

31.8 States T3

For every open U,

$$\operatorname{State}(\mathcal{A}_2(U)) \cong \varprojlim_{S \Subset C_2(U)} \operatorname{State}(\mathcal{A}(S)).$$

The transition maps are finite partial traces.

The Zariski state presheaf is not separated. Let

$$x_0 = (x, y), \qquad x_1 = (x - 1, y),$$

and let $U = X_2 \setminus \{x_0\}$, $V = X_2 \setminus \{x_1\}$. These are open and cover X_2. On the two rational q-level sites x_0, x_1, use the same maximally entangled density matrix and product density matrix as in AQM-29, tensoring with the product tracial state on every other closed point. The two global states agree after restriction to U and to V, but they differ on the two-site algebra. Thus local marginals on a Zariski cover do not determine global correlations.

31.9 Coordinate Field Profiles T4

The coordinate pair $(x, 0)$ defines the closed-point profile

$$\mathfrak{m} \longmapsto (x \bmod \mathfrak{m}, 0).$$

It is nonzero at every closed point not lying on the line $x = 0$. Hence it has infinite support: the line $x = 1$ has the same closed points as $\operatorname{Spec} k[y]$, and AQM-29 proves there are infinitely many of them. Likewise $(0, y)$ has infinite support, using the line $y = 1$. Their finite-volume restrictions define Weyl operators on finite closed supports, but the full coordinate profiles are not elements of $\mathcal{A}_2(X_2)$.

The curve-generic point (x) is also not a replacement for the missing operator: its residue field is $k(y)$, not a finite qudit field. It records the generic point of the divisor $x = 0$, while the finite Weyl theory records finite closed-point excitations on or off that divisor.

31.10 Conclusion

$\operatorname{Spec} k[x, y]$ has three geometric point types. The finite arithmetic quantum field constructed here uses only the closed points as finite qudit sites, with one $q^{d(\mathfrak{m})}$-level qudit at a closed point of degree $d(\mathfrak{m})$. Rational points give a finite q^2-site truncation, but the full closed-point quasi-local algebra is the filtered tensor product over all closed points of all degrees. Generic and curve-generic points remain essential to the Zariski topology and to divisor geometry, but they are not finite Weyl sites under the current convention.

32 Weyl-Heisenberg Representation Status over Rings and LCA Systems

32.1 Status and Source Anchors

This shard records the source status needed before defining a Weyl-Heisenberg system for $k(t)$. The LCA baseline is Prasad's `references/heisenberg_weil/prasad_0912_0574_source/SvN.tex`, lines 95–107 and 164–190. The modern LCA quantum-kinematical source is `references/heisenberg_weil/beny_crann_lee_park_youn_2204_08162_source/main.tex`, lines 305–328, 411–444, and 461–473. The locally compact ring source is `references/heisenberg_weil/bekka_2502_00387_source/CCR-GeneralRings-v5.tex`, lines 287–350, 862–881, 997–1038, and 1048–1150. The finite Frobenius-ring Pauli sources are `references/heisenberg_weil/gluesing_luerssen_pllaha_1710_09884_source/StabCodesFrob5.tex`, lines 224–249, 286–323, 353–355, 412–423, and 469–504, and `references/heisenberg_weil/sidana_kashyap_2202_00248_source/EAQECCs_over_rings_v15.tex`, lines 103–145 and 180–197. The finite generalized-Heisenberg source is `references/heisenberg_weil/solomon_2501_00650_source/TotslipVersion5_Aug_25.tex`, lines 315–318, 904–981, 1127–1136, 1637–1649, and 1677–1718. Finite Clifford/Weil splitting results are registered from Lysenko, Trias, Hashimoto–Horibe–Hayashi, Korbelar–Tolar, and Galindo in `references/heisenberg_weil/SOURCES.md`.

32.2 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: \text{ An LCA Weyl system uses } F \times \widehat{F}, \text{ not } F \times F \text{ unless } F \simeq \widehat{F}.\\
L1 &: \text{ For an LCA group } F,\ W(x,\gamma) = T_x M_\gamma \text{ is the canonical irreducible Weyl representation.}\\
D2 &: \text{ For a locally compact ring } R,\ \text{a character } \lambda \text{ defines CCR by } \lambda(a \cdot b).\\
L2 &: \text{ Bekka's strong ring Stone-von Neumann theorem requires symmetry and } R \simeq \widehat{R}.\\
D3 &: \text{ For a finite Frobenius ring } R,\ X(a), Z(b) \text{ give a Pauli/error basis.}\\
L3 &: \text{ Commutation is controlled by the character-symplectic pairing.}\\
D4 &: \text{ Finite Clifford/Weil splitting is an additional structure, not automatic from the Pauli basis.}\\
T1 &: \text{ For } K = k(t),\ \text{the honest analytic uniqueness theorem applies after choosing a full LCA phase space.}
\end{aligned}$$

32.3 LCA Weyl Systems D1–L1

Let F be a locally compact abelian group. Its Pontryagin dual $\widehat{F}$ is the group of continuous characters $F \to \mathbb{T}$. The canonical Weyl phase space is

$$G_F = F \times \widehat{F}.$$

For $(x,\gamma),(x',\gamma') \in G_F$, define

$$\sigma_{\mathrm{can}}((x,\gamma),(x',\gamma')) = \gamma(x').$$

This is the source convention of Beny–Crann–Lee–Park–Youn, lines 461–465.

The representation space is $L^2(F)$. For $f \in L^2(F)$,

$$(T_x f)(y) = f(y-x), \qquad (M_\gamma f)(y) = \gamma(y) f(y).$$

Define

$$W(x,\gamma) = T_x M_\gamma.$$

Then

$$\begin{aligned}
W(x,\gamma)W(x',\gamma') &= T_x M_\gamma T_{x'} M_{\gamma'}\\
&= \gamma(x') T_{x+x'} M_{\gamma+\gamma'}\\
&= \sigma_{\mathrm{can}}((x,\gamma),(x',\gamma')) W(x+x',\gamma+\gamma').
\end{aligned}$$

Thus W is a projective representation with multiplier σ_{can}. The commutator bicharacter is

$$\Delta((x,\gamma),(x',\gamma')) = \gamma(x')\overline{\gamma'(x)}.$$

Prasad's Stone-von Neumann-Mackey theorem, lines 181–190, states that the canonical representation on $L^2(F)$ is irreducible and that any continuous unitary representation of the corresponding Heisenberg group with the standard central character decomposes as copies of it. This is the clean theorem we may use when the phase space is $F \times \widehat{F}$.

32.4 Locally Compact Rings D2–L2

Let R be a unital second-countable locally compact ring. Let $\lambda \in \widehat{R}$ be a unitary character of the additive group of R. For $d \geq 1$, Bekka defines a pair of unitary representations U, V of R^d to satisfy

$$U(a)V(b) = \lambda(a \cdot b)V(b)U(a), \qquad a \cdot b = \sum_{i=1}^{d} a_i b_i.$$

The Schrödinger pair on $L^2(R^d)$ is

$$(U_{\mathrm{Schr}}(a)f)(x) = f(x+a), \qquad (V_{\mathrm{Schr}}(b)f)(x) = \lambda(x \cdot b)f(x).$$

The strong theorem in Bekka, lines 319–350, assumes

$$\lambda(ab) = \lambda(ba)$$

and that

$$\nabla_\lambda : R \longrightarrow \widehat{R}, \qquad a \longmapsto (x \mapsto \lambda(xa))$$

is a topological group isomorphism. Under these hypotheses, every separable CCR pair decomposes into copies of the Schrödinger pair. The source lists local fields, finite fields, finite products of finite fields, $\mathbb{Z}/N$, matrix algebras over local fields, and adeles as examples where the relevant self-duality hypotheses hold in the stated forms.

The warning for $k(t)$ appears already in the same source: for a countably infinite ring with the discrete topology, line 1014 records that no character can satisfy the isomorphism condition. Therefore a bare discrete $K = k(t)$ cannot be treated as a self-dual locally compact ring in this strong theorem. It may still support an explicit algebraic representation, but the LCA uniqueness theorem must be applied to $K_{\mathrm{disc}} \times \widehat{K_{\mathrm{disc}}}$, to a local completion, or to an adelic completion.

32.5 Finite Frobenius Rings D3–L3

Let R be a finite commutative Frobenius ring with generating character χ. On the Hilbert space with basis $\{v_x : x \in R^n\}$, define

$$X(a)v_x = v_{x+a}, \qquad Z(b)v_x = \chi(b \cdot x)v_x$$

for $a, b, x \in R^n$, where $b \cdot x = \sum_i b_i x_i$. For $P = X(a)Z(b)$ and $P' = X(a')Z(b')$, the cited sources compute

$$PP' = \chi(b \cdot a')X(a+a')Z(b+b')$$

and hence

$$PP' = \chi(b \cdot a' - b' \cdot a)P'P.$$

The symplectic inner product used for stabilizer theory is

$$\langle (a,b),(a',b')\rangle_s = b\cdot a' - b'\cdot a.$$

The Pauli group is obtained by adjoining the necessary scalar roots of unity. In characteristic 2, this scalar group is larger than the characteristic roots; this is exactly the finite-ring version of the even-phase caveat.

Thus finite Frobenius-ring literature strongly supports the Pauli/error-basis and stabilizer layer: generating characters identify the ring with its character module, the X/Z operators are unitary, and commutation is controlled by the character-symplectic form. This is not, by itself, a full Clifford or Weil representation classification over every commutative Frobenius ring.

32.6 Finite Generalized Heisenberg and Clifford Status D4

Solomon's generalized Heisenberg group starts from finite abelian groups A, B, C and a bilinear pairing $\lambda : A\times B \to C$:

$$h(a,b,c)h(a',b',c') = h(a+a', b+b', c+c'+\lambda(a,b')).$$

The commutator is

$$[h(a,b,c), h(a',b',c')] = h(0,0,\lambda(a,b') - \lambda(a',b)).$$

If A, B are finite modules over a commutative ring R and λ is R-balanced, the quotient by the center inherits R-module structure. A character $p : C \to \mathbb{C}^\times$ gives the left Schrödinger representation on functions on A:

$$(h(a,b,c)\cdot f)(x) = p(\lambda(x,b)+c)f(x+a).$$

This is a direct finite algebraic analogue of translations and modulations.

The Clifford/Weil layer is subtler. Lysenko constructs canonical representations for finite Heisenberg groups of odd exponent, explicitly leaving the even-order case open in the cited introduction. Trias constructs Weil modules over coefficient rings for finite or local nonarchimedean fields of characteristic not 2. Hashimoto–Horibe–Hayashi, Korbelar–Tolar, and Galindo analyze projective Clifford extensions and splitting. Galindo's 2026 result states that for a finite abelian group A, the Clifford extension splits iff $4 \nmid |A|$. This is important status information: the obstruction is not a defect of notation but a genuine finite-even phenomenon.

32.7 Conclusion for $k(t)$ T1

The sourced status is therefore:

LCA groups/local fields:	full Stone-von Neumann uniqueness is available.
Locally compact rings:	strong uniqueness requires $R \simeq \widehat{R}$ via λ.
Finite Frobenius rings:	Pauli/stabilizer theory is robust.
Finite Clifford/Weil:	splitting and even-order issues are genuine extra data.

For $K = k(t)$, the next shard must therefore distinguish: the algebraic discrete K-label Heisenberg group; the full discrete-LCA system $K_{\mathrm{disc}} \times \widehat{K_{\mathrm{disc}}}$; and the local or adelic completion where K becomes arithmetic input inside a self-dual LCA configuration space.

33 Weyl-Heisenberg Representations for $k(t)$

33.1 Status and Source Anchors

This shard uses convention (z). The LCA uniqueness theorem is sourced from Prasad, `references/heisenberg_weil/prasad_0912_0574_source/SvN.tex`, lines 164–190. The canonical $F \times \widehat{F}$ Weyl system is sourced from Beny–Crann–Lee–Park–Youn, `references/heisenberg_weil/beny_crann_lee_park_youn_2204_08162_source/main.tex`, lines 461–473. The locally compact ring theorem and the warning about countably infinite discrete rings are sourced from Bekka, `references/heisenberg_weil/bekka_2502_00387_source/CCR-GeneralRings-v5.tex`, lines 319–350, 862–881, and 1014. The local and adelic field background is registered in `references/heisenberg_weil/SOURCES.md` from Poonen and Kudla. The algebraic $k(t)$ calculations below are direct derivations from the displayed definitions.

33.2 Lamport Dependency Skeleton

$D1$: $K = k(t), \quad \tau : k \to \mathbb{F}_p$ nonzero, $\quad \lambda_K(f) = \tau([t^{-1}]f)$.
$L1$: $(a, b) \mapsto \chi_K(ab)$ is a nondegenerate pairing on K.
$D2$: $H_K = \mu_p \times K \times K$ with cocycle $\chi_K(ba')$.
$L2$: H_K is a class-two group with center μ_p.
$D3$: $T_a|\xi\rangle = |\xi + a\rangle, \quad M_b|\xi\rangle = \chi_K(b\xi)|\xi\rangle$.
$T1$: $\pi(z, a, b) = zT_aM_b$ is a unitary irreducible representation of H_K on $\ell^2(K)$.
$T2$: Stone-von Neumann uniqueness requires the full LCA dual or a local/adelic completion.
$E1$: $k = \mathbb{F}_3$ gives explicit ω-commutation examples.

33.3 The Coefficient Character D1

Let $k = \mathbb{F}_q$ have characteristic p, and let $K = k(t)$. Fix a nonzero $\mathbb{F}_p$-linear functional

$$\tau : k \longrightarrow \mathbb{F}_p.$$

For $f \in K$, expand f as a Laurent series at infinity:

$$f = \sum_{n \le N} c_n t^n, \qquad c_n \in k.$$

Only finitely many positive powers occur. Define

$$[t^{-1}]f = c_{-1}, \qquad \lambda_K(f) = \tau([t^{-1}]f).$$

The associated additive character is

$$\chi_K(f) = \exp\left(\frac{2\pi i}{p}\lambda_K(f)\right).$$

The trace $\mathrm{Tr}_{k/\mathbb{F}_p}$ is a standard choice for τ when a trace convention is wanted. The proofs below require only that $\tau \neq 0$.

Lemma $L1$. The pairing

$$B : K \times K \longrightarrow \mathbb{C}^\times, \qquad B(a,b) = \chi_K(ab)$$

is nondegenerate in each variable.

Proof. Let $a \in K$ be nonzero. Since $\tau \neq 0$, choose $c \in k$ with $\tau(c) \neq 0$. Put

$$b = \frac{c\,t^{-1}}{a} \in K.$$

Then $ab = c\,t^{-1}$, so

$$\lambda_K(ab) = \tau(c) \neq 0.$$

Thus $\chi_K(ab) \neq 1$. This proves that no nonzero a is killed by all right pairings. The same argument with the variables interchanged proves nondegeneracy in the second variable, because multiplication in K is commutative.

33.4 The Algebraic Heisenberg Group D2–L2

Let $\mu_p = \{z \in \mathbb{C} : z^p = 1\}$. Define

$$H_K = \mu_p \times K \times K$$

with multiplication

$$(z, a, b)(z', a', b') = \bigl(zz'\chi_K(ba'),\ a + a',\ b + b'\bigr).$$

Here a is the translation coordinate and b is the modulation coordinate.

Associativity. Let $h = (z, a, b)$, $h' = (z', a', b')$, and $h'' = (z'', a'', b'')$. The central factor in $(hh')h''$ is

$$zz'z''\chi_K(ba')\chi_K((b + b')a'') = zz'z''\chi_K(ba' + ba'' + b'a'').$$

The central factor in $h(h'h'')$ is

$$zz'z''\chi_K(b'a'')\chi_K(b(a' + a'')) = zz'z''\chi_K(b'a'' + ba' + ba'').$$

These are equal by additivity of χ_K. Therefore the multiplication is associative.

Identity and inverse. The identity is $(1, 0, 0)$. A direct multiplication gives

$$(z, a, b)^{-1} = \bigl(z^{-1}\chi_K(ba), -a, -b\bigr).$$

Commutator and center. From the multiplication law,

$$(1, a, b)(1, a', b') = \chi_K(ba')(1, a + a', b + b')$$

and

$$(1, a', b')(1, a, b) = \chi_K(b'a)(1, a + a', b + b').$$

Hence the commutator is the central element

$$[(1, a, b), (1, a', b')] = \bigl(\chi_K(ba' - b'a), 0, 0\bigr).$$

If (z, a, b) is central, then this phase is 1 for all a', b'. Taking $b' = 0$ gives $\chi_K(ba') = 1$ for all a', hence $b = 0$ by $L1$. Taking $a' = 0$ then gives $\chi_K(-b'a) = 1$ for all b', hence $a = 0$. Thus

$$Z(H_K) = \mu_p \times \{0\} \times \{0\}.$$

33.5 The Schrödinger Representation D3–T1

Let $\ell^2(K)$ have orthonormal basis $\{|\xi\rangle : \xi \in K\}$. Define

$$T_a|\xi\rangle = |\xi + a\rangle, \qquad M_b|\xi\rangle = \chi_K(b\xi)|\xi\rangle.$$

Both are unitary: T_a permutes the basis and M_b is diagonal with unit-modulus diagonal entries. Moreover,

$$\begin{aligned} M_b T_a|\xi\rangle &= M_b|\xi + a\rangle \\ &= \chi_K(b\xi + ba)|\xi + a\rangle \\ &= \chi_K(ba) T_a M_b|\xi\rangle. \end{aligned}$$

Therefore

$$T_a M_b T_{a'} M_{b'} = \chi_K(ba') T_{a+a'} M_{b+b'}.$$

The map

$$\pi_K : H_K \longrightarrow U(\ell^2(K)), \qquad \pi_K(z, a, b) = z T_a M_b$$

is a unitary representation with central character

$$\pi_K(z, 0, 0) = zI.$$

Irreducibility. Let A be a bounded operator on $\ell^2(K)$ commuting with all T_a and M_b. Write

$$A_{\xi,\eta} = \langle \xi|A|\eta\rangle.$$

Commutation with M_b gives

$$\chi_K(b\xi) A_{\xi,\eta} = \chi_K(b\eta) A_{\xi,\eta}$$

for all b, ξ, η. If $\xi \neq \eta$, then $L1$ gives b such that $\chi_K(b(\xi - \eta)) \neq 1$, so $A_{\xi,\eta} = 0$. Thus A is diagonal. Write $A_{\xi,\xi} = d_\xi$. Commutation with T_a gives

$$d_{\xi+a} = d_\xi$$

for all a, ξ. Since the additive group of K acts transitively on itself by translations, all d_ξ are equal. Thus the commutant of $\pi_K(H_K)$ is $\mathbb{C}I$. If a closed subspace were invariant, its orthogonal projection would commute with the representation, hence would be 0 or I. Therefore π_K is irreducible.

33.6 Conclusion

The concrete rational-function answer is:

$$H_{\mathbb{F}_q(t)} = \mu_p \times k(t) \times k(t), \qquad (z, a, b)(z', a', b') = (zz'\chi_K(ba'), a + a', b + b'),$$

with irreducible Schrödinger representation on $\ell^2(k(t))$. The analytic Stone-von Neumann answer is separate and is recorded in the following shard.

34 Full-Dual and Local Models for $k(t)$

34.1 Status and Source Anchors

This shard completes AQM-32. The full-dual LCA statement uses Prasad, `references/heisenberg_weil/prasad_0912_0574_source/SvN.tex`, lines 95–107 and 164–190, and Beny–Crann–Lee–Park–Youn, `references/heisenberg_weil/beny_crann_lee_park_youn_2204_08162_source/main.tex`, lines 461–473. The warning about countably infinite discrete rings uses Bekka, `references/heisenberg_weil/bekka_2502_00387_source/CCR-GeneralRings-v5.tex`, line 1014. Local and adelic background is registered in `references/heisenberg_weil/SOURCES.md` from Poonen's Tate notes and Kudla's local theta notes.

34.2 Lamport Dependency Skeleton

$D1$: K_{disc} is $K = k(t)$ with the discrete topology.
$L1$: $K_{\text{disc}} \not\cong \widehat{K_{\text{disc}}}$ as LCA groups.
$D2$: $G_{\text{full}} = K_{\text{disc}} \times \widehat{K_{\text{disc}}}$ is the full LCA Weyl phase space.
$L2$: The algebraic K-modulations form a dense subgroup of $\widehat{K_{\text{disc}}}$.
$D3$: $K_\infty = k((t^{-1}))$ is the local completion at infinity.
$T1$: Stone-von Neumann uniqueness belongs to G_{full} or to local/adelic LCA completions.
$E1$: $k = \mathbb{F}_3$ gives explicit ω-commutation examples.

34.3 The Full Discrete-Dual Model D1–L2

The algebraic representation of AQM-32 is an algebraic Schrödinger representation theorem for the concrete group H_K. It is not the full LCA Stone-von Neumann theorem. The reason is structural: if K is given the discrete topology, then its Pontryagin dual $\widehat{K_{\text{disc}}}$ is compact. Since K is countably infinite, K is not isomorphic to $\widehat{K_{\text{disc}}}$ as an LCA group. Bekka records the same obstruction for countably infinite discrete rings.

The full LCA Weyl system with configuration group K_{disc} is

$$G_{\text{full}} = K_{\text{disc}} \times \widehat{K_{\text{disc}}}.$$

It acts on $\ell^2(K)$ by

$$W(a, \gamma) = T_a M_\gamma, \qquad M_\gamma |\xi\rangle = \gamma(\xi)|\xi\rangle.$$

Prasad's theorem applies to this G_{full}. The algebraic H_K is the restriction to the subgroup of characters

$$\gamma_b(\xi) = \chi_K(b\xi), \qquad b \in K.$$

By AQM-32 $L1$, $b \mapsto \gamma_b$ is injective.

Density derivation. Let $\Gamma_K = \{\gamma_b : b \in K\} \subset \widehat{K_{\text{disc}}}$. Its annihilator in the double dual K_{disc} is

$$\Gamma_K^\perp = \{\xi \in K : \gamma_b(\xi) = 1 \text{ for all } b \in K\}.$$

By nondegeneracy of $(b, \xi) \mapsto \chi_K(b\xi)$, this annihilator is $\{0\}$. Pontryagin duality identifies the annihilator of the closure $\overline{\Gamma_K}$ with $\Gamma_K^\perp$. Hence $\overline{\Gamma_K} = \widehat{K_{\text{disc}}}$. The algebraic model therefore uses a dense rational-function family of modulations, while the full LCA theorem uses every continuous character of K_{disc}.

34.4 Local and Adelic Completions D3–T1

For analytic arithmetic quantum fields, the more natural locally compact configuration spaces are local completions such as

$$K_\infty = k((t^{-1}))$$

or, later, the adele ring $\mathbb{A}_K$. A nontrivial continuous additive character identifies a local field with its Pontryagin dual, so the canonical Weyl system has phase space $K_\infty \times K_\infty$ after that identification. The Schrödinger representation then lives on

$$L^2(K_\infty).$$

For the adelic theory, a residue character built from a differential gives the standard self-dual adelic setting; K embeds diagonally as arithmetic input, not as a replacement for the full dual phase group.

Thus there are three honest objects:

$$\begin{array}{ll} \ell^2(K): & \text{discrete algebraic rational-function representation;} \\ \ell^2(K_{\mathrm{disc}}) \text{ with } \widehat{K}: & \text{full LCA discrete-dual Stone-von Neumann system;} \\ L^2(K_\infty) \text{ or } L^2(\mathbb{A}_K): & \text{local or global analytic arithmetic system.} \end{array}$$

They should not be collapsed into one notation.

34.5 Concrete Examples for $k = \mathbb{F}_3$ E1

Let $k = \mathbb{F}_3$, $K = \mathbb{F}_3(t)$, $\tau = \mathrm{id}$, and

$$\omega = \exp(2\pi i/3).$$

Then

$$\chi_K(f) = \omega^{[t^{-1}]f}.$$

For $a = 1$ and $b = t^{-1}$,

$$M_{t^{-1}}T_1 = \omega\, T_1 M_{t^{-1}}.$$

Indeed, $ba = t^{-1}$, whose t^{-1}-coefficient is 1.

For $a = t$ and $b = t^{-1}$,

$$M_{t^{-1}}T_t = T_t M_{t^{-1}},$$

because $ba = 1$ has t^{-1}-coefficient 0. For $a = t$ and $b = t^{-2}$,

$$M_{t^{-2}}T_t = \omega\, T_t M_{t^{-2}},$$

because $ba = t^{-1}$.

On basis vectors,

$$T_{t+1}|t^{-1}\rangle = |t + 1 + t^{-1}\rangle,$$

$$M_{t^{-1}}|1\rangle = \omega|1\rangle, \qquad M_{t^{-1}}|t\rangle = |t\rangle,$$

and

$$M_{t^{-2}}|t\rangle = \omega|t\rangle.$$

Thus the rational-function Heisenberg group already contains infinitely many explicit qutrit-like commutation relations, indexed by rational functions rather than by closed points. This is a different layer from the closed-point quasi-local field of $\operatorname{Spec} \mathbb{F}_3[t]$, where each monic irreducible π contributes one finite qudit $\ell^2(\mathbb{F}_3[t]/(\pi))$.

34.6 Conclusion

The concrete rational-function construction gives an irreducible unitary representation on $\ell^2(k(t))$. Stone-von Neumann uniqueness belongs to a larger or completed LCA object: the full discrete-dual system, a local completion, or an adelic completion. The next arithmetic-field step should choose which completion is physically intended before importing Fourier transform, Gaussian, or Weil-representation statements.

35 Polynomial-Quotient Arithmetic Fields over $\mathbb{F}_3$

35.1 Status and Source Anchors

This shard is a checked finite affine example. The source-local notions are Spec, prime ideals, residue fields, and Zariski opens from `references/algebraic_geometry/stacks_project_algebra.tex`, lines 120–158, 250–274, 2972–2983, and 3104–3117. The polynomial-ring prime classification and finite residue-field count reuse the direct derivation of AQM-29. The finite Weyl representation is the closed-point construction of AQM-28, with the polynomial-quotient convention fixed in Convention (aa). The coordinate ring is written $R_p = \mathbb{F}_3[t]/(p)$.

35.2 Lamport Dependency Skeleton

$$\begin{array}{rcl} D1 & : & k = \mathbb{F}_3,\ p \in k[t] \text{ monic nonzero},\ R_p = k[t]/(p),\ X_p = \operatorname{Spec} R_p. \\ D2 & : & p = \prod_{i=1}^{r} \pi_i^{e_i}, \quad \pi_i \text{ distinct monic irreducibles},\ K_i = k[t]/(\pi_i). \\ L1 & : & X_p = \{x_i = (\pi_i)/(p) : 1 \le i \le r\},\ \kappa(x_i) = K_i, \text{ and } X_p \text{ is finite discrete.} \\ L2 & : & R_p \cong \prod_i k[t]/(\pi_i^{e_i}), \quad (R_p)_{\mathrm{red}} \cong \prod_i K_i. \\ D3 & : & E_p = \bigoplus_i K_i^2, \quad \mathcal{H}_p = \bigotimes_i \ell^2(K_i). \\ T1 & : & E_p \text{ has an explicit projective unitary Weyl representation } W_p \text{ on } \mathcal{H}_p. \\ T2 & : & \operatorname{span}\{W_p(e) : e \in E_p\} = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_p), \text{ hence the representation is irreducible.} \\ T3 & : & R_p \text{ field profiles factor through } (R_p)_{\mathrm{red}}. \end{array}$$

35.3 The Spectrum L1–L2

Assume first that $p \neq 1$, so $R_p \neq 0$. Factor

$$p = \prod_{i=1}^{r} \pi_i^{e_i}$$

in $k[t]$, where the π_i are distinct monic irreducibles and $e_i \geq 1$. Put $d_i = \deg \pi_i$ and $K_i = k[t]/(\pi_i)$.

Lemma *L*1. The points of X_p are exactly

$$x_i = (\pi_i)/(p), \qquad 1 \leq i \leq r,$$

and

$$\kappa(x_i) = K_i, \qquad |K_i| = 3^{d_i}.$$

Every point is closed, and every subset of X_p is open.

Proof. A prime ideal of R_p has inverse image a prime ideal $\mathfrak{q} \subset k[t]$ containing (p). Since $p \in \mathfrak{q}$ and $p \neq 0$, the prime $\mathfrak{q}$ is nonzero. By the AQM-29 Euclidean-domain derivation, $\mathfrak{q} = (\pi)$ for a monic irreducible π. The containment $(p) \subset (\pi)$ is equivalent to $\pi \mid p$, so $\pi = \pi_i$ for a unique i. Conversely, each $(\pi_i)/(p)$ is prime because

$$R_p/x_i \cong k[t]/(\pi_i) = K_i$$

is a field. This also proves the residue-field formula, since x_i is maximal. The residue classes $1, t, \ldots, t^{d_i - 1}$ form a k-basis of the quotient, so $|K_i| = 3^{d_i}$. Since every point is maximal, every singleton is closed. A finite space in which every singleton is closed is discrete, because the complement of a singleton is a finite union of closed singletons.

Lemma $L2$. There is a ring isomorphism

$$R_p \cong \prod_{i=1}^{r} k[t]/(\pi_i^{e_i}),$$

and the reduced quotient is

$$(R_p)_{\mathrm{red}} = R_p/\sqrt{0} \cong k[t]/(\pi_1 \cdots \pi_r) \cong \prod_{i=1}^{r} K_i.$$

Proof. If $i \neq j$, then $\pi_i^{e_i}$ and $\pi_j^{e_j}$ are coprime. Bezout therefore gives $a_{ij}\pi_i^{e_i} + b_{ij}\pi_j^{e_j} = 1$. The Chinese remainder construction gives a surjective map

$$k[t]/(p) \longrightarrow \prod_i k[t]/(\pi_i^{e_i}),$$

and its kernel is the intersection of the pairwise coprime ideals $(\pi_i^{e_i})$, equal to their product (p).

For the nilradical, a class $f \bmod p$ is nilpotent iff $p \mid f^N$ for some N. This happens iff every π_i divides f, equivalently $\pi_1 \cdots \pi_r \mid f$. Hence $\sqrt{0} = (\pi_1 \cdots \pi_r)/(p)$. Quotienting by this ideal gives $k[t]/(\pi_1 \cdots \pi_r)$, and another Chinese remainder step gives $\prod_i K_i$.

If $p = 1$, then $R_p = 0$, $\operatorname{Spec} R_p = \emptyset$, $E_p = 0$, and the observable algebra is the empty tensor product $\mathbb{C}$.

35.4 The Weyl-Heisenberg Representation D3–T2

For each closed point x_i, define the additive character

$$\psi_i(a) = \exp\!\left(\frac{2\pi i}{3}\operatorname{Tr}_{K_i/\mathbb{F}_3}(a)\right).$$

Let

$$\mathcal{H}_i = \ell^2(K_i), \qquad E_i = K_i^2.$$

On the basis $|s\rangle$, $s \in K_i$, set

$$T_i(a)|s\rangle = |s+a\rangle, \qquad R_i(b)|s\rangle = \psi_i(bs)|s\rangle, \qquad W_i(a,b) = T_i(a)R_i(b).$$

Lemma $L3$. For $a, b, a', b' \in K_i$,

$$W_i(a,b)W_i(a',b') = \psi_i(ba')W_i(a+a', b+b')$$

and

$$W_i(a,b)W_i(a',b') = \psi_i(ba' - b'a)W_i(a',b')W_i(a,b).$$

Proof. For $s \in K_i$,

$$R_i(b)T_i(a')|s\rangle = \psi_i(b(s+a'))|s+a'\rangle = \psi_i(ba')T_i(a')R_i(b)|s\rangle.$$

Substituting this into $T_i(a)R_i(b)T_i(a')R_i(b')$ gives the first formula. Dividing this product law by the same law with the two labels reversed gives the commutator formula.

For $J \subset \{1, \ldots, r\}$, write

$$E(J) = \bigoplus_{i \in J} K_i^2, \qquad \mathcal{H}(J) = \bigotimes_{i \in J} \mathcal{H}_i, \qquad \mathcal{A}(J) = \bigotimes_{i \in J} \operatorname{End}_{\mathbb{C}}(\mathcal{H}_i).$$

The empty tensor product is $\mathbb{C}$. For the global system, put

$$E_p = E(\{1,\dots,r\}), \qquad \mathcal{H}_p = \mathcal{H}(\{1,\dots,r\}).$$

For $e = ((a_i,b_i))_i \in E_p$, define

$$W_p(e) = \bigotimes_i W_i(a_i,b_i).$$

Theorem $T1$**.** The maps $W_p(e)$ form a projective unitary representation of the additive group E_p. Explicitly,

$$W_p(e)W_p(f) = c_p(e,f)W_p(e+f), \qquad c_p(e,f) = \prod_i \psi_i(b_i a_i'),$$

and

$$W_p(e)W_p(f) = B_p(e,f)W_p(f)W_p(e),$$

where

$$B_p(e,f) = \prod_i \psi_i(b_i a_i' - b_i' a_i).$$

Equivalently, the finite Heisenberg group

$$H_p^{\mathrm{WH}} = \mu_3 \times E_p$$

with multiplication

$$(\zeta,e)(\eta,f) = (\zeta\eta c_p(e,f), e+f)$$

has a unitary representation $\rho_p(\zeta,e) = \zeta W_p(e)$.

Proof. Each $T_i(a)$ is a permutation unitary and each $R_i(b)$ is a diagonal unitary, so $W_i(a,b)$ and $W_p(e)$ are unitary. Tensoring the one-site formula of $L3$ over all i gives the displayed multiplier and commutator. The multiplication law for H_p^{WH} is exactly the statement that the multiplier defect has been placed in the central μ_3-coordinate, so ρ_p is an ordinary unitary representation.

Theorem $T2$**.**

$$\mathrm{span}_{\mathbb{C}}\{W_p(e) : e \in E_p\} = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_p).$$

Hence ρ_p is irreducible on $\mathcal{H}_p$. Moreover

$$\dim_{\mathbb{C}} \mathcal{H}_p = \prod_i 3^{d_i} = 3^{\sum_i d_i} = 3^{\deg(\pi_1\cdots\pi_r)}.$$

Proof. AQM-28 proves that the one-site finite-field Weyl operators span $\mathrm{End}(\mathcal{H}_i)$. Tensor products of spanning sets span the tensor product algebra, giving the first formula. A subspace invariant under all $W_p(e)$ is invariant under their complex span, hence under the full matrix algebra on $\mathcal{H}_p$. The only such subspaces are 0 and $\mathcal{H}_p$. The dimension formula follows from $\dim \ell^2(K_i) = |K_i| = 3^{d_i}$.

35.5 Coordinate-Ring Fields and Local Nets T3

The coordinate ring maps to residues by

$$\mathrm{ev} : R_p \longrightarrow \prod_i K_i, \qquad f \bmod p \longmapsto (f \bmod \pi_i)_i.$$

By $L2$, $\ker(\mathrm{ev}) = \sqrt{0}$. Thus a pair $(f, g) \in R_p^2$ defines the Weyl label

$$e_{f,g} = ((f \bmod \pi_i,\ g \bmod \pi_i))_i \in E_p,$$

and nilpotent changes in f or g do not change $e_{f,g}$. The coordinate-ring field operator is therefore

$$W_p(f, g) := W_p(e_{f,g}),$$

with product law

$$W_p(f, g)W_p(f', g') = \left(\prod_i \psi_i((g \bmod \pi_i)(f' \bmod \pi_i))\right) W_p(f + f', g + g').$$

The commutator phase is

$$\prod_i \psi_i((g \bmod \pi_i)(f' \bmod \pi_i) - (g' \bmod \pi_i)(f \bmod \pi_i)).$$

Since X_p is discrete, every open set is $U_J = \{x_i : i \in J\}$. The open-local algebra is

$$\mathcal{A}(U_J) = \mathcal{A}(J) = \bigotimes_{i \in J} \mathrm{End}_{\mathbb{C}}(\ell^2(K_i)),$$

and $J \subset J'$ gives the inclusion $a \mapsto a \otimes \mathbf{1}$. States on $\mathcal{A}(U_J)$ are density matrices on $\mathcal{H}(J)$, and restriction along $J \subset J'$ is partial trace over $J' \setminus J$.

35.6 Conclusion

For $R_p = \mathbb{F}_3[t]/(p)$, the complete residue-qudit arithmetic field is the tensor product of one finite-field Weyl system for each distinct irreducible factor of p. The Weyl-Heisenberg representation is explicit on $\bigotimes_i \ell^2(\mathbb{F}_3[t]/(\pi_i))$, spans the full matrix algebra, and is irreducible. Repeated factors of p are not ignored: they are scheme-theoretic nilpotent thickening data in the coordinate ring. They are invisible to the present residue-qudit Weyl labels because all such labels factor through the reduced quotient. AQM-36 adds the nilpotent-sensitive thickened Weyl layer.

36 The $t^n = 1$ Arithmetic Field over $\mathbb{F}_3$

36.1 Status and Source Anchors

This shard specializes AQM-34 to

$$R_n = \mathbb{F}_3[t]/(t^n - 1).$$

The input facts are the direct polynomial derivations of AQM-29 and AQM-34: irreducible factors of $k[t]$ give closed points, residue fields are the corresponding finite quotients, and the Weyl representation is the tensor product of the one-site finite-field Weyl systems from AQM-28. No new external source is used in this specialization.

36.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & n \geq 1, \quad R_n = \mathbb{F}_3[t]/(t^n - 1), \quad X_n = \operatorname{Spec} R_n. \\
D2 & : & n = 3^s m, \quad s = v_3(n), \quad 3 \nmid m. \\
L1 & : & t^n - 1 = (t^m - 1)^{3^s}, \quad t^m - 1 \text{ is squarefree.} \\
L2 & : & \text{Closed Weyl sites are the irreducible factors of } t^m - 1. \\
L3 & : & \text{The degrees of these factors are determined by } \operatorname{ord}_d(3) \text{ for } d \mid m. \\
D3 & : & E_n = \bigoplus_{\pi \mid t^m - 1} K_\pi^2, \quad \mathcal{H}_n = \bigotimes_{\pi \mid t^m - 1} \ell^2(K_\pi). \\
T1 & : & W_n \text{ is the complete projective unitary Weyl representation on } \mathcal{H}_n. \\
T2 & : & \dim_{\mathbb{C}} \mathcal{H}_n = 3^m, \quad \mathcal{A}_n \cong M_{3^m}(\mathbb{C}).
\end{array}$$

36.3 Factorization in Characteristic 3 L1–L3

Let $n = 3^s m$, with $s = v_3(n)$ and $3 \nmid m$. In characteristic 3, Frobenius gives

$$t^n - 1 = (t^m)^{3^s} - 1 = (t^m - 1)^{3^s}.$$

The derivative of $t^m - 1$ is mt^{m-1}, and $m \neq 0$ in $\mathbb{F}_3$. Since t does not divide $t^m - 1$, the greatest common divisor of $t^m - 1$ and mt^{m-1} is 1. Thus $t^m - 1$ is squarefree. Consequently the closed points of X_n are exactly the irreducible factors of $t^m - 1$, while each such point is thickened in R_n to multiplicity 3^s.

For $d \mid m$, $d > 1$, let

$$o_d = \operatorname{ord}_d(3).$$

A primitive d-th root α lies in $\mathbb{F}_{3^r}$ exactly when $\alpha^{3^r} = \alpha$, equivalently $d \mid 3^r - 1$. Hence the degree of α over $\mathbb{F}_3$ is o_d. The Frobenius orbit of α has o_d elements, so the $\varphi(d)$ primitive d-th roots produce

$$\frac{\varphi(d)}{o_d}$$

irreducible factors of degree o_d. For $d = 1$, the factor is $t - 1$. The sum of all factor degrees is

$$\sum_{d \mid m} \varphi(d) = m.$$

36.4 The Complete Weyl-Heisenberg Representation T1–T2

Let $\mathcal{P}_n$ be the set of monic irreducible factors of $t^m - 1$. For $\pi \in \mathcal{P}_n$, write

$$K_\pi = \mathbb{F}_3[t]/(\pi), \qquad d_\pi = \deg \pi.$$

The closed point is $x_\pi = (\pi)/(t^n - 1)$, and its residue field is $\kappa(x_\pi) = K_\pi$.

Define

$$E_n = \bigoplus_{\pi \in \mathcal{P}_n} K_\pi^2, \qquad \mathcal{H}_n = \bigotimes_{\pi \in \mathcal{P}_n} \ell^2(K_\pi).$$

For $a, b, s \in K_\pi$, set

$$\psi_\pi(c) = \exp\left(\frac{2\pi i}{3} \operatorname{Tr}_{K_\pi/\mathbb{F}_3}(c)\right),$$

$$T_\pi(a)|s\rangle = |s + a\rangle, \qquad R_\pi(b)|s\rangle = \psi_\pi(bs)|s\rangle, \qquad W_\pi(a, b) = T_\pi(a)R_\pi(b).$$

For $e = ((a_\pi, b_\pi))_\pi \in E_n$, put

$$W_n(e) = \bigotimes_{\pi \in \mathcal{P}_n} W_\pi(a_\pi, b_\pi).$$

Theorem $T1$. For $e = ((a_\pi, b_\pi))$ and $f = ((a'_\pi, b'_\pi))$,

$$W_n(e)W_n(f) = \left(\prod_{\pi \in \mathcal{P}_n} \psi_\pi(b_\pi a'_\pi)\right) W_n(e + f),$$

and

$$W_n(e)W_n(f) = \left(\prod_{\pi \in \mathcal{P}_n} \psi_\pi(b_\pi a'_\pi - b'_\pi a_\pi)\right) W_n(f)W_n(e).$$

The central extension

$$H_n^{\mathrm{WH}} = \mu_3 \times E_n, \qquad (\zeta, e)(\eta, f) = (\zeta\eta c_n(e, f), e + f),$$

with

$$c_n(e, f) = \prod_{\pi \in \mathcal{P}_n} \psi_\pi(b_\pi a'_\pi),$$

is represented unitarily by $\rho_n(\zeta, e) = \zeta W_n(e)$.

Proof. This is AQM-34 applied to $p = t^n - 1$. The one-site identity

$$R_\pi(b)T_\pi(a') = \psi_\pi(ba')T_\pi(a')R_\pi(b)$$

gives the multiplier, and tensoring over all factors gives the displayed laws.

Theorem $T2$.

$$\operatorname{span}_{\mathbb{C}}\{W_n(e) : e \in E_n\} = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_n),$$

so the representation is irreducible, and

$$\dim_{\mathbb{C}} \mathcal{H}_n = \prod_{\pi \in \mathcal{P}_n} 3^{d_\pi} = 3^m, \qquad \mathcal{A}_n \cong M_{3^m}(\mathbb{C}).$$

Proof. At each π, the finite-field Weyl operators span $\operatorname{End}(\ell^2(K_\pi))$. Tensor products span the tensor-product matrix algebra. Any invariant subspace is therefore invariant under the full matrix algebra and is 0 or $\mathcal{H}_n$. The dimension formula uses $\sum_{\pi \in \mathcal{P}_n} d_\pi = \deg(t^m - 1) = m$.

The coordinate ring has 3^n elements, but the current residue-qudit Hilbert space has dimension 3^m. They agree exactly when $s = 0$, i.e. when $3 \nmid n$. If $s > 0$, the nilradical is generated by the class of $t^m - 1$, and coordinate-ring field labels factor through

$$R_n/(t^m - 1) \cong \mathbb{F}_3[t]/(t^m - 1).$$

36.5 Open Sets and States

Because X_n is a finite set of closed points, it is discrete. For $J \subset \mathcal{P}_n$,

$$U_J = \{x_\pi : \pi \in J\}$$

is open, and

$$\mathcal{A}(U_J) = \bigotimes_{\pi \in J} \mathrm{End}_{\mathbb{C}}(\ell^2(K_\pi)).$$

The inclusion $J \subset J'$ sends a to $a \otimes \mathbf{1}_{J' \setminus J}$. States on $\mathcal{A}(U_J)$ are density matrices on $\bigotimes_{\pi \in J} \ell^2(K_\pi)$, and restriction is partial trace over omitted factors.

36.6 Small Cases

For $n = 1$, $t^n - 1 = t - 1$. There is one $\mathbb{F}_3$ site:

$$\mathcal{H}_1 \cong \mathbb{C}^3, \qquad \mathcal{A}_1 \cong M_3(\mathbb{C}).$$

For $n = 2$,

$$t^2 - 1 = (t-1)(t+1),$$

so there are two $\mathbb{F}_3$ sites and

$$\mathcal{H}_2 \cong \mathbb{C}^3 \otimes \mathbb{C}^3, \qquad \mathcal{A}_2 \cong M_9(\mathbb{C}).$$

For $n = 3$,

$$t^3 - 1 = (t-1)^3.$$

The scheme is a triple thickening of the point $t = 1$, but the residue-qudit layer is one qutrit. If $u = t - 1$, then $u^3 = 0$ in R_3, and u has zero residue at the unique Weyl site.

For $n = 4$,

$$t^4 - 1 = (t-1)(t+1)(t^2+1),$$

and $t^2 + 1$ has no root in $\mathbb{F}_3$. Thus

$$\mathcal{H}_4 \cong \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^9, \qquad \mathcal{A}_4 \cong M_{81}(\mathbb{C}).$$

On the $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2+1)$ factor, Frobenius sends $u \mapsto u^3 = -u$, so

$$\mathrm{Tr}_{\mathbb{F}_9/\mathbb{F}_3}(a + bu) = 2a$$

for $a, b \in \mathbb{F}_3$. This determines the explicit phase character on that 9-level site.

For $n = 6$, $n = 3 \cdot 2$, so $m = 2$ and

$$t^6 - 1 = (t^2 - 1)^3 = (t-1)^3(t+1)^3.$$

There are still exactly two residue Weyl sites, both $\mathbb{F}_3$-sites:

$$\mathcal{H}_6 \cong \mathbb{C}^3 \otimes \mathbb{C}^3, \qquad \mathcal{A}_6 \cong M_9(\mathbb{C}),$$

although the coordinate ring has 3^6 elements.

36.7 Conclusion

The $t^n = 1$ example makes the characteristic-3 issue explicit. The residue-qudit field depends on the squarefree part $t^m - 1$, where $m = n/3^{v_3(n)}$. The full coordinate ring still records the repeated factors of $t^n - 1$ as nilpotents, but in the residue-qudit layer those nilpotents have zero residues and therefore no separate residue Weyl operators. AQM-36 adds the separate nilpotent-sensitive layer.

37 Nilpotent Thickening Weyl Fields

37.1 Status and Source Anchors

This shard adds the nilpotent-sensitive layer proposed after AQM-34 and AQM-35. The finite Frobenius-ring source anchors are `references/heisenberg_weil/gluesing_luerssen_pllaha_1710_09884_source/StabCodesFrob5.tex`, lines 224–249 for Frobenius rings and generating characters, lines 286–323 for $X(a)$ and $Z(b)$, lines 353–355 for the multiplication law, and lines 436–452 for the error-basis statement. Sidana–Kashyap, `references/heisenberg_weil/sidana_kashyap_2202_00248_source/EAQECCs_over_rings_v15.tex`, lines 103–110 and 180–197, gives the same local Frobenius-ring Pauli framework. The trace pairing input for residue fields is the Stacks Project snapshot `references/algebraic_geometry/stacks_project_algebra.tex`, lines 45278–45284. The explicit generating-character proof below is local to this shard.

37.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k=\mathbb{F}_3,\ \pi\in k[t] \text{ monic irreducible},\ e\geq 1,\ A=k[t]/(\pi^e).\\
D2 & : & \mathfrak{m}=(\pi)/(\pi^e),\ K=A/\mathfrak{m}=k[t]/(\pi),\ d=\deg\pi.\\
L1 & : & A \text{ has unique } \pi\text{-adic expansions and ideals } \mathfrak{m}^r.\\
D3 & : & \lambda(a)=\mathrm{Tr}_{K/k}(a_{e-1}) \text{ from the top } \pi\text{-coefficient},\quad \chi(a)=\exp(2\pi i\lambda(a)/3).\\
L2 & : & \chi \text{ is generating: } b\neq 0\Rightarrow (a\mapsto\chi(ba)) \text{ is nontrivial.}\\
D4 & : & \mathcal{H}_A=\ell^2(A),\quad E_A=A^2,\quad W_A(q,p)=T(q)R(p).\\
T1 & : & W_A \text{ is a projective unitary Weyl representation with multiplier } \chi(pq').\\
T2 & : & \{W_A(q,p):(q,p)\in A^2\} \text{ is an operator basis of } \mathrm{End}_{\mathbb{C}}(\mathcal{H}_A).\\
T3 & : & \mathfrak{m}^r \text{ position jets pair with } \mathfrak{m}^{e-1-r} \text{ momentum jets in the associated graded.}\\
T4 & : & \text{For } R_p=\mathbb{F}_3[t]/(p),\ \text{the thickened layer has } \dim\mathcal{H}=3^{\deg p}.
\end{array}$$

37.3 The Local Artin Ring L1–L2

Let

$$A=A_{\pi,e}=k[t]/(\pi^e),\qquad \mathfrak{m}=(\pi)/(\pi^e),\qquad K=k[t]/(\pi).$$

Every class $a\in A$ has a unique expansion

$$a=a_0+a_1\pi+\cdots+a_{e-1}\pi^{e-1},$$

where each a_j is represented by a polynomial of degree $< d=\deg\pi$. This follows by repeated Euclidean division by the monic polynomial π.

Lemma $L1$. The ideals of A are exactly

$$A=\mathfrak{m}^0\supset\mathfrak{m}\supset\cdots\supset\mathfrak{m}^{e-1}\supset\mathfrak{m}^e=0.$$

Every nonzero $a\in A$ has a unique valuation $v(a)=r$, $0\leq r<e$, with $a=\pi^r u$ for a unit $u\in A^\times$.

Proof. If $a\neq 0$, take the least r for which the coefficient a_r is nonzero. Then $a=\pi^r u$, where the residue of u in K is $a_r\neq 0$, so u is a unit. Let $I\subset A$ be a nonzero ideal and choose an element of minimal valuation r. Multiplying by the inverse of its unit part gives $\pi^r\in I$, hence $\mathfrak{m}^r\subset I$. Minimality of r gives $I\subset\mathfrak{m}^r$, so $I=\mathfrak{m}^r$.

Define

$$\lambda(a)=\mathrm{Tr}_{K/k}(a_{e-1}),\qquad \chi(a)=\exp\left(\frac{2\pi i}{3}\lambda(a)\right).$$

Here $a_{e-1}\in K$ is the top π-coefficient. For $e=1$ this is the residue-field character used in AQM-34.

Lemma $L2$. If $b \neq 0$ in A, then the additive character

$$a \longmapsto \chi(ba)$$

is nontrivial. Hence the map $b \mapsto (a \mapsto \chi(ba))$ identifies A with the additive character group $\widehat{A}$.

Proof. Write $b = \pi^r u$ with $u \in A^\times$. Let $\bar{u}$ be the residue of u in K. Since K is finite, Frobenius $x \mapsto x^3$ is injective and hence surjective; thus K is perfect and K/k is finite separable. The cited Stacks trace-pairing source says $(x, y) \mapsto \mathrm{Tr}_{K/k}(xy)$ is nondegenerate. Choose $\bar{c} \in K$ with $\mathrm{Tr}_{K/k}(\bar{u}\bar{c}) \neq 0$, and lift it to $c \in A$ of π-degree 0. Set

$$a = \pi^{e-1-r} c.$$

Then $ba = \pi^{e-1} uc$ modulo π^e, so the top coefficient of ba is $\bar{u}\bar{c}$. Therefore $\chi(ba) \neq 1$. The map $A \to \widehat{A}$ is injective, and both finite groups have cardinality $|A|$, so it is an isomorphism.

37.4 The Thickened Weyl Representation T1–T2

Let

$$\mathcal{H}_A = \ell^2(A)$$

with basis $|s\rangle$, $s \in A$. For $q, p \in A$, define

$$T(q)|s\rangle = |s + q\rangle, \qquad R(p)|s\rangle = \chi(ps)|s\rangle, \qquad W_A(q, p) = T(q)R(p).$$

Theorem $T1$. For $q, p, q', p' \in A$,

$$W_A(q, p)W_A(q', p') = \chi(pq')W_A(q + q', p + p')$$

and

$$W_A(q, p)W_A(q', p') = \chi(pq' - p'q)W_A(q', p')W_A(q, p).$$

Thus $\mu_3 \times A^2$, with multiplication

$$(\zeta, q, p)(\eta, q', p') = (\zeta\eta\chi(pq'), q + q', p + p'),$$

acts unitarily on $\mathcal{H}_A$ by $(\zeta, q, p) \mapsto \zeta W_A(q, p)$.

Proof. $T(q)$ is a permutation unitary and $R(p)$ is diagonal unitary. Also

$$R(p)T(q')|s\rangle = \chi(p(s + q'))|s + q'\rangle = \chi(pq')T(q')R(p)|s\rangle.$$

Substituting this identity into $T(q)R(p)T(q')R(p')$ gives the product law. The commutator law follows by comparing with the same product law in reversed order.

Theorem $T2$.

$$\mathrm{span}_{\mathbb{C}}\{W_A(q, p) : (q, p) \in A^2\} = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_A).$$

Consequently the thickened Weyl representation is irreducible.

Proof. Use the Hilbert–Schmidt inner product on $\mathrm{End}(\mathcal{H}_A)$. If $q \neq q'$, then $T(q' - q)$ has no fixed basis vector, so $\mathrm{Tr}(W_A(q, p)^* W_A(q', p')) = 0$. If $q = q'$ but $p \neq p'$, then the trace is

$$\sum_{s \in A} \chi((p' - p)s),$$

which is 0 because $s \mapsto \chi((p' - p)s)$ is a nontrivial character by L2. Thus the $|A|^2$ Weyl operators are pairwise orthogonal and nonzero. Since $\dim_{\mathbb{C}} \mathrm{End}(\ell^2(A)) = |A|^2$, they form a basis. Any invariant subspace is invariant under this full matrix algebra, hence is 0 or $\mathcal{H}_A$.

37.5 Jet Interpretation T3

The filtration

$$A \supset \mathfrak{m} \supset \cdots \supset \mathfrak{m}^{e-1} \supset 0$$

is the algebraic jet filtration. Its associated graded ring is

$$\mathrm{gr}_{\mathfrak{m}} A \cong K[\varepsilon]/(\varepsilon^e),$$

with $\mathfrak{m}^r/\mathfrak{m}^{r+1} \cong K\varepsilon^r$.

Lemma *L3.* For $0 \le r \le e$, the annihilator of $\mathfrak{m}^r$ under the pairing

$$A \times A \longrightarrow \mu_3, \qquad (x, y) \longmapsto \chi(xy)$$

is $\mathfrak{m}^{e-r}$.

Proof. If $y \in \mathfrak{m}^{e-r}$, then $xy \in \mathfrak{m}^e = 0$ for all $x \in \mathfrak{m}^r$, so y annihilates $\mathfrak{m}^r$. Conversely, if $y \notin \mathfrak{m}^{e-r}$, write $y = \pi^s u$ with $s < e - r$. Then $x = \pi^{e-1-s}c$ lies in $\mathfrak{m}^r$ for any $c \in A$ of degree 0. The proof of L2 chooses c so that $\chi(xy) \neq 1$. Hence y does not annihilate $\mathfrak{m}^r$.

Thus a position jet in $\mathfrak{m}^r/\mathfrak{m}^{r+1}$ pairs, in the associated graded, with a momentum jet in $\mathfrak{m}^{e-1-r}/\mathfrak{m}^{e-r}$. In particular, the ordinary residue value $A/\mathfrak{m}$ is dual to the socle momentum layer $\mathfrak{m}^{e-1}$, not to a new closed point.

The old residue qutrit embeds as the zeroth-position/socle-momentum subsystem: choose lifts $\tilde{q}, \tilde{p} \in A$ of $q, p \in K$, and use

$$q \longmapsto \tilde{q}, \qquad p \longmapsto \pi^{e-1}\tilde{p}.$$

On the subspace spanned by basis vectors with only degree-0 coefficient, these operators reproduce the residue-field Weyl law.

37.6 Global Polynomial Quotients T4

For

$$p = \prod_{i=1}^{r} \pi_i^{e_i},$$

set $A_i = \mathbb{F}_3[t]/(\pi_i^{e_i})$. By Chinese remainders,

$$R_p = \mathbb{F}_3[t]/(p) \cong \prod_i A_i.$$

The thickened arithmetic field is

$$\mathcal{H}_p^{\mathrm{thick}} = \bigotimes_i \ell^2(A_i) \cong \ell^2(R_p), \qquad E_p^{\mathrm{thick}} = \bigoplus_i A_i^2 \cong R_p^2.$$

Its Weyl operators are tensor products of the local W_{A_i}, equivalently translations and phase multiplications on $\ell^2(R_p)$ using the product character. The operator-basis theorem tensors over i, so

$$\mathrm{span}\{W(e) : e \in E_p^{\mathrm{thick}}\} = \mathrm{End}(\mathcal{H}_p^{\mathrm{thick}}).$$

Finally,

$$\dim_{\mathbb{C}} \mathcal{H}_p^{\mathrm{thick}} = \prod_i |A_i| = \prod_i 3^{e_i \deg \pi_i} = 3^{\deg p} = |R_p|.$$

This is the sense in which thickenings add jet degrees of freedom at existing closed points.

37.7 Conclusion

Nilpotent thickenings should manifest as local jet degrees of freedom. They do not create additional Zariski points. The reduced theory records point values over residue fields; the thickened theory quantizes the local Artin rings and recovers the full coordinate-ring size. The phase character detects the socle, so the jet filtration is symplectically paired from the outside in: order r position jets pair with order $e - 1 - r$ momentum jets.

38 Thickened $t^n = 1$ Examples

38.1 Status and Dependencies

This shard is the example layer for AQM-36. It uses the factorization of $t^n - 1$ from AQM-35 and the nilpotent-sensitive Weyl theorem of AQM-36. No new external source is used.

38.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & R_n = \mathbb{F}_3[t]/(t^n - 1), \quad n = 3^s m, \quad 3 \nmid m. \\
L1 & : & t^n - 1 = (t^m - 1)^{3^s}, \quad t^m - 1 \text{ squarefree.} \\
D2 & : & A_\pi = \mathbb{F}_3[t]/(\pi^{3^s}) \text{ for } \pi \mid t^m - 1. \\
T1 & : & \mathcal{H}_n^{\text{thick}} = \bigotimes_{\pi \mid t^m - 1} \ell^2(A_\pi), \quad \dim \mathcal{H}_n^{\text{thick}} = 3^n. \\
T2 & : & \mathcal{H}_n^{\text{red}} = \bigotimes_{\pi \mid t^m - 1} \ell^2(\mathbb{F}_3[t]/(\pi)), \quad \dim \mathcal{H}_n^{\text{red}} = 3^m. \\
E1 & : & n = 3 \text{ gives one 27-dimensional third-order jet qutrit.} \\
E2 & : & n = 6 \text{ gives two third-order jet qutrits, dimension } 3^6.
\end{array}
$$

38.3 General $t^n = 1$ Thickening T1–T2

AQM-35 proves that if $n = 3^s m$ with $3 \nmid m$, then

$$t^n - 1 = (t^m - 1)^{3^s}$$

and $t^m - 1$ is squarefree. Let $\mathcal{P}_m$ be the set of monic irreducible factors of $t^m - 1$. For each $\pi \in \mathcal{P}_m$, define

$$A_\pi = \mathbb{F}_3[t]/(\pi^{3^s}), \qquad K_\pi = \mathbb{F}_3[t]/(\pi).$$

The reduced layer of AQM-35 uses $\ell^2(K_\pi)$. The thickened layer of AQM-36 uses $\ell^2(A_\pi)$.

Theorem $T1$. The nilpotent-sensitive Hilbert space is

$$\mathcal{H}_n^{\text{thick}} = \bigotimes_{\pi \in \mathcal{P}_m} \ell^2(A_\pi),$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_n^{\text{thick}} = 3^n.$$

Proof. Since $|A_\pi| = 3^{3^s \deg \pi}$,

$$\dim \mathcal{H}_n^{\text{thick}} = \prod_{\pi \in \mathcal{P}_m} 3^{3^s \deg \pi} = 3^{3^s \sum_{\pi \in \mathcal{P}_m} \deg \pi}.$$

Because $t^m - 1$ is squarefree and has degree m, the sum of the degrees of its irreducible factors is m. Therefore

$$\dim \mathcal{H}_n^{\text{thick}} = 3^{3^s m} = 3^n.$$

Theorem $T2$**.** The reduced Hilbert space has dimension

$$\dim_{\mathbb{C}} \mathcal{H}_n^{\mathrm{red}} = 3^m.$$

Thus the thickened/reduced dimension ratio is

$$\frac{\dim \mathcal{H}_n^{\mathrm{thick}}}{\dim \mathcal{H}_n^{\mathrm{red}}} = 3^{n-m}.$$

Proof. AQM-35 gives

$$\mathcal{H}_n^{\mathrm{red}} = \bigotimes_{\pi \in \mathcal{P}_m} \ell^2(K_\pi),$$

so

$$\dim \mathcal{H}_n^{\mathrm{red}} = \prod_{\pi \in \mathcal{P}_m} 3^{\deg \pi} = 3^m.$$

The ratio follows from $T1$.

38.4 Example $n = 3$

Let $u = t - 1$. Then

$$R_3 = \mathbb{F}_3[t]/(t^3 - 1) \cong \mathbb{F}_3[u]/(u^3).$$

The reduced layer sees one closed point and one qutrit:

$$\mathcal{H}_3^{\mathrm{red}} \cong \mathbb{C}^3.$$

The thickened layer is

$$\mathcal{H}_3^{\mathrm{thick}} = \ell^2(\mathbb{F}_3[u]/(u^3)), \qquad \dim \mathcal{H}_3^{\mathrm{thick}} = 27.$$

Write

$$a = a_0 + a_1 u + a_2 u^2, \qquad \lambda(a) = a_2, \qquad \chi(a) = \exp(2\pi i a_2/3).$$

For $q, p \in R_3$,

$$T(q)|s\rangle = |s + q\rangle, \qquad R(p)|s\rangle = \chi(ps)|s\rangle, \qquad W(q,p) = T(q)R(p).$$

If

$$q = \sum_{j=0}^{2} q_j u^j, \qquad p = \sum_{j=0}^{2} p_j u^j,$$

then the commutator phase of (q, p) with (q', p') is

$$\exp\left(\frac{2\pi i}{3} \sum_{r+s=2} (p_r q'_s - p'_r q_s) \right).$$

Thus the value coefficient pairs with the u^2-momentum coefficient, the first jet pairs with the first-jet momentum coefficient, and the u^2 position coefficient pairs with the value momentum coefficient.

38.5 Example $n = 6$

Since $6 = 3 \cdot 2$,

$$t^6 - 1 = (t^2 - 1)^3 = (t-1)^3(t+1)^3.$$

There are two closed points, but each is third-order thickened. Let

$$A_+ = \mathbb{F}_3[t]/((t-1)^3), \qquad A_- = \mathbb{F}_3[t]/((t+1)^3).$$

Then

$$\mathcal{H}_6^{\text{thick}} = \ell^2(A_+) \otimes \ell^2(A_-) \cong \mathbb{C}^{27} \otimes \mathbb{C}^{27},$$

so $\dim \mathcal{H}_6^{\text{thick}} = 3^6$. The reduced layer is

$$\mathcal{H}_6^{\text{red}} \cong \mathbb{C}^3 \otimes \mathbb{C}^3,$$

with dimension 3^2.

38.6 Conclusion

For $t^n = 1$, the reduced layer depends only on the squarefree part $t^m - 1$, while the thickened layer recovers the full coordinate-ring cardinality. The nilpotent variables are therefore genuine local jet modes, not additional closed points.

39 Lagrangian Stabilizer Descent

39.1 Status and Source Anchors

This shard is a checked derivation from the already registered finite Weyl and local-net layers. AQM-08 and AQM-22 source and derive the prime-field Weyl commutator and the criterion that stabilizer labels commute exactly when their symplectic pairing vanishes. AQM-26 proves that the Weyl label assignment is the cosheaf-like additive layer and that the observable assignment is a local net. AQM-27 proves that quantum states do not form a sheaf because local marginals do not determine global correlations.

The new convention is `CONVENTIONS.md` (ac). We work here in the finite vector-space case over an odd prime field $k = \mathbb{F}_p$, with the explicit counterexample over $k = \mathbb{F}_3$. Mixed residue fields and finite Frobenius modules are not imported into the dimension statements in this shard.

39.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & E = k_X^n \oplus k_Z^n, \quad \omega((x,z),(x',z')) = z \cdot x' - x \cdot z'. \\
D2 & : & W(e) = T_x R_z \text{ on } \mathcal{H} = \ell^2(k^n), \quad W(e)W(f) = c(e,f)W(e+f). \\
D3 & : & L \leq E \text{ is isotropic if } \omega(L,L) = 0, \text{ and Lagrangian if } L = L^\perp. \\
D4 & : & \alpha : L \to U(1) \text{ is a phase if } \alpha(e+f) = \alpha(e)\alpha(f)c(e,f)^{-1}. \\
L1 & : & \{W(e) : e \in E\} \text{ is a basis of } \operatorname{End}(\mathcal{H}). \\
L2 & : & \text{A phased isotropic } L \text{ gives a commuting stabilizer group.} \\
T1 & : & \text{A phased Lagrangian } L \text{ gives a rank-one full stabilizer state.} \\
D5 & : & \operatorname{Loc}_U(L;\mathcal{U}) = \sum_i (L \cap E(U_i)), \quad D_U(L;\mathcal{U}) = L/\operatorname{Loc}_U(L;\mathcal{U}). \\
T2 & : & D_U(L;\mathcal{U}) = 0 \iff L \text{ is generated by stabilizers supported on } \mathcal{U}. \\
T3 & : & \text{Disjoint products of local Lagrangians have zero defect.}
\end{array}
$$

39.3 Finite Weyl Stabilizers D1–L2

Let $k = \mathbb{F}_p$ with p odd. Put

$$E = k_X^n \oplus k_Z^n, \qquad \mathcal{H} = \ell^2(k^n),$$

and write $e = (x, z)$. With a nontrivial additive character $\chi : k \to U(1)$, define

$$T_x|s\rangle = |s + x\rangle, \qquad R_z|s\rangle = \chi(z \cdot s)|s\rangle, \qquad W(x,z) = T_x R_z.$$

As in AQM-22,

$$W(e)W(f) = \chi(z \cdot x')W(e+f), \qquad W(e)W(f) = \chi(\omega(e,f))W(f)W(e).$$

Thus the multiplier in D2 is $c(e,f) = \chi(z \cdot x')$.

Lemma *L*1. The operators $\{W(e) : e \in E\}$ form a basis of $\operatorname{End}(\mathcal{H})$.

Proof. There are $|E| = p^{2n}$ such operators, and $\dim_{\mathbb{C}} \operatorname{End}(\mathcal{H}) = p^{2n}$. It is enough to prove linear independence. For $e = (x, z)$ and $f = (x', z')$, the trace of $W(e)^*W(f)$ is zero if $x \neq x'$, because the resulting translation has no fixed basis vector. If $x = x'$ but $z \neq z'$, the translation cancels and the trace is

$$\sum_{s \in k^n} \chi((z' - z) \cdot s) = 0$$

by nontriviality of the character on the nonzero linear functional $(z' - z) \cdot s$. If $e = f$, the trace is p^n. The Weyl operators are therefore orthogonal and nonzero for the Hilbert-Schmidt pairing, hence linearly independent.

Lemma $L2$. Let $L \le E$ be isotropic and let $\alpha : L \to U(1)$ satisfy D4. Then

$$S_e = \alpha(e)^{-1} W(e), \qquad e \in L,$$

is an ordinary unitary representation of the additive group L. In particular the S_e commute.

Proof. Using D4 and the Weyl product law,

$$\begin{aligned} S_e S_f &= \alpha(e)^{-1}\alpha(f)^{-1} c(e,f) W(e+f) \\ &= \alpha(e+f)^{-1} W(e+f) \\ &= S_{e+f}. \end{aligned}$$

The operators $W(e)$ are unitary, and the phases have unit modulus, so each S_e is unitary. Since L is an Abelian additive group, $S_e S_f = S_{e+f} = S_{f+e} = S_f S_e$.

39.4 Full Stabilizers Are Lagrangian T1

Theorem $T1$. If $L \le E$ is Lagrangian and phased by α, then the common $+1$-eigenspace of the stabilizer group $\{S_e : e \in L\}$ is one dimensional. Equivalently, a phased Lagrangian is a full stabilizer state at the label level.

Proof. Because E is a $2n$-dimensional nondegenerate symplectic vector space, $\dim L + \dim L^{\perp} = 2n$. Thus $L = L^{\perp}$ implies $\dim L = n$, hence $|L| = p^n$. The group average

$$P_L = \frac{1}{|L|} \sum_{e \in L} S_e$$

is the orthogonal projection onto the common $+1$-eigenspace. Indeed,

$$P_L^2 = \frac{1}{|L|^2} \sum_{e,f \in L} S_{e+f} = \frac{1}{|L|} \sum_{g \in L} S_g = P_L,$$

because every $g \in L$ has exactly $|L|$ presentations $g = e + f$. Also $P_L^* = |L|^{-1} \sum_e S_{-e} = P_L$, and $S_f P_L = P_L = P_L S_f$ by the substitution $e \mapsto f + e$. Conversely, if $S_f v = v$ for all $f \in L$, then $P_L v = v$.

Its trace is

$$\mathrm{Tr}(P_L) = \frac{1}{|L|} \sum_{e \in L} \alpha(e)^{-1} \,\mathrm{Tr}(W(e)).$$

By Lemma $L1$'s trace calculation, $\mathrm{Tr}(W(e)) = 0$ for $e \ne 0$ and $\mathrm{Tr}(W(0)) = p^n$. Therefore

$$\mathrm{Tr}(P_L) = \frac{p^n}{p^n} = 1.$$

An orthogonal projection has trace equal to its rank, so the stabilizer space is one dimensional.

The same calculation also shows maximality among Weyl observables. If an operator $A = \sum_{a \in E} \lambda_a W(a)$ commutes with every S_e, then each nonzero coefficient can occur only for labels $a \in L^{\perp} = L$, because the commutator factor is $\chi(\omega(a,e))$. Thus the centralizer of the stabilizer group inside the Weyl basis is spanned by the same labels.

39.5 Descent Defect D5–T2

Now let U be a finite region or finite open in the residue-qudit net, and let $\mathcal{U} = \{U_i\}$ be a finite cover. The label cosheaf of AQM-26 includes each $E(U_i)$ into $E(U)$ by extension by zero. For a global stabilizer label space $L \le E(U)$, define

$$\mathrm{Loc}_U(L;\mathcal{U}) = \sum_i \big(L \cap E(U_i)\big) \le L$$

and

$$D_U(L;\mathcal{U}) = L/\,\mathrm{Loc}_U(L;\mathcal{U}).$$

This quotient is the stabilizer descent defect.

Theorem $T2$. For a fixed global phased stabilizer $L \le E(U)$, the following are equivalent:

(1) $D_U(L;\mathcal{U}) = 0$,
(2) $L = \sum_i (L \cap E(U_i))$,
(3) $\{S_e : e \in L\}$ is generated by the stabilizers supported in the U_i.

Proof. The equivalence of (1) and (2) is exactly the definition of the quotient defect. If (2) holds, write any $e \in L$ as $e = e_1 + \cdots + e_m$ with each $e_j \in L \cap E(U_{i_j})$. Lemma $L2$ gives

$$S_e = S_{e_1+\cdots+e_m} = S_{e_1} \cdots S_{e_m},$$

so the global stabilizer is generated by local stabilizers. Conversely, if the stabilizer group is generated by supported stabilizers, then passing to projective labels gives every $e \in L$ as a sum of elements from $L \cap E(U_i)$, so (2) holds.

Therefore vanishing defect is the label-level local-to-global condition. It is the generation half of a cosheaf condition for a stabilizer subobject of the label cosheaf. Phase gluing is additional character compatibility on overlaps; in the disjoint examples below there is no overlap condition.

39.6 The Product Case T3

Theorem $T3$. Let $U = U_1 \sqcup U_2$, so

$$E(U) = E(U_1) \oplus E(U_2)$$

as an orthogonal symplectic direct sum. If $L_i \le E(U_i)$ is Lagrangian for $i = 1, 2$, then

$$L = L_1 \oplus L_2 \le E(U)$$

is Lagrangian and $D_U(L; \{U_1, U_2\}) = 0$. With product phases, the stabilizer projection is $P_{L_1} \otimes P_{L_2}$.

Proof. Orthogonality of the direct sum gives

$$L^{\perp} = L_1^{\perp} \oplus L_2^{\perp} = L_1 \oplus L_2 = L,$$

so L is Lagrangian. Also

$$L \cap E(U_1) = L_1, \qquad L \cap E(U_2) = L_2,$$

hence $\mathrm{Loc}_U(L; \{U_1, U_2\}) = L$ and the defect vanishes. The group average over $L_1 \oplus L_2$ factors as the product of the two group averages, which is $P_{L_1} \otimes P_{L_2}$.

39.7 Conclusion

A Lagrangian subspace is the label-level form of a full stabilizer state. A vanishing stabilizer descent defect is the separate condition that this full stabilizer is generated from local pieces of a chosen cover. In cosheaf language, E is the additive label cosheaf; a stabilizer choice has the local-to-global generation property only when its labels have zero descent defect. The next shard gives the explicit two-qutrit counterexample showing that the Lagrangian condition alone does not force this.

40 Two-Qutrit Stabilizer Descent Counterexample

40.1 Status and Source Anchors

This shard is a checked finite calculation using the convention and theorems of AQM-38. It is the promised counterexample to the overly strong claim that choosing a global Lagrangian stabilizer is the same thing as satisfying a local-to-global principle. The Weyl commutation convention is the same prime-field convention as AQM-08, AQM-22, and `CONVENTIONS.md` (l).

40.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & U = A \sqcup B, \quad E(U) = \mathbb{F}_3^2 \oplus \mathbb{F}_3^2. \\
D2 & : & \omega(e, e') = z_A x'_A - x_A z'_A + z_B x'_B - x_B z'_B. \\
D3 & : & \ell_1 = (1, 0, -1, 0), \quad \ell_2 = (0, 1, 0, 1), \quad L = \langle \ell_1, \ell_2 \rangle. \\
L1 & : & L \text{ is Lagrangian.} \\
L2 & : & L \cap E(A) = 0 \text{ and } L \cap E(B) = 0. \\
T1 & : & D_U(L; \{A, B\}) \cong L \neq 0. \\
T2 & : & \text{The stabilizer state is entangled and has maximally mixed one-site marginals.} \\
T3 & : & \text{A global Lagrangian is not the same as stabilizer descent.}
\end{array}$$

40.3 The Label Calculation D1–T1

Let $U = A \sqcup B$, with one qutrit on each site. Write labels as

$$(x_A, z_A, x_B, z_B) \in \mathbb{F}_3^2 \oplus \mathbb{F}_3^2$$

and use the direct-sum symplectic form

$$\omega(e, e') = z_A x'_A - x_A z'_A + z_B x'_B - x_B z'_B.$$

Define

$$\ell_1 = (1, 0, -1, 0), \qquad \ell_2 = (0, 1, 0, 1), \qquad L = \langle \ell_1, \ell_2 \rangle.$$

Lemma $L1$**.** The subspace L is Lagrangian.

Proof. The only nontrivial generator pairing is

$$\omega(\ell_1, \ell_2) = 0 \cdot 0 - 1 \cdot 1 + 0 \cdot 0 - (-1) \cdot 1 = -1 + 1 = 0.$$

Same-generator pairings vanish because ω is alternating, so L is isotropic. The ambient symplectic vector space has dimension 4, and ℓ_1, ℓ_2 are independent, so $\dim L = 2$. Since the form is nondegenerate, $\dim L + \dim L^\perp = 4$, so $\dim L^\perp = 2$. Isotropy gives $L \subseteq L^\perp$, and equal dimensions give $L = L^\perp$. Hence L is Lagrangian.

Lemma $L2$**.**

$$L \cap E(A) = 0, \qquad L \cap E(B) = 0.$$

Proof. A general element of L is

$$a\ell_1 + b\ell_2 = (a, b, -a, b).$$

If it is supported only on A, then $-a = 0$ and $b = 0$, hence $a = b = 0$. If it is supported only on B, then $a = 0$ and $b = 0$. These are exactly the displayed intersections.

Theorem $T1$. For the disjoint cover $\{A, B\}$,

$$D_U(L; \{A, B\}) \cong L.$$

In particular the defect has $\mathbb{F}_3$-dimension 2.

Proof. By Lemma $L2$,

$$\mathrm{Loc}_U(L; \{A, B\}) = (L \cap E(A)) + (L \cap E(B)) = 0.$$

The definition from AQM-38 gives

$$D_U(L; \{A, B\}) = L/\,\mathrm{Loc}_U(L; \{A, B\}) = L.$$

40.4 The Stabilizer State T2

Theorem $T2$. With the standard qutrit Weyl lift, the stabilizer labels L define a full global stabilizer state, but no nonzero local stabilizer label on either single site is inherited from L.

Proof. The two generator operators are

$$W(\ell_1) = X_A X_B^{-1}, \qquad W(\ell_2) = Z_A Z_B.$$

The Weyl multiplier is trivial on all of L, because

$$(b, b) \cdot (a', -a') = 0$$

for $a\ell_1 + b\ell_2$ and $a'\ell_1 + b'\ell_2$. Thus the phase $\alpha = 1$ satisfies the AQM-38 phase condition. The two displayed operators commute by Lemma $L1$. The vector

$$|\Phi_-\rangle = \frac{1}{\sqrt{3}} \sum_{s \in \mathbb{F}_3} |s, -s\rangle$$

is fixed by both operators. Indeed, $X_A X_B^{-1}$ sends $|s, -s\rangle$ to $|s+1, -s-1\rangle = |s+1, -(s+1)\rangle$, only reindexing the sum. Also $Z_A Z_B$ multiplies $|s, -s\rangle$ by $\chi(s - s) = 1$. Since L is Lagrangian, AQM-38 Theorem $T1$ says the common stabilizer space is one-dimensional, so this is the full stabilizer state.

The one-site density matrix on A is

$$\mathrm{Tr}_B(|\Phi_-\rangle\langle\Phi_-|) = \frac{1}{3} \sum_{s \in \mathbb{F}_3} |s\rangle\langle s| = I_3/3,$$

and the same computation gives $I_3/3$ on B. Thus the one-site restrictions are maximally mixed. This agrees with Lemma $L2$: the global Lagrangian has no nonzero labels supported on a single site.

40.5 Corrected Local-to-Global Statement T3

Theorem $T3$. Choosing a full stabilizer is not equivalent to local-to-global gluing. The correct label-level statement is:

$$\text{full global stabilizer state} \iff \text{global phased Lagrangian } L \le E(U),$$

whereas

$$\text{full global stabilizer generated from the cover } \mathcal{U}$$
$$\Longleftrightarrow$$
$$\text{global phased Lagrangian } L \leq E(U) \text{ with } D_U(L; \mathcal{U}) = 0,$$

plus phase compatibility on overlaps.

Proof. The first equivalence is AQM-38 Theorem $T1$. The second equivalence is AQM-38 Theorem $T2$ applied to a Lagrangian L, with phase compatibility added because labels alone do not choose eigenvalues. AQM-38 Theorem $T3$ proves that product local stabilizer states satisfy the condition. Theorem $T1$ above proves that the Lagrangian condition alone does not imply it.

40.6 Conclusion

The broad equivalence must be rejected in this form. A Lagrangian is a full global stabilizer. A zero descent defect is the extra condition that this stabilizer is generated from the local pieces of a chosen cover. Nonzero defect is the stabilizer-label analogue of the state-presheaf phenomenon from AQM-27: local restrictions can miss global correlations.

41 Product-Field Spectra and n-Qudit Stabilizers

41.1 Status and Source Anchors

This shard studies $X = \operatorname{Spec} R$ for

$$R = k^n = k \times \cdots \times k, \qquad k = \mathbb{F}_q.$$

It is a finite reduced spectrum example in the residue-qudit convention of AQM-23, AQM-24, and AQM-28. The Weyl operator formulas use the same finite field sources as AQM-22 and AQM-23: Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313 for finite-field shift and phase operators, and Bombin–Martin–Delgado, `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 1208–1259 for qudit Paulis and symplectic commutators.

The new product-field convention is `CONVENTIONS.md` (ad). The key point is that $\operatorname{Spec}(k^n)$ has n points, not q^n points.

41.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k = \mathbb{F}_q,\ R = k^n,\ e_i = (0, \ldots, 1, \ldots, 0). \\
L1 & : & \operatorname{Spec} R = \{\mathfrak{m}_i = \ker(\mathrm{pr}_i)\}_{i=1}^n. \\
L2 & : & \kappa(\mathfrak{m}_i) = k, \quad D(e_i) = \{\mathfrak{m}_i\}. \\
D2 & : & U \subset X \text{ corresponds to } I_U \subset \{1, \ldots, n\}. \\
D3 & : & E(U) = \bigoplus_{i \in I_U} k^2 = k_X^{I_U} \oplus k_Z^{I_U}. \\
D4 & : & \mathcal{H}(U) = \bigotimes_{i \in I_U} \ell^2(k). \\
L3 & : & W_U(x, z) = T_x R_z \text{ satisfies the finite-field Weyl law.} \\
T1 & : & \operatorname{span}\{W_U(e) : e \in E(U)\} = \bigotimes_{i \in I_U} M_q(\mathbb{C}). \\
D5 & : & U \subset V \text{ gives extension by zero and tensoring by identities.} \\
T2 & : & X = \operatorname{Spec}(k^n) \text{ gives the ambient } n\text{-qudit Weyl net.} \\
T3 & : & \text{Stabilizer codes are extra isotropic labels plus phases.}
\end{array}$$

41.3 The Product-Field Spectrum D1–L2

Let $R = k^n$. Write $\mathrm{pr}_i : R \to k$ for the i-th projection and $e_i \in R$ for the idempotent with 1 in the i-th coordinate and 0 elsewhere.

Lemma $L1$. The prime ideals of R are exactly

$$\mathfrak{m}_i = \ker(\mathrm{pr}_i) = k \times \cdots \times k \times 0 \times k \times \cdots \times k.$$

Hence $\operatorname{Spec} R$ has n points.

Proof. First, $R/\mathfrak{m}_i \cong k$, so each $\mathfrak{m}_i$ is maximal and therefore prime.

Conversely, let $\mathfrak{p} \subset R$ be prime. Since

$$e_i e_j = 0 \quad (i \neq j), \qquad 1 = e_1 + \cdots + e_n,$$

at most one idempotent e_i can be outside $\mathfrak{p}$: if $e_i, e_j \notin \mathfrak{p}$ for $i \neq j$, primeness would be contradicted by $e_i e_j = 0 \in \mathfrak{p}$. At least one e_i is outside $\mathfrak{p}$, because otherwise $1 = \sum_i e_i \in \mathfrak{p}$. Let this unique index be i. Then $e_j \in \mathfrak{p}$ for $j \neq i$. Since $e_i R \cong k$ is a field and $\mathfrak{p} \cap e_i R$ is a proper ideal, it is 0. Thus $\mathfrak{p}$ is exactly the direct product of the $j \neq i$ factors and 0 in the i-th factor, namely $\mathfrak{m}_i$.

Lemma $L2$. For every i,

$$\kappa(\mathfrak{m}_i) \cong k, \qquad D(e_i) = \{\mathfrak{m}_i\}.$$

Consequently $X = \operatorname{Spec} R$ is a finite discrete topological space.

Proof. The quotient $R/\mathfrak{m}_i$ is k, so the residue field at $\mathfrak{m}_i$ is k. The standard open $D(e_i)$ consists of prime ideals not containing e_i. By the proof of Lemma $L1$, $e_i \notin \mathfrak{m}_i$, while $e_i \in \mathfrak{m}_j$ for $j \neq i$. Hence $D(e_i) = \{\mathfrak{m}_i\}$. Since every singleton is open, every subset is open.

41.4 The Residue-Qudit Field D2–L3

Let $U \subset X$, and let $I_U \subset \{1, \ldots, n\}$ be the corresponding index set. Define

$$E(U) = \bigoplus_{i \in I_U} \kappa(\mathfrak{m}_i)^2 = k_X^{I_U} \oplus k_Z^{I_U}.$$

Write $e = (x, z)$, with $x, z : I_U \to k$. Let

$$\chi(a) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{k/\mathbb{F}_p}(a)\right)$$

be the standard additive character. Define

$$\Omega_U(e, f) = \sum_{i \in I_U} (z_i x_i' - x_i z_i') \in k$$

and the commutator bicharacter

$$B_U(e, f) = \chi(\Omega_U(e, f)).$$

The Hilbert space is

$$\mathcal{H}(U) = \bigotimes_{i \in I_U} \ell^2(k) \cong \ell^2(k^{I_U}).$$

On the basis $|s\rangle$, $s : I_U \to k$, define

$$T_x|s\rangle = |s + x\rangle, \qquad R_z|s\rangle = \chi\left(\sum_{i \in I_U} z_i s_i\right)|s\rangle,$$

and

$$W_U(x, z) = T_x R_z.$$

Lemma $L3$. For $e = (x, z)$ and $f = (x', z')$,

$$W_U(e) W_U(f) = \chi\left(\sum_{i \in I_U} z_i x_i'\right) W_U(e + f),$$

and

$$W_U(e) W_U(f) = B_U(e, f) W_U(f) W_U(e).$$

Proof. Translations add and phases add:

$$T_x T_{x'} = T_{x+x'}, \qquad R_z R_{z'} = R_{z+z'}.$$

Also

$$\begin{aligned} R_z T_{x'}|s\rangle &= R_z|s+x'\rangle \\ &= \chi\left(\sum_i z_i(s_i+x'_i)\right)|s+x'\rangle \\ &= \chi\left(\sum_i z_i x'_i\right) T_{x'}R_z|s\rangle. \end{aligned}$$

Multiplying $T_x R_z T_{x'} R_{z'}$ gives the first formula. Dividing this formula by the same formula with e and f exchanged gives the commutator factor $\chi(\Omega_U(e,f))$.

41.5 Operator Algebra T1–T2

Theorem $T1$.

$$\operatorname{span}_{\mathbb{C}}\{W_U(e) : e \in E(U)\} = \operatorname{End}(\mathcal{H}(U)) \cong \bigotimes_{i\in I_U} M_q(\mathbb{C}).$$

Proof. There are $|E(U)| = q^{2|I_U|}$ Weyl operators and

$$\dim_{\mathbb{C}} \operatorname{End}(\mathcal{H}(U)) = (q^{|I_U|})^2 = q^{2|I_U|}.$$

It is enough to prove linear independence. If $e = (x,z)$ and $f = (x',z')$, the trace of $W_U(e)^* W_U(f)$ is zero when $x \neq x'$, because the resulting translation has no fixed basis vector. If $x = x'$ but $z \neq z'$, the trace is

$$\sum_{s\in k^{I_U}} \chi\left(\sum_i (z'_i - z_i)s_i\right) = 0,$$

because $z' - z$ is a nonzero linear functional and χ is nontrivial. If $e = f$, the trace is $q^{|I_U|}$. Thus the Weyl operators are orthogonal and nonzero for the Hilbert-Schmidt pairing, hence form a basis. The tensor-product matrix identification follows from $\mathcal{H}(U) = \bigotimes_{i\in I_U} \ell^2(k)$.

Local net D5. If $U \subset V$, include $E(U)$ in $E(V)$ by extension by zero. The operator inclusion is

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(V), \qquad a \longmapsto a \otimes \mathbf{1}_{I_V \setminus I_U}.$$

The identity and composition laws are immediate from tensoring with identity operators on the missing tensor factors. Since X is discrete, this is the usual tensor-factor local net over all subsets of $\{1,\dots,n\}$.

Theorem $T2$. For $X = \operatorname{Spec}(k^n)$, the residue-qudit arithmetic field is the ambient standard n-qudit Weyl/Pauli net for q-level qudits:

$$E(X) = k_X^n \oplus k_Z^n, \qquad \mathcal{H}(X) = \ell^2(k)^{\otimes n}, \qquad \mathcal{A}(X) \cong M_{q^n}(\mathbb{C}).$$

Proof. By Lemma $L1$, X has exactly the n points $\mathfrak{m}_1,\dots,\mathfrak{m}_n$. By Lemma $L2$, every residue field is k, and the topology is discrete. Therefore the full open X has label space

$$E(X) = \bigoplus_{i=1}^n k^2 = k_X^n \oplus k_Z^n$$

and Hilbert space $\bigotimes_{i=1}^n \ell^2(k)$. Lemma $L3$ is exactly the finite-field multi-qudit Weyl law, and Theorem $T1$ gives the full matrix algebra on n q-level tensor factors.

41.6 Stabilizer Comparison T3

Theorem *T*3. The ring k^n supplies the ambient n-qudit Weyl/Pauli system. A stabilizer code is additional data: an isotropic additive subgroup

$$L \leq E(X)$$

for the bicharacter B_X, together with compatible phases/eigenvalues. If one restricts L to be k-linear and requires $\Omega_X(L, L) = 0$ in k, one gets the usual k-linear q-ary stabilizer subclass. Allowing $\mathbb{F}_p$-additive L with

$$\operatorname{Tr}_{k/\mathbb{F}_p} \Omega_X(L, L) = 0$$

gives the standard finite-field additive stabilizer convention.

Proof. By Lemma $L3$, two Weyl operators commute exactly when

$$B_X(e, f) = \chi(\Omega_X(e, f)) = 1.$$

Thus an additive subgroup L gives commuting projective stabilizer labels exactly when $B_X(e, f) = 1$ for all $e, f \in L$, equivalently $\operatorname{Tr}_{k/\mathbb{F}_p} \Omega_X(e, f) = 0$ for all $e, f \in L$. The phases/eigenvalues are not determined by R; as in AQM-22 and AQM-38, they are a separate lift of the commuting projective labels to actual operators with chosen joint eigenvalues.

If L is a k-linear subspace and the stronger equality $\Omega_X(L, L) = 0$ holds in k, then the trace condition holds automatically. This is the k-linear stabilizer subclass. Conversely, for general $q = p^r$, the trace condition may hold for an $\mathbb{F}_p$-linear additive subgroup that is not closed under multiplication by k. Such subgroups are part of the standard additive finite-field stabilizer formalism but are not k-linear subspaces. When $q = p$ is prime, these two notions coincide.

41.7 Conclusion

For $R = k^n$, with $X = \operatorname{Spec} R$, the hoped-for statement is correct with the standard stabilizer caveat. The product ring is the coordinate ring of a finite discrete n-point k-scheme. Its residue-qudit arithmetic field is exactly the ambient n q-level qudit Weyl/Pauli system. Stabilizer codes do not come for free from the ring: they are later choices of isotropic label subgroups and compatible phases.

42 Cech Cohomology and Quantum Descent

42.1 Status and Source Anchors

This shard adds the cover-complex layer requested after AQM-26–AQM-40. Stacks Cohomology, `references/algebraic_geometry/stacks_project_cohomology.tex`, lines 868–924, 926–946, 2438–2517, and 4071–4165, supplies the Cech complex, H^0 sheaf criterion, refinement colimit, and ordered Cech complex. Curry, `references/cosheaves/curry_1303_3255_source/ch_abstract_sheaves.tex`, lines 246–259 and 270–335, supplies the dual vector-space exactness and Cech homology convention. The quantum input is local: AQM-26 for labels and observable nets, AQM-27 for states, and AQM-38–AQM-39 for stabilizer descent. The new convention is `CONVENTIONS.md` (ae).

42.2 Lamport Dependency Skeleton

$D1$: $\mathcal{U} = \{U_i\}_{i\in I}$ is a finite ordered cover of U.
$D2$: $\check{C}^p(\mathcal{U}, F) = \prod_{i_0<\cdots<i_p} F(U_{i_0\cdots i_p})$.
$L1$: $d^2 = 0$ for the Cech differential.
$D3$: $\check{H}^p(\mathcal{U}, F) = \ker d^p / \operatorname{im} d^{p-1}$.
$L2$: $\check{H}^0(\mathcal{U}, F)$ is compatible local section data.
$D4$: $\check{H}^p(U, F) = \varinjlim_{\mathcal{U}} \check{H}^p(\mathcal{U}, F)$ when refinements are declared.
$D5$: $\check{C}_p(\mathcal{U}, G) = \bigoplus_{i_0<\cdots<i_p} G(U_{i_0\cdots i_p})$ for a precosheaf G.
$T1$: $\check{H}_0(\mathcal{U}, E) \cong E(U)$, $\check{H}_p(\mathcal{U}, E) = 0$ $(p > 0)$.
$D6$: $\check{Z}^0(\mathcal{U}, \mathsf{S})$ is the set of compatible local states.
$T2$: $\mathsf{S}(U) \to \check{Z}^0(\mathcal{U}, \mathsf{S})$ may fail injectivity and surjectivity.
$D7$: $L_{\leq U}(W) = L \cap E(W)$ for a global stabilizer $L \leq E(U)$.
$T3$: $D_U(L; \mathcal{U})$ is the cokernel of stabilizer Cech H_0 generation.
$P1$: $\mathcal{A}$ has an observable-net comparison map, not ordinary abelian Cech cohomology.

42.3 Standard Cover Cohomology D1–D4

Let X be a topological space and let $\mathcal{U} = \{U_i\}_{i\in I}$ be a finite open cover of $U \subset X$. Fix a total order on I, and write

$$U_{i_0\cdots i_p} = U_{i_0} \cap \cdots \cap U_{i_p}.$$

For an abelian presheaf F, define

$$\check{C}^p(\mathcal{U}, F) = \prod_{i_0<\cdots<i_p} F(U_{i_0\cdots i_p}).$$

If $c \in \check{C}^p(\mathcal{U}, F)$, set

$$(dc)_{i_0\cdots i_{p+1}} = \sum_{j=0}^{p+1} (-1)^j \, c_{i_0\cdots\widehat{i_j}\cdots i_{p+1}}\big|_{U_{i_0\cdots i_{p+1}}}.$$

Lemma $L1$. The differential satisfies $d^2 = 0$.

Proof. Fix $i_0 < \cdots < i_{p+2}$. In $(d^2c)_{i_0\cdots i_{p+2}}$, the term obtained by deleting i_a and i_b, with $a < b$, appears exactly twice. The two restriction maps are equal by functoriality of the presheaf. The corresponding signs are

$$(-1)^a(-1)^{b-1} \quad \text{and} \quad (-1)^b(-1)^a.$$

Their sum is $(-1)^{a+b-1} + (-1)^{a+b} = 0$. Summing over all pairs (a, b) gives $d^2c = 0$.

Thus

$$\check{H}^p(\mathcal{U}, F) = \ker\bigl(d : \check{C}^p \to \check{C}^{p+1}\bigr) / \operatorname{im}\bigl(d : \check{C}^{p-1} \to \check{C}^p\bigr)$$

is well-defined.

Lemma *L*2. $\check{H}^0(\mathcal{U}, F)$ is the group of compatible local sections:

$$\check{H}^0(\mathcal{U}, F) = \left\{ (s_i) \in \prod_i F(U_i) : s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \text{ for all } i, j \right\}.$$

If F is a sheaf, the restriction map $F(U) \to \check{H}^0(\mathcal{U}, F)$ is bijective.

Proof. Since $\check{C}^{-1} = 0$, $\check{H}^0 = \ker(d : \check{C}^0 \to \check{C}^1)$. The displayed formula for d says exactly that all pairwise restrictions agree. The second sentence is precisely the sheaf gluing axiom, as in Stacks Cohomology, lines 926–946.

When a cover-independent group is needed, we follow Stacks Cohomology, lines 2438–2517, and define

$$\check{H}^p(U, F) = \varinjlim_{\mathcal{U}} \check{H}^p(\mathcal{U}, F),$$

over open covers ordered by refinement. In the arithmetic-field questions below, the fixed cover is part of the physical question, so we keep the cover in the notation unless this direct limit is explicitly invoked.

42.4 Dual Cech Homology for Labels D5–T1

Let G be a covariant precosheaf of abelian groups or vector spaces. Define

$$\check{C}_p(\mathcal{U}, G) = \bigoplus_{i_0 < \cdots < i_p} G(U_{i_0 \cdots i_p}).$$

For $s \in G(U_{i_0 \cdots i_p})$, put

$$\partial s = \sum_{j=0}^{p} (-1)^j r_{U_{i_0 \cdots i_p}, U_{i_0 \cdots \widehat{i_j} \cdots i_p}}(s),$$

where r is the covariant extension map. The same sign cancellation as in Lemma $L1$ gives $\partial^2 = 0$, hence $\check{H}_p(\mathcal{U}, G)$.

For the residue-qudit label layer, $E(W) = \bigoplus_{x \in W} E_x$, with extension by zero along inclusions. Assume the cover $\mathcal{U}$ is finite and $E(U)$ is the finite-support direct sum used in AQM-24–AQM-26.

Theorem *T*1. For every finite cover $\mathcal{U}$ of U,

$$\check{H}_0(\mathcal{U}, E) \cong E(U), \qquad \check{H}_p(\mathcal{U}, E) = 0 \quad (p > 0).$$

Proof. For a point $x \in U$, let $I_x = \{i \in I : x \in U_i\}$. This set is nonempty because $\mathcal{U}$ covers U. Since direct sums distribute over the finite Cech chain groups,

$$\check{C}_p(\mathcal{U}, E) \cong \bigoplus_{x \in U} \left(\bigoplus_{i_0 < \cdots < i_p,\ i_j \in I_x} E_x \right).$$

The differential preserves the summand indexed by x, because all extension maps are the identity on the E_x component. Therefore the x-summand is E_x tensored with the simplicial chain complex of the full simplex on the vertex set I_x.

Choose $a \in I_x$. The cone operator $h(\sigma) = [a, \sigma]$, with the orientation sign determined by inserting a into the ordered simplex, satisfies $\partial h + h\partial = \mathrm{id} - \epsilon$, where ϵ is the augmentation to one copy of E_x in degree 0. Thus the full simplex has homology E_x in degree 0 and 0 in higher degrees. Taking the direct sum over x gives the displayed result.

So the full label cosheaf carries no higher cover obstruction: every label is assembled pointwise from the cover, and all higher cycles are simplex boundaries.

42.5 Quantum Counterparts D6–P1

States. For the state presheaf $\mathsf{S}(W) = \mathrm{State}(\mathcal{A}(W))$, define only the zeroth compatible-family object

$$\check{Z}^0(\mathcal{U}, \mathsf{S}) = \left\{ (\omega_i) \in \prod_i \mathsf{S}(U_i) : \omega_i|_{U_i \cap U_j} = \omega_j|_{U_i \cap U_j} \right\}.$$

There is a natural restriction map

$$\rho_{\mathcal{U}} : \mathsf{S}(U) \longrightarrow \check{Z}^0(\mathcal{U}, \mathsf{S}).$$

Theorem $T2$. $\rho_{\mathcal{U}}$ is not generally injective and not generally surjective.

Proof. Non-injectivity is AQM-27 Theorem $T2$: on $\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$, an entangled state and a product state have the same one-factor restrictions. Non-surjectivity is AQM-27 Theorem $T3$: the AB and BC Bell marginals agree on B, but no ABC state has both as marginals.

Thus the state-side quantum counterpart is the marginal comparison map. Failure of injectivity measures hidden global correlations. Failure of surjectivity measures incompatible marginal data. These are not cohomology groups yet, because S is a convex set-valued presheaf, not an abelian presheaf.

Stabilizers. Let $L \leq E(U)$ be a global stabilizer label subspace. Define the supported sublabel precosheaf on opens $W \subset U$ by

$$L_{\leq U}(W) = L \cap E(W).$$

The augmentation $\check{C}_0(\mathcal{U}, L_{\leq U}) \to L$ sends a tuple of local stabilizer labels to its sum in L. Its image is

$$\mathrm{Loc}_U(L; \mathcal{U}) = \sum_i (L \cap E(U_i)).$$

The boundary from $L \cap E(U_i \cap U_j)$ maps to zero under this augmentation, because the two copies are the same global label with opposite signs. Hence the augmentation factors through $\check{H}_0(\mathcal{U}, L_{\leq U})$.

Theorem $T3$. The stabilizer descent defect of AQM-38 is the cokernel of this zeroth Cech homology generation map:

$$\mathrm{coker}\big(\check{H}_0(\mathcal{U}, L_{\leq U}) \to L\big) \cong L / \mathrm{Loc}_U(L; \mathcal{U}) = D_U(L; \mathcal{U}).$$

Proof. The image of $\check{C}_0 \to L$ is exactly $\mathrm{Loc}_U(L; \mathcal{U})$. Passing through the quotient $\check{H}_0$ does not change the image, because boundaries already map to zero. The cokernel is therefore the displayed quotient. AQM-38 Theorem $T2$ proves that its vanishing is equivalent to local generation of the stabilizer labels; AQM-39 gives the two-qutrit nonzero-defect example.

Observable nets. For a finite cover, let $\operatorname{colim}^{*\mathrm{alg}}_{\mathcal{U}}$ denote the ordinary colimit of the cover diagram of unital $*$-algebras. The observable net supplies a canonical comparison map

$$\pi_{\mathcal{U}} : \operatorname{colim}^{*\mathrm{alg}}_{\mathcal{U}} \mathcal{A} \longrightarrow \mathcal{A}(U).$$

AQM-26 proves that this ordinary algebra colimit is not the physical gluing: for disjoint pieces it is a free-product gluing, while the Weyl net glues by commuting tensor-local subalgebras inside the ambient algebra. Therefore the observable quantum counterpart is the comparison map $\pi_{\mathcal{U}}$: its image is the additive generated algebra, and its kernel records the extra locality and commutation relations imposed by the physical representation. Without abelianization or another declared linearization, this is not an ordinary Cech cohomology group.

42.6 Conclusion

Classical Cech cohomology is the abelian cover-complex layer. In AQF it splits into layered descent tests: label Cech homology, stabilizer cokernels, state marginal comparison maps, and observable-net colimit comparison maps. Only the first two are cohomology or homology groups at this stage.

43 Gaussian and Clifford Dynamics

43.1 Status and Source Anchors

This shard adds the first dynamics layer for the finite residue-qudit arithmetic fields. The new convention is `CONVENTIONS.md` (af). Gross `references/symplectic_qecc/gross_0602001_source/poswig.tex` is now registered locally: lines 340–453 define odd-dimensional Weyl operators, the symplectic form, the Weyl composition law, the trace, and the Heisenberg representation; lines 458–537 define the Clifford group and state its symplectic/metaplectic structure theorem; lines 794–802 give Clifford covariance of Wigner functions; lines 846–903 define stabilizer states from maximally isotropic subspaces; lines 935–976 compute their Wigner functions; lines 202–214 state the discrete Hudson theorem; and lines 1633–1639 state that positivity-preserving unitaries are Clifford. AQM-22, AQM-24, and AQM-40 supply the finite Weyl nets on arithmetic spectra. When $k = \mathbb{F}_{p^r}$, the intrinsic algebraic statements below use the k-symplectic form, while the Gross-Hudson state dictionary is imported after choosing an $\mathbb{F}_p$-basis and viewing $\ell^2(k^m) \cong \ell^2(\mathbb{F}_p^{rm})$.

43.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & (E,\Omega) \text{ is a finite nondegenerate symplectic } k\text{-space, } \operatorname{char} k \neq 2. \\
D2 & : & W_E(e)W_E(f) = \chi(\tfrac{1}{2}\Omega(e,f))W_E(e+f), \quad W_E(e)^* = W_E(-e). \\
D3 & : & \mathcal{A}(E) = \operatorname{span}_{\mathbb{C}}\{W_E(e) : e \in E\}. \\
L1 & : & \{W_E(e) : e \in E\} \text{ is a basis of } \mathcal{A}(E). \\
D4 & : & S : E \to F \text{ is symplectic if } \Omega_F(Se,Sf) = \Omega_E(e,f). \\
L2 & : & \text{A symplectic morphism from nondegenerate } E \text{ is injective.} \\
T1 & : & \alpha_S(W_E(e)) = W_F(Se) \text{ is a unital } *\text{-monomorphism.} \\
T2 & : & S \in \operatorname{Sp}(E) \text{ gives a } *\text{-automorphism of } \mathcal{A}(E). \\
T3 & : & \text{For odd qudits, } S \text{ has a projective Clifford lift } U_S. \\
D5 & : & \alpha_{a,S} = \operatorname{Ad}_{W(a)U_S} \text{ is affine Clifford dynamics.} \\
T4 & : & \text{Gross-Hudson identifies pure nonnegative-Wigner states with stabilizer states.}
\end{array}
$$

43.3 Half-Weyl Systems D1–L1

Let $k = \mathbb{F}_q$ have odd characteristic and let $\chi : k \to U(1)$ be a nontrivial additive character. Let (E,Ω) be a finite-dimensional nondegenerate symplectic k-vector space. A half-Weyl system for E is a map $W_E : E \to U(\mathcal{H}_E)$ such that

$$W_E(e)W_E(f) = \chi\left(\frac{1}{2}\Omega(e,f)\right)W_E(e+f), \qquad W_E(e)^* = W_E(-e).$$

Here $1/2 \in k$ is defined because $\operatorname{char} k \neq 2$. Set

$$\mathcal{A}(E) = \operatorname{span}_{\mathbb{C}}\{W_E(e) : e \in E\}.$$

In the residue-qudit examples this is the full finite matrix algebra on $\mathcal{H}_E$. Gross's trace calculation for odd qudits, `poswig.tex`, lines 437–453, is the local source for the corresponding orthogonality statement.

Lemma *L*1. The operators $W_E(e)$ are linearly independent. Hence they form a basis of $\mathcal{A}(E)$ by definition of the span.

Proof. In the concrete residue-qudit representation, the trace pairing satisfies

$$\mathrm{Tr}\big(W_E(e)^* W_E(f)\big) = \dim(\mathcal{H}_E)\,\delta_{e,f},$$

as in the finite Weyl trace computation already used in AQM-22 and AQM-38. If $\sum_e c_e W_E(e) = 0$, multiply on the left by $W_E(f)^*$ and take the trace. The displayed orthogonality gives $\dim(\mathcal{H}_E)c_f = 0$. Since f was arbitrary, all c_f vanish.

43.4 Symplectic Morphisms and Algebra Maps D4–T2

Let (E, Ω_E) and (F, Ω_F) be finite nondegenerate symplectic k-spaces. A symplectic morphism is a k-linear map

$$S : E \to F$$

such that

$$\Omega_F(Se, Sf) = \Omega_E(e, f) \qquad (e, f \in E).$$

Lemma $L2$**.** Every symplectic morphism $S : E \to F$ with E nondegenerate is injective.

Proof. If $Se = 0$, then for every $f \in E$,

$$\Omega_E(e, f) = \Omega_F(Se, Sf) = 0.$$

Nondegeneracy of Ω_E gives $e = 0$.

Theorem $T1$**.** For every symplectic morphism $S : E \to F$, the formula

$$\alpha_S(W_E(e)) = W_F(Se)$$

extends uniquely to a unital $*$-monomorphism $\alpha_S : \mathcal{A}(E) \to \mathcal{A}(F)$.

Proof. Lemma $L1$ makes the linear extension unique. It is multiplicative because

$$\begin{aligned}
\alpha_S(W_E(e)W_E(f)) &= \chi\left(\frac{1}{2}\Omega_E(e, f)\right) W_F(S(e + f)) \\
&= \chi\left(\frac{1}{2}\Omega_F(Se, Sf)\right) W_F(Se + Sf) \\
&= W_F(Se)W_F(Sf) = \alpha_S(W_E(e))\alpha_S(W_E(f)).
\end{aligned}$$

It preserves the involution since

$$\alpha_S(W_E(e)^*) = \alpha_S(W_E(-e)) = W_F(-Se) = W_F(Se)^*.$$

It is unital because $S0 = 0$. Finally, S is injective by Lemma $L2$, so the labels Se are distinct; Lemma $L1$ for F then implies that the images $W_F(Se)$ are linearly independent. Thus α_S has zero kernel.

Theorem $T2$**.** If $S : E \to E$ is a symplectic automorphism, then α_S is a $*$-automorphism of $\mathcal{A}(E)$. Moreover

$$\alpha_S\alpha_T = \alpha_{ST}.$$

Proof. The first statement follows from Theorem $T1$ applied to S and S^{-1}. The composition law follows on the Weyl basis:

$$\alpha_S\alpha_T(W_E(e)) = \alpha_S(W_E(Te)) = W_E(STe) = \alpha_{ST}(W_E(e)).$$

Thus a discrete one-parameter Gaussian or quasi-free dynamics is simply a homomorphism $t \mapsto S_t \in \mathrm{Sp}(E)$, with $\alpha_t = \alpha_{S_t}$.

43.5 Clifford Implementations T3–D5

For odd qudits, Gross's Clifford structure theorem `poswig.tex`, lines 458–537, states that every symplectic S has a unitary $\mu(S)$, unique projectively, such that

$$\mu(S)W_E(e)\mu(S)^* = W_E(Se),$$

and that $S \mapsto \mu(S)$ is a projective representation. Therefore the observable automorphism α_S is implemented by a Clifford unitary. This is the unitary version of Theorem $T2$.

Allowing a displacement $a \in E$, define

$$U_{a,S} = W_E(a)\mu(S).$$

Then

$$\begin{aligned} U_{a,S}W_E(e)U_{a,S}^* &= W_E(a)W_E(Se)W_E(-a) \\ &= \chi(\Omega_E(a, Se))W_E(Se). \end{aligned}$$

The second equality uses the half-Weyl law twice:

$$\frac{1}{2}\Omega(a, Se) + \frac{1}{2}\Omega(a + Se, -a) = \Omega(a, Se).$$

These affine Clifford maps are the finite odd-qudit analogue of affine Gaussian phase-space dynamics. They move Wigner functions by affine symplectic transformations, as sourced by Gross's Clifford covariance theorem, `poswig.tex`, lines 794–802.

43.6 Local Nets and Support

For a finite open U, the arithmetic-field label space $E(U)$ is an orthogonal direct sum of local residue label spaces. The inclusion $E(U) \hookrightarrow E(V)$ for $U \subset V$, given by extension by zero, is a symplectic morphism. Theorem $T1$ therefore recovers the open-local observable inclusion

$$\mathcal{A}(U) \hookrightarrow \mathcal{A}(V)$$

from AQM-24 as a Gaussian morphism.

A global symplectic automorphism $S : E(X) \to E(X)$ gives a global dynamics on $\mathcal{A}(X)$. It is a dynamics of the open-local net only if it respects the support structure being used. For example, if $S(E(U)) = E(U)$ for each open U under consideration, then every $\mathcal{A}(U)$ is invariant. If instead $S(E(U)) = E(\varphi(U))$ for a declared open-set permutation φ, then S gives a covariant net dynamics $\mathcal{A}(U) \to \mathcal{A}(\varphi(U))$. Otherwise it is a nonlocal global automorphism, not an automorphism of the chosen local net.

43.7 Gaussian States and Gross-Hudson T4

Gross defines stabilizer states from maximally isotropic subspaces and Weyl eigenvalue equations, `poswig.tex`, lines 846–903. He proves that their Wigner functions are normalized indicators of affine Lagrangian subspaces, lines 935–976. His discrete Hudson theorem, lines 202–214, says that for odd d, a pure state in $L^2(\mathbb{Z}_d^n)$ has nonnegative Wigner function if and only if it is a stabilizer state; for full support, this is equivalent to a quadratic-phase form

$$\psi(q) \propto \exp\left(\frac{2\pi i}{d}(q\theta q + xq)\right), \qquad \theta^T = \theta.$$

In the present report, this fixes the finite odd-qudit meaning of "Gaussian state": pure Gaussian states are stabilizer states, equivalently phased Lagrangian label data. Clifford dynamics sends them to Gaussian states because it sends Lagrangian subspaces to Lagrangian subspaces and transports the phase character by conjugation. Gross's positivity-preservation theorem, lines 1633–1639, gives the converse in Wigner language: a unitary preserving nonnegative Wigner states is Clifford.

43.8 Conclusion

Dynamics in the finite odd residue-qudit AQF layer is therefore label-first. Symplectic morphisms give the functorial algebra maps; symplectic automorphisms give Gaussian/quasi-free automorphisms; Clifford unitaries are their projective Hilbert-space implementations; and Gross-Hudson identifies the pure finite Gaussian states with stabilizer states. Even characteristic and mixed-residue dynamics remain separate convention problems.

44 Finite-Field Scales and Continuum Limits

44.1 Status and Source Anchors

This shard adds finite-field extension rank as the first arithmetic "scale" parameter. The closed-point Weyl net is AQM-28 and the dynamics language is AQM-42. The scheme facts use the Stacks Project snapshot: `references/algebraic_geometry/stacks_project_algebra.tex`, lines 250–274, 2972–2983, 3104–3117, 3310–3350, 6520–6665, and 45278–45284. Frobenius is sourced from `references/algebraic_geometry/stacks_project_varieties.tex`, lines 7005–7112. The finite-field trace formula also appears in `references/heisenberg_weil/sidana_kashyap_2202_00248_source/EAQECCs_over_rings_v15.tex`, lines 789–801.

44.2 Lamport Dependency Skeleton

$$\begin{array}{rcl} D1 & : & k = \mathbb{F}_q,\ k_r = \mathbb{F}_{q^r} \subset \bar{k},\ k_r \subset k_s \Longleftrightarrow r \mid s. \\ D2 & : & X_r = X_0 \times_{\operatorname{Spec} k} \operatorname{Spec} k_r,\ \pi_{s,r} : X_s \to X_r. \\ D3 & : & \mathcal{A}_r(U) \text{ is the AQM-28 closed-point Weyl net on } X_r. \\ L1 & : & \text{A closed point } x \in X_r \text{ splits over } k_s \text{ through } B_x = \kappa(x) \otimes_{k_r} k_s. \\ L2 & : & \text{The naive diagonal label map multiplies the trace commutator by } n = s/r. \\ L3 & : & \text{A trace-one element } c_x \in B_x \text{ gives a Weyl-compatible label embedding.} \\ T1 & : & \text{Trace-normalized local embeddings induce monomorphisms of open-local Weyl nets.} \\ T2 & : & \text{Prime-to-}p\text{ scales have canonical coherent embeddings.} \\ T3 & : & X_0 = \operatorname{Spec} \mathbb{F}_3 \text{ gives } \varinjlim_{3 \nmid r} M_{3^r}(\mathbb{C}). \\ T4 & : & \text{Frobenius acts by } W(q,p) \mapsto W(q^3, p^3). \end{array}$$

44.3 Scales and Closed-Point Nets D1–L1

Fix an algebraic closure $\bar{k}$ of $k = \mathbb{F}_q$, and let $k_r = \mathbb{F}_{q^r}$ be the unique subfield of $\bar{k}$ with q^r elements. Then $k_r \subset k_s$ exactly when $r \mid s$. For an affine k-scheme $X_0 = \operatorname{Spec} R$, define the rank-$r$ scale

$$X_r = X_0 \times_{\operatorname{Spec} k} \operatorname{Spec} k_r = \operatorname{Spec}(R \otimes_k k_r).$$

If $r \mid s$, base change along $k_r \subset k_s$ gives $\pi_{s,r} : X_s \to X_r$.

For an open $U \subset X_r$, $\mathcal{A}_r(U)$ denotes the AQM-28 closed-point Weyl algebra:

$$\mathcal{A}_r(U) = \varinjlim_{S \Subset U \cap |X_r|_{\mathrm{cl}}} \bigotimes_{x \in S} \operatorname{End}_{\mathbb{C}} \ell^2(\kappa(x)).$$

Thus a scale change should compare $\mathcal{A}_r(U)$ with $\mathcal{A}_s(\pi_{s,r}^{-1} U)$.

Let $x \in |X_r|_{\mathrm{cl}}$, put $K = \kappa(x)$, and write $B_x = K \otimes_{k_r} k_s$. Since finite fields are separable, $B_x \cong \prod_{y \mapsto x} \kappa(y)$, where y ranges over the closed points of X_s above x. This is the finite algebraic form of site splitting at finer scale.

44.4 The Trace Obstruction L2

The one-site character at x is

$$\psi_x(a) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{K/\mathbb{F}_p}(a) \right).$$

The naive diagonal map sends $(q, p_0) \in K^2$ to $\Delta_x(q, p_0) = ((q_y, p_{0,y}))_{y \mapsto x}$, where $q_y, p_{0,y}$ are the images in $\kappa(y)$. Let $a = p_0 q' - q p_0'$. The fine-scale commutator phase of the diagonal images is

$$\prod_{y \mapsto x} \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{\kappa(y)/\mathbb{F}_p}(a_y)\right).$$

The exponent is

$$\operatorname{Tr}_{B_x/\mathbb{F}_p}(a \otimes 1) = \operatorname{Tr}_{K/\mathbb{F}_p}\big(\operatorname{Tr}_{B_x/K}(a \otimes 1)\big).$$

As a K-algebra, B_x has dimension $n = s/r$, and $a \otimes 1$ acts on it as scalar multiplication by a. Hence

$$\operatorname{Tr}_{B_x/K}(a \otimes 1) = na,$$

so the fine phase is the coarse phase raised to the n-th power:

$$\prod_{y \mapsto x} \psi_y(a_y) = \psi_x(a)^n.$$

Therefore the naive diagonal label map is Weyl-compatible for all labels only when $n \equiv 1 \pmod p$. The geometry is canonical, but the Weyl phases are not automatically compatible.

44.5 Trace-Normalized Embeddings L3–T1

Choose $c_x = (c_y)_{y \mapsto x} \in B_x$ with $\operatorname{Tr}_{B_x/K}(c_x) = 1$. Such an element exists: for a finite separable field extension the trace pairing is nondegenerate, so the trace map is a nonzero K-linear map to the one-dimensional K-space K, hence is surjective; products are handled componentwise.

Define

$$\tau_{s,r,x}(q, p_0) = ((q_y, c_y p_{0,y}))_{y \mapsto x}.$$

For $e = (q, p_0)$ and $f = (q', p_0')$, the fine exponent becomes

$$\begin{aligned}\sum_{y \mapsto x} \operatorname{Tr}_{\kappa(y)/\mathbb{F}_p}\big(c_y(p_{0,y} q_y' - q_y p_{0,y}')\big) &= \operatorname{Tr}_{K/\mathbb{F}_p}\big(\operatorname{Tr}_{B_x/K}(c_x(a \otimes 1))\big) \\ &= \operatorname{Tr}_{K/\mathbb{F}_p}(a).\end{aligned}$$

Thus $\tau_{s,r,x}$ preserves the character-valued Weyl commutator. It is injective because $c_x \neq 0$, so some component c_y is nonzero.

Taking the direct sum of $\tau_{s,r,x}$ over the finite support of a label gives

$$\tau_{s,r,U} : E_r(U) \to E_s(\pi_{s,r}^{-1} U).$$

The Weyl relations are preserved on every finite support, so

$$\alpha_{s,r,U}(W_r(e)) = W_s(\tau_{s,r,U} e)$$

extends to a unital $*$-monomorphism

$$\alpha_{s,r,U} : \mathcal{A}_r(U) \hookrightarrow \mathcal{A}_s(\pi_{s,r}^{-1} U).$$

For $U \subset V$, these maps commute with the open-local inclusions because both constructions are defined pointwise on closed supports.

44.6 Canonical and Noncanonical Continuum Systems T2

If $p \nmid n = s/r$, then

$$c_x = n^{-1} \cdot 1_{B_x}$$

is canonical and has trace 1, because $\mathrm{Tr}_{B_x/K}(1_{B_x}) = n$. These choices are coherent: for $r \mid s \mid t$, the normalizing factors multiply as

$$(t/r)^{-1} = (s/r)^{-1}(t/s)^{-1}.$$

Hence the scales r with $p \nmid r$, ordered by divisibility, form a canonical directed subsystem of Weyl nets. Its algebraic continuum net is the filtered colimit of these nets.

For the full tower of all r, the current trace-character convention requires extra data. If $p \mid n$, then n^{-1} does not exist. A trace-one element still exists, but it is not canonical in the above data. The report therefore does not call the full all-rank tower canonical until a coherent trace-one system, or an alternative compatible-character convention, has been fixed.

44.7 The Point over $\mathbb{F}_3$ T3

Let

$$X_0 = \operatorname{Spec} \mathbb{F}_3, \qquad k_r = \mathbb{F}_{3^r}.$$

Then $X_r = \operatorname{Spec} k_r$ has one closed point and

$$E_r = k_r^2, \qquad \mathcal{H}_r = \ell^2(k_r), \qquad \mathcal{A}_r \cong M_{3^r}(\mathbb{C}).$$

The character is

$$\psi_r(a) = \exp\left(\frac{2\pi i}{3} \mathrm{Tr}_{k_r/\mathbb{F}_3}(a)\right).$$

For $r \mid s$, put $n = s/r$. The naive inclusion $(q,p) \mapsto (q,p)$ gives

$$\psi_s(pq' - qp') = \psi_r(pq' - qp')^n.$$

Already $r = 1, s = 2$ fails: $\mathrm{Tr}_{\mathbb{F}_9/\mathbb{F}_3}(1) = 2$, so the phase of 1 becomes $\exp(4\pi i/3)$, not $\exp(2\pi i/3)$. The canonical corrected map is

$$S_{s,r}(q,p) = (q, n^{-1}p)$$

whenever $3 \nmid n$. Consequently the canonical prime-to-3 continuum algebra of the arithmetic point is

$$\mathcal{A}_\infty^{(3')} = \varinjlim_{\substack{r \geq 1 \\ 3 \nmid r}} M_{3^r}(\mathbb{C}),$$

with transition maps $W_r(q,p) \mapsto W_s(q, n^{-1}p)$. This is an algebraic colimit: every element is represented at a finite extension rank. No Hilbert space $\ell^2(\bar{\mathbb{F}}_3)$ or C*-completion is part of the definition yet.

For the first missing full-tower step, $r = 1, s = 3$, the naive trace is zero on $\mathbb{F}_3$ because $\mathrm{Tr}_{\mathbb{F}_{27}/\mathbb{F}_3}(1) = 3 = 0$. A trace-one element exists but is a choice. For example, if $\theta^3 + 2\theta^2 + 1 = 0$, then this irreducible cubic has root trace $\mathrm{Tr}(\theta) = 1$, and $(q,p) \mapsto (q, \theta p)$ is a valid embedding $\mathcal{A}_1 \hookrightarrow \mathcal{A}_3$. Choosing θ is extra structure, not a canonical consequence of $k \subset k_3$.

44.8 Frobenius on the Point Net T4

At scale r, define

$$F_r : E_r \to E_r, \qquad F_r(q,p) = (q^3, p^3).$$

The trace is Frobenius-invariant:

$$\mathrm{Tr}_{k_r/\mathbb{F}_3}(a^3) = \sum_{i=0}^{r-1} a^{3^{i+1}} = \sum_{i=0}^{r-1} a^{3^i} = \mathrm{Tr}_{k_r/\mathbb{F}_3}(a).$$

Therefore F_r preserves the Weyl commutator and induces

$$\Phi_r(W_r(q,p)) = W_r(q^3, p^3).$$

This is a Clifford/Gaussian automorphism of order r. For $k_2 = \mathbb{F}_3[u]/(u^2+1)$, $u^3 = -u$, so

$$\Phi_2(W(a+bu, c+du)) = W(a-bu, c-du).$$

The prime-to-3 embeddings are Frobenius-equivariant. Indeed

$$S_{s,r}F_r(q,p) = (q^3, n^{-1}p^3) = F_s S_{s,r}(q,p),$$

because $n^{-1} \in \mathbb{F}_3$ is fixed by Frobenius. Thus Frobenius acts on the canonical prime-to-3 continuum algebra $\mathcal{A}_\infty^{(3')}$. In contrast, a noncanonical trace-one choice for a 3-divisible step is generally moved by Frobenius, so full-tower Frobenius equivariance is additional structure.

44.9 Conclusion

Finite-field extension rank gives a precise arithmetic scale system. Base change of schemes is canonical, and closed points split by the finite etale algebra $\kappa(x) \otimes_{k_r} k_s$. The Weyl net sees an extra phase constraint: absolute trace characters force trace-normalized embeddings. Therefore canonical continuum limits exist immediately on the prime-to-p subtower; the full tower exists only after coherent trace-one or compatible character data are added. For the arithmetic point over $\mathbb{F}_3$, the canonical continuum object currently justified by the report is the prime-to-3 algebraic colimit of the matrix algebras $M_{3^r}(\mathbb{C})$, and Frobenius acts on it by the compatible Clifford automorphisms $W(q,p) \mapsto W(q^3, p^3)$.

45 Trace Gauges for Full Continuum Towers

45.1 Status and Source Anchors

AQM-43 proved that pairwise trace-one elements repair the Weyl phase obstruction for finite-field scale embeddings. This shard proves that the pairwise choices can be made coherently for the whole divisibility tower, and records exactly what is gauge-like about the arbitrariness. The source anchors are Stacks Algebra, `references/algebraic_geometry/stacks_project_algebra.tex`, lines 45278–45284 for nondegenerate separable trace pairing, and Stacks Varieties, `references/algebraic_geometry/stacks_project_varieties.tex`, lines 7005–7112 for Frobenius. All coherence statements below are local derivations from those facts.

45.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & k_r = \mathbb{F}_{q^r} \subset \bar{k}, \quad r \mid s \Rightarrow k_r \subset k_s. \\
D2 & : & \text{A trace density is } a_r \in k_r \text{ with } \operatorname{Tr}_{k_s/k_r}(a_s) = a_r. \\
L1 & : & \operatorname{Tr}_{k_s/k_r} : k_s \to k_r \text{ is surjective.} \\
T1 & : & \text{A coherent trace-density system } (a_r)_r \text{ with } a_1 = 1 \text{ exists.} \\
D3 & : & c_{s,r} = a_s/a_r. \\
L2 & : & \operatorname{Tr}_{k_s/k_r}(c_{s,r}) = 1, \quad c_{t,r} = c_{t,s}c_{s,r}. \\
T2 & : & (c_{s,r}) \text{ gives coherent full-tower Weyl-net embeddings.} \\
T3 & : & \text{The choice of } (a_r) \text{ is trace gauge data.} \\
T4 & : & p \mid s/r \text{ forbids a relative-Frobenius-fixed trace-one scalar.}
\end{array}
$$

45.3 Coherent Trace Densities D1–T1

Fix $k = \mathbb{F}_q$ and $k_r = \mathbb{F}_{q^r} \subset \bar{k}$. A coherent trace-density system is a family $a_r \in k_r$ such that $a_1 = 1$ and, whenever $r \mid s$,

$$\operatorname{Tr}_{k_s/k_r}(a_s) = a_r.$$

This condition forces $a_r \neq 0$, because

$$\operatorname{Tr}_{k_r/k}(a_r) = a_1 = 1.$$

Lemma $L1$. For $r \mid s$, the trace map

$$\operatorname{Tr}_{k_s/k_r} : k_s \to k_r$$

is surjective.

Proof. Finite fields are perfect, hence k_s/k_r is separable. The sourced separable trace pairing is nondegenerate:

$$(x, y) \longmapsto \operatorname{Tr}_{k_s/k_r}(xy).$$

If the trace map itself were zero, then $\operatorname{Tr}_{k_s/k_r}(1 \cdot y) = 0$ for every y, contradicting nondegeneracy with $x = 1$. Thus the trace is a nonzero k_r-linear map from k_s to the one-dimensional k_r-space k_r, so it is surjective.

Theorem $T1$. Coherent trace-density systems exist.

Proof. Let $N_m = m!$. Start with $a_{N_1} = a_1 = 1$. Having chosen a_{N_m}, Lemma $L1$ gives $a_{N_{m+1}} \in k_{N_{m+1}}$ with

$$\mathrm{Tr}_{k_{N_{m+1}}/k_{N_m}}(a_{N_{m+1}}) = a_{N_m}.$$

For an arbitrary $r \geq 1$, choose m with $r \mid N_m$, and define

$$a_r = \mathrm{Tr}_{k_{N_m}/k_r}(a_{N_m}).$$

This is independent of m: if $n \geq m$, then transitivity of trace and the recursive construction give

$$\mathrm{Tr}_{k_{N_n}/k_r}(a_{N_n}) = \mathrm{Tr}_{k_{N_m}/k_r}\big(\mathrm{Tr}_{k_{N_n}/k_{N_m}}(a_{N_n})\big) = \mathrm{Tr}_{k_{N_m}/k_r}(a_{N_m}).$$

If $r \mid s$, choose m with $s \mid N_m$. Then

$$\mathrm{Tr}_{k_s/k_r}(a_s) = \mathrm{Tr}_{k_s/k_r}\big(\mathrm{Tr}_{k_{N_m}/k_s}(a_{N_m})\big) = \mathrm{Tr}_{k_{N_m}/k_r}(a_{N_m}) = a_r.$$

Thus $(a_r)_r$ is coherent.

45.4 Coherent Weyl Embeddings D3–T2

Given a coherent trace-density system, define

$$c_{s,r} = a_s/a_r \in k_s$$

for $r \mid s$, where a_r is embedded in k_s.

Lemma $L2$.

$$\mathrm{Tr}_{k_s/k_r}(c_{s,r}) = 1, \qquad c_{t,r} = c_{t,s}c_{s,r} \quad (r \mid s \mid t).$$

Proof. Since $a_r \in k_r^\times$,

$$\mathrm{Tr}_{k_s/k_r}(a_s/a_r) = a_r^{-1}\,\mathrm{Tr}_{k_s/k_r}(a_s) = 1.$$

The multiplicative identity is immediate:

$$c_{t,s}c_{s,r} = (a_t/a_s)(a_s/a_r) = a_t/a_r = c_{t,r}.$$

Let $X_0 = \operatorname{Spec} R$ be affine over k, and put $X_r = X_0 \times_k k_r$. If $x \in |X_r|_{\mathrm{cl}}$ has residue field $K = \kappa(x)$, then the points of X_s above x are encoded by

$$B_{s,r,x} = K \otimes_{k_r} k_s.$$

Use $1 \otimes c_{s,r} \in B_{s,r,x}$ as the trace-normalizing scalar. Its trace to K is 1: choose a k_r-basis of k_s; the multiplication matrix of $c_{s,r}$ has trace $\mathrm{Tr}_{k_s/k_r}(c_{s,r}) = 1$, and base change of that matrix from k_r to K leaves the trace equal to 1.

Theorem $T2$. The coherent trace-density system gives coherent monomorphisms of all open-local Weyl nets

$$\alpha_{s,r,U} : \mathcal{A}_r(U) \hookrightarrow \mathcal{A}_s(\pi_{s,r}^{-1}U), \qquad r \mid s.$$

Proof. AQM-43 proves that any trace-one normalizing scalar gives a Weyl-compatible monomorphism at one transition. The preceding paragraph gives trace-one scalars at every closed point, and Lemma $L2$ gives

$$(1 \otimes c_{t,s})(1 \otimes c_{s,r}) = 1 \otimes c_{t,r}.$$

Therefore the label embeddings compose strictly, hence the induced Weyl-net monomorphisms compose strictly.

45.5 Gauge Meaning and Frobenius T3–T4

The family (a_r), equivalently $(c_{s,r})$, is called a trace gauge. It is not extra geometry of X_0; it is extra presentation data needed to compare Weyl phases across scales. The prime-to-p subsystem has the canonical gauge $c_{s,r} = (s/r)^{-1}$. The full tower has many gauges, by Theorem $T1$, but no canonical one is singled out by the finite-field embeddings alone.

Frobenius acts on gauges. If $c_{s,r}$ has trace 1, then

$$\mathrm{Tr}_{k_s/k_r}(c_{s,r}^q) = \mathrm{Tr}_{k_s/k_r}(c_{s,r})^q = 1.$$

Thus Frobenius sends one trace gauge to another. For $p \mid s/r$, however, there is no trace-one scalar fixed by the relative Frobenius group. If $c \in k_s$ is fixed by $\mathrm{Gal}(k_s/k_r)$, then $c \in k_r$, and

$$\mathrm{Tr}_{k_s/k_r}(c) = (s/r)c = 0$$

in characteristic p. This cannot equal 1.

For $k = \mathbb{F}_3$, the first such case is $\mathbb{F}_3 \subset \mathbb{F}_{27}$. A trace-one c exists, but it cannot lie in $\mathbb{F}_3$. Hence the full tower supports Frobenius only gauge-covariantly unless an additional Frobenius-compatible gauge convention is supplied. This is the precise sense in which the arbitrariness is gauge freedom rather than a mathematical obstruction to the continuum limit.

46 The Arithmetic Quantum Field on $\operatorname{Spec}\mathbb{Z}$

46.1 Status and Source Anchors

This shard extends `CONVENTIONS.md` (u)–(y) and (ai). The source-local scheme facts are from `references/algebraic_geometry/stacks_project_algebra.tex`: lines 120–158, 250–274, 2972–2983, 3104–3160, 3251–3298, 3310–3342, and 4657–4703 cover prime ideals, residue fields, Spec, Zariski opens, functoriality, quasi-compact opens, and generic points. AQM-28 supplies the finite closed-residue Weyl net, and AQM-42 supplies finite-support dynamics.

46.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & X = \operatorname{Spec}\mathbb{Z}, \quad \eta = (0), \quad x_\ell = (\ell). \\
L1 & : & \operatorname{Prime}(\mathbb{Z}) = \{0\} \cup \{(\ell) : \ell \text{ prime}\}. \\
L2 & : & \kappa(\eta) = \mathbb{Q}, \quad \kappa(x_\ell) = \mathbb{F}_\ell. \\
L3 & : & \text{Nonempty opens are } \{\eta\} \cup \{x_\ell : \ell \notin S\} \text{ with } S \text{ finite.} \\
D2 & : & E_{\mathrm{cl}}(U) = \bigoplus_{x_\ell \in U} \mathbb{F}_\ell^2, \quad \mathcal{A}_{\mathrm{cl}}(U) = \varinjlim_{S \Subset U_{\mathrm{cl}}} \bigotimes_{\ell \in S} M_\ell(\mathbb{C}). \\
T1 & : & U \mapsto \mathcal{A}_{\mathrm{cl}}(U) \text{ is the closed-prime open-local AQF.} \\
D3 & : & \mathcal{H}_{\mathrm{cl}}^{\Omega} = \ell^2(\bigoplus_\ell \mathbb{F}_\ell). \\
L4 & : & \eta \text{ is a } \mathbb{Q}\text{-Weyl sector, not a finite qudit.} \\
T2 & : & \operatorname{End}_{\mathrm{Sch}}(\operatorname{Spec}\mathbb{Z}) = \{\mathrm{id}\}. \\
D4 & : & \alpha_n = \prod_\ell^{\mathrm{loc}} \operatorname{Ad} T_\ell(n \bmod \ell). \\
T3 & : & \alpha : \mathbb{Z} \curvearrowright \mathcal{A}_{\mathrm{cl}}(X) \text{ is faithful and extends profinitely.} \\
T4 & : & \alpha_n,\ n \neq 0, \text{ is not implemented in the } |0\rangle \text{ sector.}
\end{array}$$

46.3 The Point Set and Topology D1–L3

Every ideal of $\mathbb{Z}$ has the form (n). If (n) is prime, then $\mathbb{Z}/(n)$ is a domain by the source-local prime-ideal criterion. This happens exactly for $n = 0$ or for $n = \pm\ell$ with ℓ a rational prime. Hence

$$X = \operatorname{Spec}\mathbb{Z} = \{\eta\} \cup \{x_\ell : \ell \text{ prime}\}, \qquad \eta = (0), \quad x_\ell = (\ell).$$

The point η is generic because $\overline{\{\eta\}} = V(0) = X$, using the Stacks closure formula $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$.

Lemma $L2$. The residue fields are

$$\kappa(\eta) = \mathbb{Q}, \qquad \kappa(x_\ell) = \mathbb{F}_\ell.$$

Proof. By the Stacks residue-field definition, $\kappa(\mathfrak{p})$ is the fraction field of $\mathbb{Z}/\mathfrak{p}$. For $\mathfrak{p} = (0)$ this is $\operatorname{Frac}(\mathbb{Z}) = \mathbb{Q}$. For $\mathfrak{p} = (\ell)$ this is $\operatorname{Frac}(\mathbb{Z}/\ell\mathbb{Z}) = \mathbb{F}_\ell$.

Lemma $L3$. The nonempty Zariski opens of X are exactly

$$U_S = X \setminus \{x_\ell : \ell \in S\} = \{\eta\} \cup \{x_\ell : \ell \notin S\}, \qquad S \Subset \{\text{rational primes}\}.$$

Proof. Closed subsets are $V(n)$. If $n = 0$, then $V(n) = X$. If $n = \pm 1$, then $V(n) = \emptyset$. If $n \neq 0, \pm 1$, then $x_\ell \in V(n)$ exactly when $\ell \mid n$, and $\eta \notin V(n)$. Thus every proper nonempty closed subset is a finite set of closed prime points. Taking complements gives the displayed opens.

46.4 Closed-Prime Qudit Field D2–T1

For a prime ℓ, attach

$$E_\ell = \mathbb{F}_\ell^2, \qquad \mathcal{H}_\ell = \ell^2(\mathbb{F}_\ell), \qquad \mathcal{A}_\ell = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_\ell) \cong M_\ell(\mathbb{C}).$$

Write a label $e_\ell = (q_\ell, r_\ell)$. The local Weyl operators are

$$T_\ell(q)|s\rangle = |s+q\rangle, \qquad R_\ell(r)|s\rangle = \exp(2\pi i\, rs/\ell)|s\rangle, \qquad W_\ell(q,r) = T_\ell(q)R_\ell(r).$$

For a finite set S of primes,

$$E(S) = \bigoplus_{\ell\in S} \mathbb{F}_\ell^2, \quad \mathcal{H}(S) = \bigotimes_{\ell\in S} \mathcal{H}_\ell, \quad \mathcal{A}(S) = \bigotimes_{\ell\in S} \mathcal{A}_\ell.$$

The Weyl commutator is

$$W_S(e)W_S(f) = \prod_{\ell\in S} \exp(2\pi i(r_\ell q'_\ell - q_\ell r'_\ell)/\ell)\, W_S(f)W_S(e),$$

by the one-site calculation of AQM-28.

For an open U, let $U_{\mathrm{cl}} = U \cap \{x_\ell\}$, set $E_{\mathrm{cl}}(U) = \bigoplus_{x_\ell\in U_{\mathrm{cl}}} \mathbb{F}_\ell^2$ with finite support, and define

$$\mathcal{A}_{\mathrm{cl}}(U) = \varinjlim_{S\Subset U_{\mathrm{cl}}} \mathcal{A}(S), \qquad a \mapsto a\otimes \mathbf{1} \text{ for } S\subset S'.$$

Theorem *T*1**.** The assignment $U \mapsto \mathcal{A}_{\mathrm{cl}}(U)$ is the closed-prime open-local arithmetic quantum field on $\mathrm{Spec}\,\mathbb{Z}$. In particular,

$$\mathcal{A}_{\mathrm{cl}}(X) = \varinjlim_{S\Subset\{\ell\}} \bigotimes_{\ell\in S} M_\ell(\mathbb{C}),$$

the algebraic infinite tensor product of one ℓ-level qudit for each rational prime ℓ.

Proof. This is AQM-28 applied to the finite-residue closed points of X. If $U\subset V$, then $U_{\mathrm{cl}}\subset V_{\mathrm{cl}}$, so extension by identity gives a unital injective $*$-homomorphism $\mathcal{A}_{\mathrm{cl}}(U)\hookrightarrow\mathcal{A}_{\mathrm{cl}}(V)$. Disjoint finite supports commute because their Weyl commutator has no nontrivial local factor. If $U = \bigcup_i U_i$, every label has finite support, so each support point may be assigned to some U_i; the same additive proof as AQM-28 shows that $\mathcal{A}_{\mathrm{cl}}(U)$ is generated by the images of the $\mathcal{A}_{\mathrm{cl}}(U_i)$.

46.5 Incomplete Tensor Representation D3

The algebra $\mathcal{A}_{\mathrm{cl}}(X)$ is representation-independent. To get a Hilbert space, choose reference vectors $\Omega_\ell = |0\rangle \in \mathcal{H}_\ell$. Let

$$\Gamma_{\mathrm{fin}} = \bigoplus_\ell \mathbb{F}_\ell = \{s=(s_\ell) : s_\ell = 0 \text{ for all but finitely many } \ell\}.$$

Define

$$\mathcal{H}_{\mathrm{cl}}^{\Omega} = \ell^2(\Gamma_{\mathrm{fin}}), \qquad |s\rangle = \bigotimes_\ell |s_\ell\rangle$$

with the understood convention that all but finitely many tensor factors are $|0\rangle$. A finite-support algebra $\mathcal{A}(S)$ acts on the coordinates in S and fixes the others. These actions are compatible, so $\mathcal{H}_{\mathrm{cl}}^{\Omega}$ represents $\mathcal{A}_{\mathrm{cl}}(X)$. Changing the reference product state changes this Hilbert-space realization; it is not extra geometry of $\mathrm{Spec}\,\mathbb{Z}$.

46.6 The Generic Point L4

The generic point has residue field $\mathbb{Q}$, so it is not a finite qudit. If one includes it, the separate rational Weyl sector is

$$E_\eta = \mathbb{Q}^2, \qquad \mathcal{H}_\eta = \ell^2(\mathbb{Q}), \qquad \chi_\mathbb{Q}(a) = \exp(2\pi i a).$$

With

$$T_\mathbb{Q}(q)|s\rangle = |s+q\rangle, \qquad R_\mathbb{Q}(r)|s\rangle = \chi_\mathbb{Q}(rs)|s\rangle,$$

the same calculation gives $R_\mathbb{Q}(r)T_\mathbb{Q}(q) = \chi_\mathbb{Q}(rq)T_\mathbb{Q}(q)R_\mathbb{Q}(r)$. The pairing is nondegenerate: if $q \neq 0$, choose $r = (2q)^{-1}$; if $r \neq 0$, choose $q = (2r)^{-1}$.

Because every nonempty open contains η, the generic-sector local rule is

$$\mathcal{A}_{\mathrm{gen}}(U) = \begin{cases} \mathbb{C}, & U = \emptyset, \\ \mathcal{A}_\eta \otimes \mathcal{A}_{\mathrm{cl}}(U), & U \neq \emptyset. \end{cases}$$

Thus the generic sector is present in every nonempty local algebra. It is a global generic-fibre layer, not another prime-local qudit.

46.7 No Scheme Translation $z \mapsto z+1$ on X T2

Theorem *T*2. Every scheme endomorphism of $\operatorname{Spec}\mathbb{Z}$ is the identity. Therefore there is no nontrivial scheme-theoretic translation $z \mapsto z+1$ on $\operatorname{Spec}\mathbb{Z}$.

Proof. By affine functoriality, scheme endomorphisms of $\operatorname{Spec}\mathbb{Z}$ correspond contravariantly to unital ring endomorphisms $\varphi : \mathbb{Z} \to \mathbb{Z}$. Such a φ satisfies $\varphi(1) = 1$, hence $\varphi(n) = n\varphi(1) = n$ for all $n \in \mathbb{Z}$. Thus $\varphi = \mathrm{id}$, and the induced scheme map is the identity.

Consequently the expression $z \mapsto z+1$ belongs to $\mathbb{A}^1_\mathbb{Z} = \operatorname{Spec}\mathbb{Z}[z]$, where z is a coordinate. It does not define a self-map of the base scheme $\operatorname{Spec}\mathbb{Z}$.

46.8 Internal Integer Translation D4–T4

There is nevertheless a canonical internal residue translation on the closed-prime qudit algebra. For $n \in \mathbb{Z}$, define at prime ℓ

$$U_\ell(n) = T_\ell(n \bmod \ell).$$

For finite support S, set

$$\alpha_n^S(a) = \left(\bigotimes_{\ell \in S} U_\ell(n)\right) a \left(\bigotimes_{\ell \in S} U_\ell(n)\right)^*.$$

These maps are compatible with $a \mapsto a \otimes \mathbf{1}$, hence define an automorphism α_n of $\mathcal{A}_{\mathrm{cl}}(X)$. On one Weyl operator,

$$\alpha_n(W_\ell(q,r)) = \exp(-2\pi i\, nr/\ell)\, W_\ell(q,r).$$

Theorem *T*3. The maps $n \mapsto \alpha_n$ define a faithful action of $\mathbb{Z}$ on $\mathcal{A}_{\mathrm{cl}}(X)$. On $\mathcal{A}(S)$ the action factors through $\mathbb{Z}/N_S\mathbb{Z}$, where $N_S = \prod_{\ell \in S} \ell$. Therefore it extends to the compact residue product $\prod_\ell \mathbb{F}_\ell$, and hence to $\widehat{\mathbb{Z}}$ after reducing each p-adic coordinate modulo p.

Proof. The group law follows from $T_\ell(n)T_\ell(m) = T_\ell(n+m)$. If α_n is the identity, then for every prime ℓ and every $r \in \mathbb{F}_\ell$ the factor $\exp(-2\pi i nr/\ell)$ is 1. Hence $n \equiv 0 \pmod{\ell}$ for every ℓ, so $n = 0$. On finite S, only the residues $n \bmod \ell$ for $\ell \in S$ matter, which is the same as $n \bmod N_S$ by the Chinese remainder theorem. The same finite-coordinate dependence gives the continuous extension to the product of residue fields.

Theorem $T4$. For $n \neq 0$, the automorphism α_n is not implemented by a unitary in the reference-vector incomplete tensor representation $\mathcal{H}_{\mathrm{cl}}^{\Omega}$.

Proof. Let $P_\ell(a)$ be the rank-one projection onto $|a\rangle$. If a unitary U implemented α_n, then

$$UP_\ell(0)U^* = P_\ell(n \bmod \ell).$$

Since $P_\ell(0)\Omega = \Omega$, the vector $U\Omega$ would satisfy

$$P_\ell(n \bmod \ell)U\Omega = U\Omega$$

for every prime ℓ. For $n \neq 0$, $n \bmod \ell \neq 0$ for all but finitely many ℓ. But $\mathcal{H}_{\mathrm{cl}}^{\Omega}$ has basis indexed by configurations that are zero at all but finitely many primes. No nonzero vector can be supported only on configurations with $s_\ell = n \bmod \ell$ for infinitely many ℓ. Hence $U\Omega = 0$, impossible for a unitary.

46.9 Speculative Directions

The checked surprise is that the scheme has no translation, while the quantum closed-prime field has a faithful arithmetic translation action by residues. Finite experiments see only $\mathbb{Z}/N_S\mathbb{Z}$, while the infinite algebra sees the profinite completion. The global translation by 1 is an automorphism of the quasi-local algebra but not a unitary in the finite-excitation vacuum sector: a large arithmetic symmetry, not a local observable.

Source-gated questions now become visible. Should the generic $\mathbb{Q}$ sector be replaced by an adelic self-dual sector? Can a sourced C*-dynamical system with prime-indexed dimensions and $\log \ell$ energy weights recover Euler-product or zeta data without inserting them by hand? Is the nonimplementability of α_1 the first arithmetic superselection sector in this programme? These are questions, not conclusions.

47 Affine-Line Symmetries and Weyl-Net Automorphisms

47.1 Status and Source Anchors

This shard extends AQM-29. It fixes the word "symmetry" to mean a base-preserving k-scheme automorphism of $\mathbf{A}^1_k = \operatorname{Spec} k[x]$, where $k = \mathbb{F}_q$. The affine-space definition is source-local from the Stacks Project snapshot `references/algebraic_geometry/stacks_project_constructions.tex`, lines 733–758. The affine-scheme anti-equivalence and morphism-to-affine theorem are in `references/algebraic_geometry/stacks_project_schemes.tex`, lines 841–963. Spectrum, $D(f)$, and functoriality are in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 2972–2983 and 3123–3160. The finite Weyl operators use Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313. Odd-characteristic Clifford language is grounded by Gross, `references/symplectic_qecc/gross_0602001_source/poswig.tex`, lines 340–537.

47.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & k = \mathbb{F}_q, \quad X = \mathbf{A}^1_k = \operatorname{Spec} k[x]. \\
D2 & : & \operatorname{Aut}_k(X) \text{ means automorphisms over } \operatorname{Spec} k. \\
L1 & : & \operatorname{Aut}_k(X) = \{g_{a,b} : x \mapsto ax + b,\ a \in k^\times,\ b \in k\}. \\
D3 & : & x_\pi = (\pi), \quad K_\pi = k[x]/(\pi), \quad \pi \text{ monic irreducible.} \\
L2 & : & g_{a,b}(x_\pi) = x_{\pi^g}, \quad \pi^g(X) = a^{\deg \pi}\pi((X-b)/a). \\
D4 & : & \tau_{g,\pi} : K_{\pi^g} \to K_\pi, \quad \tau_{g,\pi}([X]) = a[X] + b. \\
L3 & : & \tau_{g,\pi} \text{ is a } k\text{-field isomorphism and preserves absolute traces.} \\
D5 & : & E(U) = \bigoplus_{x_\pi \in U} K_\pi^2, \quad \mathcal{A}(U) = \operatorname{span}\{W_U(e) : e \in E(U)\}. \\
D6 & : & S_g : E(U) \to E(gU) \text{ is residue pushforward by } \tau^{-1}. \\
T1 & : & S_g \text{ preserves the Weyl multiplier and commutator.} \\
T2 & : & \alpha_g^U(W_U(e)) = W_{gU}(S_g e) \text{ is a covariant net } *\text{-isomorphism.} \\
T3 & : & \text{On finite supports, } \alpha_g \text{ has a tensor-permutation Clifford implementation.}
\end{array}
$$

47.3 The Base-Preserving Symmetry Group D1–L1

Let $k = \mathbb{F}_q$. By the Stacks affine-space source,

$$\mathbf{A}^1_k = \operatorname{Spec} k[x].$$

A base-preserving symmetry means an automorphism of this affine scheme over $\operatorname{Spec} k$. By the affine-scheme anti-equivalence, this is the same data as a k-algebra automorphism

$$\theta : k[x] \longrightarrow k[x].$$

Lemma $L1$. Every k-scheme automorphism of $\mathbf{A}^1_k$ is uniquely

$$g_{a,b} : x \longmapsto ax + b, \qquad a \in k^\times,\ b \in k.$$

The group law is

$$g_{a,b} g_{c,d} = g_{ac,ad+b}.$$

Proof. Let $\theta : k[x] \to k[x]$ be a k-algebra automorphism. Put $h = \theta(x)$. Since $k[x]$ is generated as a k-algebra by x, θ is determined by h. Let $\psi = \theta^{-1}$ and put $r = \psi(x)$. Then

$$x = (\theta\psi)(x) = \theta(r) = r(h).$$

If $\deg h = m$ and $\deg r = n$, then $\deg r(h) = mn$. Since $\deg x = 1$, we have $mn = 1$. Thus $m = n = 1$, so $h = ax + b$ with $a \neq 0$. Conversely, every polynomial $ax + b$ with $a \neq 0$ gives the inverse $x \mapsto a^{-1}(x - b)$. Composition of functions gives the displayed law.

This proves the user's proposed group exactly in the base-preserving sense. Absolute automorphisms that are allowed to move the coefficient field are a different question and are not used in this shard.

47.4 Closed Points and Residue Fields D3–L3

By AQM-29, the closed points of X are

$$x_\pi = (\pi), \qquad \pi \in k[x] \text{ monic irreducible.}$$

The residue field at x_π is

$$K_\pi = \kappa(x_\pi) = k[x]/(\pi).$$

Let $g = g_{a,b}$. Its coordinate pullback is

$$\theta_g(f)(x) = f(ax + b).$$

Lemma *L*2. If π is monic irreducible of degree d, then

$$g(x_\pi) = x_{\pi^g}, \qquad \pi^g(X) = a^d \pi((X - b)/a).$$

The polynomial π^g is monic and irreducible.

Proof. Let $\alpha = [x] \in K_\pi$. The point image is the prime $\theta_g^{-1}((\pi))$. A polynomial f lies in that prime exactly when

$$f(a\alpha + b) = 0 \quad \text{in } K_\pi.$$

Thus the image closed point is the minimal polynomial over k of $\beta = a\alpha + b$. The displayed π^g is monic, because the leading coefficient is $a^d a^{-d} = 1$, and $\pi^g(\beta) = 0$. Since $\alpha = a^{-1}(\beta - b)$, the fields $k(\alpha)$ and $k(\beta)$ are equal. Hence β has degree d, so π^g is its irreducible minimal polynomial.

Definition *D*4. The residue pullback at x_π is

$$\tau_{g,\pi} : K_{\pi^g} \longrightarrow K_\pi, \qquad \tau_{g,\pi}([X]) = a[X] + b.$$

Lemma *L*3. $\tau_{g,\pi}$ is a k-field isomorphism. For every $u \in K_{\pi^g}$,

$$\mathrm{Tr}_{K_{\pi^g}/\mathbb{F}_p}(u) = \mathrm{Tr}_{K_\pi/\mathbb{F}_p}(\tau_{g,\pi} u).$$

Proof. The relation $\pi^g(X) = 0$ is sent to $\pi^g(a\alpha + b) = 0$, so the displayed formula defines a homomorphism. It is onto because $\alpha = a^{-1}(\tau([X]) - b)$ lies in its image. The two finite fields have the same size q^d, hence the map is an isomorphism. As a k-linear isomorphism, it conjugates the k-linear multiplication map m_u on K_{π^g} to $m_{\tau u}$ on K_π. Conjugate linear maps have the same trace over k; composing with $\mathrm{Tr}_{k/\mathbb{F}_p}$ gives the absolute-trace identity.

47.5 The Weyl Label Pushforward D5–T1

For an open $U \subset X$, define

$$E(U) = \bigoplus_{x_\pi \in U} K_\pi^2$$

with finite support. A label is written

$$e = (q_\pi, p_\pi)_\pi.$$

At x_π, use

$$T_\pi(q)|s\rangle = |s+q\rangle, \qquad R_\pi(p)|s\rangle = \exp\left(\frac{2\pi i}{p_0} \operatorname{Tr}_{K_\pi/\mathbb{F}_{p_0}}(ps)\right)|s\rangle,$$

where $p_0 = \operatorname{char} k$, and $W_\pi(q,p) = T_\pi(q)R_\pi(p)$. For finite support, $W_U(e)$ is the tensor product of the local operators.

Definition $D6$. The pushforward

$$S_g : E(U) \longrightarrow E(gU)$$

is defined by

$$(S_g e)_{\pi^g} = \big(\tau_{g,\pi}^{-1}(q_\pi), \tau_{g,\pi}^{-1}(p_\pi)\big).$$

Theorem $T1$. S_g preserves the Weyl multiplier and commutator. In particular,

$$W_{gU}(S_g e) W_{gU}(S_g f)$$

has the same scalar multiplier as $W_U(e)W_U(f)$.

Proof. It is enough to check one closed point. Let $(Q,P) = \tau^{-1}(q,p)$ and $(Q',P') = \tau^{-1}(q',p')$. By Lemma $L3$,

$$\operatorname{Tr}_{K_{\pi^g}/\mathbb{F}_{p_0}}(PQ') = \operatorname{Tr}_{K_\pi/\mathbb{F}_{p_0}}(pq').$$

The same identity with P,Q and P',Q' exchanged gives preservation of

$$\operatorname{Tr}(PQ' - QP').$$

These are exactly the local multiplier and commutator exponents. Multiplying over the finite support proves the claim.

47.6 Net Automorphism and Clifford Implementation T2–T3

Theorem $T2$. For every $g \in \operatorname{Aut}_k(X)$ and open U, the rule

$$\alpha_g^U(W_U(e)) = W_{gU}(S_g e)$$

extends uniquely to a unital $*$-isomorphism

$$\alpha_g^U : \mathcal{A}(U) \longrightarrow \mathcal{A}(gU).$$

These maps are covariant for inclusions and satisfy

$$\alpha_g^{hU}\alpha_h^U = \alpha_{gh}^U.$$

Proof. The Weyl operators form a basis on every finite support, and $\mathcal{A}(U)$ is the union of the finite-support algebras. Theorem $T1$ shows that the assignment respects products and adjoints on each finite support, hence on the union. It is bijective because $S_{g^{-1}}$ is the inverse label map. If $U \subset V$, extension by zero commutes with pushforward of finite supports, so the square

$$\mathcal{A}(U) \to \mathcal{A}(V), \qquad \mathcal{A}(gU) \to \mathcal{A}(gV)$$

commutes. The residue maps τ compose because the coordinate pullbacks θ_g compose; therefore $S_g S_h = S_{gh}$, and the displayed group law follows on the Weyl basis.

Theorem $T3$. For a finite closed support S, the finite-volume map

$$\alpha_g^S : \mathcal{A}(S) \to \mathcal{A}(gS)$$

is implemented by a unitary tensor relabelling. If

$$\mathcal{H}(S) = \bigotimes_{x_\pi \in S} \ell^2(K_\pi),$$

define

$$U_g^S \left(\bigotimes_{x_\pi \in S} |s_\pi\rangle \right) = \bigotimes_{x_\pi g \in gS} |\tau_{g,\pi}^{-1}(s_\pi)\rangle.$$

Then

$$U_g^S W_S(e) (U_g^S)^* = W_{gS}(S_g e).$$

Proof. At one site, residue relabelling sends $|s\rangle$ to $|\tau^{-1}s\rangle$. It carries translation by q to translation by $\tau^{-1}q$. It carries the phase $\exp(2\pi i \, \mathrm{Tr}(ps)/p_0)$ to the phase with $\tau^{-1}p$ and $\tau^{-1}s$, by Lemma $L3$. Tensoring the one-site calculation over S proves the formula.

Thus the affine-line symmetry acts on the Weyl local net by symplectic finite-support label maps. In odd characteristic, Gross's Clifford structure theorem makes each finite-volume implementation a Clifford unitary. In even characteristic, the algebra automorphism above is still proved, but the half-Weyl Clifford convention is not imported.

48 The $\mathbb{F}_3$ Rational-Support Affine-Line Symmetry Example

48.1 Status and Source Anchors

This shard is the finite rational-support line-by-line example for

$$k = \mathbb{F}_3, \qquad X = \mathbf{A}^1_{\mathbb{F}_3} = \operatorname{Spec} \mathbb{F}_3[x].$$

It uses the definitions and proofs of AQM-29 and AQM-46. The full countable closed-point Hilbert representation is AQM-48. No new general claim is introduced here. The source anchors remain Stacks for Spec, affine space, affine-scheme functoriality, and standard opens; Ashikhmin–Knill for finite-field shift and phase operators; and Gross for the odd-qudit Clifford interpretation.

48.2 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: \; k = \mathbb{F}_3, \quad X = \operatorname{Spec} k[x]. \\
D2 &: \; G = \operatorname{Aut}_k(X) = \{g_{a,b} : x \mapsto ax + b\}. \\
L1 &: \; G = \{x, x+1, x+2, 2x, 2x+1, 2x+2\}. \\
D3 &: \; x_r = (x - r), \; r = 0, 1, 2, \text{ are the rational closed points.} \\
L2 &: \; g_{a,b}(x_r) = x_{ar+b}. \\
D4 &: \; S_{\mathrm{rat}} = \{x_0, x_1, x_2\}, \quad E_{\mathrm{rat}} = \mathbb{F}_3^3 \oplus \mathbb{F}_3^3. \\
L3 &: \; S_g(q_0, q_1, q_2, p_0, p_1, p_2) = (q_{g^{-1}0}, q_{g^{-1}1}, q_{g^{-1}2}, p_{g^{-1}0}, p_{g^{-1}1}, p_{g^{-1}2}). \\
T1 &: \; S_g \text{ preserves } \Omega(e, f) = \textstyle\sum_r (p_r q'_r - q_r p'_r). \\
T2 &: \; \alpha_g(W(e)) = W(S_g e) \text{ is a Weyl algebra automorphism.} \\
T3 &: \; U_g |s_0, s_1, s_2\rangle = |s_{g^{-1}0}, s_{g^{-1}1}, s_{g^{-1}2}\rangle \text{ implements } \alpha_g.
\end{aligned}$$

48.3 Step 1: The Space D1

The coordinate ring is

$$R = \mathbb{F}_3[x].$$

The affine line is

$$X = \operatorname{Spec} R.$$

The closed points of X are ideals (π), where π is monic irreducible in $\mathbb{F}_3[x]$. The residue field is

$$K_\pi = \mathbb{F}_3[x]/(\pi).$$

At a degree d point, K_π has 3^d elements, so the site carries one 3^d-level qudit.

48.4 Step 2: The Symmetry Group D2–L1

By AQM-46,

$$G = \operatorname{Aut}_{\mathbb{F}_3}(X) = \{g_{a,b} : x \mapsto ax + b, \; a \in \mathbb{F}_3^\times, \; b \in \mathbb{F}_3\}.$$

Since

$$\mathbb{F}_3^\times = \{1, 2\},$$

there are $2 \cdot 3 = 6$ symmetries:

name	(a, b)	map
e	$(1, 0)$	x
τ	$(1, 1)$	$x + 1$
τ^2	$(1, 2)$	$x + 2$
ρ	$(2, 0)$	$2x$
$\rho\tau$	$(2, 1)$	$2x + 1$
$\rho\tau^2$	$(2, 2)$	$2x + 2$

The group law is

$$g_{a,b} g_{c,d} = g_{ac, ad+b}.$$

For example,

$$\tau\rho = g_{1,1} g_{2,0} = g_{2,1}, \qquad \rho\tau = g_{2,0} g_{1,1} = g_{2,2}.$$

Thus the group is nonabelian and has six elements.

48.5 Step 3: Rational Closed Points D3–L2

For $r \in \mathbb{F}_3$, define

$$x_r = (x - r).$$

The residue field is

$$K_r = \mathbb{F}_3[x]/(x - r) \cong \mathbb{F}_3.$$

For $g_{a,b}$, AQM-46 gives

$$g_{a,b}(x_r) = x_{ar+b}.$$

The six rational-point permutations are

g	x_0	x_1	x_2
x	x_0	x_1	x_2
$x + 1$	x_1	x_2	x_0
$x + 2$	x_2	x_0	x_1
$2x$	x_0	x_2	x_1
$2x + 1$	x_1	x_0	x_2
$2x + 2$	x_2	x_1	x_0

At rational points the residue-field isomorphism is the identity map on $\mathbb{F}_3$, because every K_r is canonically $\mathbb{F}_3$ after evaluation at r.

48.6 Step 4: Quadratic Closed Points

The monic irreducible quadratics over $\mathbb{F}_3$ are

$$q_0 = x^2 + 1, \qquad q_1 = x^2 + x + 2, \qquad q_2 = x^2 + 2x + 2.$$

They are irreducible because none has a root in $\mathbb{F}_3$. For a quadratic π, the transformation rule of AQM-46 is

$$\pi^g(X) = a^2 \pi((X - b)/a).$$

In $\mathbb{F}_3$, $a^2 = 1$ for $a = 1, 2$. The action is

g	q_0	q_1	q_2
x	q_0	q_1	q_2
$x+1$	q_1	q_2	q_0
$x+2$	q_2	q_0	q_1
$2x$	q_0	q_2	q_1
$2x+1$	q_1	q_0	q_2
$2x+2$	q_2	q_1	q_0

Example: for $g = x + 1$,

$$q_0^g(X) = q_0(X-1) = (X-1)^2 + 1 = X^2 + X + 2 = q_1.$$

48.7 Step 5: The Rational Three-Site Label Space D4

Restrict now to the finite rational closed support

$$S_{\text{rat}} = \{x_0, x_1, x_2\}.$$

The label space is

$$E_{\text{rat}} = \bigoplus_{r=0}^{2} \mathbb{F}_3^2 = \mathbb{F}_{3,q}^3 \oplus \mathbb{F}_{3,p}^3.$$

Write a label as

$$e = (q_0, q_1, q_2, p_0, p_1, p_2).$$

The symplectic form is

$$\Omega(e, f) = \sum_{r=0}^{2} (p_r q'_r - q_r p'_r).$$

This is the direct sum of the three one-qutrit forms.

48.8 Step 6: The Symplectic Map L3–T1

For $g \in G$, the label pushforward is

$$(S_g e)_{g(r)} = e_r$$

on both the q-coordinates and the p-coordinates. Equivalently,

$$S_g(q_0, q_1, q_2, p_0, p_1, p_2) = (q_{g^{-1}0}, q_{g^{-1}1}, q_{g^{-1}2}, p_{g^{-1}0}, p_{g^{-1}1}, p_{g^{-1}2}).$$

For $\tau : x \mapsto x + 1$,

$$\tau^{-1}(0) = 2, \quad \tau^{-1}(1) = 0, \quad \tau^{-1}(2) = 1,$$

so

$$S_\tau(e) = (q_2, q_0, q_1, p_2, p_0, p_1).$$

For $\rho : x \mapsto 2x$,

$$\rho^{-1} = \rho, \qquad S_\rho(e) = (q_0, q_2, q_1, p_0, p_2, p_1).$$

Check.

$$\begin{aligned}\Omega(S_g e, S_g f) &= \sum_y \big((S_g p)_y (S_g q')_y - (S_g q)_y (S_g p')_y\big) \\ &= \sum_y \big(p_{g^{-1}y} q'_{g^{-1}y} - q_{g^{-1}y} p'_{g^{-1}y}\big) \\ &= \sum_r (p_r q'_r - q_r p'_r) = \Omega(e, f).\end{aligned}$$

Thus S_g is symplectic.

48.9 Step 7: The Weyl Operators T2

The Hilbert space is

$$\mathcal{H}_{\mathrm{rat}} = \ell^2(\mathbb{F}_3)^{\otimes 3}$$

with basis $|s_0, s_1, s_2\rangle$. At site r,

$$T_r(q)|s_r\rangle = |s_r + q\rangle, \qquad R_r(p)|s_r\rangle = \omega^{ps_r}|s_r\rangle, \qquad \omega = e^{2\pi i/3}.$$

For $e = (q_0, q_1, q_2, p_0, p_1, p_2)$,

$$W(e) = \bigotimes_{r=0}^{2} T_r(q_r) R_r(p_r).$$

The Weyl law is

$$W(e)W(f) = \omega^{\sum_r p_r q'_r} W(e + f).$$

Because S_g preserves the exponent, the formula

$$\alpha_g(W(e)) = W(S_g e)$$

defines an algebra automorphism. For τ,

$$\alpha_\tau(W(q_0, q_1, q_2, p_0, p_1, p_2)) = W(q_2, q_0, q_1, p_2, p_0, p_1).$$

48.10 Step 8: The Clifford Unit T3

Define

$$U_g|s_0, s_1, s_2\rangle = |s_{g^{-1}0}, s_{g^{-1}1}, s_{g^{-1}2}\rangle.$$

For τ,

$$U_\tau|s_0, s_1, s_2\rangle = |s_2, s_0, s_1\rangle.$$

For ρ,

$$U_\rho|s_0, s_1, s_2\rangle = |s_0, s_2, s_1\rangle.$$

These are permutation matrices, hence unitaries. Directly,

$$U_g T_r(q) U_g^* = T_{g(r)}(q), \qquad U_g R_r(p) U_g^* = R_{g(r)}(p).$$

Multiplying over $r = 0, 1, 2$ gives

$$U_g W(e) U_g^* = W(S_g e) = \alpha_g(W(e)).$$

Since 3 is odd, Gross's finite Clifford theorem applies: this finite rational-support implementation is a Clifford unitary. This is only a finite-volume truncation; AQM-48 gives the full countable closed-point net, its infinite Hilbert representation, and the generic-point caveat.

49 The Full $\mathbb{F}_3$ Affine-Line Net and Hilbert Action

49.1 Status and Source Anchors

This shard corrects the scope of AQM-47. The three-rational-point calculation there is a finite closed-support truncation. It is not the full local net on $\mathbf{A}^1_{\mathbb{F}_3}$. The full closed-point net uses every monic irreducible polynomial in $\mathbb{F}_3[x]$. AQM-28 supplies the closed-point quasi-local convention, AQM-29 supplies the point and open-set calculation for $\operatorname{Spec} k[t]$, AQM-45 supplies the reference-vector incomplete tensor representation pattern, and AQM-46 supplies the affine symmetry action.

49.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & k = \mathbb{F}_3, \quad R = k[x], \quad X = \operatorname{Spec} R. \\
D2 & : & \eta = (0), \quad \mathcal{P} = \{\pi \in k[x] : \pi \text{ monic irreducible}\}. \\
L1 & : & X = \{\eta\} \cup \{x_\pi = (\pi) : \pi \in \mathcal{P}\}, \quad \mathcal{P} \text{ is countably infinite.} \\
D3 & : & K_\pi = R/(\pi), \quad \mathcal{H}_\pi = \ell^2(K_\pi), \quad \mathcal{A}_\pi = \operatorname{End}(\mathcal{H}_\pi). \\
D4 & : & E(U) = \bigoplus_{x_\pi \in U} K_\pi^2, \quad \mathcal{A}(U) = \varinjlim_{S \Subset C(U)} \bigotimes_{x_\pi \in S} \mathcal{A}_\pi. \\
L2 & : & \text{Every nonempty open } U \text{ has cofinite } C(U) \subset \mathcal{P}. \\
T1 & : & \mathcal{A}(X) \text{ is an algebraic infinite tensor product over } \mathcal{P}. \\
D5 & : & \Gamma_{\text{fin}} = \bigoplus_{\pi \in \mathcal{P}} K_\pi, \quad \mathcal{H}^0 = \ell^2(\Gamma_{\text{fin}}). \\
L3 & : & \mathcal{H}^0 \text{ is countably infinite-dimensional and represents } \mathcal{A}(X). \\
D6 & : & G = \{g_{a,b} : x \mapsto ax + b,\ a \in \mathbb{F}_3^\times,\ b \in \mathbb{F}_3\}. \\
D7 & : & (V_g s)_{\pi^g} = \tau_{g,\pi}^{-1}(s_\pi), \quad U_g|s\rangle = |V_g s\rangle. \\
T2 & : & U_g \mathcal{A}(U) U_g^* = \mathcal{A}(gU), \quad U_g W_U(e) U_g^* = W_{gU}(S_g e). \\
L4 & : & g(\eta) = \eta, \quad \eta \in U \text{ for every nonempty open } U. \\
T3 & : & \eta \text{ is not a finite-residue qudit site in this net.}
\end{array}
$$

49.3 The Countable Closed-Point Set D1–L1

Let

$$R = \mathbb{F}_3[x], \qquad X = \operatorname{Spec} R.$$

The zero ideal is prime because R is a domain; write

$$\eta = (0).$$

For each monic irreducible $\pi \in R$, write

$$x_\pi = (\pi).$$

Lemma *L*1. The points of X are exactly

$$\{\eta\} \cup \{x_\pi : \pi \in \mathcal{P}\}.$$

The set $\mathcal{P}$ is countably infinite.

Proof. The point classification is AQM-29 with $k = \mathbb{F}_3$. Directly: $\mathbb{F}_3[x]$ is Euclidean, so every nonzero prime ideal has a monic generator. The generator must be irreducible, and every monic irreducible generates a maximal, hence prime, ideal.

For countability, R is the union over degrees $n \geq 0$ of the finite sets of polynomials of degree at most n. Hence R, and therefore $\mathcal{P}$, is countable. For infinitude, suppose $\pi_1, \ldots, \pi_m$ were all monic irreducibles. Then

$$F = \pi_1 \cdots \pi_m + 1$$

is nonconstant. Choose a monic nonconstant divisor ρ of F with least degree; then ρ is irreducible. No π_i divides F, because π_i divides $F-1$. Thus ρ is a new monic irreducible, a contradiction.

49.4 The Full Closed-Point Net D3–T1

At the closed point x_π, put

$$K_\pi = R/(\pi).$$

If $\deg \pi = d$, then K_π has 3^d elements. Define

$$\mathcal{H}_\pi = \ell^2(K_\pi), \qquad \mathcal{A}_\pi = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_\pi) \cong M_{3^d}(\mathbb{C}).$$

For a finite closed support $S \Subset \mathcal{P}$, set

$$\mathcal{A}(S) = \bigotimes_{\pi \in S} \mathcal{A}_\pi.$$

For an open $U \subset X$, define

$$C(U) = \{\pi \in \mathcal{P} : x_\pi \in U\}.$$

The label group and local algebra are

$$E(U) = \bigoplus_{\pi \in C(U)} K_\pi^2, \qquad \mathcal{A}(U) = \varinjlim_{S \Subset C(U)} \mathcal{A}(S),$$

where the transition maps are $a \mapsto a \otimes \mathbf{1}$.

Lemma $L2$. If $U \neq \emptyset$, then $C(U)$ is cofinite in $\mathcal{P}$. If $U = \emptyset$, then $C(U) = \emptyset$.

Proof. Every closed subset of $\operatorname{Spec} R$ has the form $V(I)$. Since R is a principal ideal domain, $I = (h)$. Thus every open is $D(h)$. If $h = 0$, then $D(h) = \emptyset$. If $h \neq 0$, then

$$x_\pi \notin D(h) \iff h \in (\pi) \iff \pi \mid h.$$

A nonzero polynomial has only finitely many monic irreducible divisors. Hence the closed points omitted by $D(h)$ are finite.

Theorem $T1$. The global closed-point algebra is

$$\mathcal{A}(X) = \varinjlim_{S \Subset \mathcal{P}} \bigotimes_{\pi \in S} M_{3^{\deg \pi}}(\mathbb{C}).$$

It is an algebraic infinite tensor product over all closed points. The rational three-site algebra of AQM-47 is only the finite subalgebra for

$$S_{\mathrm{rat}} = \{x, x-1, x-2\} \subset \mathcal{P}.$$

Proof. For $X = D(1)$, $C(X) = \mathcal{P}$, so the displayed formula is Definition $D4$. Since $\mathcal{P}$ is infinite by Lemma $L1$, this is not a finite matrix algebra. The rational support is a finite subset of $\mathcal{P}$, so $\mathcal{A}(S_{\mathrm{rat}})$ maps into the filtered colimit. It does not equal the colimit, because any irreducible $\pi \notin S_{\mathrm{rat}}$ contributes the nontrivial one-site algebra $\mathcal{A}_\pi$.

49.5 The Infinite Hilbert Representation D5–L3

The algebraic net above is defined before choosing any Hilbert-space completion. To get the standard reference representation, choose the reference vector $|0\rangle_\pi \in \mathcal{H}_\pi$ at every closed point. Define

$$\Gamma_{\text{fin}} = \bigoplus_{\pi\in\mathcal{P}} K_\pi = \{s = (s_\pi)_\pi : s_\pi = 0 \text{ for all but finitely many } \pi\}.$$

Set

$$\mathcal{H}^0 = \ell^2(\Gamma_{\text{fin}}), \qquad |s\rangle = \bigotimes_{\pi\in\mathcal{P}} |s_\pi\rangle,$$

with all but finitely many tensor factors equal to $|0\rangle_\pi$.

Lemma *L*3. $\mathcal{H}^0$ is countably infinite-dimensional, and it represents $\mathcal{A}(X)$.

Proof. The set of finite subsets of the countable set $\mathcal{P}$ is countable. For a fixed finite support S, the set $\prod_{\pi\in S} K_\pi$ is finite. Thus Γ_{fin}, a countable union of finite sets, is countable. It is infinite because the configurations supported at one closed point with value 1 are distinct as π varies. Hence $\ell^2(\Gamma_{\text{fin}})$ has a countably infinite orthonormal basis.

For finite S, $\mathcal{A}(S)$ acts on the coordinates in S and fixes all other coordinates. If $S \subset T$, this action agrees with the action of $a \otimes \mathbf{1}_{\mathcal{H}(T\setminus S)}$. Therefore the compatible finite-support actions define a representation of the filtered colimit $\mathcal{A}(X)$.

49.6 The Full Affine Symmetry Implementation D6–T2

Let

$$G = \mathrm{Aut}_{\mathbb{F}_3}(X) = \{g_{a,b} : x \mapsto ax + b,\ a \in \mathbb{F}_3^\times,\ b \in \mathbb{F}_3\}.$$

For $\pi \in \mathcal{P}$, AQM-46 gives

$$g(x_\pi) = x_{\pi^g}, \qquad \pi^g(X) = a^{\deg \pi}\pi((X - b)/a),$$

and a residue-field isomorphism

$$\tau_{g,\pi} : K_{\pi^g} \to K_\pi, \qquad \tau_{g,\pi}([X]) = a[X] + b.$$

Define a map on finite-support configurations by

$$(V_g s)_{\pi^g} = \tau_{g,\pi}^{-1}(s_\pi).$$

Because $\tau_{g,\pi}^{-1}(0) = 0$, V_g sends finite-support configurations to finite-support configurations. It is bijective, with inverse $V_{g^{-1}}$. Hence

$$U_g|s\rangle = |V_g s\rangle$$

defines a unitary on $\mathcal{H}^0$.

Theorem *T*2. For every open $U \subset X$ and every $e \in E(U)$,

$$U_g W_U(e) U_g^* = W_{gU}(S_g e).$$

Consequently

$$U_g \mathcal{A}(U) U_g^* = \mathcal{A}(gU),$$

and $g \mapsto U_g$ is a unitary implementation of the full closed-point net action in the zero-reference representation.

Proof. At one closed point, the basis relabelling $|s\rangle \mapsto |\tau_{g,\pi}^{-1}s\rangle$ carries translation by q to translation by $\tau_{g,\pi}^{-1}q$. It carries the phase labelled by p to the phase labelled by $\tau_{g,\pi}^{-1}p$, because AQM-46 proves absolute-trace preservation for $\tau_{g,\pi}$. Thus the formula holds on a one-site Weyl operator. Tensoring over the finite support of e proves the displayed Weyl formula. The Weyl operators span each finite-support algebra, so the algebra statement follows. The maps V_g compose because the residue maps $\tau_{g,\pi}$ compose, again by AQM-46; hence the U_g compose on the basis of $\mathcal{H}^0$.

49.7 The Generic Point L4–T3

Lemma *L*4. The generic point $\eta = (0)$ is fixed by every $g \in G$, and every nonempty open $U \subset X$ contains η.

Proof. The coordinate pullback of g is an automorphism of the domain $\mathbb{F}_3[x]$, so the inverse image of the zero ideal is the zero ideal. Thus $g(\eta) = \eta$. By Lemma *L*2, every nonempty open is $D(h)$ with $h \neq 0$. Since $h \notin (0)$, the point η lies in $D(h)$.

Theorem *T*3. The generic point is not a tensor factor of the finite-residue Weyl net above.

Proof. Its residue field is
$$\kappa(\eta) = \mathbb{F}_3(x),$$
the fraction field of $\mathbb{F}_3[x]$. This field is infinite. Convention (y) attaches finite Weyl sites only to closed points, whose residue fields are finite. Therefore η affects the topology of opens, but it contributes no factor K_η^2, no finite qudit $\ell^2(K_\eta)$, and no coordinate of Γ_{fin} in the closed-point net.

If one adds the rational-function sector of convention (z), an additional choice of additive character is required. The closed-point rule $(q, p) \mapsto (\tau^{-1}q, \tau^{-1}p)$ must not be copied blindly. For the default functional λ that extracts the coefficient of x^{-1} at infinity, let $\rho : x \mapsto 2x$. Then $\tau_\rho^{-1}(x^{-1}) = 2x^{-1}$, so
$$\lambda(1 \cdot x^{-1}) = 1, \qquad \lambda(\tau_\rho^{-1}(1)\tau_\rho^{-1}(x^{-1})) = 2.$$
The multiplier changes. Thus a generic-sector affine action needs a separate dual-label or character-invariance convention before it can be called part of the Weyl-net symmetry theorem.

50 Regular Functions, Supports, and Weyl Labels on the Affine Line

50.1 Status and Source Anchors

This shard fixes the current meaning of "regular function" in the closed-point Weyl-net layer. It refines AQM-28 and AQM-29. The source-local facts are Stacks: affine-scheme global sections in `references/algebraic_geometry/stacks_project_schemes.tex`, lines 670–719; Spec, $D(f)$, and $V(I)$ in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 2972–3023 and 3104–3211; residue fields in the same file, lines 250–274 and 3310–3350. Finite-field Weyl operators are AQM-28, sourced from Ashikhmin–Knill as recorded there. The Artin-ring layer is AQM-36.

50.2 Lamport Dependency Skeleton

$D1$: $k = \mathbb{F}_Q, \quad R = k[x], \quad X = \operatorname{Spec} R, \quad \mathcal{P} = \{\pi \in R : \pi \text{ monic irreducible}\}$.
$D2$: $x_\pi = (\pi), \quad K_\pi = R/(\pi)$.
$D3$: $h \neq 0, \quad D(h) = \{\mathfrak{p} : h \notin \mathfrak{p}\}, \quad \Gamma(D(h), \mathcal{O}_X) = R_h$.
$D4$: $\mathrm{ev}_U(F,G)_\pi = (F \bmod \pi, G \bmod \pi) \in K_\pi^2, \quad x_\pi \in U$.
$L1$: A nonzero regular-function pair on nonempty $D(h)$ has cofinite closed support in $D(h)$.
$T1$: $\mathrm{ev}_U(F,G) \in E(U)$ iff $(F,G) = (0,0)$.
$D5$: $Z_{\mathrm{cl}}(h) = \{x_\pi : \pi \mid h\}, \quad \mathrm{rad}(h) = \prod_{\pi \mid h} \pi$.
$L2$: $Z_{\mathrm{cl}}(h)$ is finite and depends only on $\mathrm{rad}(h)$.
$D6$: $E_h^{\mathrm{red}} = \bigoplus_{\pi \mid h} K_\pi^2, \quad W_h^{\mathrm{red}}(F,G) = W_{Z_{\mathrm{cl}}(h)}((F \bmod \pi, G \bmod \pi)_\pi)$.
$L3$: $W_h^{\mathrm{red}}(F,G)$ depends only on $F, G \bmod \mathrm{rad}(h)$.
$T2$: Every finite closed-support Weyl label is represented by polynomial pairs modulo the squarefree support
$D7$: $h = \prod_i \pi_i^{e_i}$ also defines the thickened rings $R/(\pi_i^{e_i})$.
$T3$: Multiplicities e_i are invisible in E_h^{red} and visible only in the Artin-ring layer.

50.3 Regular Functions as Residue Profiles D1–T1

Let $k = \mathbb{F}_Q$, $R = k[x]$, and $X = \operatorname{Spec} R$. By the Stacks affine source,

$$\Gamma(X, \mathcal{O}_X) = R.$$

More generally, on the nonempty standard open $D(h)$, $h \neq 0$, the regular functions are the localized ring

$$\Gamma(D(h), \mathcal{O}_X) = R_h.$$

For $h = 1$, this is the global case.

Let $U = D(h) \neq \emptyset$. A closed point of U is x_π with $\pi \nmid h$. For $F = u/h^m \in R_h$, define

$$F(x_\pi) = (u \bmod \pi)(h \bmod \pi)^{-m} \in K_\pi.$$

This is well-defined because $h \bmod \pi \neq 0$ in the field K_π. For a pair $(F,G) \in R_h^2$, define the residue profile

$$\mathrm{ev}_U(F,G)_\pi = (F(x_\pi), G(x_\pi)) \in K_\pi^2.$$

Lemma $L1$. If $(F,G) \neq (0,0)$ in R_h^2, then the set of closed points

$$\{\pi : \pi \nmid h,\ \mathrm{ev}_U(F,G)_\pi \neq (0,0)\}$$

is cofinite in $\{\pi : \pi \nmid h\}$.

Proof. Write $F = u/h^m$ and $G = v/h^n$. The profile is zero at $x_\pi \in U$ exactly when $u \bmod \pi = 0$ and $v \bmod \pi = 0$. This means that π divides both u and v. If $(F,G) \neq (0,0)$, then at least one of u, v is nonzero after localization. A nonzero polynomial has only finitely many monic irreducible divisors. Hence only finitely many $\pi \nmid h$ can make both residues zero.

Theorem $T1$. Under the current closed-point Weyl-net convention,

$$\mathrm{ev}_U(F,G) \in E(U) \quad \Longleftrightarrow \quad (F,G) = (0,0).$$

Proof. By AQM-28,

$$E(U) = \bigoplus_{x_\pi \in U} K_\pi^2$$

means finite closed support. If $(F,G) = (0,0)$, the profile has empty support. If $(F,G) \neq (0,0)$, Lemma $L1$ gives cofinite support in the infinite closed-point set of U; AQM-29 proves that $D(h)$ omits only finitely many closed points from the infinite set $\mathcal{P}$. Hence the profile is not finite-support and is not an element of $E(U)$.

Thus a regular function is a classical global or local residue profile. It is not, by itself, a quasi-local Weyl observable.

50.4 Polynomials as Finite Closed Supports D5–L2

Now let $h \in R$ be nonzero and factor

$$h = c \prod_{i=1}^{r} \pi_i^{e_i}, \qquad c \in k^\times, \quad e_i \geq 1,$$

with distinct monic irreducibles π_i. Define

$$Z_{\mathrm{cl}}(h) = \{x_{\pi_i} : 1 \leq i \leq r\}, \qquad \mathrm{rad}(h) = \prod_{i=1}^{r} \pi_i.$$

Lemma $L2$. $Z_{\mathrm{cl}}(h)$ is finite and depends only on $\mathrm{rad}(h)$, not on the exponents e_i.

Proof. A closed point x_π lies in $V(h)$ exactly when $h \in (\pi)$, which is equivalent to $\pi \mid h$. The factorization of a nonzero polynomial has only finitely many irreducible factors. Replacing h by $\mathrm{rad}(h)$ preserves the same divisibility relation $\pi \mid h$.

This is the first quantum use of a polynomial: it selects a finite closed support. It does not yet choose a Weyl operator.

50.5 Reduced Smearing over a Support D6–T2

Define the reduced support label space

$$E_h^{\mathrm{red}} = \bigoplus_{\pi \mid h} K_\pi^2.$$

For polynomial pairs $(F, G) \in R^2$, define the reduced smeared label

$$e_h^{\mathrm{red}}(F, G) = ((F \bmod \pi,\ G \bmod \pi))_{\pi | h} \in E_h^{\mathrm{red}}$$

and the reduced smeared Weyl operator

$$W_h^{\mathrm{red}}(F, G) = W_{Z_{\mathrm{cl}}(h)}(e_h^{\mathrm{red}}(F, G)).$$

Lemma $L3$. $e_h^{\mathrm{red}}(F, G)$, hence $W_h^{\mathrm{red}}(F, G)$, depends only on the classes of F and G modulo $\mathrm{rad}(h)$.

Proof. The label at x_π is obtained by reducing modulo π. Two polynomials have the same reduction modulo every $\pi \mid h$ iff their difference is divisible by every $\pi \mid h$. Since these irreducibles are pairwise coprime, this is equivalent to divisibility by $\mathrm{rad}(h)$.

Theorem $T2$. For every finite closed support $S = \{x_{\pi_1}, \ldots, x_{\pi_r}\}$ and every label

$$e = ((q_i, p_i))_{i=1}^r \in \bigoplus_{i=1}^r K_{\pi_i}^2,$$

there exist polynomials $F, G \in R$ with

$$F \bmod \pi_i = q_i, \qquad G \bmod \pi_i = p_i \quad \text{for every } i.$$

Thus distinct closed-point qudits have no finite compatibility condition in the reduced Weyl net.

Proof. The ideals (π_i) are pairwise coprime. The Chinese remainder theorem therefore gives a surjection

$$R \longrightarrow \prod_{i=1}^r R/(\pi_i) = \prod_{i=1}^r K_{\pi_i}.$$

Choose F mapping to $(q_i)_i$ and G mapping to $(p_i)_i$. These polynomials have the required residues. Since every finite label is obtained this way, the qudit factors indexed by distinct irreducibles are independent inside the finite-support Weyl algebra.

50.6 Multiplicity and the Artin-Ring Layer D7–T3

The same polynomial $h = \prod_i \pi_i^{e_i}$ may also be read as a closed subscheme with multiplicity. Then the local rings

$$A_i = R/(\pi_i^{e_i})$$

are finite local Artin rings supported at the same closed points x_{π_i}. AQM-36 quantizes this thickened layer by

$$E_h^{\mathrm{thick}} = \bigoplus_i A_i^2, \qquad \mathcal{H}_h^{\mathrm{thick}} = \bigotimes_i \ell^2(A_i).$$

Theorem $T3$. The reduced support operator $W_h^{\mathrm{red}}(F, G)$ forgets all multiplicities e_i. Multiplicities become quantum degrees of freedom only after replacing the residue fields K_{π_i} by the Artin rings A_i.

Proof. By Lemma $L3$, the reduced label only uses F, G modulo $\mathrm{rad}(h) = \prod_i \pi_i$. Therefore h and $\mathrm{rad}(h)$ give the same reduced label space, the same Hilbert space $\bigotimes_i \ell^2(K_{\pi_i})$, and the same reduced Weyl operators. AQM-36 instead uses $R/(\pi_i^{e_i})$, whose cardinality is $Q^{e_i \deg \pi_i}$. These larger rings record nilpotent jet coefficients at the same closed point. They are not additional Zariski points.

50.7 Conclusion

In the present framing, regular functions have three separate roles:

$$\begin{array}{rcl} \text{profile} & : & (F,G) \in \Gamma(U,\mathcal{O}_X)^2 \mapsto \mathrm{ev}_U(F,G), \text{ usually infinite support;} \\ \text{support selector} & : & h \neq 0 \mapsto Z_{\mathrm{cl}}(h), \text{ a finite closed support;} \\ \text{smeared operator} & : & (h,F,G) \mapsto W_h^{\mathrm{red}}(F,G), \text{ or } W_h^{\mathrm{thick}}(F,G) \text{ after Artin thickening.} \end{array}$$

Only the last line is a Weyl observable, and only after a finite support or a thickened finite scheme has been chosen.

51 Regular Functions on $\mathbb{F}_3[x]$ Line by Line

51.1 Status and Source Anchors

This is the explicit $k = \mathbb{F}_3$ companion to AQM-49. It uses AQM-29 for the point set of $\operatorname{Spec}\mathbb{F}_3[x]$, AQM-28 for finite-support Weyl operators, AQM-34 for polynomial quotients, and AQM-36 for Artin thickenings.

51.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k = \mathbb{F}_3, \quad R = k[x], \quad X = \operatorname{Spec} R. \\
D2 & : & x_r = (x-r) \quad (r = 0,1,2), \quad q_0 = x^2+1. \\
L1 & : & x \text{ as a function has infinite closed support.} \\
L2 & : & x^{-1} \in \Gamma(D(x), \mathcal{O}_X) \text{ has infinite support in } D(x). \\
D3 & : & h = x(x-1), \quad Z_{\mathrm{cl}}(h) = \{x_0, x_1\}. \\
T1 & : & W_h^{\mathrm{red}}(F,G) \text{ is a two-qutrit operator determined by values at 0 and 1.} \\
T2 & : & \text{Every two-qutrit label on } Z_{\mathrm{cl}}(h) \text{ is represented by polynomial pairs modulo } h. \\
D4 & : & q_0 = x^2+1, \quad K_{q_0} = k[\alpha]/(\alpha^2+1). \\
T3 & : & Z_{\mathrm{cl}}(q_0) \text{ gives one 9-level qudit.} \\
D5 & : & m = x^2 q_0^3. \\
T4 & : & m \text{ has reduced Hilbert dimension 27 and thickened dimension } 3^8.
\end{array}$$

51.3 Step 1: The Space and Its First Closed Points

Let

$$R = \mathbb{F}_3[x], \qquad X = \operatorname{Spec} R.$$

The closed points are $x_\pi = (\pi)$ for monic irreducible π. The rational closed points are

$$x_0 = (x), \qquad x_1 = (x-1), \qquad x_2 = (x-2).$$

Their residue fields are all $\mathbb{F}_3$. The three monic irreducible quadratics are

$$q_0 = x^2+1, \qquad q_1 = x^2+x+2, \qquad q_2 = x^2+2x+2.$$

For q_0, write

$$K_{q_0} = \mathbb{F}_3[x]/(q_0) = \mathbb{F}_3[\alpha], \qquad \alpha^2 = -1 = 2.$$

51.4 Step 2: A Global Regular Function Is Usually Infinite Support

Take the regular function $F = x \in R$. At a closed point x_π,

$$F(x_\pi) = x \bmod \pi \in K_\pi.$$

At rational points,

$$F(x_0) = 0, \qquad F(x_1) = 1, \qquad F(x_2) = 2.$$

At x_{q_0},

$$F(x_{q_0}) = \alpha \in K_{q_0}.$$

Lemma $L1$. The profile of x is nonzero at every closed point except x_0.

Proof. $x \bmod \pi = 0$ iff $x \in (\pi)$, equivalently $\pi \mid x$. Since x is monic irreducible, this happens only for $\pi = x$. Thus all closed points except x_0 lie in the support.

Therefore x is not a quasi-local Weyl label. It is a classical residue profile with cofinite support.

51.5 Step 3: A Local Regular Function Still Has Infinite Support

Let

$$U = D(x) = X \setminus \{x_0\}.$$

The function x^{-1} is regular on U. Its residue value at $x_\pi \in U$ is

$$x^{-1}(x_\pi) = (x \bmod \pi)^{-1} \in K_\pi.$$

At rational points in U,

$$x^{-1}(x_1) = 1, \qquad x^{-1}(x_2) = 2,$$

because $2^{-1} = 2$ in $\mathbb{F}_3$. At q_0,

$$x^{-1}(x_{q_0}) = \alpha^{-1} = 2\alpha,$$

since $\alpha(2\alpha) = 2\alpha^2 = 4 = 1$.

Lemma $L2$. The profile of x^{-1} is nonzero at every closed point of $D(x)$.

Proof. For $x_\pi \in D(x)$, $\pi \nmid x$. Hence $x \bmod \pi \neq 0$ in the field K_π, so it has an inverse. An inverse is never zero.

Thus local regularity on $D(x)$ does not make x^{-1} a finite-support Weyl label.

51.6 Step 4: A Polynomial as a Finite Support Selector

Now take

$$h = x(x-1).$$

The closed zero set is

$$Z_{\mathrm{cl}}(h) = \{x_0, x_1\}.$$

The reduced support Hilbert space is

$$\mathcal{H}_h^{\mathrm{red}} = \ell^2(\mathbb{F}_3) \otimes \ell^2(\mathbb{F}_3),$$

so it has dimension 9. The reduced label space is

$$E_h^{\mathrm{red}} = \mathbb{F}_3^2 \oplus \mathbb{F}_3^2.$$

51.7 Step 5: A Concrete Smeared Two-Qutrit Operator

Choose

$$F = 1, \qquad G = x + 1.$$

At x_0,

$$(F, G) = (1, 1).$$

At x_1,

$$(F, G) = (1, 2).$$

Therefore

$$e_h^{\text{red}}(F,G) = ((1,1),(1,2)).$$

If $T_r(a)$ and $R_r(b)$ are the qutrit shift and phase at x_r, then

$$W_h^{\text{red}}(F,G) = T_0(1)R_0(1) \otimes T_1(1)R_1(2).$$

This is a genuine quasi-local observable because the support has two closed points.

51.8 Step 6: No Compatibility Between the Two Qutrits

Prescribe the label

$$q_0 = 2, \quad q_1 = 1, \qquad p_0 = 1, \quad p_1 = 0.$$

We seek $F, G \in \mathbb{F}_3[x]$ with

$$F(0) = 2, \quad F(1) = 1, \qquad G(0) = 1, \quad G(1) = 0.$$

Take

$$F = 2 + 2x, \qquad G = 1 + 2x.$$

Then

$$F(0) = 2, \quad F(1) = 2 + 2 = 1,$$

and

$$G(0) = 1, \quad G(1) = 1 + 2 = 0.$$

Theorem *T*2. Every label in $\mathbb{F}_3^2 \oplus \mathbb{F}_3^2$ is obtained by polynomial pairs modulo $x(x-1)$.

Proof. The ideals (x) and $(x-1)$ are coprime, since

$$1 = x - (x-1).$$

The Chinese remainder theorem gives

$$\mathbb{F}_3[x]/(x(x-1)) \cong \mathbb{F}_3[x]/(x) \times \mathbb{F}_3[x]/(x-1) \cong \mathbb{F}_3 \times \mathbb{F}_3.$$

Apply this once to the position labels and once to the momentum labels.

51.9 Step 7: A Quadratic Support Is One Nine-Level Qudit

Take

$$h = q_0 = x^2 + 1.$$

Then

$$Z_{\text{cl}}(h) = \{x_{q_0}\}.$$

The residue field is

$$K_{q_0} = \mathbb{F}_3[\alpha]/(\alpha^2 + 1),$$

which has 9 elements. Thus the reduced Hilbert space is

$$\mathcal{H}_{q_0}^{\text{red}} = \ell^2(K_{q_0}).$$

Choose

$$F = x + 1, \qquad G = 2x.$$

Then the reduced label is

$$(F \bmod q_0,\ G \bmod q_0) = (\alpha + 1,\ 2\alpha) \in K_{q_0}^2.$$

The smeared operator is the one-site 9-level Weyl operator

$$W_{q_0}^{\mathrm{red}}(F, G) = W_{q_0}(\alpha + 1, 2\alpha).$$

51.10 Step 8: Repeated Factors and Artin Data

Take

$$m = x^2 q_0^3.$$

Its reduced closed support is

$$Z_{\mathrm{cl}}(m) = \{x_0, x_{q_0}\}.$$

Therefore the reduced Hilbert dimension is

$$\dim \mathcal{H}_m^{\mathrm{red}} = |\mathbb{F}_3| \cdot |K_{q_0}| = 3 \cdot 9 = 27.$$

The exponents 2 and 3 have disappeared.

In the thickened layer, the local Artin rings are

$$A_0 = \mathbb{F}_3[x]/(x^2), \qquad A_q = \mathbb{F}_3[x]/(q_0^3).$$

Their sizes are

$$|A_0| = 3^2, \qquad |A_q| = 3^{3 \deg q_0} = 3^6.$$

Thus

$$\dim \mathcal{H}_m^{\mathrm{thick}} = |A_0| \cdot |A_q| = 3^8 = 6561.$$

Theorem *T*4. For $m = x^2 q_0^3$, the reduced theory sees two closed-point qudits, while the thickened theory sees jet degrees of freedom of total Hilbert dimension $3^{\deg m} = 3^8$.

Proof. The reduced theory uses only the distinct irreducible factors x and q_0, so its dimension is $3^1 \cdot 3^2 = 27$. The thickened theory uses the Artin rings $R/(x^2)$ and $R/(q_0^3)$, whose sizes are 3^2 and 3^6. Tensoring the two local Hilbert spaces gives dimension 3^8. This equals $3^{\deg m}$, since $\deg m = 2 + 3 \cdot 2 = 8$.

51.11 Conclusion

For $\mathbf{A}^1_{\mathbb{F}_3}$, a regular function such as x or x^{-1} on $D(x)$ is a residue profile with infinite closed support. A nonzero polynomial h becomes finite only when it is used as an equation: it selects $Z_{\mathrm{cl}}(h)$. A Weyl observable then needs labels on that finite support. Repeated factors of h are invisible in the reduced support net and become visible exactly in the Artin-ring thickened layer.

52 Affine Symmetry Action on Regular Functions

52.1 Status and Source Anchors

This shard extends AQM-46 and AQM-49. The group $\mathrm{Aut}_k(\mathbf{A}^1_k)$, its action on closed points, and the residue-field isomorphisms $\tau_{g,\pi}$ are those of AQM-46. Regular functions, residue profiles, reduced supports, and reduced smearing are those of AQM-49. The Artin-ring layer is AQM-36. No new external source is used; the new assertions below are checked algebraic consequences of those shards.

52.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k=\mathbb{F}_q, \quad R=k[x], \quad X=\operatorname{Spec} R, \quad g=g_{a,b}: x\mapsto ax+b. \\
D2 & : & \theta_g(f)(x)=f(ax+b), \quad \sigma_g=\theta_{g^{-1}}, \quad F^g=\sigma_g(F). \\
L1 & : & \sigma_g: R_h\to R_{\sigma_g(h)} \text{ is an isomorphism and } gD(h)=D(\sigma_g(h)). \\
D3 & : & \mathrm{Prof}(U)=\prod_{x_\pi\in U} K_\pi^2. \\
D4 & : & (S_g^{\Pi}e)_{\pi^g}=(\tau_{g,\pi}^{-1}q_\pi,\tau_{g,\pi}^{-1}p_\pi). \\
T1 & : & \mathrm{ev}_{gU}(F^g,G^g)=S_g^{\Pi}\,\mathrm{ev}_U(F,G). \\
T2 & : & \text{The affine action preserves the distinction between profiles and finite Weyl labels.} \\
D5 & : & h^g=\sigma_g(h), \quad W_h^{\mathrm{red}}(F,G)=W_{Z_{\mathrm{cl}}(h)}(e_h^{\mathrm{red}}(F,G)). \\
T3 & : & \alpha_g(W_h^{\mathrm{red}}(F,G))=W_{h^g}^{\mathrm{red}}(F^g,G^g). \\
D6 & : & A_{\pi,e}=R/(\pi^e), \quad A_{\pi^g,e}=R/((\pi^g)^e). \\
T4 & : & \sigma_g \text{ induces } A_{\pi,e}\simeq A_{\pi^g,e} \text{ and preserves the multiplicity } e.
\end{array}$$

52.3 Pushforward of Regular Functions D1–L1

Let $g=g_{a,b}$, with $a\in k^\times$ and $b\in k$. Its pullback on the coordinate ring is

$$\theta_g: R\longrightarrow R, \qquad \theta_g(f)(x)=f(ax+b).$$

For regular functions we use the covariant, or pushforward, notation

$$F^g=g_\# F:=\sigma_g(F), \qquad \sigma_g=\theta_{g^{-1}}.$$

Thus, as a function on points,

$$(g_\# F)(y)=F(g^{-1}y).$$

Lemma $L1$**.** If $h\neq 0$, then σ_g extends to a ring isomorphism

$$\sigma_g: R_h\longrightarrow R_{\sigma_g(h)}$$

by

$$\sigma_g(u/h^m)=\sigma_g(u)/\sigma_g(h)^m.$$

Moreover

$$g(D(h))=D(\sigma_g(h)).$$

Proof. The map σ_g is a k-algebra automorphism of R, with inverse θ_g. It sends the multiplicative set $\{1,h,h^2,\dots\}$ bijectively to $\{1,\sigma_g(h),\sigma_g(h)^2,\dots\}$, so it extends uniquely to the displayed localization isomorphism.

For a point $y\in X$,

$$y\in g(D(h))\iff g^{-1}y\in D(h)\iff h(g^{-1}y)\neq 0\iff \sigma_g(h)(y)\neq 0.$$

This is exactly $y \in D(\sigma_g(h))$.

Composition has the expected order:

$$(gh)_\# = g_\# h_\#,$$

because $(gh)^{-1} = h^{-1}g^{-1}$.

52.4 Residue Profiles D3–T2

For an open U, define the full residue-profile set

$$\mathrm{Prof}(U) = \prod_{x_\pi \in U} K_\pi^2.$$

The finite-support Weyl label group $E(U)$ of AQM-46 and AQM-49 is the subgroup of $\mathrm{Prof}(U)$ with finite closed support.

Define

$$S_g^\Pi : \mathrm{Prof}(U) \longrightarrow \mathrm{Prof}(gU)$$

by

$$(S_g^\Pi e)_{\pi^g} = \big(\tau_{g,\pi}^{-1}(q_\pi), \tau_{g,\pi}^{-1}(p_\pi)\big),$$

where $e_\pi = (q_\pi, p_\pi)$. On finite-support profiles this is the finite Weyl-label map S_g from AQM-46.

Theorem $T1$**.** For $F, G \in \Gamma(U, \mathcal{O}_X)$,

$$\mathrm{ev}_{gU}(F^g, G^g) = S_g^\Pi\, \mathrm{ev}_U(F, G).$$

Proof. It is enough to check one closed point $x_\pi \in U$. Let $x_{\pi^g} = g(x_\pi)$. The residue map $\tau_{g,\pi} : K_{\pi^g} \to K_\pi$ is induced by θ_g. Therefore, for every $H \in \Gamma(gU, \mathcal{O}_X)$,

$$\tau_{g,\pi}(H(x_{\pi^g})) = (\theta_g H)(x_\pi).$$

Apply this to $H = F^g = \sigma_g(F)$. Since $\theta_g \sigma_g = \mathrm{id}$,

$$\tau_{g,\pi}(F^g(x_{\pi^g})) = F(x_\pi).$$

Thus $F^g(x_{\pi^g}) = \tau_{g,\pi}^{-1}(F(x_\pi))$. The same argument applies to G, proving the displayed profile identity.

Theorem $T2$**.** The affine action never turns a nonzero regular-function profile into a finite Weyl label, and never turns the zero profile into a nonzero one.

Proof. The map S_g^Π is a bijection of closed-point profiles. It sends the zero profile to the zero profile. It sends a finite-support profile to a finite-support profile because $x_\pi \mapsto x_{\pi^g}$ is a bijection of closed points. It sends an infinite-support profile to an infinite-support profile for the same reason. AQM-49 proves that a nonzero regular-function pair on a nonempty standard open has cofinite, hence infinite, closed support. Therefore the profile/operator distinction is invariant under g.

52.5 Reduced Smeared Operators D5–T3

For a nonzero polynomial h, put

$$h^g = \sigma_g(h).$$

This polynomial need not be monic. Multiplying it by a nonzero scalar does not change $D(h^g)$, $Z_{\mathrm{cl}}(h^g)$, or the reduced Weyl support. If π is monic irreducible, then $\pi \mid h$ exactly when $\pi^g \mid h^g$, so

$$g(Z_{\mathrm{cl}}(h)) = Z_{\mathrm{cl}}(h^g).$$

Theorem $T3$. For $F, G \in R$,

$$\alpha_g\Big(W_h^{\mathrm{red}}(F,G)\Big) = W_{h^g}^{\mathrm{red}}(F^g, G^g).$$

Proof. At the source point $x_\pi \in Z_{\mathrm{cl}}(h)$, the reduced label is

$$(F \bmod \pi,\ G \bmod \pi) \in K_\pi^2.$$

At the target point x_{π^g}, Theorem $T1$ gives the label

$$(F^g \bmod \pi^g,\ G^g \bmod \pi^g) = \big(\tau_{g,\pi}^{-1}(F \bmod \pi), \tau_{g,\pi}^{-1}(G \bmod \pi)\big).$$

Thus

$$e_{h^g}^{\mathrm{red}}(F^g, G^g) = S_g e_h^{\mathrm{red}}(F,G).$$

AQM-46 proves

$$\alpha_g(W(e)) = W(S_g e)$$

on finite supports. Substituting the displayed labels proves the theorem.

Consequently the rule is simple: move the support by g, and precompose the regular functions by g^{-1}.

52.6 Artin-Ring Transport D6–T4

Let $h = \prod_i \pi_i^{e_i}$. AQM-49 records that the reduced theory sees only the distinct π_i, while the thickened layer uses

$$A_{\pi_i,e_i} = R/(\pi_i^{e_i}).$$

Theorem $T4$. For each (π, e), the automorphism σ_g induces a ring isomorphism

$$\bar\sigma_{g,\pi,e} : R/(\pi^e) \longrightarrow R/((\pi^g)^e).$$

It sends the class of F to the class of F^g, and it preserves the exponent e.

Proof. AQM-46 gives

$$\pi^g = a^{\deg \pi}\sigma_g(\pi).$$

Hence the ideals $(\sigma_g(\pi)^e)$ and $((\pi^g)^e)$ are equal, because they differ by a unit. Therefore σ_g descends to the displayed quotient isomorphism. The exponent remains e because an isomorphism sends the e-th power of the source ideal to the e-th power of the target ideal.

This is the canonical thickened support transport. A thickened Weyl automorphism also needs a character statement. If the target generating character is transported through $\bar\sigma_{g,\pi,e}$, the same proof as AQM-36 gives Weyl covariance. If one insists on separately chosen default top-coefficient characters at source and target, a dual momentum-label correction must be supplied before claiming character preservation.

52.7 Conclusion

The affine symmetry action on regular functions is:

$$(h, F, G) \longmapsto (h^g, F^g, G^g) = (\sigma_g(h), \sigma_g(F), \sigma_g(G)).$$

On profiles it is the product residue relabelling S_g^{Π}. On finite reduced Weyl smearings it is exactly the local-net automorphism of AQM-46:

$$\alpha_g W_h^{\mathrm{red}}(F,G) = W_{h^g}^{\mathrm{red}}(F^g, G^g).$$

Thus the symmetry does not add compatibility conditions between closed-point qudits. It only permutes the closed supports and relabels each residue field by the induced field isomorphism.

53 The $\mathbb{F}_3$ Regular-Function Symmetry Action Line by Line

53.1 Status and Source Anchors

This is the $k = \mathbb{F}_3$ companion to AQM-51. It uses AQM-46 for the affine symmetry group, AQM-48 for the full closed-point Hilbert action, AQM-49 for regular functions and reduced smearing, and AQM-50 for the concrete regular-function examples.

53.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & R = \mathbb{F}_3[x],\ X = \operatorname{Spec} R,\ G = \{g_{a,b} : x \mapsto ax + b\}. \\
D2 & : & g_\# F = F(a^{-1}(x - b)). \\
L1 & : & g_\# x = a^{-1}(x - b) \text{ for all six } g. \\
D3 & : & \tau : x \mapsto x + 1, \quad \rho : x \mapsto 2x. \\
T1 & : & \operatorname{ev}(g_\# F) = S_g^{\Pi} \operatorname{ev}(F) \text{ at rational points.} \\
D4 & : & h = x(x - 1), \quad F = 1, \quad G = x + 1. \\
T2 & : & \alpha_\tau W_h^{\mathrm{red}}(F, G) = W_{\tau_\# h}^{\mathrm{red}}(\tau_\# F, \tau_\# G). \\
D5 & : & q_0 = x^2 + 1, \quad q_1 = x^2 + x + 2. \\
T3 & : & \tau_\# q_0 = q_1 \text{ and quadratic labels transform by } \tau_{g,\pi}^{-1}. \\
D6 & : & m = x^2 q_0^3. \\
T4 & : & \tau_\# m = (x - 1)^2 q_1^3, \text{ preserving reduced and thickened dimensions.}
\end{array}$$

53.3 Step 1: The Six Symmetries D1–L1

For $k = \mathbb{F}_3$,

$$G = \operatorname{Aut}_{\mathbb{F}_3}(X) = \{g_{a,b} : x \mapsto ax + b,\ a \in \{1, 2\},\ b \in \mathbb{F}_3\}.$$

By AQM-51,

$$g_\# F = F \circ g^{-1}, \qquad g_\# F(x) = F(a^{-1}(x - b)).$$

Since $1^{-1} = 1$ and $2^{-1} = 2$ in $\mathbb{F}_3$,

g	(a, b)	$g_\# x$
x	$(1, 0)$	x
$x + 1$	$(1, 1)$	$x - 1 = x + 2$
$x + 2$	$(1, 2)$	$x - 2 = x + 1$
$2x$	$(2, 0)$	$2x$
$2x + 1$	$(2, 1)$	$2(x - 1) = 2x + 1$
$2x + 2$	$(2, 2)$	$2(x - 2) = 2x + 2$

Lemma $L1$. The table gives $g_\# x = g^{-1}(x)$ in every case.

Proof. The inverse of $g_{a,b}$ is $g_{a^{-1}, -a^{-1}b}$. Therefore $g_\# x = a^{-1}(x - b)$. Substituting the six pairs (a, b) gives the table.

53.4 Step 2: Translation and Reflection D3

Write

$$\tau : x \mapsto x + 1, \qquad \rho : x \mapsto 2x.$$

Then $\tau_\# F = F(x - 1)$ and $\rho_\# F = F(2x)$. For $F = x$,

$$\tau_\# x = x - 1, \qquad \rho_\# x = 2x.$$

For the local regular function $x^{-1} \in \Gamma(D(x), \mathcal{O}_X)$,

$$\tau_{\#}(x^{-1}) = (x-1)^{-1} \in \Gamma(D(x-1), \mathcal{O}_X), \rho_{\#}(x^{-1}) = (2x)^{-1} = 2x^{-1} \in \Gamma(D(x), \mathcal{O}_X).$$

The open $D(x)$ moves to $D(x-1)$ under τ, because x_0 moves to x_1. It is fixed by ρ, because x_0 is fixed by ρ.

53.5 Step 3: Rational-Point Profile Check T1

The rational closed points are $x_r = (x-r)$, $r = 0, 1, 2$. The translation τ sends $x_r \mapsto x_{r+1}$. At rational points the residue fields are all $\mathbb{F}_3$, and the residue-field isomorphism is the identity after evaluation.

Check $F = x$. The source value at x_r is r. The transformed function is $\tau_{\#}F = x - 1$. At x_{r+1}, $(\tau_{\#}F)(x_{r+1}) = (r+1) - 1 = r$. Line by line,

source	F	target	$\tau_{\#}F$
x_0	0	x_1	$1-1=0$
x_1	1	x_2	$2-1=1$
x_2	2	x_0	$0-1=2$

This is exactly $\mathrm{ev}(\tau_{\#}F) = S_\tau^{\Pi}\,\mathrm{ev}(F)$.

For x^{-1} on $D(x)$, the source points are x_1, x_2. They move to x_2, x_0. The values are

source	x^{-1}	target	$(x-1)^{-1}$
x_1	1	x_2	$(2-1)^{-1}=1$
x_2	2	x_0	$(0-1)^{-1}=2$

Thus local regular functions satisfy the same covariance rule.

53.6 Step 4: A Reduced Two-Qutrit Smearing D4–T2

Use the AQM-50 support and labels

$$h = x(x-1), \qquad F = 1, \qquad G = x+1.$$

The support is $Z_{\mathrm{cl}}(h) = \{x_0, x_1\}$. The source labels are $x_0 : (1,1)$ and $x_1 : (1,2)$. Under τ,

$$\tau_{\#}h = (x-1)(x-2), \qquad \tau_{\#}F = 1, \qquad \tau_{\#}G = x.$$

The target support is $Z_{\mathrm{cl}}(\tau_{\#}h) = \{x_1, x_2\}$. At x_1 and x_2, the target labels are $(1,1)$ and $(1,2)$.

Theorem $T2$.

$$\alpha_\tau W^{\mathrm{red}}_{x(x-1)}(1, x+1) = W^{\mathrm{red}}_{(x-1)(x-2)}(1, x).$$

Proof. The source label $((1,1),(1,2))$ at (x_0, x_1) has just been carried to the same ordered values at (x_1, x_2). AQM-46 says $\alpha_\tau W(e) = W(S_\tau e)$, and AQM-51 identifies $S_\tau e$ with the reduced label of $(1, x)$ on the transformed support.

53.7 Step 5: Quadratic Closed Point D5–T3

Let $q_0 = x^2 + 1$ and $q_1 = x^2 + x + 2$. For τ,

$$\tau_{\#} q_0 = q_0(x-1) = (x-1)^2 + 1 = x^2 + x + 2 = q_1.$$

Let $K_{q_0} = \mathbb{F}_3[\alpha]$, $\alpha^2 = 2$, and $K_{q_1} = \mathbb{F}_3[\beta]$, $\beta^2 + \beta + 2 = 0$. The residue pullback is

$$\tau_{\tau,q_0} : K_{q_1} \to K_{q_0}, \qquad \beta \longmapsto \alpha + 1.$$

Hence $\tau_{\tau,q_0}^{-1}(\alpha) = \beta - 1 = \beta + 2$.

Use the AQM-50 labels

$$F = x + 1, \qquad G = 2x$$

on q_0. The source label is $(\alpha + 1,\ 2\alpha)$. The transformed functions are $\tau_{\#} F = x$ and $\tau_{\#} G = 2x + 1$, giving $(\beta,\ 2\beta + 1)$ at q_1.

Theorem $T3$.

$$(\beta,\ 2\beta + 1) = \big(\tau_{\tau,q_0}^{-1}(\alpha + 1), \tau_{\tau,q_0}^{-1}(2\alpha)\big).$$

Proof. Since $\tau_{\tau,q_0}^{-1}(\alpha) = \beta + 2$,

$$\tau_{\tau,q_0}^{-1}(\alpha + 1) = \beta, \qquad \tau_{\tau,q_0}^{-1}(2\alpha) = 2(\beta + 2) = 2\beta + 1.$$

These are exactly the target residues of x and $2x + 1$ at q_1.

53.8 Step 6: Repeated Factors D6–T4

Take the AQM-50 repeated-factor polynomial $m = x^2 q_0^3$. Under τ,

$$\tau_{\#} m = (x-1)^2 q_1^3.$$

The reduced source support is $\{x_0, x_{q_0}\}$, and the reduced target support is $\{x_1, x_{q_1}\}$. The reduced Hilbert dimension is unchanged:

$$3 \cdot 9 = 27.$$

The thickened source rings are $\mathbb{F}_3[x]/(x^2)$ and $\mathbb{F}_3[x]/(q_0^3)$. The target rings are $\mathbb{F}_3[x]/((x-1)^2)$ and $\mathbb{F}_3[x]/(q_1^3)$. The map $F \mapsto F(x-1)$ gives ring isomorphisms from source to target. Therefore the thickened dimension is also unchanged:

$$3^2 \cdot 3^6 = 3^8 = 6561.$$

Theorem $T4$. The affine action preserves the exponents 2 and 3. It moves the closed support and the thickened Artin rings, but it does not turn a repeated factor into extra closed points.

Proof. The equality $\tau_{\#} m = (x-1)^2 q_1^3$ shows that exponents are carried with their irreducible factors. The reduced support keeps only the distinct factors, giving one 3-level site and one 9-level site before and after translation. The thickened rings keep the powers, so their sizes remain 3^2 and 3^6. Thus multiplicity is preserved as Artin-ring thickness, not converted into a larger closed-point set.

53.9 Step 7: The Full Hilbert-Space Statement

The full closed-point Hilbert representation from AQM-48 is

$$\mathcal{H}^0 = \ell^2\left(\bigoplus_{\pi} K_\pi\right),$$

where π runs over all monic irreducibles and vectors have finite closed support. The affine unitary satisfies

$$U_g W_h^{\mathrm{red}}(F, G) U_g^* = W_{g_\# h}^{\mathrm{red}}(g_\# F, g_\# G)$$

for every finite support selector h and polynomial labels F, G.

There is no operator $W(\mathrm{ev}(x))$ in the algebraic finite-support Weyl net, because AQM-50 proves that x has infinite closed support. The symmetry acts on that infinite profile by relabelling it, but the Weyl operator exists only after choosing a finite support such as h, or after adding a new infinite-volume completion convention.

54 Minimal Fermion Geometry Catalogue

54.1 Status and Source Anchors

This shard is a definition catalogue for later fermion constructions. It does not assert that ordinary arithmetic quantum fields automatically contain fermions. The CAR input is AQM-19 and `references/canonical_fields/SOURCES.md`. Supergeometric odd variables, reduced spaces, parity reversal, and Clifford quantization are sourced from `references/supergeometry/SOURCES.md`. Kähler differentials use `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33853, 33856–33866, 34103–34120, 34310–34315, and 34392–34415. Spin structures as square roots of the canonical line bundle use `references/spin_geometry/SOURCES.md`. Logarithmic differentials use `references/log_geometry/SOURCES.md`. The nilpotent Weyl layer is AQM-36. The identity $\Gamma(\mathbf{P}^1, \mathcal{O}) = k$ is sourced by `references/algebraic_geometry/SOURCES.md`.

54.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & \text{One complex fermion mode is the one-mode CAR representation.} \\
L1 & : & D1 \text{ satisfies } c^2 = (c^*)^2 = 0,\ \{c, c^*\} = I. \\
D2 & : & \text{A supercommutative } k\text{-algebra is } \mathbb{Z}/2\text{-graded.} \\
L2 & : & \operatorname{char} k \neq 2 \text{ and } \theta \text{ odd imply } \theta^2 = 0. \\
D3 & : & \mathsf{Spt}_k = (\operatorname{Spec} k, k[\theta]) \text{ is the algebraic superpoint.} \\
D4 & : & \Pi\Omega^1_{A/k} \text{ is the cotangent parity-shift mode source.} \\
L3 & : & \Omega^1_{k/k} = 0. \\
L4 & : & \Omega^1_{k[\epsilon]/(\epsilon^2)/k} \cong k\,\mathrm{d}\epsilon \text{ when } \operatorname{char} k \neq 2. \\
D5 & : & (C, D, S, \mu : S^{\otimes 2} \cong \Omega^1_C(\log D)) \text{ is a log-spin datum.} \\
L5 & : & (\mathbf{P}^1_k, \{0, \infty\}, \mathcal{O}, \mathrm{d}\log x) \text{ is a log-spin datum.} \\
L6 & : & \mathbb{F}_3\text{-vector spaces do not canonically scalar-extend to } \mathbb{C}\text{-Hilbert spaces.} \\
T1 & : & \operatorname{Spec} k \text{ has no forced fermion in this catalogue.} \\
T2 & : & \mathsf{Spt}_{\mathbb{F}_3},\ \Pi\Omega^1_{\mathbb{F}_3[\epsilon]/(\epsilon^2)/\mathbb{F}_3}, \text{ and } (\mathbf{P}^1_{\mathbb{F}_3}, \{0, \infty\}, \mathcal{O}) \text{ give one mode each.}
\end{array}
$$

54.3 The One-Mode CAR D1–L1

Let

$$\mathcal{H}_1 = \mathbb{C}|0\rangle \oplus \mathbb{C}|1\rangle.$$

Define linear maps $c, c^* : \mathcal{H}_1 \to \mathcal{H}_1$ by

$$c|0\rangle = 0, \quad c|1\rangle = |0\rangle, \quad c^*|0\rangle = |1\rangle, \quad c^*|1\rangle = 0.$$

The catalogue phrase "one fermion mode" means this CAR representation, or a unitarily isomorphic copy. For r modes it means the antisymmetric Fock space $\bigwedge^\bullet \mathbb{C}^r$, as in AQM-19.

Lemma $L1$. The operators c, c^* satisfy

$$c^2 = (c^*)^2 = 0, \qquad cc^* + c^*c = I_{\mathcal{H}_1}.$$

Proof. Both c^2 and $(c^*)^2$ kill $|0\rangle$ and $|1\rangle$, so they are zero. Also

$$(cc^* + c^*c)|0\rangle = c|1\rangle + 0 = |0\rangle$$

and

$$(cc^* + c^*c)|1\rangle = 0 + c^*|0\rangle = |1\rangle.$$

The two basis vectors span $\mathcal{H}_1$, hence $cc^* + c^*c = I$.

54.4 Odd Variables and the Superpoint D2–D3

Assume $\operatorname{char} k \neq 2$ in this subsection. A supercommutative k-algebra is a $\mathbb{Z}/2$-graded algebra

$$A = A_{\bar{0}} \oplus A_{\bar{1}}$$

such that homogeneous elements satisfy

$$uv = (-1)^{|u||v|} vu.$$

Elements of $A_{\bar{0}}$ are even; elements of $A_{\bar{1}}$ are odd.

Lemma *L*2. If $\theta \in A_{\bar{1}}$, then $\theta^2 = 0$.

Proof. Supercommutativity gives $\theta\theta = (-1)\theta\theta$. Hence $2\theta^2 = 0$. Since $2 \in k^\times$, multiplication by 2 is injective, so $\theta^2 = 0$.

Define

$$k[\theta] = k \oplus k\theta, \qquad |\theta| = \bar{1}, \qquad \theta^2 = 0.$$

The algebraic superpoint is the locally ringed graded object

$$\mathsf{Spt}_k = (\operatorname{Spec} k, k[\theta]).$$

Its reduced ordinary point is obtained by quotienting by the odd ideal:

$$k[\theta]/(\theta) \cong k.$$

For $k = \mathbb{F}_3$, the ungraded ring underlying $k[\theta]$ is isomorphic to the even dual-number ring $k[\epsilon]/(\epsilon^2)$, by $\theta \mapsto \epsilon$. The difference is the parity assignment, not the ungraded multiplication table.

The CAR quantization convention for Spt_k assigns the odd line $k\theta$ one complex fermion mode *D*1. This is a catalogue convention: the standard Hilbert-space CAR is complex, while the odd geometric line is a k-vector space.

54.5 Cotangent Parity Shift D4–L4

Let A be a commutative k-algebra. The module of Kähler differentials $\Omega^1_{A/k}$ is the universal target for k-derivations $A \to M$, as recorded in the Stacks source. The cotangent parity-shift catalogue entry is

$$\Pi\Omega^1_{A/k},$$

meaning that the module $\Omega^1_{A/k}$ is treated as odd. If $\dim_k \Omega^1_{A/k} = r < \infty$, the standard complex CAR realization uses r complex fermion modes.

Lemma *L*3. $\Omega^1_{k/k} = 0$.

Proof. Every k-derivation $D : k \to M$ kills all elements of k by definition. Thus the functor $M \mapsto \operatorname{Der}_k(k, M)$ is the zero functor. The zero module represents this functor, so the universal module $\Omega^1_{k/k}$ is zero.

Lemma *L*4. Let $A = k[\epsilon]/(\epsilon^2)$ and assume $\operatorname{char} k \neq 2$. Then

$$\Omega^1_{A/k} \cong k\,\mathrm{d}\epsilon.$$

Proof. Let $S = k[t]$ and $I = (t^2)$, so $A = S/I$. The Stacks polynomial-ring lemma gives $\Omega^1_{S/k} = S\,\mathrm{d}t$. The quotient exact sequence for differentials gives

$$I/I^2 \longrightarrow \Omega^1_{S/k} \otimes_S A \longrightarrow \Omega^1_{A/k} \longrightarrow 0,$$

where $t^2 \mapsto \mathrm{d}(t^2) = 2t\,\mathrm{d}t$. Therefore

$$\Omega^1_{A/k} \cong A\,\mathrm{d}\epsilon/(2\epsilon\,\mathrm{d}\epsilon).$$

Since $2 \in k^\times$, this is

$$A\,\mathrm{d}\epsilon/(\epsilon\,\mathrm{d}\epsilon) \cong (A/(\epsilon))\,\mathrm{d}\epsilon \cong k\,\mathrm{d}\epsilon.$$

For $k = \mathbb{F}_3$, the even thickened point $\operatorname{Spec}(\mathbb{F}_3[\epsilon]/(\epsilon^2))$ has two distinct catalogue layers:

$$\ell^2(\mathbb{F}_3[\epsilon]/(\epsilon^2))$$

is the 9-dimensional even Artin-Weyl layer of AQM-36, while $\Pi\Omega^1_{A/k} \cong \mathbb{F}_3\,\mathrm{d}\epsilon$ supplies one optional fermion mode by the cotangent parity-shift convention.

54.6 Log-Spin Datum D5–L5

Let C be a smooth curve over k, and let D be an effective divisor for which the logarithmic one-forms $\Omega^1_C(\log D)$ are defined. A log-spin datum in this catalogue is

$$(C, D, S, \mu), \qquad \mu : S^{\otimes 2} \xrightarrow{\sim} \Omega^1_C(\log D),$$

where S is a line bundle on C. Its algebraic mode source is

$$\Gamma(C, S).$$

If $\dim_k \Gamma(C, S) = r < \infty$, the standard complex CAR realization uses r complex fermion modes. Atiyah's source supports the square-root spin dictionary over compact complex curves; the logarithmic variant is local.

Lemma *L*5. Let $C = \mathbf{P}^1_k$ with affine coordinate x, and let

$$D = \{0, \infty\}.$$

Then $\Omega^1_C(\log D) \cong \mathcal{O}_C$, generated by

$$\omega = \mathrm{d}\log x.$$

Consequently $(C, D, \mathcal{O}_C, \mu)$, with $\mu(1 \otimes 1) = \omega$, is a log-spin datum.

Proof. On $U_0 = \operatorname{Spec} k[x]$, the divisor point 0 has local equation x. The Stacks log source says logarithmic one-forms are generated by ordinary forms and $\mathrm{d}\log x = x^{-1}\mathrm{d}x$. Since $\mathrm{d}x = x\,\mathrm{d}\log x$, the module is generated by $\mathrm{d}\log x$ on U_0. On $U_\infty = \operatorname{Spec} k[u]$, with $u = x^{-1}$, the divisor point ∞ has local equation u, and

$$\mathrm{d}\log x = \mathrm{d}\log(u^{-1}) = -\mathrm{d}\log u.$$

Thus the same form generates the logarithmic module on U_∞. The two descriptions agree on $U_0 \cap U_\infty$, so they glue to a global basis of $\Omega^1_C(\log D)$. Therefore $\Omega^1_C(\log D) \cong \mathcal{O}_C$. The map $\mu : \mathcal{O}_C^{\otimes 2} \to \Omega^1_C(\log D)$ sending $1 \otimes 1$ to ω is an isomorphism.

54.7 Finite-Field Scalar Caveat L6

Lemma *L*6. There is no unital field homomorphism $\mathbb{F}_3 \to \mathbb{C}$.

Proof. If $\iota : \mathbb{F}_3 \to \mathbb{C}$ were unital, then

$$0 = \iota(0) = \iota(1 + 1 + 1) = 1 + 1 + 1 = 3,$$

which is false in $\mathbb{C}$.

Therefore an $\mathbb{F}_3$-vector-space mode source V does not canonically become a complex one-particle Hilbert space by scalar extension. The catalogue records $r = \dim_{\mathbb{F}_3} V$ and then uses a chosen complex Hilbert space $\mathbb{C}^r$ for the standard CAR realization. A genuinely $\mathbb{F}_3$-linear Clifford theory is a different algebraic layer and is not defined here.

54.8 Catalogue Consequences T1–T2

Theorem *T*1. The ordinary point $\operatorname{Spec} k$ has no forced fermion mode in this catalogue.

Proof. The superpoint entry requires an odd structure sheaf $k[\theta]$; the ordinary structure sheaf k has no odd part. The cotangent parity-shift entry gives no mode by Lemma *L*3. The log-spin entry requires a curve, a divisor, and a square-root line bundle; none is part of the ordinary point $\operatorname{Spec} k$. Thus no catalogue mechanism above assigns a nonzero fermion mode to $\operatorname{Spec} k$ without extra data.

Theorem *T*2. For $k = \mathbb{F}_3$, the following are the minimal one-mode entries:

geometric datum	mode source	$\mathbb{F}_3$-dimension	complex CAR Hilbert space
$\mathsf{Spt}_{\mathbb{F}_3}$	$\mathbb{F}_3\theta$	1	$\mathcal{H}_1$
$\operatorname{Spec}(\mathbb{F}_3[\epsilon]/(\epsilon^2))$ with $\Pi\Omega^1$	$\mathbb{F}_3\, \mathrm{d}\epsilon$	1	$\mathcal{H}_1$
$(\mathbf{P}^1_{\mathbb{F}_3}, \{0, \infty\}, \mathcal{O})$	$\Gamma(\mathbf{P}^1, \mathcal{O}) = \mathbb{F}_3$	1	$\mathcal{H}_1$

The ordinary closed point $\operatorname{Spec} \mathbb{F}_3$ contributes zero fermion modes.

Proof. The superpoint row is Definition *D*3. The dual-number row is Lemma *L*4. The log-spin row is Lemma *L*5 plus the cited projective-space cohomology calculation $\Gamma(\mathbf{P}^1, \mathcal{O}) = \mathbb{F}_3$. Lemma *L*6 explains why each row determines the standard complex CAR Hilbert space only through the dimension-one mode count. The last sentence is Theorem *T*1 with $k = \mathbb{F}_3$.

55 Derivative Dynamics from Kähler Differentials over $\mathbb{F}_3$

55.1 Status and Source Anchors

This shard is a checked definition-and-example shard plus one labelled proposal. The definitions of derivation, Kähler differentials, their universal property, the quotient exact sequence, polynomial differentials, and the de Rham complex are sourced from the Stacks algebra snapshot `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33872, 34103–34120, 34310–34340, and 34388–34524. Tangent space by dual numbers and the cotangent-tangent duality are sourced from `references/algebraic_geometry/stacks_project_varieties.tex`, lines 2915–3019. Residue fields are sourced from `references/algebraic_geometry/stacks_project_algebra.tex`, lines 250–278. AQM-08 and AQM-09 supply the CSS/stabilizer dictionary, AQM-36 supplies the even Artin-Weyl layer for dual numbers, and AQM-53 supplies the cotangent parity-shift comparison. The convention for this shard is `CONVENTIONS.md` (ao).

55.2 Lamport Dependency Skeleton

$$
\begin{aligned}
D1 &: \; k = \mathbb{F}_3, \quad A \text{ is a finite commutative } k\text{-algebra.}\\
D2 &: \; \mathrm{Der}_k(A, M) \text{ is the } A\text{-module of } k\text{-derivations } A \to M.\\
D3 &: \; \Omega^1_{A/k} \text{ represents } M \mapsto \mathrm{Der}_k(A, M).\\
D4 &: \; T^*_{A/k,x} = \Omega^1_{A/k} \otimes_A \kappa(x), \quad T_{A/k,x} = \mathrm{Hom}_A(\Omega^1_{A/k}, \kappa(x)).\\
D5 &: \; A \xrightarrow{d_0} \Omega^1_{A/k} \xrightarrow{d_1} \Omega^2_{A/k} \to \cdots \text{ is the algebraic de Rham complex.}\\
P1 &: \; \text{The de Rham CSS test puts qudits on } C^1 = \Omega^1_{A/k}.\\
L1 &: \; \Omega^1_{\mathbb{F}_3/\mathbb{F}_3} = 0.\\
L2 &: \; \Omega^1_{\mathbb{F}_3[\epsilon]/(\epsilon^2)/\mathbb{F}_3} \cong \mathbb{F}_3\, d\epsilon.\\
T1 &: \; \text{For } A = \mathbb{F}_3[\epsilon]/(\epsilon^2), \text{ the de Rham CSS test has one } X\text{-check and no logical qudit.}
\end{aligned}
$$

55.3 Definitions D1–D5

Let $k \to A$ be a ring map and let M be an A-module. A k-derivation is an additive map $D : A \to M$ satisfying

$$D(c) = 0 \quad (c \in k), \qquad D(ab) = aD(b) + bD(a).$$

The Stacks source denotes the resulting A-module by $\mathrm{Der}_k(A, M)$.

The module of Kähler differentials $\Omega^1_{A/k}$, with universal derivation $d : A \to \Omega^1_{A/k}$, is characterized by the functorial identity

$$\mathrm{Hom}_A(\Omega^1_{A/k}, M) \cong \mathrm{Der}_k(A, M).$$

Thus $\Omega^1_{A/k}$ is the algebraic cotangent object before any quantization choice.

If $x \in \mathrm{Spec}\, A$, with residue field $\kappa(x)$, the cotangent fibre is

$$T^*_{A/k,x} = \Omega^1_{A/k} \otimes_A \kappa(x),$$

and the tangent space is

$$T_{A/k,x} = \mathrm{Hom}_A(\Omega^1_{A/k}, \kappa(x)).$$

This is the derivative geometry missing from the bare Weyl-Heisenberg kinematics: Weyl labels give conjugate translations and phases, while Ω^1 records which first-order directions are geometric.

The algebraic de Rham complex is

$$A = \Omega^0_{A/k} \xrightarrow{d_0} \Omega^1_{A/k} \xrightarrow{d_1} \Omega^2_{A/k} \xrightarrow{d_2} \cdots ,$$

where $\Omega^i_{A/k} = \wedge^i_A \Omega^1_{A/k}$. The Stacks construction gives $d_{i+1}d_i = 0$. For finite k-algebras, each term is a finite k-vector space in the examples below.

55.4 Calculable Dynamical Proposals P1

The first proposal to test is the *de Rham CSS test*. Choose finite k-bases for

$$C^0 = A, \qquad C^1 = \Omega^1_{A/k}, \qquad C^2 = \Omega^2_{A/k}.$$

Put qudits on C^1. With the basis dot pairing, form

$$R_X = \operatorname{im}(d_0) \subset C^1, \qquad R_Z = \operatorname{im}(d_1^T) \subset C^1.$$

Since $d_1 d_0 = 0$, every element of R_X is orthogonal to every element of R_Z. By AQM-08, this is exactly the CSS isotropy condition. After choosing phases, this gives commuting stabilizer equations. This is a candidate dynamical principle: exact one-forms and coexact one-forms become local checks, and the remaining logical labels are controlled by the finite de Rham cohomology

$$H^1_{\mathrm{dR}}(A/k) = \ker d_1 / \operatorname{im} d_0.$$

This test deliberately does not claim that the ring alone chooses all phases, a Hamiltonian normalization, or a metric. Three further calculable branches are therefore left as proposals:

cotangent-Weyl test:	quantize $T^*_{A/k,x}$ or $\Omega^1_{A/k}$ directly;
Artin-jet comparison:	compare Ω^1 with the even A-Weyl layer of AQM-36;
Hodge/Laplacian test:	add a pairing or trace density and form $d^*d + dd^*$.

The last branch is least canonical because d^* requires extra metric or pairing data.

55.5 Example 1: The Field $A = \mathbb{F}_3$ L1

Let $A = k = \mathbb{F}_3$. A k-derivation $D : k \to M$ must annihilate every element of the image of k. Since the image is all of A,

$$D(0) = D(1) = D(2) = 0.$$

Therefore $\mathrm{Der}_k(k, M) = 0$ for every k-module M. By the universal property,

$$\mathrm{Hom}_k(\Omega^1_{k/k}, M) = 0 \quad \text{for all } M,$$

so

$$\Omega^1_{k/k} = 0.$$

The unique point is $x = (0)$, with $\kappa(x) = k$. Hence

$$T^*_{k/k,x} = 0 \otimes_k k = 0, \qquad T_{k/k,x} = \mathrm{Hom}_k(0, k) = 0.$$

The de Rham complex begins and ends as

$$k \longrightarrow 0 \longrightarrow 0.$$

The de Rham CSS test has no C^1 site. Thus a reduced one-point field contributes no intrinsic derivative stabilizer. This agrees with AQM-40's caveat: the ambient Weyl system may exist, but stabilizer data do not come for free from a geometrically discrete point.

55.6 Example 2: $A = \mathbb{F}_3[\epsilon]/(\epsilon^2)$ L2–T1

Let

$$A = k[\epsilon]/(\epsilon^2), \qquad k = \mathbb{F}_3.$$

Write $A = k \oplus k\epsilon$, with $\epsilon^2 = 0$. Present A as

$$S/I, \qquad S = k[t], \quad I = (t^2),$$

and let ϵ be the image of t. The Stacks polynomial-ring lemma gives

$$\Omega^1_{S/k} = S\,dt.$$

The quotient exact sequence gives

$$I/I^2 \longrightarrow \Omega^1_{S/k} \otimes_S A \longrightarrow \Omega^1_{A/k} \longrightarrow 0,$$

where $t^2 \in I$ maps to

$$d(t^2) \otimes 1 = 2t\,dt \otimes 1 = 2\epsilon\,d\epsilon.$$

In $\mathbb{F}_3$, the scalar 2 is invertible. Thus the relation is

$$\epsilon\,d\epsilon = 0.$$

Consequently

$$\Omega^1_{A/k} \cong A\,d\epsilon/(\epsilon\,d\epsilon) \cong (A/(\epsilon))\,d\epsilon \cong k\,d\epsilon.$$

The universal derivation is explicit:

$$d(a + b\epsilon) = b\,d\epsilon, \qquad a, b \in k.$$

The spectrum has one prime, $\mathfrak{m} = (\epsilon)$, and its residue field is

$$\kappa(\mathfrak{m}) = A/\mathfrak{m} \cong k.$$

Therefore the cotangent fibre is

$$T^*_{A/k,\mathfrak{m}} = \Omega^1_{A/k} \otimes_A k \cong k\,d\epsilon,$$

and the tangent space is one-dimensional:

$$T_{A/k,\mathfrak{m}} = \mathrm{Hom}_A(\Omega^1_{A/k}, k) \cong k.$$

The tangent vector v with $v(d\epsilon) = 1$ corresponds to the derivation

$$D_v(a + b\epsilon) = b.$$

Since $\Omega^1_{A/k}$ is generated by one element, its exterior square is zero:

$$\Omega^2_{A/k} = 0.$$

Thus the de Rham complex is

$$A \xrightarrow{d_0} k\,d\epsilon \longrightarrow 0, \qquad d_0(a + b\epsilon) = b\,d\epsilon.$$

The kernel is k, the image is all of $k\,d\epsilon$, and hence

$$H^0_{\mathrm{dR}}(A/k) = k, \qquad H^1_{\mathrm{dR}}(A/k) = 0.$$

Now apply the de Rham CSS test. Choose the basis $d\epsilon$ of $C^1 = k\,d\epsilon$. Then

$$R_X = \operatorname{im} d_0 = k\,d\epsilon, \qquad R_Z = 0.$$

The Pauli label space for the single C^1-qutrit is $k \oplus k$, and the candidate stabilizer label subspace is

$$L = (R_X \oplus 0) = k \oplus 0.$$

It is isotropic because all X-type labels commute with each other. Its rank is 1, so the corresponding all-+1 stabilizer space has dimension

$$3^{1-1} = 1.$$

This is the first toy calculation where the nilpotent first-order geometry selects a nontrivial stabilizer equation. It is not yet a general dynamical law; it is a checked finite test case for convention (ao).

56 More $\mathbb{F}_3$ Examples for the de Rham CSS Test

56.1 Status, Sources, and Common Rule

This shard is a follow-up calculation shard for the proposal in AQM-54 and convention `CONVENTIONS.md` (ao). All rings below are finite-dimensional coordinate rings over

$$k = \mathbb{F}_3.$$

The derivative input is the module of Kähler differentials $\Omega^1_{A/k}$; the dynamical test puts qutrits on $C^1 = \Omega^1_{A/k}$ and selects CSS support spaces

$$R_X = \mathrm{im}(d_0 : C^0 \to C^1), \qquad R_Z = \mathrm{im}(d_1^T : C^{2*} \to C^{1*}),$$

with $d_1 d_0 = 0$ giving the CSS commutation check. The source anchors are the same as AQM-54: Stacks algebra lines 33793–33872 for derivations and Kähler differentials, lines 34103–34120 for the quotient exact sequence, lines 34310–34340 for polynomial differentials, and lines 34388–34524 for the de Rham complex and $d^2 = 0$. AQM-08 supplies the CSS stabilizer dictionary. The common one-variable calculation is this. Let $A = k[t]/(f)$, $\tau = t \bmod (f)$. For $S = k[t]$, the polynomial differential rule gives $\Omega^1_{S/k} = S\,dt$. The quotient exact sequence sends f to $df = f'(t)\,dt$, hence

$$\Omega^1_{A/k} \cong A\,d\tau/(f'(\tau)d\tau).$$

Because $\Omega^1_{A/k}$ is generated by one symbol in all one-variable examples, $\Omega^2_{A/k} = 0$ there and $R_Z = 0$.

56.2 Lamport Registers for the Five Examples

$$\begin{array}{lll}
E0 & : & A_0 = k[t]/(t) \cong k. \\
E1 & : & A_1 = k[t]/(t^2-1), \quad \text{the split quadratic two-point ring.} \\
E2 & : & A_2 = k[t]/(t^2+1), \quad \text{the irreducible quadratic field point.} \\
E3 & : & A_3 = k[t]/(t^2), \quad \text{the double point at } 0. \\
E4 & : & A_4 = k[u,v]/(u^2,v^2), \quad \text{the quadratic fat plane point.}
\end{array}$$

The examples are ordered so that $E0, E1, E2$ test the reduced squarefree cases, while $E3, E4$ test nilpotent quadratic directions.

56.3 E0: The Linear Point

Let $A_0 = k[t]/(t)$, and let $\tau = 0$ be the image of t. Here $f = t$, $f' = 1$. The common rule gives

$$\Omega^1_{A_0/k} = A_0\,d\tau/(1 \cdot d\tau) = 0.$$

Thus

$$C^1 = 0, \qquad R_X = 0, \qquad R_Z = 0.$$

The de Rham CSS test selects no qutrit and no stabilizer generator. This is the same result as AQM-54 Example 1, now recorded as the linear hypersurface case $t = 0$.

56.4 E1: The Split Quadratic Two-Point Ring

Let

$$A_1 = k[t]/(t^2 - 1), \qquad \tau = t \bmod (t^2 - 1).$$

The relation is $\tau^2 = 1$. The defining polynomial and derivative are

$$f = t^2 - 1, \qquad f' = 2t.$$

In A_1,

$$(2\tau)(2\tau) = 4\tau^2 = \tau^2 = 1,$$

because $4 = 1$ in k. Hence 2τ is a unit. The quotient relation is therefore

$$2\tau \, d\tau = 0,$$

and multiplication by the inverse of 2τ gives

$$d\tau = 0.$$

Consequently

$$\Omega^1_{A_1/k} = 0.$$

The de Rham CSS data are

$$C^1 = 0, \qquad R_X = 0, \qquad R_Z = 0.$$

Thus the split reduced quadratic coordinate ring contributes two reduced points but no intrinsic derivative qutrit. In this test, reduced squarefree finite geometry has kinematics from the Weyl layer but no derivative stabilizer selected by Ω^1.

56.5 E2: The Irreducible Quadratic Field Point

Let

$$A_2 = k[t]/(t^2 + 1), \qquad \tau = t \bmod (t^2 + 1).$$

Since $0^2 + 1 = 1$, $1^2 + 1 = 2$, and $2^2 + 1 = 5 = 2$ in k, the polynomial $t^2 + 1$ has no k-root. The ring A_2 is the quadratic residue-field extension represented by the closed point $t^2 + 1$. Its relation is

$$\tau^2 = -1 = 2.$$

Again

$$f = t^2 + 1, \qquad f' = 2t.$$

In A_2,

$$(2\tau)\tau = 2\tau^2 = 2 \cdot 2 = 4 = 1.$$

Thus 2τ is a unit, now with inverse τ. The quotient relation

$$2\tau \, d\tau = 0$$

forces

$$d\tau = 0, \qquad \Omega^1_{A_2/k} = 0.$$

Therefore

$$C^1 = 0, \qquad R_X = 0, \qquad R_Z = 0.$$

The irreducible quadratic closed point behaves like the split squarefree quadratic for this derivative test: separability kills first differentials.

56.6 E3: The Quadratic Double Point

Let

$$A_3 = k[t]/(t^2), \qquad \epsilon = t \bmod (t^2).$$

Then $A_3 = k \oplus k\epsilon$ and $\epsilon^2 = 0$. Here

$$f = t^2, \qquad f' = 2t.$$

The common rule gives

$$\Omega^1_{A_3/k} = A_3\, d\epsilon/(2\epsilon\, d\epsilon).$$

Since 2 is a unit in k, this is

$$\Omega^1_{A_3/k} = A_3\, d\epsilon/(\epsilon\, d\epsilon) \cong k\, d\epsilon.$$

The universal derivation is computed on the basis $1, \epsilon$:

$$d_0(a + b\epsilon) = b\, d\epsilon.$$

Thus

$$C^1 = k\, d\epsilon, \qquad R_X = \operatorname{im} d_0 = k\, d\epsilon.$$

Because C^1 has one generator, $C^2 = \Omega^2_{A_3/k} = 0$, so

$$R_Z = 0.$$

With one qutrit labelled by $e_1 = d\epsilon$, the selected stabilizer is

$$X(e_1).$$

There is no Z-check. The stabilizer rank is 1 on one qutrit, so the all-$+1$ stabilizer space has dimension $3^{1-1} = 1$.

56.7 E4: The Quadratic Fat Plane Point

Let

$$A_4 = k[u, v]/(u^2, v^2),$$

where u, v also denote the residue classes. A k-basis is

$$1, \quad u, \quad v, \quad uv,$$

with $u^2 = v^2 = 0$. Start from $S = k[U, V]$. The polynomial differential rule gives

$$\Omega^1_{S/k} = S\, dU \oplus S\, dV.$$

The quotient ideal is (U^2, V^2). The two differential relations are

$$d(U^2) = 2U\, dU, \qquad d(V^2) = 2V\, dV.$$

Passing to A_4, and using that 2 is a unit, gives

$$u\, du = 0, \qquad v\, dv = 0.$$

Therefore

$$\Omega^1_{A_4/k} = (A_4\, du \oplus A_4\, dv)/(u\, du,\ v\, dv).$$

The $A_4\,du$ part has basis $du, v\,du$, since $u\,du = 0$ and hence $uv\,du = v(u\,du) = 0$. The $A_4\,dv$ part has basis $dv, u\,dv$, since $v\,dv = 0$ and hence $uv\,dv = u(v\,dv) = 0$. Set

$$e_1 = du, \quad e_2 = v\,du, \quad e_3 = dv, \quad e_4 = u\,dv.$$

Thus $C^1 = \Omega^1_{A_4/k} \cong k^4$, so the test has four qutrits. The derivation $d_0 : A_4 \to C^1$ is fixed on the basis:

$$\begin{aligned} d_0(1) &= 0, \\ d_0(u) &= e_1, \\ d_0(v) &= e_3, \\ d_0(uv) &= u\,dv + v\,du = e_4 + e_2. \end{aligned}$$

Hence

$$R_X = \langle e_1,\ e_3,\ e_2 + e_4 \rangle_k.$$

Now compute $C^2 = \Omega^2_{A_4/k}$. It is generated by

$$\omega = du \wedge dv.$$

Wedging $u\,du = 0$ with dv gives $u\omega = 0$, and wedging $v\,dv = 0$ with du gives $v\omega = 0$. Hence

$$C^2 = \Omega^2_{A_4/k} \cong k\,\omega.$$

The de Rham differential $d_1 : C^1 \to C^2$ is

$$\begin{aligned} d_1(e_1) &= d(du) = 0, \\ d_1(e_2) &= d(v\,du) = dv \wedge du = -\omega = 2\omega, \\ d_1(e_3) &= d(dv) = 0, \\ d_1(e_4) &= d(u\,dv) = du \wedge dv = \omega. \end{aligned}$$

With the ordered bases e_1, e_2, e_3, e_4 and ω, the matrix of d_1 is

$$\begin{bmatrix} 0 & 2 & 0 & 1 \end{bmatrix}.$$

Thus the Z-support selected by the transpose is

$$R_Z = \langle 2e_2 + e_4 \rangle_k.$$

The CSS orthogonality is visible without abbreviation:

$$e_1 \cdot (2e_2 + e_4) = 0, \quad e_3 \cdot (2e_2 + e_4) = 0, \quad (e_2 + e_4) \cdot (2e_2 + e_4) = 2 + 1 = 0.$$

Therefore the selected stabilizer generators on the four qutrits are

$$X_1, \qquad X_3, \qquad X_2X_4, \qquad Z_2^2 Z_4.$$

Their rank is 4, so the all-+1 stabilizer space has dimension

$$3^{4-4} = 1.$$

56.8 End Summary: Qudits and Selected Stabilizers

For C^1-basis vector e_i, write $X_i = X(e_i)$ and $Z_i = Z(e_i)$. The de Rham CSS test gives:

example	C^1-qutrits	X-checks from R_X	Z-checks from R_Z
$E0: \ k[t]/(t)$	none	none	none
$E1: \ k[t]/(t^2-1)$	none	none	none
$E2: \ k[t]/(t^2+1)$	none	none	none
$E3: \ k[t]/(t^2)$	$e_1 = d\epsilon$	X_1	none
$E4: \ k[u,v]/(u^2,v^2)$	$e_1 = du,\ e_2 = vdu,\ e_3 = dv,\ e_4 = udv$	$X_1,\ X_3,\ X_2X_4$	$Z_2^2 Z_4$

Thus the first sharp lesson is not that every finite ring supplies dynamics. It is that the de Rham CSS proposal detects nilpotent or singular first-order geometry and is silent on these squarefree finite reduced examples.

57 $\mathbb{F}_2$ Qubit Examples for the de Rham CSS Test

57.1 Status, Sources, and Characteristic-Two Warning

This shard continues AQM-54 and AQM-55 with $k = \mathbb{F}_2$. The source anchors are unchanged: Stacks algebra lines 33793–33872 for derivations and Kähler differentials, 34103–34120 for the quotient exact sequence, 34310–34340 for polynomial differentials, and 34388–34524 for the de Rham complex. AQM-08 supplies the CSS/stabilizer dictionary, and `CONVENTIONS.md` (ao) names the `de_rham_css_test`. The characteristic-two edge case is immediate from the polynomial rule:

$$d(t^2) = 2t\,dt = 0.$$

Thus a square-zero equation $t^2 = 0$ imposes no relation on $t\,dt$. This is the opposite of the $\mathbb{F}_3$ dual-number calculation in AQM-54, where 2 is invertible and $\epsilon\,d\epsilon = 0$. A squarefree control still behaves normally: for $k[t]/(t^2 + t)$, $d(t^2 + t) = dt$, hence $\Omega^1 = 0$. The examples below deliberately use nilpotent quadratic relations, because those are where the qubit CSS test sees structure.

57.2 Lamport Registers

$$\begin{array}{lcl}
F0 & : & A_0 = k[t]/(t^6), \quad 6 \text{ qubits.} \\
F1 & : & A_1 = k[t]/(t^{10}), \quad 10 \text{ qubits.} \\
F2 & : & A_2 = k[x,y]/(x^2, y^2, xy), \quad 5 \text{ qubits.} \\
F3 & : & A_3 = k[x,y]/(x^2, y^2), \quad 8 \text{ qubits.} \\
F4 & : & A_4 = k[x,y,z]/(x,y,z)^2, \quad 9 \text{ qubits.}
\end{array}$$

For a C^1-basis vector e_i, write $X_i = X(e_i)$ and $Z_i = Z(e_i)$. All stabilizers below have $+1$ phase; the ring calculation fixes only the support spaces.

57.3 F0: The Length-Six Fat Line

Let $A_0 = k[t]/(t^6)$, and write $\tau = t \bmod (t^6)$. The k-basis is

$$1, \tau, \tau^2, \tau^3, \tau^4, \tau^5.$$

The quotient rule gives

$$\Omega^1_{A_0/k} = A_0\,d\tau/(d(\tau^6)).$$

Since $d(\tau^6) = 6\tau^5\,d\tau = 0$ in k, no relation is imposed. Set

$$e_i = \tau^{i-1}d\tau, \qquad 1 \le i \le 6.$$

Then $C^1 = \Omega^1_{A_0/k} \cong k^6$, so there are six qubits. The universal derivation is

$$d_0(\tau^j) = j\,\tau^{j-1}d\tau,$$

so only odd j contribute:

$$d_0(\tau) = e_1, \quad d_0(\tau^3) = e_3, \quad d_0(\tau^5) = e_5.$$

Hence

$$R_X = \langle e_1, e_3, e_5\rangle, \qquad R_Z = 0,$$

because a cyclic one-form module has $\Omega^2 = 0$. The selected stabilizers are

$$X_1, \qquad X_3, \qquad X_5.$$

The rank is 3 on 6 qubits, leaving $6 - 3 = 3$ logical qubits.

57.4 F1: The Length-Ten Fat Line

Let $A_1 = k[t]/(t^{10})$, with $\tau = t \bmod (t^{10})$. Its basis is

$$1, \tau, \tau^2, \tau^3, \tau^4, \tau^5, \tau^6, \tau^7, \tau^8, \tau^9.$$

Again

$$d(\tau^{10}) = 10\tau^9 \, d\tau = 0,$$

so $\Omega^1_{A_1/k} = A_1 \, d\tau$. Set

$$e_i = \tau^{i-1} d\tau, \qquad 1 \leq i \leq 10.$$

Thus $C^1 \cong k^{10}$, giving ten qubits. The same parity calculation gives

$$d_0(\tau) = e_1, \quad d_0(\tau^3) = e_3, \quad d_0(\tau^5) = e_5, \quad d_0(\tau^7) = e_7, \quad d_0(\tau^9) = e_9,$$

while $d_0(1), d_0(\tau^2), d_0(\tau^4), d_0(\tau^6), d_0(\tau^8)$ vanish. Therefore

$$R_X = \langle e_1, e_3, e_5, e_7, e_9 \rangle, \qquad R_Z = 0.$$

The selected stabilizers are

$$X_1, \qquad X_3, \qquad X_5, \qquad X_7, \qquad X_9.$$

The stabilizer rank is 5 on 10 qubits, so the code space has $10 - 5 = 5$ logical qubits.

57.5 F2: The Two-Variable Square-Zero Point

Let

$$A_2 = k[x, y]/(x^2, y^2, xy).$$

The basis is $1, x, y$. The square relations give no differential relation:

$$d(x^2) = 0, \qquad d(y^2) = 0.$$

The mixed relation gives

$$d(xy) = x \, dy + y \, dx = 0.$$

Thus $x \, dy = y \, dx$. A C^1-basis is

$$e_1 = dx, \quad e_2 = x \, dx, \quad e_3 = dy, \quad e_4 = y \, dy, \quad e_5 = x \, dy = y \, dx.$$

There are five qubits. The derivation on the basis of A_2 is

$$d_0(1) = 0, \qquad d_0(x) = e_1, \qquad d_0(y) = e_3,$$

so

$$R_X = \langle e_1, e_3 \rangle.$$

Let $\omega = dx \wedge dy$. Wedging $x \, dy + y \, dx = 0$ with dx and dy gives $x\omega = 0$ and $y\omega = 0$, so $C^2 = k\omega$. Now

$$d_1(e_1) = d_1(e_2) = d_1(e_3) = d_1(e_4) = 0, \qquad d_1(e_5) = \omega.$$

Hence $R_Z = \langle e_5 \rangle$. The selected stabilizers are

$$X_1, \qquad X_3, \qquad Z_5.$$

The rank is 3 on 5 qubits, leaving 2 logical qubits.

57.6 F3: The Two-Variable Quadratic Fat Point

Let

$$A_3 = k[x, y]/(x^2, y^2).$$

The basis is $1, x, y, xy$. Since $d(x^2) = d(y^2) = 0$, the module $\Omega^1_{A_3/k}$ is freely generated over A_3 by dx, dy. Use

$$\begin{aligned} &e_1 = dx, \quad e_2 = x\,dx, \quad e_3 = y\,dx, \quad e_4 = xy\,dx, \\ &e_5 = dy, \quad e_6 = x\,dy, \quad e_7 = y\,dy, \quad e_8 = xy\,dy. \end{aligned}$$

Thus there are eight qubits. The derivation is

$$d_0(x) = e_1, \qquad d_0(y) = e_5, \qquad d_0(xy) = x\,dy + y\,dx = e_6 + e_3.$$

Therefore

$$R_X = \langle e_1, e_5, e_3 + e_6 \rangle.$$

Let $\omega = dx \wedge dy$; then $C^2 = A_3\omega$ has basis $\omega, x\omega, y\omega, xy\omega$. The nonzero d_1-values are

$$d_1(e_3) = \omega, \quad d_1(e_4) = x\omega, \quad d_1(e_6) = \omega, \quad d_1(e_8) = y\omega.$$

Hence

$$R_Z = \langle e_3 + e_6,\ e_4,\ e_8 \rangle.$$

The shared support $e_3 + e_6$ is not a mistake: in characteristic two its self-dot-product is $1 + 1 = 0$. The selected stabilizers are

$$X_1, \qquad X_5, \qquad X_3X_6, \qquad Z_3Z_6, \qquad Z_4, \qquad Z_8.$$

The rank is 6 on 8 qubits, leaving 2 logical qubits.

57.7 F4: The Three-Variable Square-Zero Tangent Space

Let

$$A_4 = k[x, y, z]/(x, y, z)^2.$$

The basis is $1, x, y, z$. The relations $x^2 = y^2 = z^2 = xy = xz = yz = 0$ give

$$x\,dy = y\,dx, \qquad x\,dz = z\,dx, \qquad y\,dz = z\,dy,$$

while the square relations again give no $x\,dx$, $y\,dy$, or $z\,dz$ relation. Choose

$$\begin{array}{lll} e_1 = dx, & e_2 = dy, & e_3 = dz, \\ e_4 = x\,dx, & e_5 = y\,dy, & e_6 = z\,dz, \\ e_7 = x\,dy = y\,dx, & e_8 = x\,dz = z\,dx, & e_9 = y\,dz = z\,dy. \end{array}$$

Thus $C^1 \cong k^9$, so there are nine qubits. The image of d_0 is

$$R_X = \langle e_1, e_2, e_3 \rangle.$$

For C^2, set

$$\omega_{xy} = dx \wedge dy, \quad \omega_{xz} = dx \wedge dz, \quad \omega_{yz} = dy \wedge dz.$$

The wedged mixed relations make the coefficient terms compatible; the only extra coefficient direction needed below is

$$\eta = x\omega_{yz} = y\omega_{xz} = z\omega_{xy}.$$

Indeed, wedging $x\,dy + y\,dx = 0$, $x\,dz + z\,dx = 0$, and $y\,dz + z\,dy = 0$ with dx, dy, dz gives

$$\begin{aligned} &x\omega_{xy} = 0, \quad y\omega_{xy} = 0, \quad x\omega_{yz} = y\omega_{xz}, \\ &x\omega_{xz} = 0, \quad z\omega_{xz} = 0, \quad x\omega_{yz} = z\omega_{xy}, \\ &y\omega_{yz} = 0, \quad z\omega_{yz} = 0, \quad y\omega_{xz} = z\omega_{xy}. \end{aligned}$$

The d_1-values relevant for R_Z are

$$d_1(e_7) = \omega_{xy}, \qquad d_1(e_8) = \omega_{xz}, \qquad d_1(e_9) = \omega_{yz},$$

and $d_1(e_i) = 0$ for $1 \le i \le 6$. Therefore

$$R_Z = \langle e_7, e_8, e_9 \rangle.$$

The selected stabilizers are

$$X_1, \qquad X_2, \qquad X_3, \qquad Z_7, \qquad Z_8, \qquad Z_9.$$

The rank is 6 on 9 qubits, leaving 3 logical qubits.

57.8 End Summary: Qubits and de Rham CSS Stabilizers

ex.	A	$\#C^1$	X-checks	Z-checks
$F0$	$k[t]/(t^6)$	6	X_1, X_3, X_5	none
$F1$	$k[t]/(t^{10})$	10	X_1, X_3, X_5, X_7, X_9	none
$F2$	$k[x,y]/(x^2, y^2, xy)$	5	X_1, X_3	Z_5
$F3$	$k[x,y]/(x^2, y^2)$	8	X_1, X_5, X_3X_6	Z_3Z_6, Z_4, Z_8
$F4$	$k[x,y,z]/(x,y,z)^2$	9	X_1, X_2, X_3	Z_7, Z_8, Z_9

The qubit lesson is sharper than the odd-characteristic lesson: in characteristic two, quadratic nilpotent equations can leave more one-form directions alive, so the de Rham CSS test naturally produces medium-sized qubit stabilizer systems from very small finite coordinate rings.

58 Entanglement and Code Properties in $\mathbb{F}_2$ de Rham CSS Examples

58.1 Status and Local Test for Entanglement

This shard continues AQM-54–AQM-56. It uses the same source anchors: Stacks algebra lines 33793–33872, 34103–34120, 34310–34340, and 34388–34524 for Kähler differentials and the de Rham complex, AQM-08 for the CSS/stabilizer dictionary, and `CONVENTIONS.md` (ao) for the de Rham CSS test. The new claims in this shard are not imported code facts: they are direct solutions of the displayed stabilizer equations. The local entanglement test used below is the following. If two qubits p, q are constrained by

$$X_p X_q = +1, \qquad Z_p Z_q = +1,$$

and no other displayed generator touches them, then their joint state is

$$|\Phi^+\rangle_{pq} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Indeed, $Z_p Z_q = +1$ removes the $|01\rangle$ and $|10\rangle$ amplitudes, while $X_p X_q = +1$ makes the $|00\rangle$ and $|11\rangle$ amplitudes equal. The reduced density matrix on either qubit is

$$\mathrm{diag}(1/2, 1/2),$$

so the state is not a product state. This is the only entanglement criterion used in the examples.

58.2 Example C0: The Eight-Qubit Square Fat Point

Let

$$A_0 = \mathbb{F}_2[x, y]/(x^2, y^2).$$

Use the $C^1 = \Omega^1_{A_0/\mathbb{F}_2}$ basis

$$\begin{aligned} &e_1 = dx, \quad e_2 = x\,dx, \quad e_3 = y\,dx, \quad e_4 = xy\,dx, \\ &e_5 = dy, \quad e_6 = x\,dy, \quad e_7 = y\,dy, \quad e_8 = xy\,dy. \end{aligned}$$

This is the eight-qubit example from AQM-56. The de Rham CSS images are

$$R_X = \langle e_1,\ e_5,\ e_3 + e_6 \rangle, \qquad R_Z = \langle e_3 + e_6,\ e_4,\ e_8 \rangle.$$

Thus the selected stabilizer generators are

$$X_1, \qquad X_5, \qquad X_3 X_6, \qquad Z_3 Z_6, \qquad Z_4, \qquad Z_8.$$

The six generators are independent and commute; independence is visible from the private supports $1, 5, 4, 8$ plus the common two-qubit support $3, 6$. Therefore the code parameters are

$$[[8,\ 8 - 6,\ d]] = [[8, 2, d]].$$

The pair $3, 6$ is forced to be

$$|\Phi^+\rangle_{3,6}.$$

The remaining pinned qubits are

$$|+\rangle_1, \quad |+\rangle_5, \quad |0\rangle_4, \quad |0\rangle_8.$$

Qubits 2 and 7 are untouched by every generator, so a code basis is

$$|+\rangle_1 \, |\alpha\rangle_2 \, |\Phi^+\rangle_{3,6} \, |0\rangle_4 \, |+\rangle_5 \, |\beta\rangle_7 \, |0\rangle_8,$$

where $\alpha, \beta \in \{0,1\}$. The logical operators may be chosen as

$$\bar{X}_1 = X_2, \quad \bar{Z}_1 = Z_2, \qquad \bar{X}_2 = X_7, \quad \bar{Z}_2 = Z_7.$$

Hence $d = 1$. This is an entangled stabilizer code, but not an error-correcting code.

58.3 Example C1: A Ten-Qubit Fat Rectangle

Let

$$A_1 = \mathbb{F}_2[x,y]/(x^3, y^2).$$

Since $d(x^3) = x^2 dx$ and $d(y^2) = 0$, the dx-part has coefficients $1, x, y, xy$, while the dy-part has coefficients $1, x, x^2, y, xy, x^2y$. Use

$$\begin{aligned}
&a_0 = dx, && a_1 = x\,dx, && b_0 = y\,dx, && b_1 = xy\,dx,\\
&c_0 = dy, && c_1 = x\,dy, && c_2 = x^2 dy, && d_0 = y\,dy,\\
&d_1 = xy\,dy, && d_2 = x^2 y\,dy.
\end{aligned}$$

There are ten qubits. The d_0-image is

$$d(x) = a_0, \quad d(y) = c_0, \quad d(xy) = b_0 + c_1, \quad d(x^2y) = c_2,$$

so

$$R_X = \langle a_0,\ c_0,\ b_0 + c_1,\ c_2 \rangle.$$

For C^2, write $\omega = dx \wedge dy$. The relation $x^2 dx = 0$ implies $x^2\omega = 0$. The nonzero d_1-values are

$$d_1(b_0) = \omega, \quad d_1(c_1) = \omega, \quad d_1(b_1) = x\omega, \quad d_1(d_1) = y\omega.$$

Thus

$$R_Z = \langle b_0 + c_1,\ b_1,\ d_1 \rangle.$$

The selected stabilizers are

$$X_{a_0},\ X_{c_0},\ X_{b_0}X_{c_1},\ X_{c_2}, \qquad Z_{b_0}Z_{c_1},\ Z_{b_1},\ Z_{d_1}.$$

Their rank is $4 + 3 = 7$, so the code is

$$[[10,\ 10 - 7,\ d]] = [[10, 3, d]].$$

The pair b_0, c_1 is forced to be a Bell pair. The pinned single-qubit states are

$$|+\rangle_{a_0}, \quad |+\rangle_{c_0}, \quad |+\rangle_{c_2}, \quad |0\rangle_{b_1}, \quad |0\rangle_{d_1}.$$

The untouched qubits a_1, d_0, d_2 give logical operators

$$X_{a_1}, Z_{a_1}, \qquad X_{d_0}, Z_{d_0}, \qquad X_{d_2}, Z_{d_2}.$$

Therefore $d = 1$. The code has a real Bell sector but no single-qubit error protection.

58.4 Example C2: An Eighteen-Qubit Fat Rectangle

Let

$$A_2 = \mathbb{F}_2[x,y]/(x^5,y^2).$$

The relation $d(x^5) = x^4dx$ removes x^4dx and $x^4y\,dx$, while $d(y^2) = 0$. Use the 18 basis one-forms

$$\begin{array}{ll} a_i = x^i dx \ (0 \le i \le 3), & b_i = x^i y\, dx \ (0 \le i \le 3), \\ c_i = x^i dy \ (0 \le i \le 4), & d_i = x^i y\, dy \ (0 \le i \le 4). \end{array}$$

The nonzero d_0-images needed for R_X are

$$\begin{array}{rclrcl} d(x) &=& a_0, & d(x^3) &=& a_2, \\ d(y) &=& c_0, & d(xy) &=& b_0 + c_1, \\ d(x^2y) &=& c_2, & d(x^3y) &=& b_2 + c_3, \\ d(x^4y) &=& c_4. & & & \end{array}$$

Hence

$$R_X = \langle a_0, a_2, c_0, c_2, c_4, b_0 + c_1, b_2 + c_3 \rangle.$$

Again let $\omega = dx \wedge dy$, with $x^4\omega = 0$. The nonzero d_1-rows are

$$\begin{array}{rclrcl} d_1(b_0) &=& \omega, & d_1(c_1) &=& \omega, \\ d_1(b_1) &=& x\omega, & d_1(b_2) &=& x^2\omega, \\ d_1(c_3) &=& x^2\omega, & d_1(b_3) &=& x^3\omega, \\ d_1(d_1) &=& y\omega, & d_1(d_3) &=& x^2y\omega. \end{array}$$

Thus

$$R_Z = \langle b_0 + c_1,\ b_1,\ b_2 + c_3,\ b_3,\ d_1,\ d_3 \rangle.$$

The selected stabilizers are therefore

$$\begin{array}{l} X_{a_0}, X_{a_2}, X_{c_0}, X_{c_2}, X_{c_4}, X_{b_0}X_{c_1}, X_{b_2}X_{c_3}, \\ Z_{b_0}Z_{c_1}, Z_{b_1}, Z_{b_2}Z_{c_3}, Z_{b_3}, Z_{d_1}, Z_{d_3}. \end{array}$$

The rank is $7 + 6 = 13$, so this is an

$$[[18,\ 18 - 13,\ d]] = [[18, 5, d]]$$

CSS code. The two disjoint pairs b_0, c_1 and b_2, c_3 are forced to be

$$|\Phi^+\rangle_{b_0,c_1} \otimes |\Phi^+\rangle_{b_2,c_3}.$$

The untouched qubits

$$a_1, \quad a_3, \quad d_0, \quad d_2, \quad d_4$$

give five weight-one logical X/Z pairs, so $d = 1$. This example reaches the requested 15+-qubit scale and shows the same diagnosis: the de Rham CSS test naturally creates entangled stabilizer sectors, but these fat rectangles still carry unprotected differential logicals.

58.5 Summary

The three examples have parameters

ring	qubits	Bell sectors	$[[n,k,d]]$
$\mathbb{F}_2[x,y]/(x^2,y^2)$	8	$(y\,dx,\ x\,dy)$	$[[8,2,1]]$
$\mathbb{F}_2[x,y]/(x^3,y^2)$	10	$(y\,dx,\ x\,dy)$	$[[10,3,1]]$
$\mathbb{F}_2[x,y]/(x^5,y^2)$	18	$(y\,dx,\ x\,dy),\ (x^2y\,dx,\ x^3dy)$	$[[18,5,1]]$

So the first qubit-code conclusion is mixed. Entanglement is already present and exactly derivable from the de Rham CSS stabilizers. However, the same characteristic-two pathology that creates many one-form qubits also leaves weight-one logical operators. These examples are entangled stabilizer codes, not useful distance codes.